

THALES
Component Maintenance Manual
C16221K

TRANSMITTAL SHEET

TO : HOLDER OF INTEGRATED STANDBY FLIGHT DISPLAY

PNR: C16221KA01/Boeing PN:S231A120xx

Component Maintenance Manual with Illustrated Parts List

ATA No 34-22-45

The Component Maintenance Manual with Illustrated Parts List is fully re-printed.

REVISION No. 14 DATED JUL 04/2018

CHAPTER/SECTION	DESIGNATION	DESCRIPTION OF CHANGE
TITLE PAGE	R	New presentation and new date.
RECORD OF REVISION	R	Revision No. 14 added.
SERVICE BULLETIN LIST	R	Incorporation of the SBs C16221-34-006 and C16221-34-010.
LIST OF EFFECTIVE PAGE	R	Revised to reflect this revision.
LIST OF ILLUSTRATIONS	R	Revised to reflect this revision.
INTRODUCTION	R	
TASK 34-22-45-879-801-A01	R	Minor non technical correction.
DESCRIPTION AND OPERATION	R	
TASK 34-22-45-870-801-A01	R	Standard Overhaul Practices Manual reference updated.
TESTING AND FAULT ISOLATION	R	
TASK 34-22-45-750-802-A01	R	Standard Overhaul Practices Manual reference updated.
TASK 34-22-45-750-803-A01	R	Minor non technical correction.
SUBTASK 34-22-45-750-001-A01	R	Minor non technical correction.
TASK 34-22-45-750-804-A01	R	
SUBTASK 34-22-45-750-025-A01	R	Minor non technical correction.
TASK 34-22-45-750-805-A01	R	
SUBTASK 34-22-45-750-049-A01	R	Minor non technical correction.
TASK 34-22-45-750-901-A01	R	

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CHAPTER/SECTION	DESIGNATION	DESCRIPTION OF CHANGE
SUBTASK 34-22-45-210-050-A01	R	Standard Overhaul Practices Manual reference updated.
DISASSEMBLY	R	
TASK 34-22-45-000-801-A01	R	Consumable list updated.
SUBTASK 34-22-45-000-008-A01	R	Consumable name updated. Standard Overhaul Practices Manual reference updated.
TASK 34-22-45-000-804-A01	R	Note added.
CHECK	R	
TASK 34-22-45-200-801-A01	R	Standard Overhaul Practices Manual reference updated.
REPAIR	R	
TASK 34-22-45-300-801-A01	R	Operation conditions added. Standard Overhaul Practices Manual reference updated.
SUBTASK 34-22-45-300-005-A01	R	Standard Overhaul Practices Manual reference updated.
TASK 34-22-45- 300-805-A01	N	New procedure added.
SUBTASK 34-22-45-350-001-A01	N	
SUBTASK 34-22-45-350-002-A01	N	
SUBTASK 34-22-45-350-003-A01	N	
SUBTASK 34-22-45-350-004-A01	N	
ASSEMBLY	R	
TASK 34-22-45-400-801-A01	R	Laminar air flow hood characteristics added. Consumable list updated.
TASK 34-22-45-400-802-A01	R	Note added.
TASK 34-22-45-400-803-A01	R	Note added.
TASK 34-22-45-400-804-A01	R	Note added.
TASK 34-22-45-400-805-A01	R	Note added.
TASK 34-22-45-400-806-A01	R	Note added.
SUBTASK 34-22-45-400-002-A01	R	Consumable name updated.
TASK 34-22-45-400-807-A01	R	Note added.

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CHAPTER/SECTION	DESIGNATION	DESCRIPTION OF CHANGE
TASK 34-22-45-400-809-A01	R	Note added.
TASK 34-22-45-400-810-A01	R	Note added.
TASK 34-22-45-400-811-A01	R	Note added.
TASK 34-22-45-400-812-A01	R	Consumable list updated. Condition procedure added.
TASK 34-22-45-400-813-A01	R	Note added.
SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES	R	
TASK 34-22-45-940-802-A01	R	List of Special Tools table updated. Note deleted
TASK 34-22-45-940-803-A01	R	List of Consumable table updated.
SPECIAL PROCEDURES	R	
TASK 34-22-45-650-802-A01	R	
SUBTASK 34-22-45-650-004-A01	R	Minor non technical correction.
TASK 34-22-45-650-804-A01	R	Three-hours test updated.
SUBTASK 34-22-45-650-011-A01	R	
SUBTASK 34-22-45-650-013-A01	R	
STORAGE (INCLUDING TRANSPORTATION)	R	
TASK 34-22-45-500-801-A01	R	Minor non technical correction.
SUBTASK 34-22-45- 550-002- A01	R	Standard Overhaul Practices Manual reference updated.
ILLUSTRATED PARTS LIST	R	
VENDOR CODE LIST	R	Vendor Code address updated.
EQUIPMENT DESIGNATOR INDEX	R	
NUMERICAL INDEX	R	
DETAIL PARTS LIST		
FIGURE-34-22-45-950-020-A01	R	
GRAPHIC-34-22-45-991-902-A01	R	
PRTLST-34-22-45-02	R	
FIGURE-34-22-45-950-030-A01	R	

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CHAPTER/SECTION	DESIGNATION	DESCRIPTION OF CHANGE
GRAPHIC-34-22-45-991-906-A01	R	
PRTLST-34-22-45-03	R	

NOTE: N: New - D: Deleted - R: Revised - U: Unchanged

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COMPONENT MAINTENANCE MANUAL WITH ILLUSTRATED PARTS LIST

INTEGRATED STANDBY FLIGHT DISPLAY (ISFD)

C16221KA01/Boeing
PN:S231A120-1

C16221KA02/Boeing
PN:S231A120-5

C16221LA01/Boeing
PN:S231A120-2

C16221LA02/Boeing
PN:S231A120-3

C16221LA03/Boeing
PN:S231A120-6

C16221MA01/Boeing
PN:S231A120-4

C16221QA01/Boeing
PN:S231A120-7

FOR AUTOMATIC TEST REQUIREMENTS, REFER TO 34-22-45A

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Revision N. 14
First Issue: Jun 25/2001

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SERVICE BULLETIN LIST

SERVICE BULLETIN/ REVISION NUMBER	ISSUE DATE	DATE INCORPORATED OR NO EFFECT	TITLE
C16221-34-001	JUL 31/2003	NONE	Inspection of the output of the inertial measurement unit
C16221-34-001 Rev 1	FEB 16/2007	NONE	Inspection of the output of the inertial measurement unit
C16221-34-001 Rev 2	NO EFFECT	NONE	Inspection of the output of the inertial measurement unit
C16221-34-001 Rev 3	NO EFFECT	NONE	Inspection of the output of the inertial measurement unit
C16221-34-001 Rev 4	NO EFFECT	NONE	Inspection of the output of the inertial measurement unit
C16221-34-002	JUL 31/2003	NONE	Inspection of the pressure sensors assemblies
C16221-34-002 Rev1	NO EFFECT	NONE	Inspection of the pressure sensors assemblies
C16221-34-003	JUL 31/2003	NONE	Inspection and replacement of the accelerometers
C16221-34-003 Rev1	NO EFFECT	NONE	No longer applicable and superseded by SB C16221-34-001
C16221-34-004	NO EFFECT	NONE	Inspection of the connectors installed on the programmed motherboard
C16221-34-005	MAR 26/2015	NONE	Inspection of the oscillator of the programmed interface
C16221-34-006	NO EFFECT	NONE	Inspection of the assembly process used to install pushbuttons switches
C16221-34-006 Rev1	NO EFFECT	NONE	Inspection of the assembly process used to install pushbuttons switches
C16221-34-010	JUL 04/2018	C16221KA02/C16221LA03/C16221MA01/C16221QA01	Inspection of the switches assembly and keys

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SERVICE BULLETIN/ REVISION NUMBER	ISSUE DATE	DATE INCORPORATED OR NO EFFECT	TITLE
C16221K-34-001	JUL 31/2003	C16221KA02	Eliminate display of incorrect attitude due to airplane motion during initialization
C16221K-34-001 Rev1	MAY 02/2005	C16221KA02	Eliminate display of incorrect attitude due to airplane motion during initialization
C16221K-34-002	JUL 31/2003	C16221KA02 AmdtA	Eliminate display luminance flicker in dark ambient condition
C16221K-34-003	JUL 31/2003	NONE	Inspection & replacement of defective control processor graphic display units
C16221K-34-004	MAY 02/2005	C16221KA02 AmdtB	Replacement of the no longer manufactured hybrid converter DC/DC module (Eracle)
C16221K-34-005	JUL 31/2003	C16221KA02 AmdtC	Replacement of a no longer available connector
C16221K-34-005 Rev1	NO EFFECT	NONE	Replacement of a no longer available connector
C16221K-34-006	MAY 02/2005	C16221KA02 AmdtD	Improvement of the P90-3 pressure sensor
C16221K-34-007	MAY 02/2005	C16221KA01/KA 02 AmdtE	Replacement of no longer manufactured tubes
C16221K-34-007 Rev1	NO EFFECT	C16221KA0X AmdtE	Replacement of no longer manufactured tubes
C16221K-34-008	NOV 30/2005	C16221KA02 AmdtF	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221K-34-008 Rev1	NO EFFECT	C16221KA02 AmdtF	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221K-34-009	JUN 15/2006	C16221KA02 AmdtG	Improvement of integrated standby flight display robustness
C16221K-34-010	JUN 15/2006	C16221KA02 AmdtH	Encoder robustness improvement

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SERVICE BULLETIN/ REVISION NUMBER	ISSUE DATE	DATE INCORPORATED OR NO EFFECT	TITLE
C16221K-34-010 Rev1	NO EFFECT	C16221KA01/KA 02 AmdtH	Encoder robustness improvement
C16221K-34-010 Rev2	NO EFFECT	C16221KA01/KA 02 Amdts H + N	Encoder robustness improvement
C16221K-34-010 Rev3	NO EFFECT	C16221KA01/KA 02 Amdts H + N	Encoder robustness improvement
C16221K-34-011	JUN 15/2006	C16221KA01 AmdtJ	Improvement of the power supply robustness
C16221K-34-012	JAN 25/2008	C16221KA02 AmdtK	To overcome obsolescence on the programmed interface board and on filter board
C16221K-34-012 Rev1	OCT 30/2009	C16221KA02 Amdts K + L	To overcome obsolescence on the programmed interface board and on filter board
C16221K-34-013	OCT 30/2008	C16221KA02	To overcome obsolescence of the display drivers
C16221K-34-014	OCT 30/2009	C16221KA02 Amdt L	Harmonization of the push-buttons luminance law
C16221K-34-014 Rev1	NO EFFECT	C16221KA02 Amdt L	Harmonization of the push-buttons luminance law
C16221K-34-015	OCT 30/2009	C16221KA02 Amdt M	To overcome obsolescence of the initial measurement unit accelerometer
C16221L-34-001	FEB 15/2003	C16221LA02	Introduction of the ssec laws parameters for 767-200 and 767-300 aircrafts configuration
C16221L-34-001 Rev1	NO EFFECT	C16221LA02	Introduction of the ssec laws parameters for 767-200 and 767-300 aircrafts configuration
C16221L-34-001 Rev2	NO EFFECT	C16221LA02	Introduction of the ssec laws parameters for 767-200 and 767-300 aircrafts configuration

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SERVICE BULLETIN/ REVISION NUMBER	ISSUE DATE	DATE INCORPORATED OR NO EFFECT	TITLE
C16221L-34-002	JUL 31/2003	C16221LA03	Eliminate display of incorrect attitude due to airplane motion during initialization
C16221L-34-002 Rev1	MAY 02/2005	C16221LA03	Eliminate display of incorrect attitude due to airplane motion during initialization
C16221L-34-003	JUL 31/2003	C16221LA03 AmdtA	Eliminate display luminance flicker in dark ambient condition
C16221L-34-004	MAY 02/2005	C16221LA03 AmdtB	Replacement of a longer manufactured hybrid converter DC/DC module (Eracle)
C16221L-34-005	JUL 31/2003	C16221LA03 AmdtC	Replacement of a no longer available connector
C16221L-34-005 Rev1	MAY 02/2005	C16221LA03 AmdtC	Replacement of a no longer available connector
C16221L-34-005 Rev2	NO EFFECT	C16221LA03 AmdtC	Replacement of a no longer available connector
C16221L-34-006	MAY 02/2005	C16221LA03 AmdtD	Improvement of the P90-3 pressure sensor
C16221L-34-007	MAY 02/2005	C16221LA01/LA 02/ LA03 AmdtE	Replacement of no longer manufactured tubes
C16221L-34-007 Rev1	NO EFFECT	C16221LA0X AmdtE	Replacement of no longer manufactured tubes
C16221L-34-008	NOV 30/2005	C16221LA03 AmdtF	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221L-34-008 Rev1	NO EFFECT	C16221LA03 AmdtF	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221L-34-009	JUN 15/2006	C16221LA03 AmdtG	Improvement of integrated standby flight display robustness
C16221L-34-010	JUN 15/2006	C16221LA03 AmdtH	Encoder robustness improvement

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SERVICE BULLETIN/ REVISION NUMBER	ISSUE DATE	DATE INCORPORATED OR NO EFFECT	TITLE
C16221L-34-010 Rev1	NO EFFECT	C16221LA01/LA 02/ LA03 AmdtH	Encoder robustness improvement
C16221L-34-010 Rev2	NO EFFECT	C16221LA01/LA 02/ LA03 Amdts H + N	Encoder robustness improvement
C16221L-34-010 Rev3	NO EFFECT	C16221LA01/LA 02/ LA03 Amdts H + N	Encoder robustness improvement
C16221L-34-011	JUN 15/2006	C16221LA03 AmdtJ	Improvement of the power supply robustness
C16221L-34-012	JAN 25/2008	C16221LA03 AmdtK	To overcome obsolescence on the programmed interface board and on filter board
C16221L-34-012 Rev1	OCT 30/2009	C16221LA03 Amdts K + L	To overcome obsolescence on the programmed interface board and on filter board
C16221L-34-013	OCT 30/2008	C16221LA03	To overcome obsolescence of the display drivers
C16221L-34-014	OCT 30/2009	C16221LA03 Amdt L	Harmonization of the push-buttons luminance law
C16221L-34-014 Rev1	NO EFFECT	C16221LA03 Amdt L	Harmonization of the push-buttons luminance law
C16221L-34-015	OCT 30/2009	C16221LA03 Amdt M	To overcome obsolescence of the initial measurement unit accelerometer
C16221M-34-001	MAY 02/2005	C16221MA01 AmdtA	Improvement of the P90-3 pressure sensor
C16221M-34-002	MAY 02/2005	C16221MA01 AmdtB	Replacement of a no longer manufactured tubes
C16221M-34-002 Rev1	NO EFFECT	C16221MA01 AmdtB	Replacement of a no longer manufactured tubes
C16221M-34-003	NOV 30/2005	C16221MA01 AmdtC	Improvement of the EPLD robustness to cyclic ratio variation of clock input

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SERVICE BULLETIN/ REVISION NUMBER	ISSUE DATE	DATE INCORPORATED OR NO EFFECT	TITLE
C16221M-34-003 Rev1	NO EFFECT	C16221MA01 AmdtC	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221M-34-004	JUN 15/2006	C16221MA01 AmdtD	Improvement of integrated standby flight display robustness
C16221M-34-005	JUN 15/2006	C16221MA01 AmdtE	Encoder robustness improvement
C16221M-34-005 Rev1	NO EFFECT	C16221MA01 AmdtE+G	Encoder robustness improvement
C16221M-34-005 Rev2	NO EFFECT	C16221MA01 AmdtE+G	Encoder robustness improvement
C16221M-34-006	JUN 15/2006	C16221MA01 AmdtF	Improvement of the power supply robustness
C16221M-34-007	OCT 30/2008	C16221MA01	To overcome obsolescence of the display drivers
C16221Q-34-001 Rev1	MAY 02/2005	C16221QA01 AmdtB	Improvement of the P90-3 pressure sensor
C16221Q-34-002	MAY 02/2005	C16221QA01 AmdtC	Replacement of a longer manufactured tubes
C16221Q-34-002 Rev1	NO EFFECT	C16221QA01 AmdtC	Replacement of a longer manufactured tubes
C16221Q-34-003	NOV 30/2005	C16221QA01 AmdtD	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221Q-34-003 Rev1	NO EFFECT	C16221QA01 AmdtD	Improvement of the EPLD robustness to cyclic ratio variation of clock input
C16221Q-34-004	JUN 15/2006	C16221QA01 AmdtE	Improvement of integrated standby flight display robustness
C16221Q-34-004 Rev1	NO EFFECT	C16221QA01 AmdtE	Improvement of integrated standby flight display robustness
C16221Q-34-005	JUN 15/2006	C16221QA01 AmdtF	Encoder robustness improvement

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C16221Q-34-005 Rev1	OCT 30/2009	C16221QA01 Amdts F + K	Encoder robustness improvement
C16221Q-34-005 Rev2	NO EFFECT	C16221QA01 Amdts F + K	Encoder robustness improvement
C16221Q-34-006	JUN 15/2006	C16221QA01 AmdtG	Improvement of the power supply robustness
C16221Q-34-007	JAN 25/2008	C16221QA01 AmdtH	To overcome obsolescence on the programmed interface board and on filter board
C16221Q-34-008	OCT 30/2008	C16221QA01	To overcome obsolescence of the display drivers
C16221Q-34-009	OCT 30/2009	C16221QA01 AmdtJ	To overcome obsolescence of the initial measurement unit accelerometer

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INTRODUCTION

TASK 34-22-45-871-801-A01

1. General

A. Purpose of the Manual

- (1) The Component Maintenance Manual (CMM) conforms to ATA Specification No. 2200.
- (2) The CMM provides shop verified procedures that enable a mechanic to restore the component to its serviceable condition in the overhaul shop.
- (3) The Abbreviated Component Maintenance Manual (ACMM) is a brief version of the CMM, for simple components which require periodic maintenance and/or testing.
- (4) The ACMM/CMM is prepared by the Manufacturer for each Repairable or Overhaulable unit.
- (5) THALES Avionics Technical Publications use both metric and non-metric system of measurement. The system used in the original reference documents is given first, followed by the conversion into the other system in brackets.

B. Manual Breakdown Arrangement

- (1) The information contained in the CMM has been divided into three main categories:
 - Description and Operation (DO)
 - Maintenance Procedure (MP)
 - Illustrated Parts List (IPL).

C. Pageblock (PAGEBLOCK) Assignment

- (1) The table that follows gives the identifications (Pageblock) used for the three categories of data.

Table INTRO 1 : Pageblock Assignment

Pageblock Type	Pageblock Number	Pageblock Pagination
INTRODUCTION	INTRO 1	1-99
DESCRIPTION AND OPERATION	1	1-999
TESTING AND FAULT ISOLATION	1000	1001-1999
SCHEMATIC AND WIRING DIAGRAMS	2000	2001-2999
DISASSEMBLY	3000	3001-3999
CLEANING	4000	4001-4999
CHECK	5000	5001-5999
REPAIR	6000	6001-6999
ASSEMBLY	7000	7001-7999
FITS AND CLEARANCES	8000	8001-8999
SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES	9000	9001-9999
SPECIAL PROCEDURES	11000	11001-11999

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Pageblock Type	Pageblock Number	Pageblock Pagination
REMOVAL	12000	12001-12999
INSTALLATION	13000	13001-13999
SERVICING	14000	14001-14999
STORAGE (including TRANSPORTATION)	15000	15001-15999
REWORK (Service Bulletin accomplishment)	16000	16001-16999
ILLUSTRATED PARTS LIST	10000	10001-10999

- (2) The Introduction explains the organization of the data in the CMM and the use of the data (DO and MP parts only)
 - For IPL part: refer to IPL INTRO (PAGEBLOCK 10000)
 - The introduction includes a glossary of non standard abbreviations.
- (3) Description and Operation
 - (a) This pageblock gives general overall information.
 - (b) It gives a mechanical and an electrical description.
 - (c) It explains the configuration and the operation of the component.
 - (d) It explains external connector interconnections in the component.
- (4) Testing and Fault Isolation
 - (a) This pageblock contains specific tests and procedures required to determine the condition of a component.
 - (b) It includes fault isolation data.
- (5) Schematic and Wiring Diagrams
 - (a) This pageblock contains all schematic diagrams, wiring and interconnecting diagrams of cards, chassis wiring, modules, subassemblies.
- (6) Disassembly
 - (a) This pageblock gives step-by-step disassembly instructions in a logical sequence (in the order of IPL item numbers). It includes references to the figure item numbers from the IPL illustration(s) where applicable.
- (7) Cleaning
 - (a) This pageblock gives step-by-step description of the cleaning operations for the component and its parts.
- (8) Check
 - (a) This pageblock gives all the check procedures that are necessary during shop maintenance.
- (9) Repair
 - (a) This pageblock contains detailed repair procedures and specifications necessary to restore a worn or damaged part to serviceable condition.
- (10) Assembly
 - (a) This pageblock gives step-by-step assembly instructions in a logical sequence, references to the figure item numbers from the IPL illustration(s) where applicable.

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(11) Fits and Clearances

- (a) This pageblock gives the fits and clearances data, in one or more tables, and torque tightening figures if applicable.

(12) Special Tools, Fixtures, Equipment and Consumables

- (a) This pageblock gives all special tools, fixtures, equipment and consumables necessary for maintenance actions given in the other pageblocks.

(13) Illustrated Parts List

- (a) This pageblock contains the complete list of all the parts of the component (For detailed description of the IPL, refer to IPL INTRO).

(14) Special Procedures

- (a) This pageblock contains tasks which are not part of the other pageblocks, but are necessary as a result of other maintenance tasks.
- (b) These tasks are self-contained repairs or preventive maintenance.

(15) Removal

- (a) Not applicable.

(16) Installation

- (a) Not applicable.

(17) Servicing

- (a) Not applicable.

(18) Storage

- (a) This pageblock contains requirements related to preparation for transportation, special handling, packaging, storage and preservation requirements of the component after assembly or test.

(19) Rework

- (a) Not applicable.

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2. General Arrangement

A. Preliminary Pages at the Beginning of the CMM

- (1) Title Page (TP)
- (2) Record of Revisions (ROR)
- (3) Highlights
 - (a) Gives revised chapter/section list and a brief description of the chapter/section modifications.
- (4) Record of Temporary Revisions (RTR)
 - (a) Gives the temporary revisions which are to be removed from the manual or which remain still effective.
- (5) Service Bulletin List (SBL)
 - (a) Gives the Service Bulletins and the Service Information Letters which are applicable for the equipment.
- (6) List of Effective Pages (LEP) (printed CMMs only)
 - (a) Gives all effective pages for a given revision issue.
- (7) Table of Contents (TOC)
 - (a) Gives all major sub-divisions of each page block contained in the manual.
- (8) List of Illustrations
 - (a) Gives all illustrations contained in the manual.

B. Introduction (INTRO) Structure

- (1) General
 - (a) The introduction is broken down into Tasks.
 - (b) This Task gives general information related to the CMM content.
- (2) General arrangement
- (3) Maintenance Task Oriented Support System (MTOSS) Task/Subtask numbering
 - (a) This Task explains MTOSS Task/Subtask numbering system.
- (4) Symbols and Definitions
- (5) Glossary of Abbreviations

C. Description and Operation (DO) Structure

- (1) General
 - (a) The Description and Operation is broken down into Tasks.
 - (b) This Task gives general overall information.
- (2) Description
 - (a) This Task gives a mechanical and electrical description of the component.
- (3) Operation
 - (a) This Task explains the configuration and the operation of the component and its sub-assemblies.

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D. Maintenance Procedure (MP) Structure

(1) General

(a) The MP is broken down into Tasks and Subtasks.

(b) Each Task is broken down into five Topics:

- Reason for the job
- Job set-up information
- Job set-up
- Procedure
- Close-up (if applicable).

(2) Reason for the Job

(a) This topic is used to explain the reason for the Task, but if the Task title gives sufficient information, then the statement "Self explanatory" is used.

(3) Job Set-up information

(a) This Topic is broken down into four Tables:

- Fixtures, Tools and Support Equipment
- Consumable Materials
- Expendable Parts
- Referenced Information.

(b) Fixtures, Tools and Support Equipment

1 This Table gives the fixtures, tools and support equipment necessary for the maintenance procedure. They are of types:

- Special Tools and equipment identified by their Part/Number
- Standard Tools and Equipment identified by technical description. In this case the mention "Not specific" is given in the reference column.

(c) Consumable Materials

1 This Table gives all consumable materials used during the maintenance procedure.

(d) Expendable Parts

1 This Table gives the IPC CSN (Catalogue Sequence Number) of all expendable parts used in the maintenance procedure.

2 The IPL gives the related Part/Numbers.

(e) Referenced Information

1 This Table gives all the cross-references to other documents.

(4) Job Set-up

(a) This Topic gives all the instructions necessary to prepare for the procedure.

(b) The instructions are broken down into Subtasks.

(5) Procedure

(a) This Topic gives all the instructions necessary to do the maintenance procedure.

(b) The instructions are broken down into Subtasks.

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- (6) Close-up
 - (a) This Topic gives all the instructions necessary to put the component back to its initial configuration.
 - (b) The instructions are broken down into Subtasks.

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TASK 34-22-45-879-802-A01

3. Maintenance Task Oriented Support System Task/Subtask Numbering

A. General

(1) The functional arrangement of data and the numbering system form the basis for the assignment of maintenance Task/Subtask numbers to each maintenance operation (Task) described in the CMM. Task/Subtask numbers are mainly for the use of the Data management and have no significance for the reader of the CMM. The Introduction (INTRO) and the Description and Operation (DO) are broken down into Tasks. The Maintenance Procedures (MP) are broken down into Tasks and Subtasks.

(2) TASK

(a) A TASK is defined as:

- A maintenance work procedure which contains a sequence of work steps, which may be organised into Subtasks
- A block of data e.g. Description and Operation
- A table, an illustration or a multi-sheet illustration e.g. Figure x Sheets 1 through to 4 will all be part of one Task.

(3) SUBTASK

(a) A SUBTASK is a single-skill work step or group of work steps within a Task, which provides detail of a significant part of the procedure. The completion of a SUBTASK must be possible in one location at one time.

B. Structure of Task/Subtask Numbers

(1) Table of Task/Subtask numbers

Table INTRO 2 : Structure of Task/Subtask Numbers

Form	Tag Name	-xx-xx-xx	-xxx	-xxx	-xxx	-xxx
Element	1	2	3	4	5	6

(2) Element 1

(a) The tag name is the word TASK, SUBTASK or GRAPHIC.

(3) Element 2

(a) ATA number (six digits).

(4) Element 3

(a) MTOSS function code (three digits).

(b) It is used to indicate the particular maintenance function involved.

(c) For each tag name, this code is selected from numbers 000 through 999 as given in the MTOSS Function Codes.

(5) Element 4

(a) Sequential number of the function code (three digits).

(b) It enables a unique identification number to be allocated for all Tasks/Subtasks which are similarly numbered throughout the preceding elements.

(c) Tasks and Subtasks are numbered differently:

- Task idents begin at 801 and rise, sequentially, to 999 (maximum) within the procedure

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- Subtask idents begin at 001 and rise, sequentially, to 800 (maximum) within the procedure
- Illustrations and Tables are considered as Tasks. These Task idents begin at 001 and rise, sequentially, to 999 (maximum) within the procedure.

NOTE: If a particular Task or Subtask is deleted, the Task/Subtask number must not be re-used.

(6) Element 5

(a) Alphanumeric indicator (three characters):

- The first character is a letter to indicate a different configuration due to different criteria configuration (modification, Service Bulletin(s), variations of work practices, ...)
- The second and third characters are digits to indicate alternative methods/techniques of maintenance.

(7) Element 6

(a) Alphanumeric indicator (three characters).

(b) It is to be assigned by an airline to highlight unique airline data.

(8) Example

(a) The example below shows the task and the elements.

Table INTRO 3 : Structure of Task

Example	TASK	-xx-yy-zz	-000	-801	-A01
Element	Tag Name	1	2	3	4

Table INTRO 4 : Structure of Task

Element	Definition
1	ATA number of the equipment
2	MTOSS function code (ex: 000 = REMOVAL AND DISASSEMBLY)
3	Sequential number of the function code (ex: 801 = first Task of the REMOVAL AND DISASSEMBLY type)
4	A = first configuration, 01 = first method

C. MTOSS Function Codes

(1) MTOSS Function Codes

(a) The first two digits of the numbering system fourth element are in accordance with the ATA Specification No. 2200, although in some instances there is variation by using a third digit for further breakdown of the basic function.

Table INTRO 5 : Removal and Disassembly

Function Code	Function
000	Removal and Disassembly

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Function Code	Function
010	Removal of complete component from transportation/work stand
020	Disassembly of unit into modular units
030	Disassembly of modules into sub-assemblies
040	Disassembly of sub-assemblies into individual parts
050	Removal of accessories/components not part of subject unit/opening panels for access
060	Disassembly of accessories into sub-assemblies
070	Disassembly of accessory into sub-assemblies into individual parts
080	Remove test equipment
090	Disassemble support equipment

Table INTRO 6 : Cleaning

Function Code	Function
100	Cleaning
110	Chemical cleaning/power flushing
120	Wet or dry abrasive cleaning
130	Ultrasonic cleaning
140	Mechanical/hand cleaning
150	Unassigned
160	Compressed air/miscellaneous hand cleaning or combination of procedures
170	Foam/water wash
180	Electro-mechanical test of solutions to identify materials
190	Unassigned

Table INTRO 7 : Inspection

Function Code	Function
200	Inspection
210	Visual examination to determine serviceability/unserviceability of unit
220	Visual/dimensional comparison with specifications/acceptance
230	Fluorescent penetrant NDT
240	Magnetic particle NDT

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Function Code	Function
250	Eddy current NDT
260	X-ray NDT
270	Ultrasonic NDT
280	Any special inspection
290	Unassigned

Table INTRO 8 : Repair

Function Code	Function
300	Repair
310	Welding, brazing and soldering
320	Machining
330	Metallic coating removal/application
340	Plasma/flame spraying of coating
350	Repairs using hand tools, including necessary disassembly/assembly
360	Bonding and moulding
370	Heat treating
380	Surface treatment-painting, aluminizing, etc., including masking
390	Machine riveting and flaring

Table INTRO 9 : Installation and Assembly

Function Code	Function
400	Installation and Assembly
410	Installation of complete component into transportation/work stand
420	Assembly of complete unit from modular units
430	Assembly of modules
440	Assembly of subassembly from parts
450	Installation of accessories/closing of panels, removed/opened for access
460	Assembly of accessory from sub-assemblies
470	Assembly of accessory subassembly from individual parts
480	Install test equipment

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Function Code	Function
490	Assemble support equipment

Table INTRO 10 : Material Handling

Function Code	Function
500	Material Handling
510	Shipping
520	Receiving
530	Packing
540	Unpacking
550	Storage
560	Collecting parts required for task (s) in holding area prior to release (Marshalling)
570	Work before and after air transportation of a unit
580	Unassigned
590	Unassigned

Table INTRO 11 : Servicing/Preserving/Lubricating

Function Code	Function
600	Servicing/Preserving/Lubricating
610	Servicing
620	Preserving
630	De-preserving
640	Lubricating
650	Unassigned
660	Unassigned
670	Unassigned
680	Unassigned
690	Unassigned

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Table INTRO 12 : Testing/Checking

Function Code	Function
700	Testing/Checking
710	Oil flow measurement
720	Air flow measurement
730	Fuel flow measurement/functional check
740	Water flow measurement/functional check
750	Electrical functional checks/measurement of parameters to determine serviceability, may include fault isolation
760	Functional test of systems/performance check
770	Functional test of an accessory carte
780	Pressure check of system/component normally pressurized
790	Leak check

Table INTRO 13 : Miscellaneous

Function Code	Function
800	Miscellaneous
810	Fault Isolation
820	Adjusting/aligning/calibration
830	Rigging/adjusting/component linkage
840	Service Bulletin incorporation
850	Part number change/application
860	Special procedure
870	Description and Operation
871	General
872	Description
873	Operation
874	Effectivity of cards and amendments
875	Unassigned
876	Unassigned
877	Unassigned

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Function Code	Function
878	Location
879	How to use
880	Approved vendor processes
890	Airline maintenance program (customer use)

D. Relationship between PAGEBLOCKS and TASK/SUBTASK Numbers

Table INTRO 14 : Miscellaneous

PAGEBLOCK	TASK Number	SUBTASK Number
Introduction	871, 879	None
Description and Operation	871, 872, 873	None
Testing and Fault Isolation	871, 700, 750	None
Schematic and Wiring Diagrams	878	None
Disassembly	000	020
Cleaning	100	160
Check	200	210, 220
Repair	300	350
Assembly	400	420
Fits and Clearances	NA	NA
Special Tools, Fixtures, Equipment and Consumables	940	None
Special Procedures	860, 300	None
Removal	NA	NA
Installation	NA	NA
Servicing	NA	NA
Storage	550	530, 550
Rework	NA	NA
Illustrated Parts List	871, 874, 879, 950, 952	None

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TASK 34-22-45-879-803-A01

4. Symbols and Definitions

A. Units and Measurement

(1) Symbols

Table INTRO 15 : Symbols

SI System		US System	
Abbreviation	Definition	Abbreviation	Definition
bar	bar	psi	pound square inch
deg. C	degrees Celsius	deg. F	degrees Fahrenheit
g	gram	oz	ounce
kg	kilogram	lb	pound
m	meter	ft	foot/feet
N.m	Newton.meter	lbf.in, lbf.ft	pound force.inch, pound force.foot
mm	millimeter	in	inch

(2) Conversion table

Table INTRO 16 : Conversion

From SI System	To US System
1 bar	14.5037 psi
1 g	0.0353 oz
1 kg	2.2046 lb
1 m	3.2809 ft
1 N.m	8.85075 lbf.in, 0.737562 lbf.ft
1 mm	0.0394 in.

NOTE: Temperature conversion from degrees Celsius (deg. C) to degrees Fahrenheit (deg. F): $\text{deg. F} = (1.8 \times \text{deg. C}) + 32$.

B. Reference Designator Index (RDI)

(1) The designation of a component (RDI) includes:

- A code letter in relation to Table INTRO-17
- A sequential number.

Table INTRO 17 : Reference Designator Index

Type of Component	Code Letter
Battery	BT
Capacitor	C
Connector	J or P
Coil	L
Diode, zener diode, bridge diode	D or CR
Fuse	F
Integrated circuit, microcircuit	IC
Jumper, strap	CV
Light Emitting Diode, light	DS
Quartz, oscillator	Y
Relay	K
Resistor, potentiometer, resistor network	R
Switch, push-button switch	S
Test point	TP
Thermal resistor	RT
Transformer	T
Transistor, thyristor	Q
Varistor	RV

C. Definition of Terms

(1) The terms that follow are used in the CMM and are defined as follows:

WARNING: CALLS ATTENTION TO USE OF MATERIAL, PROCESSES, METHODS, PROCEDURES OR LIMITS WHICH MUST BE FOLLOWED PRECISELY TO PREVENT INJURY OR DEATH TO PERSONS.

CAUTION: CALLS ATTENTION TO METHODS AND PROCEDURES WHICH MUST BE FOLLOWED TO PREVENT DAMAGE TO EQUIPMENT.

NOTE: Calls attention to methods which make the job easier or provide supplementary or explanatory information.

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- The term "TIGHTEN" is generally used in locations where no torque value is specified in the text
- The term "TORQUE" is used in the text together with a given specific torque value.

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TASK 34-22-45-879-805-A01

5. List of Abbreviations

A. General

- (1) A number of abbreviations are used in the document to make the text and illustrations shorter. These abbreviations and their definitions are given in the table that follows.

B. Abbreviations

(1) List :

ALT:	Altitude
ATT:	Attitude
BARO:	Barometric Pressure
CPGDU:	Control Processor Graphic Display Unit
EMI:	Electromagnetic Interference
ESDS:	Electrostatic Discharge Sensitive
FAIL:	Failure
FFA:	Front Face Assembly
FIB:	Filtering Interconnection Board
FPI:	Flat Panel Instrument, Flat Panel Interface
GND:	Ground
G/S:	Glide Slope
HDG:	Heading
hPa:	Hecto pascal
HS:	High Speed
ICS:	Instrument Combine Secours (Combined standby instrument)
IESI:	Integrated Electronic Standby Instrument
ILS:	Instrument Landing System
IMU:	Inertial Measurement Unit
IN:	Input
in.Hg:	Inches of Mercury
init:	Initialization
ISFD:	Integrated Standby Flight Display
LBB:	Light Box Board
LOC:	Localizer
LS:	Low Speed
LSP:	Lightning Strike Protection
M:	Mach
MAINT:	Maintenance
MMO:	Mach Maximum in Operation
OA:	Optical Assembly

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OC:	Open Circuit
OUT:	Output
PIB:	Programmed Interface Board
PMB:	Programmed Motherboard
PSA:	Pressure Sensor Assembly
PSU:	Power Supply Unit
QAC:	Qualification Aviation Civile (Commercial aircraft certification)
RST:	Reset
SIMU:	Simulation
SOPM:	Standard Overhaul Practices Manual
SPD:	Speed
SRU:	Shop Replaceable Unit
TSO:	Technical Standard Order
VMO:	Maximum Velocity in Operation

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TASK 34-22-45-879-804-A01

6. Shop Verification

A. Verification of the Procedures in the Manual

- (1) Testing and Fault Isolation: shop verification done by performance JUL 31/03.
- (2) Disassembly: shop verification done by performance FEB 15/00.
- (3) Assembly: shop verification done by performance FEB 15/00.

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DESCRIPTION AND OPERATION

TASK 34-22-45-870-801-A01

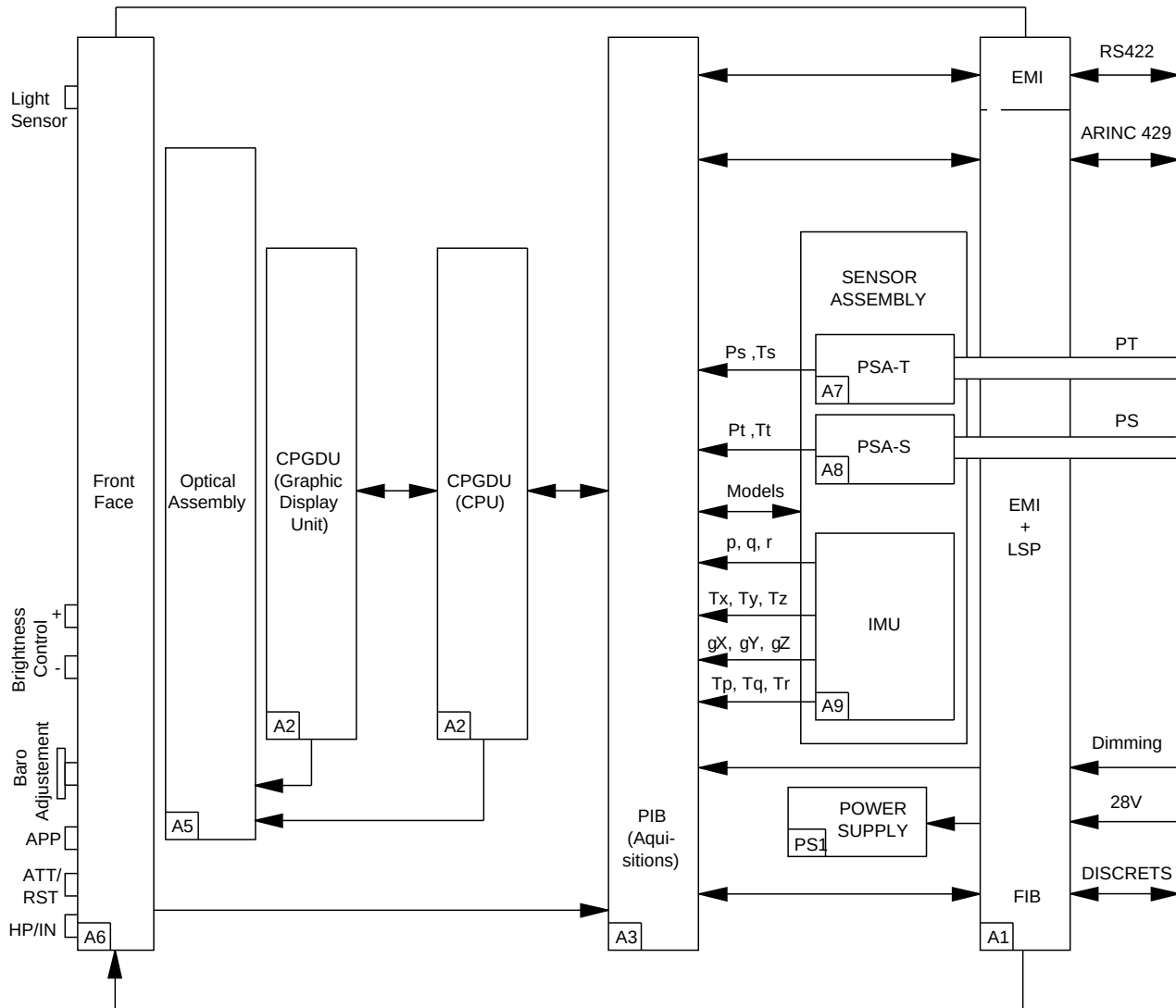
1. General

A. Purpose

- (1) The purpose of the Integrated Standby Flight Display ISFD is to give a standby display of the parameters that follow:
 - (a) Pitch and roll attitude
 - (b) Standard or baro-corrected altitude and the related barometric pressure in hPa or in.Hg
 - (c) Indicated airspeed
- (2) The options that follow can be selected by pin programming:
 - (a) ILS or Back Course data
 - (b) Lateral acceleration
 - (c) Magnetic heading data.
 - (d) Altitude in meters
 - (e) V-bar airplane symbol
 - (f) Mach number display.
- (3) In addition, two other functions are available:
 - (a) Initiated built-in test (IBIT). This test is run on the ground only. It is activated by simultaneous pressure on the two "APP" and "HP/IN" buttons on the front panel and displays the maintenance menus.
 - (b) Simulator or simulation mode. This mode is selected by pin programming.
- (4) [\(Refer to Figure 1\)](#) for the general architecture of the instruments.

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NOTE : For C16221QA..., the informations PT and PS are transmitted by two ARINC 429 inputs.

D_CM_TAV_342245_00_AAG0_01_00

Figure 1/Graphic 34-22-45-991-001-A01
Architecture Diagram

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B. Identification - Modification State - Other Labels

CAUTION: THE EQUIPMENT CONTAINS ELECTROSTATIC DISCHARGE SENSITIVE (ESDS) DEVICES. YOU MUST BE VERY CAREFUL WITH THESE DEVICES. UNWANTED VOLTAGES (STATIC OR INDUCED VOLTAGES, LEAK VOLTAGES, ETC.) BETWEEN THE INPUT AND OUTPUT TERMINALS OF ESDS DEVICES CAN EASILY CAUSE DAMAGE TO THEM. TO KEEP THE RISK OF FAILURE TO A MINIMUM:

- DO NOT LET THE TERMINALS OF AN ESDS DEVICE STAY IN OPEN CIRCUIT.
- OBEY THE SPECIAL PRECAUTIONS GIVEN IN THE STANDARD OVERHAUL PRACTICES MANUAL ([REFER TO SOPM 20-02-00](#)) FOR ALL OPERATIONS ON ESDS DEVICES AND THEIR BOARDS.

(1) Identification ([Refer to Figure 2](#)) ([Refer to Figure 3](#)) ([Refer to Figure 4](#))

- The identification is given by the permanent and modifiable bar code and amendment labels installed on the lateral face of the equipment.
- The permanent bar code label ([Refer to Figure 2](#)) gives this information:
 - MFR : Manufacturer code in alphanumeric and bar code form
 - DMF : Date of final factory inspection (month and year) in numeric and bar code form
 - Cle (INSP) : Inspection stamp
 - SER : Basic part number and serial number in alphanumeric and bar code form.
- The modifiable bar code label ([Refer to Figure 3](#)), ([Refer to Table 1](#)) gives this information:
 - Manufacturer's name
 - PNR : Equipment Part Number in alphanumeric and bar code form

Table 1 : Part Number Element signification

Part Number Element	Signification
C	Type of equipment
16221	Model of equipment in the type
K	Version of equipment
A	Hardware modification state
01	Software modification state

- Designation of the equipment
- Customer code numbers.

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Component Maintenance Manual

C16221K

MFR F9111	Cle (INSP)
DMF	
SER	

D_CM_TAV_342245_00_AAA0_01_00

Figure 2/Graphic 34-22-45-991-002-A01
Permanent Bar Code Label

THOMSON-CSF SEXTANT PNR	THALES AVIONICS PNR

D_CM_TAV_342245_00_AAB0_01_00

Figure 3/Graphic 34-22-45-991-003-A01
Modifiable Bar Code Label

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Amdt	A	C	E	G	J	L	N
	B	D	F	H	K	M	P

D_CM_TAV_342245_00_AAC0_01_00

Figure 4/Graphic 34-22-45-991-004-A01
Amendment Label

(2) Modification State

(a) Change to the equipment are shown as follows:

- 1 For important changes (called "Modifications"), the equipment version or the hardware and/or software modification state is increased by one increment: A becomes B, 01 becomes 02 and so on.
- 2 For less important changes (called "Amendments") which have no effect on the interchangeability of the equipment, the letter related the amendment is put on the amendment label (Refer to Figure 4).

(3) Other Labels

- (a) One label on the rear face (Refer to Figure 5) gives a warning that the equipment contains ESDS devices.
- (b) One TSO/QAC label on the lateral face (Refer to Figure 6) gives the reference of the QAC, TSO and JTSO standards, the weight of the equipment, and the range of the ALT and IAS functions of the equipment.



D_CM_TAV_342245_00_AAD0_01_00

Figure 5/Graphic 34-22-45-991-005-A01
Static Sensitive Devices Label

QAC	J/TSO	A S	WEIGHT: SOFTWARE LEVEL: POWER SUPPLY: ENVIRONMENT:
RANGE: ALT to ft IAS to kt			

D_CM_TAV_342245_00_AAE0_01_00

Figure 6/Graphic 34-22-45-991-006-A01
TSO/QAC Label

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C. Detailed Breakdown

(Refer to Figure 7)

(Refer to Table 2)

(Refer to Table 3)

(Refer to Table 4)

(Refer to Table 5)

(Refer to Table 6)

(Refer to Table 7)

(1) Composition Table

(a) The table shows the subassemblies installed in the equipment (Title and Part Number), the changes made to the equipment (Modifications and Amendments) with the related subassembly changes, and the equipment designators for the installation of each subassembly in the equipment.

(b) IESI

- This part used in the service of a civil aircraft is controlled by the U.S. Export Administration Regulations (EAR) and subject to ECCN 7A994. In any application other than a civil aircraft, this part is subject to the US ITAR regulations (International Traffic in Arms Regulations) and requires a license or retransfer approval from the U.S. Department of State. Other uses include simulator (civil or military) military aircraft, and parts removed from a civil aircraft and used in other applications etc.

Table 2 : Composition Table for C16221K.(sheet 1)

DESIGNATION	PART NUMBER	MODIF. / AMEND. STATES								EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221K	A01	A02 *	A02 A*	A02 B*	A02 C*	A02 D*	A02 E*	A02 F*		34-22-45
Subassemblies											
REAR FACE ASSY	E16387E	A	A	A	A	A	A	A	A	A1	
. FILTER INTER. BOARD	E16368C	A	A	A	A	A	A	A	A	A11	
FRONT FACE ASSY	E16364J	A	A	A	A	B	B	B	B	A6	
. FRONT FACE BOARD	E16389J	A	A	A	A	B	B	B	B	A61	
PRESSURE SENSOR ASSY	E17397A	A	A	A	A	A				A7,A8	
	E17397C						A	A	A	A7,A8	
	or E17397D						A	A	A	A8	
PROG. INTERFACE BOARD	E16485B	B01	B01							A3	34-22-26
	E16485E			A01	A01	A01	A01	A01		A3	
	E16485K								A01	A3	

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DESIGNATION	PART NUMBER	MODIF. / AMEND. STATES								EQUIP. DESIG.	CMM ATA NUMBER
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453H	A01	A01	A02	A02	A02	A02	A02	A02	A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371E	A	A	A	A	A	A	A	A	A9	
OPTICAL ASSY	E24119C	C	C	C	C	C	C			A5	
	E24119F							A	A	A5	
. LCD MATRIX	E19947B	A	A	A	A	A	A	A	A	DS1	
. LIGHT BOX BOARD	E24100B	B	B	B	B	B	B			A51	
	E24100E							A	A	A51	
. TUBES	F1252154	X	X	X	X	X	X			(DS2, (DS3, (DS4	
	F1405513							X	X		
PROGRAMMED MOTHERBOARD	E16486B	A01	A01	A01	A01	B01	B01	B01	B01	TB1	
POWER SUPPLY	E17395A	A	A	A						PS1	34-22-28
	E17395D				A	A	A	A	A	PS1	
Software											
PARAMETERING	G3116AA	G01	G03	G03	G03	G03	G03	G03	G03		
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01	B01	B01	B01	B01	B01		
CPU	G5614AA	K01	K02	K02	K02	K02	K02	K02	K02		
GSM	G5615AA	K01	K02	K02	K02	K02	K02	K02	K02		

NOTE: * For the ISFD C16221KA02 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221K-34-013.

NOTE:

- Configuration for production
- For operation, for all combinations of amendments refer to Service Bulletins.

Table 3 : Composition Table for C16221K. (sheet 2)

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DESIGNATION	PART NUMBER	MODIF. / AMEND. STATES							EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221K	A02 G*	A02 H*	A02 J*	A02 K*	A02L *	A02 M*	A02N *		34-22-45
Subassemblies										
REAR FACE ASSY	E16387E or E16387J	A	A	A					A1	
. FILTER INTER. BOARD	E16368C or E16368F	A	A	A					A1 A11 A11	
FRONT FACE ASSY	E16364J E38062A	B	C	C	C	C	C	A	A6 A6	
. FRONT FACE BOARD	E16389J	B	B	B	B	B	B	A	A61	
PRESSURE SENSOR ASSY	E17397C or E17397D or F1407830 or F1407831	A	A	A	A	A			A7,A8 A8 A7,A8 A8	
PROG. INTERFACE BOARD	E16485E E16485K or E16485R or E16485U								A3 A3 A3 A3	34-22-26
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453H	B03	B03	B03	B03	B03	B03	B03	A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371E E18520E	A	A	A	A	A			A9 A9	

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DESIGNATION	PART NUMBER	MODIF. / AMEND. STATES								EQUIP. DESIG.	CMM ATA NUMBER
OPTICAL ASSY . LCD MATRIX . LIGHT BOX BOARD . TUBES	E24119F	A	A	A	A	A	A	A	A	A5	34-22-28
	E19947B	A	A	A	A					DS1	
	or E19947G					A	A	A		DS1	
	E24100E	A	A	A	A	A	A	A		A51	
	F1405513	X	X	X	X	X	X	X		(DS2, (DS3, (DS4	
PROGRAMMED MOTHERBOARD	E16486B	B01	B01	B01	B01	B01				TB1	
	or E16486J	A01	A01	A01	A01	A01				TB1	
	E16981H						A01	A01		TB1	
POWER SUPPLY	E17395D	A	A							PS1	
	or E17395E			A	A	A	A	A		PS1	
Software											
PARAMETERING	G3116AA	G03	G03	G03	G03	G03	G03	G03			
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01	B01	B01	B01	B01			
CPU	G5614AA	K02	K02	K02	K02	K02	K02	K02			
GSM	G5615AA	K02	K02	K02	K02	K02	K02	K02			

NOTE: * For the ISFD C16221KA02 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221K-34-013.

NOTE:

- Configuration for production
- For operation, for all combinations of amendments refer to Service Bulletins.

Table 4 : Composition Table for C16221L. (sheet 1)

DESIGNATION	PART NUMBER	MODIF. / AMDT STATES								EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221L	A01	A02	A03*	A03 A*	A03 B*	A03 C*	A03 D*	A03 E*	A03 F*	34-22-45

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DESIGNATION	PART NUMBER	MODIF. / AMDT STATES										EQUIP. DESIG.	CMM ATA NUMBER
Subassemblies													
REAR FACE ASSY	E16387E	A	A	A	A	A	A	A	A	A	A	A1	
. FILTER INTER. BOARD	E16368C	A	A	A	A	A	A	A	A	A	A	A11	
FRONT FACE ASSY	E16364K	A	A	A	A	A	B	B	B	B	B	A6	
. FRONT FACE BOARD	E16389J	A	A	A	A	A	B	B	B	B	B	A61	
PRESSURE SENSOR ASSY	E17397A	A	A	A	A	A	A					A7,A8	
	E17397C							A	A	A		A7,A8	
	or												
	E17397D							A	A	A		A8	
PROG. INTERFACE BOARD	E16485B	B01	B01	B01								A3	34-22-26
	E16485E				A01	A01	A01	A01	A01	A01		A3	
	E16485K										A01	A3	
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453H	A01	A01	A01	A02	A02	A02	A02	A02	A02	A02	A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371D	A	A	A	A	A	A	A	A	A	A	A9	
OPTICAL ASSY	E24119C	C	C	C	C	C	C	C				A5	
	E24119F								A	A		A5	
. LCD MATRIX	E19947B	A	A	A	A	A	A	A	A	A		DS1	
. LIGHT BOX BOARD	E24100B	B	B	B	B	B	B	B				A51	
	E24100E								A	A		A51	
. TUBES	F1252154	X	X	X	X	X	X	X				(DS2,	
	F1405513								X	X		(DS3,	
												(DS4	
PROGRAMMED MOTHERBOARD	E16486B	A01	A01	A01	A01	A01	B01	B01	B01	B01	B01	TB1	
POWER SUPPLY	E17395A	A	A	A	A							PS1	34-22-28
	E17395D					A	A	A	A	A		PS1	
Software													
PARAMETERING	G3116AA	G01	G02	G03	G03	G03	G03	G03	G03	G03	G03		

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DESIGNATION	PART NUMBER	MODIF. / AMDT STATES									EQUIP. DESIG.	CMM ATA NUMBE R
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01	B01	B01	B01	B01	B01	B01		
CPU	G5614AA	K01	K01	K02	K02	K02	K02	K02	K02	K02		
GSM	G5615AA	K01	K01	K02	K02	K02	K02	K02	K02	K02		

NOTE: * For the ISFD C16221LA03 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221L-34-013.

NOTE:

- Configuration for production
- For operation, for all combinations of amendments refer to Service Bulletins.

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Table 5 : Composition Table for C16221L. (sheet 2)

DESIGNATION	PART NUMBER	MODIF. / AMDT STATES							EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221L	A03 G*	A03 H*	A03 J*	A03 K*	A03 L*	A03 M*	A03 N*		34-22-45
Subassemblies										
REAR FACE ASSY	E16387E or E16367J	A	A	A				A	A1	
. FILTER INTER. BOARD	E16368C or E16368F	A	A	A				A	A11	
FRONT FACE ASSY	E16364K	B	C	B	B	B	C	C**	A6	
. FRONT FACE BOARD	E16389J	B	B	B	B	B	B	B	A61	
PRESSURE SENSOR ASSY	E17397C or E17397D or F1407830 or F1407831	A	A	A	A	A			A7,A8	
PROG. INTERFACE BOARD	E16485K or E16485R or E16485U	A01	A01	A01					A3	34-22-26
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453H	B03	A02	A02	A02	A02	A02	A02	A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371D E18520F	A	A	A	A	A		A	A9 A9	

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DESIGNATION	PART NUMBER	MODIF. / AMDT STATES								EQUIP. DESIG.	CMM ATA NUMBER
OPTICAL ASSY . LCD MATRIX . LIGHT BOX BOARD . TUBES PROGRAMMED MOTHERBOARD POWER SUPPLY	E24119F	A	A	A	A	A	A	A	A	A5	34-22-28
	E19947B	A	A	A	A		A			DS1	
	or E19947G					A	A	A		DS1	
	E24100B	B	B	B	B	B	B	B		A51	
	F1405513	X	X	X	X	X	X	X		(DS2, (DS3, (DS4	
	E16486B	B01	B01	B01	B01	B01				TB1	
	E16486J	A01	A01	A01	A01	A01				TB1	
	E16981H						A01	A01		TB1	
	E17395D	A	A							PS1	
	or E17395E			A	A	A	A	A		PS1	
Software											
PARAMETERING	G3116AA	G03	G03	G03	G03	G03	G03	G03			
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01	B01	B01	B01	B01			
CPU	G5614AA	K02	K02	K02	K02	K02	K02	K02			
GSM	G5615AA	K02	K02	K02	K02	K02	K02	K02			

NOTE: * For the ISFD C16221LA03 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221L-34-013.

NOTE:

- Configuration for production
- For operation, for all combinations of amendments refer to Service Bulletins.

Table 6 : Composition Table for C16221M.

DESIGNATION	PART NUMBER	MODIF./AMDT STATES								EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221M	A01**	A01A**	A01B**	A01C**	A01D**	A01E**	A01F**	A01G**		34-22-45
Subassemblies											

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DESIGNATION	PART NUMBER	MODIF./AMDT STATES								EQUIP. DESIG.	CMM ATA NUMBER
REAR FACE ASSY	E16387E	A	A	A	A	A	A	A	A	A1	
. FILTER INTER. BOARD	E16368C	A	A	A	A	A	A	A	A	A11	
FRONT FACE ASSY	E16364J	B	B	B	B	B	C	C	C***	A6	
. FRONT FACE BOARD	E16389J	B	B	B	B	B	B	B	B	A61	
PRESSURE SENSOR ASSY	E17397A	A								A7,A8	
	E17397C		A	A	A	A	A	A	A	A7,A8	
	E17397B	A								A8	
	E17397D		A	A	A	A	A	A	A	A8	
PROG. INTERFACE BOARD	E16485E	A01	A01	A01						A3	34-22-26
	E16485K				A01	A01	A01	A01	A01	A3	
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453H	A02	A02	A02	A02	B03	B03	B03	B03	A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371G	A	A	A	A	A	A	A	A	A9	
OPTICAL ASSY	E24119C	C	C							A5	
	E24119F			A	A	A	A	A	A	A5	
. LCD MATRIX	E19947B	A	A							DS1	
. LIGHT BOX BOARD	E24100B	B	B							A51	
	E24100E			A	A	A	A	A	A	A51	
. TUBES	F1252154	X	X							(DS2,	
	F1405513			X	X	X	X	X	X	(DS3,	
										(DS4	
PROGRAMMED MOTHERBOARD	E16486B	B01	B01	B01	B01					TB1	
	E16486J					A01	A01	A01	A01	TB1	
POWER SUPPLY	E17395D	A	A	A	A	A	A			PS1	34-22-28
	E17395E							A	A	PS1	
Software											
PARAMETERING	G3116AA	G03	G03	G03	G03	G03	G03	G03	G03		
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01	B01	B01	B01	B01	B01		
CPU	G5614AA	K02	K02	K02	K02	K02	K02	K02	K02		
GSM	G5615AA	K02	K02	K02	K02	K02	K02	K02	K02		

NOTE:

* For C16221MA01, the PRESSURE SENSOR ASSY (A8) with the PN E17397AA or E17397BA can be installed.

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* For C16221MA01 with amendments, the PRESSURE SENSOR ASSY (A8) with the PN E17397CA or E17397DA can be installed.

NOTE: ** For the ISFD C16221MA01 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221M-34-007.

*** For the ISFD C16221MA01 with amendment G, the front face assy E16364JC is with amendment B as indicated in SB C16221M-34-005.

NOTE:

- Configuration for production
- For operation, for all combinations of amendments refer to Service Bulletins.

Table 7 : Composition Table for C16221Q. (sheet 1)

DESIGNATION	PART NUMBER	MODIF./AMDT STATES							EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221Q	A01 A**	A01 B**	A01 C**	A01 D**	A01 E**	A01 F**	A01 G**		34-22-45
Subassemblies										
REAR FACE ASSY	E16387E	A	A	A	A	A	A	A	A1	
. FILTER INTER. BOARD	E16368C	A	A	A	A	A	A	A	A11	
FRONT FACE ASSY	E16364K	B	B	B	B	B	C	C	A6	
. FRONT FACE BOARD	E16389J	B	B	B	B	B	B	B	A61	
PRESSURE SENSOR ASSY	E17397A	A							A7,A8	
	E17397C		A	A	A	A	A	A	A7,A8	
	E17397B	A							A8	
	E17397D		A	A	A	A	A	A	A8	
PROG. INTERFACE BOARD	E16485G	A01	A01	A01	A02	A02	A02	A02	A3	34-22-26
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453M	A01	A01	A01	A01	A02	A02	A02	A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371D	A	A	A	A	A	A	A	A9	

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DESIGNATION	PART NUMBER	MODIF./AMDT STATES							EQUIP. DESIG.	CMM ATA NUMBER
OPTICAL ASSY	E24119C	C	C						A5	
	E24119F			A	A	A	A	A	A5	
. LCD MATRIX	E19947B	A	A	A	A	A	A	A	DS1	
. LIGHT BOX BOARD	E24100B	B	B						A51	
	E24100E			A	A	A	A	A	A51	
. TUBES	F1252154	X	X						(DS2, (DS3, (DS4	
	F1405513			X	X	X	X	X		
PROGRAMMED MOTHERBOARD	E16486B	B01	B01	B01	B01				TB1	
	E16486J					A01	A01	A01	TB1	
POWER SUPPLY	E17395D	A	A	A	A	A	A		PS1	34-22-28
	E17395E							A	PS1	
Software										
PARAMETERING	G3116AA	K01	K01	K01	K01	K01	K01	K01		
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01	B01	B01	B01	B01		
CPU	G5614AA	L02	L02	L02	L02	L02	L02	L02		
GSM	G5615AA	L01	L01	L01	L01	L01	L01	L01		

NOTE: * For C16221QA01 with amendment A, the PRESSURE SENSOR ASSY (A8) with the PN E17397AA or E17397BA can be installed.

* For C16221MA01 with other amendments, the PRESSURE SENSOR ASSY (A8) with the PN E17397CA or E17397DA can be installed.

NOTE: ** For the ISFD C16221QA01 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221Q-34-008.

NOTE:

- Configuration for production
- For operation, for all combinations of amendments refer to Service Bulletins.

Table 8 : Composition Table for C16221Q. (sheet 2)

DESIGNATION	PART NUMBER	MODIF./AMDT STATES							EQUIP. DESIG.	CMM ATA NUMBER
INTEGRATED STANDBY FLIGHT DISPLAY	C16221Q	A01 H**	A01 J**	A01 K**						34-22-45
Subassemblies										

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DESIGNATION	PART NUMBER	MODIF./AMDT STATES							EQUIP. DESIG.	CMM ATA NUMBER
REAR FACE ASSY	E16387J	A	A	A					A1	
. FILTER INTER. BOARD	E16368F	A	A	A					A11	
FRONT FACE ASSY	E16364K	B	B	C					A6	
. FRONT FACE BOARD	E16389J	B	B	B					A61	
PRESSURE SENSOR ASSY	E17397C	A							A7,A8	
	E17397D	A							A8	
	F1407830		A	A					A7,A8	
	F1407831		A	A					A8	
PROG. INTERFACE BOARD	E16485H	A01	A01	A01					A3	34-22-26
CONTROL PROCESSOR GRAPHIC DISPLAY UNIT	E16453M	A02	A02	A02					A2	34-22-27
INERTIAL MEASUREMENT UNIT	E16371D	A							A9	
	E18520F		A	A					A9	
OPTICAL ASSY	E24119F	A	A	A					A5	
. LCD MATRIX	E19947B	A	A	A					DS1	
. LIGHT BOX BOARD	E24100E	A	A	A					A51	
. TUBES	F1405513	X	X	X					(DS2, (DS3, (DS4	
PROGRAMMED MOTHERBOARD	E16486B	B01							TB1	
	E16981H		A01	A01					TB1	
POWER SUPPLY	E17395E	A	A	A					PS1	34-22-28
Software										
PARAMETERING	G3116AA	K01	K01	K01						
SOFT. IDENTIFICATION	G3156AA	B01	B01	B01						
CPU	G5614AA	L02	L02	L02						
GSM	G5615AA	L01	L01	L01						

NOTE:

* For C16221QA01 with amendment A, the PRESSURE SENSOR ASSY (A8) with the PN E17397AA or E17397BA can be installed.

* For C16221MA01 with other amendments, the PRESSURE SENSOR ASSY (A8) with the PN E17397CA or E17397DA can be installed.

NOTE:

** For the ISFD C16221QA01 with or without amendments, the decomposition of the optical assemblies E24119CC or E24119FA with amendment A is indicated in SB C16221Q-34-008.

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- NOTE:**
- Configuration for production
 - For operation, for all combinations of amendments refer to Service Bulletins.

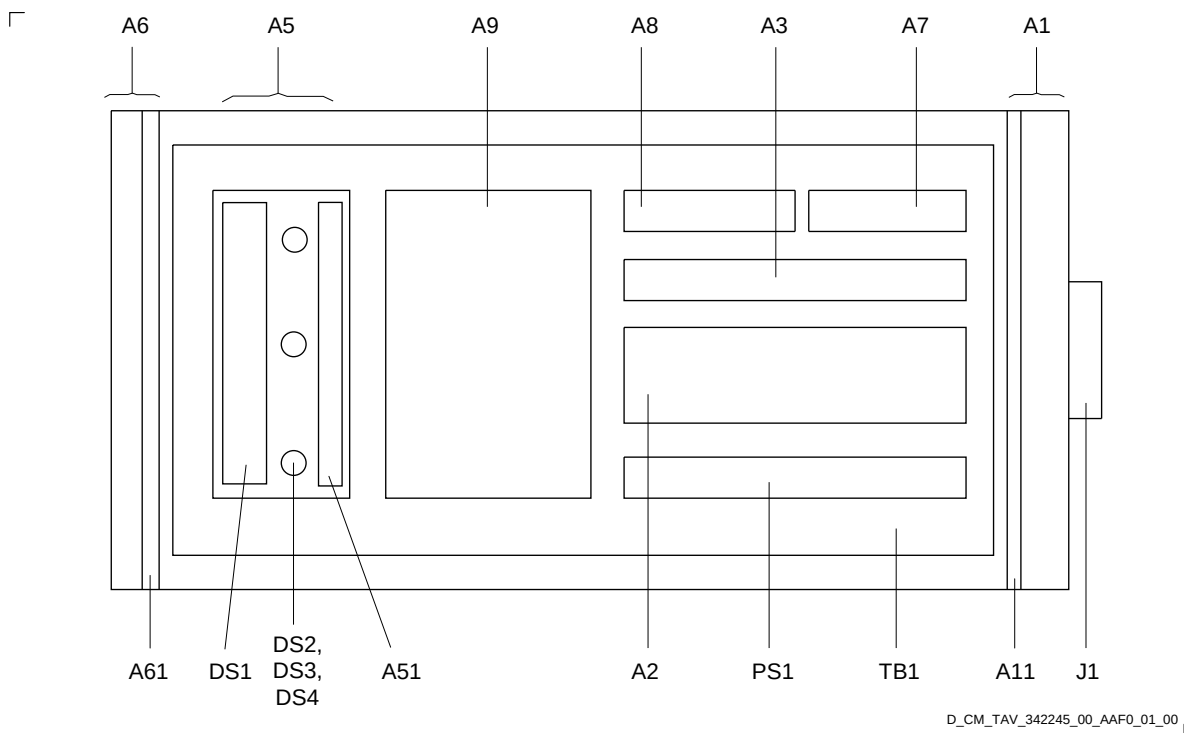


Figure 7/Graphic 34-22-45-991-007-A01
Component Layout

D. Technical Data

(1) Physical Data

- (a) Overall dimensions: [\(Refer to Figure 8\)](#).
- (b) Weight: 5 lbs (2.04 kg) +/-2% (for C16221KA.., C16221LA.. and C16221MA..) and 5.08 lbs (2.12 kg) +/-2% (for C16221QA..).

(2) Electrical Data

- (a) Power Supply:
 - +28VDC (maximum power 20W in operational mode)
 - 0-5VAC/400Hz (pushbutton lighting control voltage) (maximum power 1W) .
- (b) Lightning strike/EMI protection: The RS422 input and output, one output discrete and three mode configuration discretes have EMI protection but not lightning strike protection. All the other inputs and outputs have EMI protection and lightning strike protection.

(c) Input/Output Summary

1 Inputs

Table 9 : Input Signals

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TYPE	CHARACTERISTICS	NUMBER
DISCRETES	OPEN/GND	16
ARINC	ARINC 429 HS	1
	ARINC 429 LS	3 (Refer to Table 16) for C16221QA.. indicated by an asterix *) and 1(other versions)
RS422		1

NOTE: The 16 discretes include three mode selection discretes.
One supplementary discrete and Arinc inputs are spare.

2 Outputs

Table 10 : Output Signals

TYPE	CHARACTERISTICS	NUMBER
DISCRETES	OPEN/GND	1
ARINC	ARINC 429 HS	1
	ARINC 429 LS	1
RS422		1

(3) Pneumatic Data

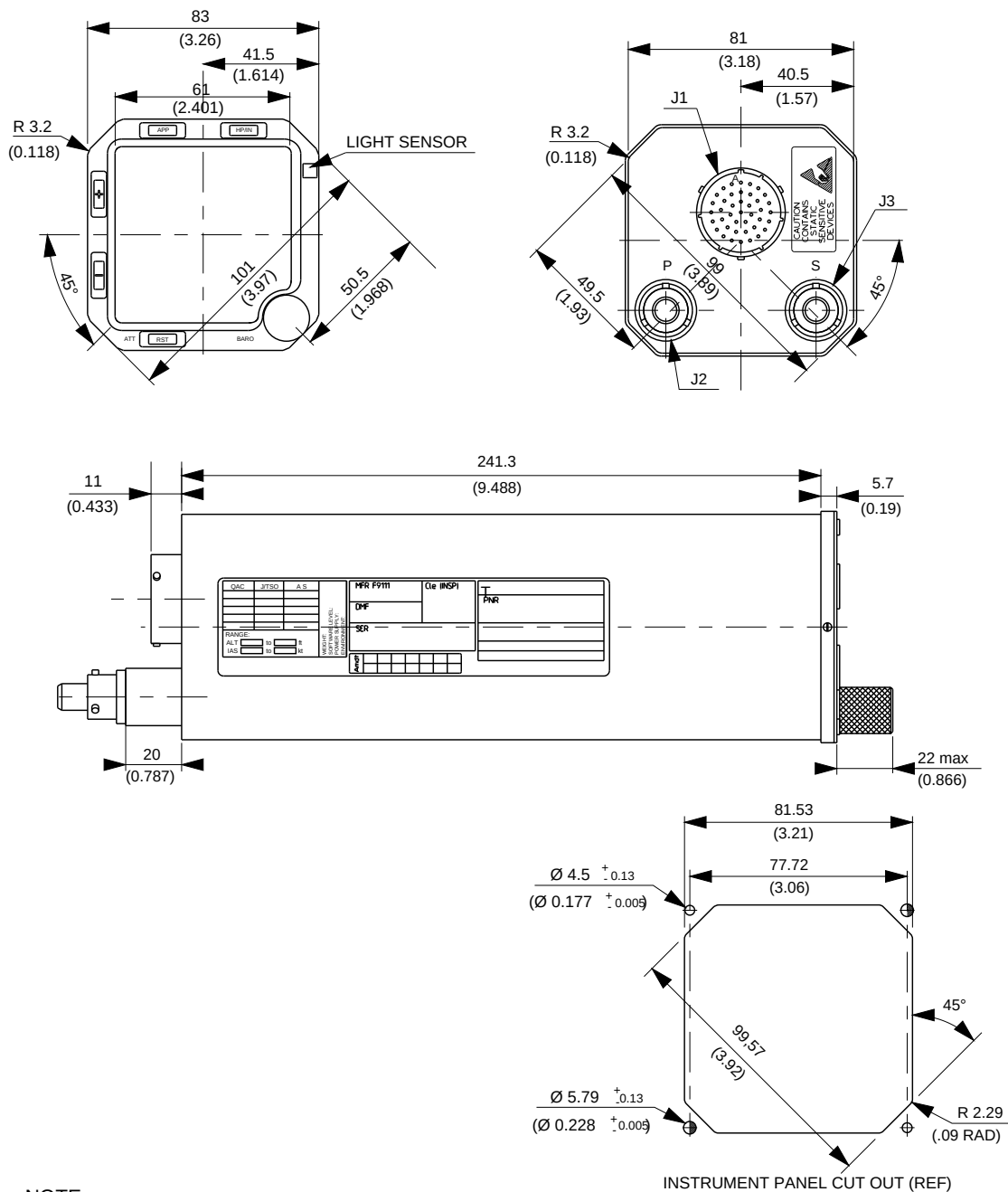
NOTE: These pressure unions are equipped with pressure caps (Refer to Table 16) for C16221QA.. indicated by an asterix *

- (a) The instrument has two pressure unions: J2 for pitot pressure and J3 for static pressure.
- (b) Leakage: for a pressure of 500 hPa in the pneumatic circuits, the leakage is not more than 1.7 hPa.

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NOTE

ALL DIMENSIONS ARE GIVEN IN MILLIMETERS WITH THEIR EQUIVALENTS IN INCHES IN BRACKETS

J1 CONNECTOR 41 PINS PER UTE C 93422 FAB:SOURIAU-REF:8525-02H-20-41PN

J2 TOTAL PRESSURE PNEUMATIC CONNECTOR PER MS 33649-4

+ QUICK DISCONNECT (AMERICAN SAFETY REF 40006-1 A 44)

J3 STATIC PRESSURE PNEUMATIC CONNECTOR PER MS 33649-6

+ QUICK DISCONNECT (AMERICAN SAFETY REF 40006-1 B 46)

D_CM_TAV_342245_00_AAHO_01_00

Figure 8/Graphic - Sheet 01 / 34-22-45-991-008-A01
Overall Dimensions (For C16221KA../LA../MA..)

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NOTE : ALL DIMENSIONS ARE GIVEN IN INCHES WITH THEIR EQUIVALENTS IN MILLIMETERS IN BRACKETS

J1 CONNECTOR 41 PINS PER UTE C 93422 FAB: SOURIAU-REF: 8525-02H-20-41PN

J2 TOTAL PRESSURE PNEUMATIC CONNECTOR PER MS 33649-4

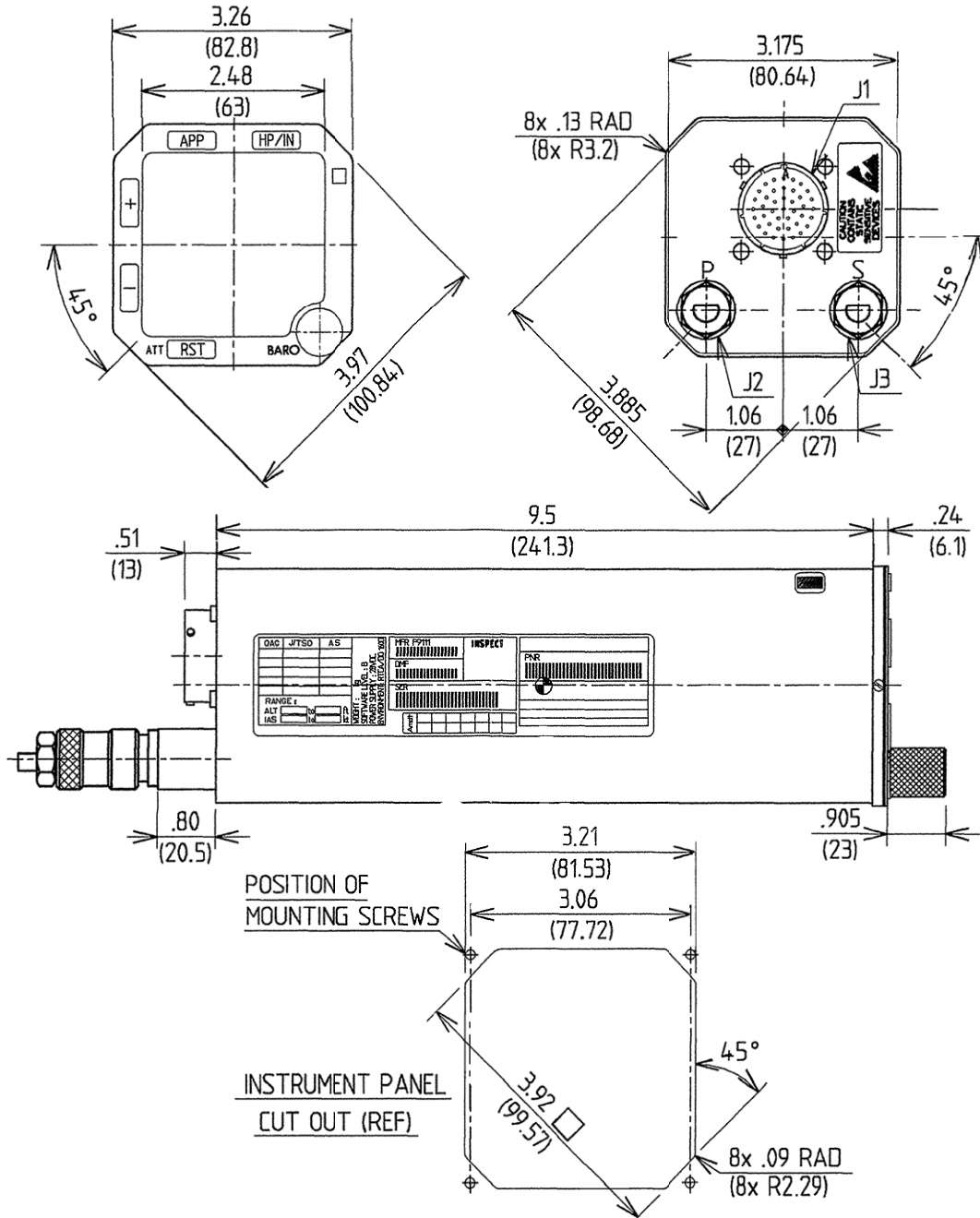
+ QUICK DISCONNECT (AMERICAN SAFETY REF 40006-1 A 44)

+ FEMALE CAP ASSEMBLY QUICK DISCONNECT (HYDRAFLOW REF 1QF3-2)

J3 STATIC PRESSURE PNEUMATIC CONNECTOR PER MS 33649-6

+ QUICK DISCONNECT (AMERICAN SAFETY REF 40006-1 B 46)

+ FEMALE CAP ASSEMBLY QUICK DISCONNECT (HYDRAFLOW REF 1QF3-3)



D_CM_TAV_342245_00_AA0_02_01

Figure 8/Graphic - Sheet 02 / 34-22-45-991-008-A01
Overall Dimensions (For C16221QA..)

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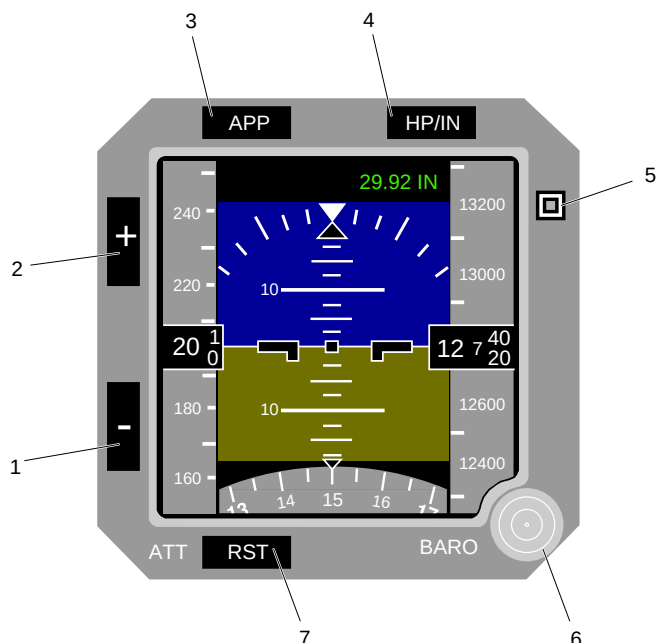
TASK 34-22-45-870-802-A01

2. Description

A. External Description

- (1) (Refer to Figure 8) for a general view of the instrument.
- (2) (Refer to Figure 9) and the related legend for the elements on the front face of the instrument.

NOTE: For C16221KA..MA., the color of the front face is grey and for C16221LA../QA., the color of front face is brown.



D_CM_TAV_342245_00_AA10_01_00

Figure 9/Graphic 34-22-45-991-009-A01
Front Face Controls

Table 11 : Legend for Figure 9

1,2	"+" and "-" brightness adjustment pushbuttons.
3	"APP" pushbutton for the display of the ILS (Localizer and Glideslope) and Back Course approach modes.
4	"HP/IN" pushbutton for the display of the barometric pressure in inches of mercury (in.Hg) or hectopascals (hPa).
5	Light sensor .
6	"BARO" rotary knob/pushbutton. The pushbutton function is used to change the barometric pressure (which can be adjusted by the rotary knob function of the pushbutton) to the standard pressure and back again).
7	"ATT/RST" pushbutton to reset the attitude function.

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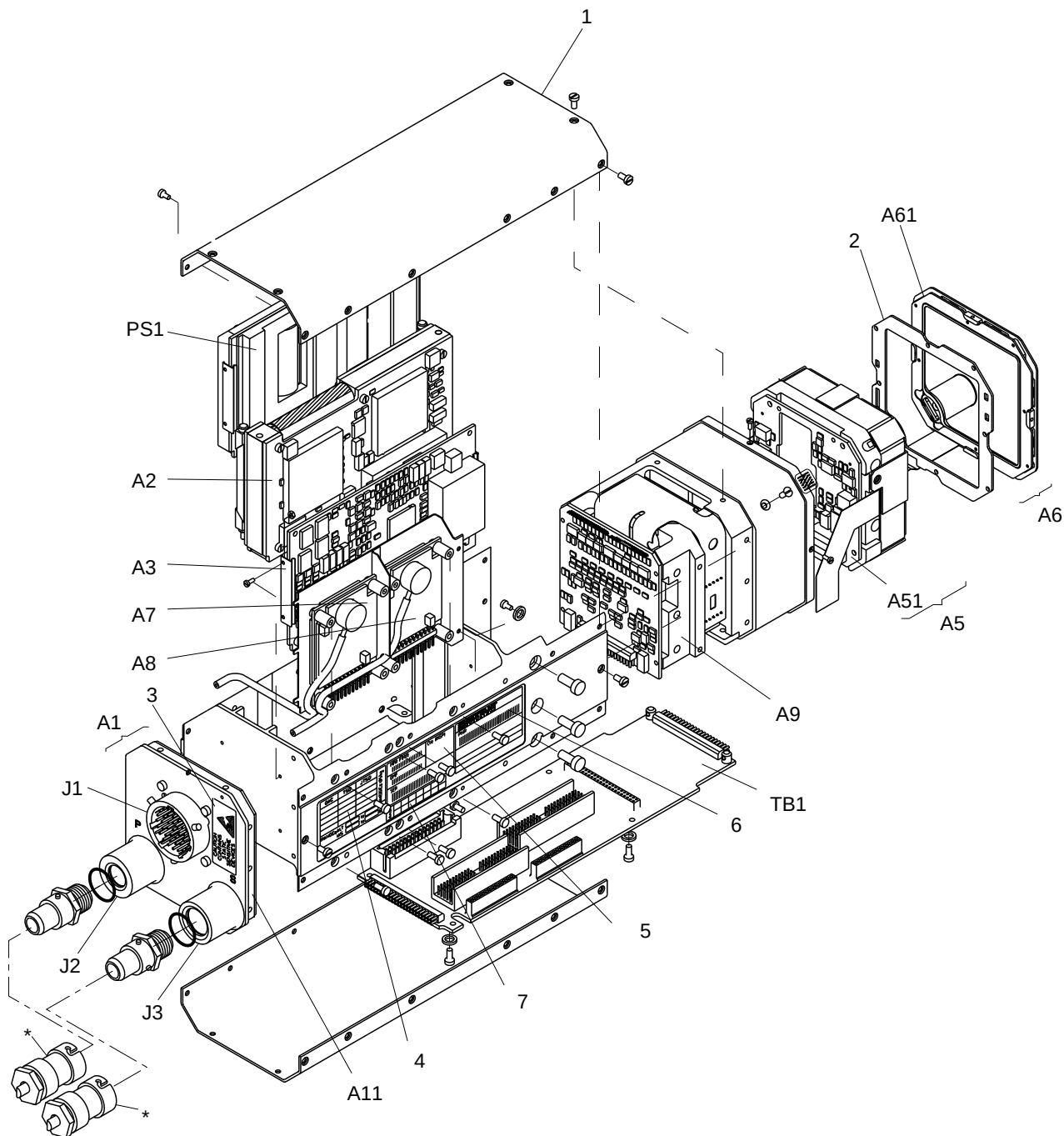
B. Internal Description

- (1) [\(Refer to Figure 10\)](#) and the related legend for the parts and subassemblies of the instrument.

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Figure 10/Graphic 34-22-45-991-010-A01
Exploded View

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Table 12 : Legend for Figure 10

1 :	Covers attached by screws
2 :	Gasket
3 :	ESDS label
4 :	TSO/QAC label
5 :	Permanent bar code label
6 :	Modifiable bar code label
7 :	Amendment label
A1:	Rear face assembly. It includes the rear plate with the pressure unions J2 (pitot) , J3 (static) and the quick disconnects and only for C16221QA.. the two pressure caps (indicated by an *) , the Filter Interconnection board A11 and the ESDS label (3).
A2:	Control Processor Graphic Display Unit. It is connected to the Programmed Motherboard TB1 by one 125-pin connector.
A3:	Programmed Interface board. It is connected to the Programmed Motherboard TB1 by one 125-pin connector.
A5:	Optical Assembly (OA). It includes a liquid crystal display (LCD) matrix DS1, the light box that contains three tubes and the CBL board A51. This assembly is connected to the Programmed Motherboard TB1 by two flex circuits.
A6:	Front face assembly. It includes the Front Face board A61, the control pushbuttons and knobs and the photoelectric cell. This assembly is connected to the Programmed Motherboard TB1 by one 30-pin connector.
A7, A8:	Pressure sensor assembly. It includes two boards A7 and A8 that are electrically connected to the Programmed Motherboard TB1 by one 12-pin connector. The boards are mechanically connected to the pressure unions J2 and J3 by hoses attached by ties.
A9:	Inertial Measurement Unit (IMU). It contains three single-axis rate sensors and their temperature probes, two acceleration sensors and their temperature probes, one IMU board and one mechanical structure. This assembly is connected to the Programmed Motherboard TB1 by one 36-pin connector.
A11:	Filter Interconnection board. It includes the 41-pin external connector J1 and is connected to the Programmed Motherboard TB1 by two 23-pin strip connectors.
PS1:	Power supply unit. It is connected to the Programmed Motherboard TB1 by one 30-pin connector.
TB1:	Programmed Motherboard. It makes the interconnection between all the subassemblies of the instrument.

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TASK 34-22-45-870-803-A01

3. Operation

A. General Operation

SUBTASK 34-22-45-870-001-A01

(1) Operating Modes

(a) The instrument has four principal modes:

- FLIGHT mode: operational mode used on aircraft
- SIMULATOR or SIMULATION mode: mode used during simulation
- SHOP MAINTENANCE mode: mode used for maintenance
- DOWNLOADING: mode used for software downloading.

(b) These modes are accessible after the power-up and initialization phases in relation to the combination of the three configuration discretes 1 to 3 ([Refer to Table 13](#)).

Table 13 : Mode Selection Discretes

	Conf. 1	Conf. 3	Conf. 4	Conf. 2
	Flight mode	Maintenance mode	Simulator or simulation mode	Downloading mode
Discrete 1 (pin X)	Open	Ground	Ground	Ground
Discrete 2 (pin h)	Open	Open	Ground	Ground
Discrete 3 (pin a)	Open	Ground	Open	Ground

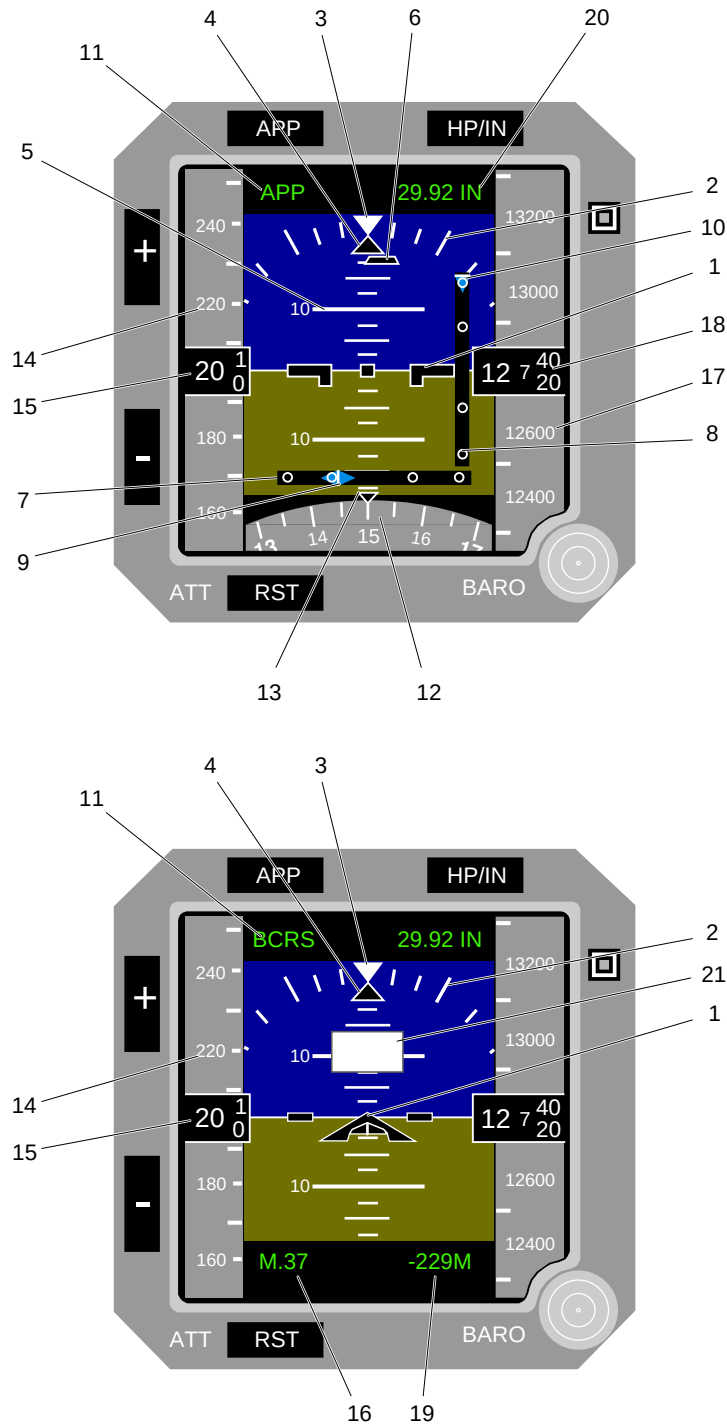
(c) FLIGHT Mode

- 1 The instrument goes into this mode at the end of the initialization phase when the discretes are in configuration 1. It displays the attitude, altitude, airspeed, approach and heading data (option).
- 2 Flight mode stays active while the 28V supply voltage is applied.

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Figure 11/Graphic 34-22-45-991-011-A01
FLIGHT Mode Symbology

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Table 14 : Legend for Figure 11

ATTITUDE DISPLAY	
1:	Airplane symbol (conventional or V-bar)
2:	Fixed white scale with roll graduations at 10-degree intervals between -30 and +30 degrees and at -45, +45, -60 and +60 degrees. The roll angle is shown by a reference triangle.
3:	Fixed white roll triangle.
4:	Reference roll triangle painted in black and surrounded in white when roll angle is more less 35 degrees, otherwise it is white.
5:	Moving pitch scale with graduations at 2.5-degree intervals between -30 and +30 degrees (larger ones every 10°), and at 10-degree intervals from -/+30 to -/+50 degrees and at -/+ 70° and -/+90°. At a pitch angle of more than 30 degrees, large red arrow heads (in the shape of a "V") are displayed to show an excessive attitude and the direction to follow to decrease it. At a pitch angle of 0 degrees, the top half of the screen is custom shade sky (blue) and the bottom half is custom shade ground (brown).
LATERAL ACCELERATION DISPLAY (in option)	
6:	Lateral acceleration symbol
ILS DISPLAY	
7-8:	Horizontal (LOC) and vertical (G/S) scales.
9-10:	Magenta diamonds (LOC and G/S). If the ILS data is in the scale range, the diamonds are displayed. If the ILS data is out of the scale range, only a half-diamond is displayed in the pointing toward the center of the scale in accordance with the overtaking direction.
11:	"APP", "BCRS" green annunciator in case the ILS is in front course or backcourse approach.
NOTE:	For C16221QA01, the color is green + white.
MAGNETIC HEADING DISPLAY	
12:	Moving heading tape with graduations at 5-degree intervals and numerals at 10-degree intervals large numerals at 30 degree intervals.
13:	Fixed white heading triangle.
AIRSPEED DISPLAY	
14:	Moving airspeed tape with graduations from 30 to 500 Knots at 10-Knot intervals and numerals at 20-Knot intervals. The airspeed indication is grey tape positioned vertically up on the left of the horizon area. This tape is fitted with a digital read-out located in the center of the tape.
15:	Digital Airspeed number display with three white digits indicated the speed from 30 kts to 500kts. The resolution is 1 knot.
MACH NUMBER DISPLAY	

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- 16: The digital Mach number display with the letter M, decimal point and two digits. The characters are in white.

ALTITUDE DISPLAY

- 17: Moving altitude scale with graduations from -2000 ft to +50,000 ft at 200-ft intervals and numerals at 200-ft intervals.
- 18: A 5-digit readout in feet shows the altitude from -2000 ft to 50,000 ft. If the altitude is negative, the indication "-" is shown in white adjacent to the digital readout. For positive altitude less than 10,000 ft, the fifth significant figure position contains a solid filled green box. The resolution is 50 ft.
- 19: Digital readout with its "M" unit label that shows the altitude in meters if this option is selected. If the altitude is negative, the indication "-" is shown in white in front of the altitude value.

BAROMETRIC PRESSURE DISPLAY

- 20: Digital readout that shows the barometric pressure in hectopascals (hPa) from 745 to 1100 hPa and inches of mercury (in.Hg) from 22 to 32.48 in.Hg. If the standard barometric pressure is selected, the green letters "STD" are displayed in place of the digital readout.

NOTE: For C16221QA01, the letters "STD" are green + white.

ATTITUDE RESET ANNUNCIATION

- 21: The indication "ATT 10s" is shown if the instrument is in the attitude reset (caging) phase. In the caging phase, the roll and pitch scales and the lateral acceleration symbol are not displayed.

(d) ON-AIRCRAFT MAINTENANCE Mode

- 1 This mode can be engaged on the aircraft at the end of the alignment phase. To enter the mode, press the "APP" and "HP/IN" buttons together for two seconds.
- 2 Eleven maintenance pages are available ([Refer to Figure 12](#)) To go from one page to another, press the "+" or "-" buttons.

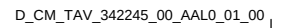


Figure 12/Graphic 34-22-45-991-012-A01
ON-AIRCRAFT MAINTENANCE Display

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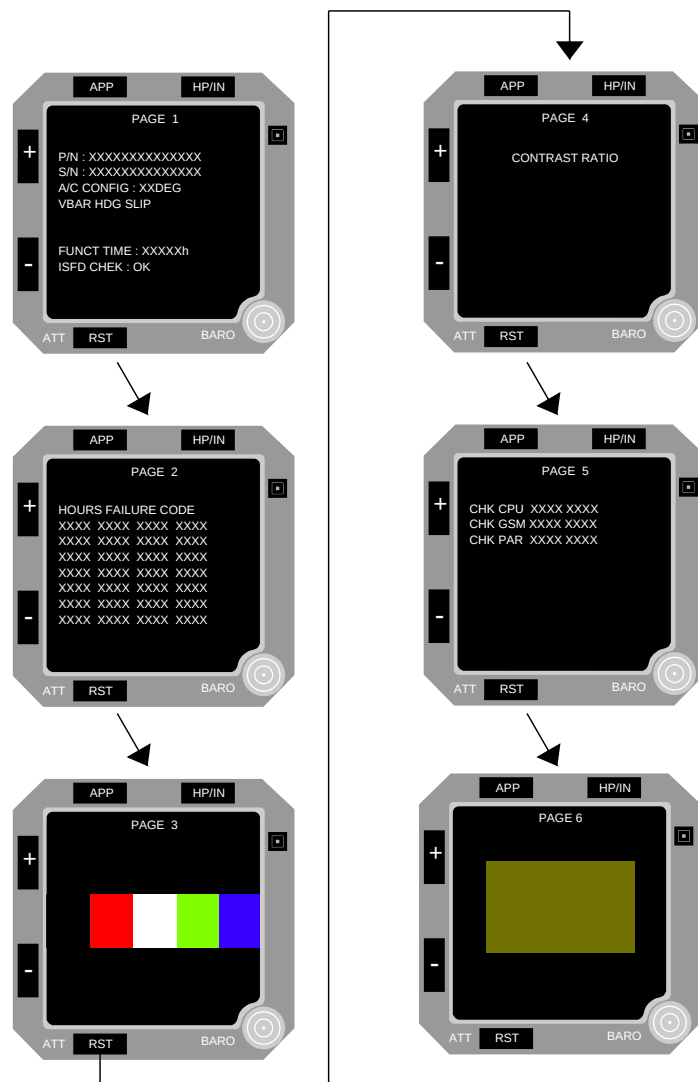
(e) SHOP MAINTENANCE Mode

- 1 The instrument goes into this mode at the end of the initialization phase when the discretes are in configuration 3.
- 2 Six maintenance pages are available ([Refer to Figure 13](#)) To go from one page to another, press the "APP" button or transmit the "ATT" word on the RS422 input bus.

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Figure 13/Graphic 34-22-45-991-013-A01
SHOP MAINTENANCE Display

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(f) DOWNLOADING Mode

- 1 The instrument goes into this mode at the end of the power-up built-in test when the discretes are in configuration 2.
- 2 In this mode, the instrument can dialog with the external systems through the RSS422 bus to download software. The data exchange rate on the bus is 115.2 Kb/s.

(g) SIMU Mode

- 1 The instrument goes into this mode at the end of the power-up built-in test when the discretes are in configuration 4.
- 2 SIMU mode includes two submodes: Simulator submode (use on flight simulator) and simulation submode (used during maintenance). The difference between these two submodes is as follows:
 - In simulation submode, the instrument receives the body rates and body accelerations (labels 326, 327, 330, 331 and 322) and the pitot and static pressures (labels 242 and 245) on the ARINC input buses No. 3 and No. 1.
 - In simulator submode, the instrument receives the pitch and roll angles (labels 325 and 324). Simulator submode is selected automatically if these labels are received.

NOTE: The pitot and static pressures are the same on the simulation submode.

(h) Other Modes

1 DOWNGRADED (LOW POWER) Mode

- a If there is an interruption of the 28VDC supply voltage for more than 50ms, the instrument changes to DOWNGRADED mode. In this mode, the display is lost but the computation of the attitude and air data parameters continues. In the case of power transients between 50ms and 200ms, the current configuration is stored in memory.
- b When the 28VDC supply voltage comes back:
 - If the power transient was shorter than 200ms, the instrument goes back to the same mode.
 - If the power transient was longer than 200ms, the instrument goes back to the initialization phase. It starts again with the same configuration as before the power transient.

2 INITIALIZATION Mode

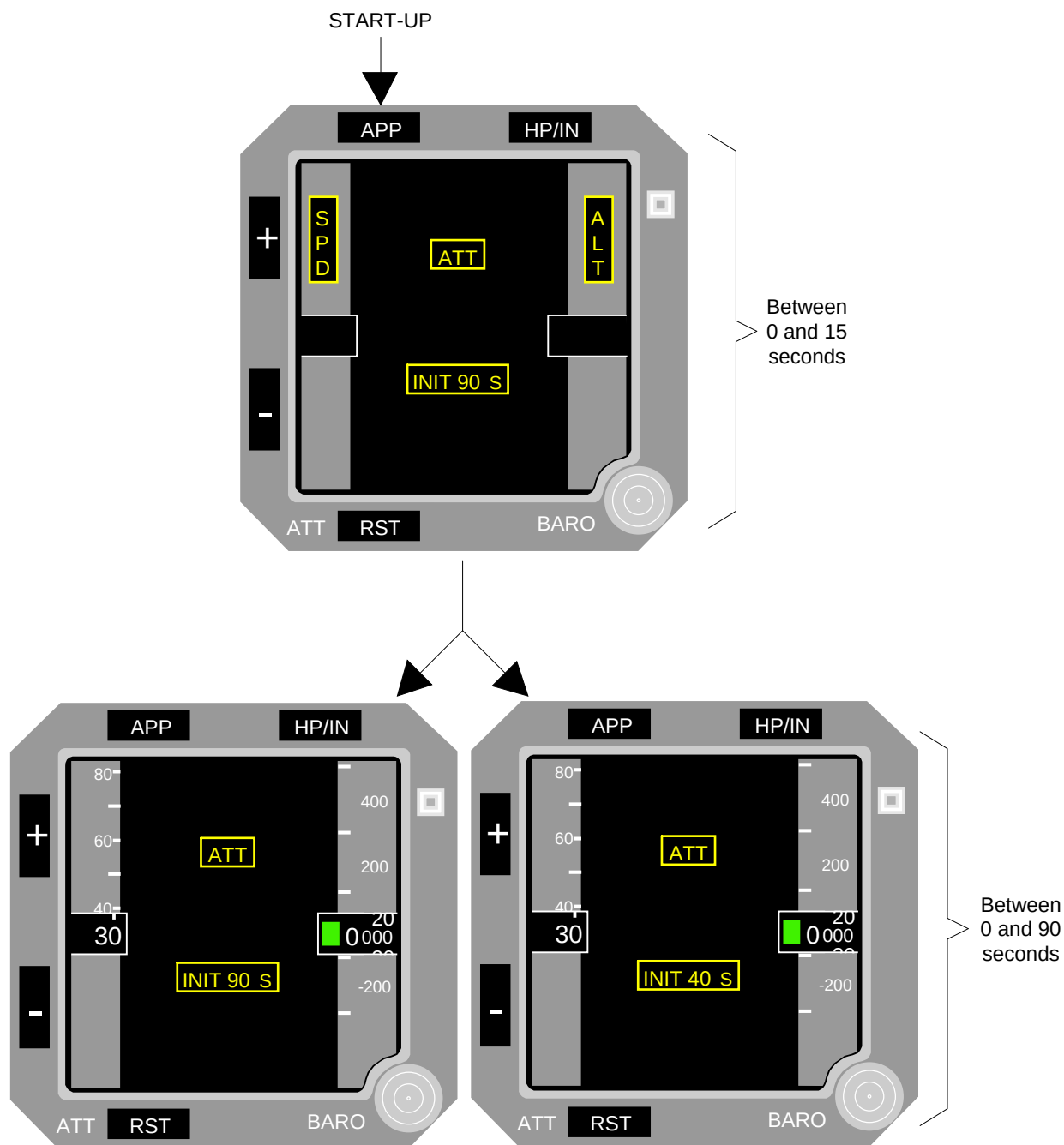
- a The instrument goes into this mode at STARTUP when the discretes are in configurations 1, 3 and 4.
- b The duration of initialization phase is 90 seconds (Refer to Figure 14) and is divided as follows:
 - The power-up built-in test (PBIT) sequence. The PBIT takes 8 seconds and does a test of the integrity of the instrument. If there is a failure of one of the functions, the related flag is displayed (but not for the external data: ILS and heading).
 - A phase during which the instrument receives the input discrete that sets the operating mode. At the end of the PBIT sequence, the instrument displays the message "INIT90s". For the versions C16221KA02/LA03/MA01, the indication "INIT 90s" decreases until zero.

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Figure 14/Graphic 34-22-45-991-014-A01
INIT Screen Mode

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(i) Submodes

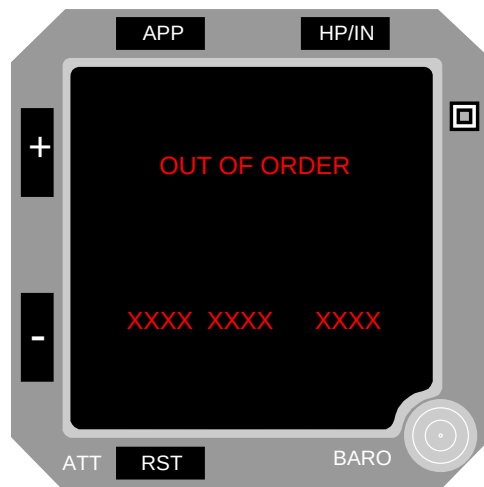
1 CAGING

- a This submode is engaged by action on the ATT/RST pushbutton and resets the attitude function to the true value in stabilized straight and level flight.

2 ILS

- a This submode is engaged/disengaged by action on the LS pushbutton. It displays the LOC and G/S scales and diamonds.

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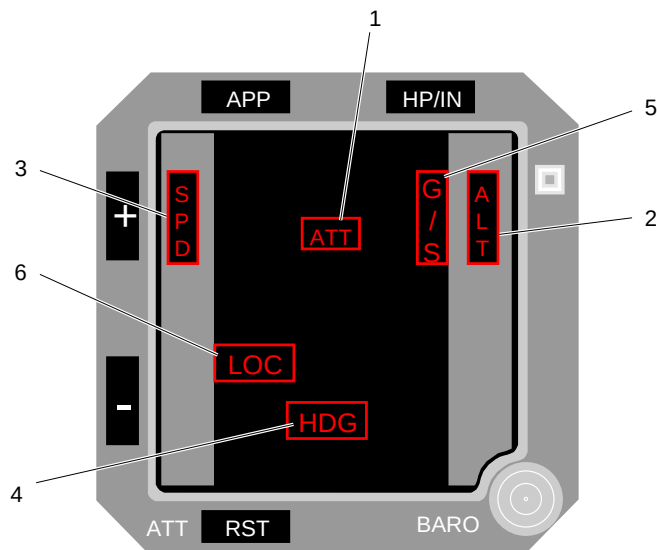
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Figure 15/Graphic 34-22-45-991-015-A01
FAIL Display

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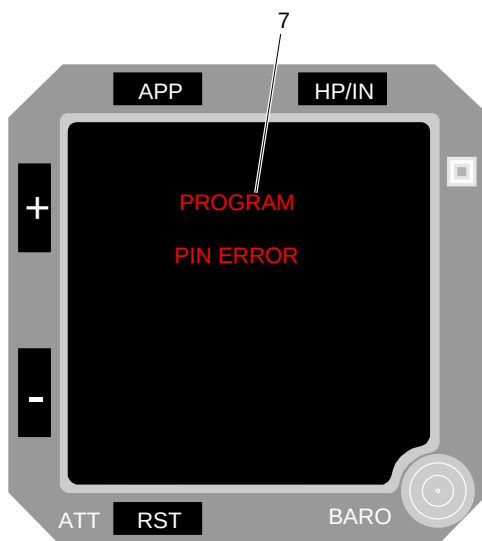
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Figure 16/Graphic - Sheet 01 / 34-22-45-991-016-A01
ERROR Display



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Figure 16/Graphic - Sheet 02 / 34-22-45-991-016-A01
ERROR Display

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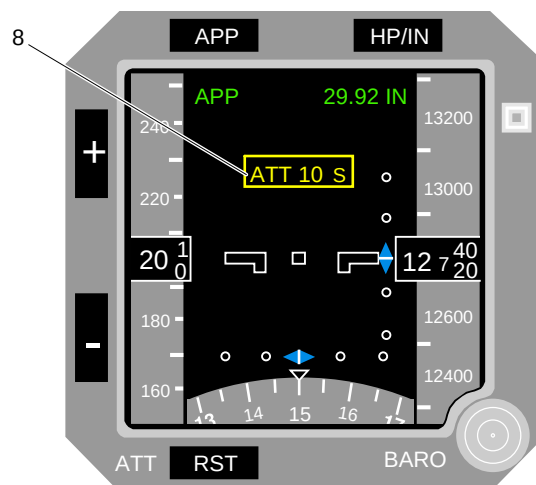
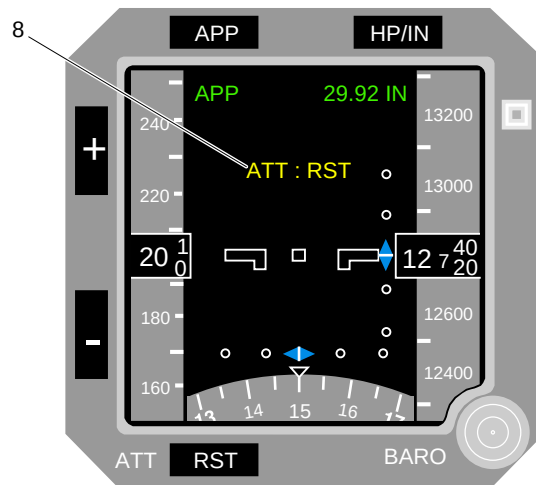
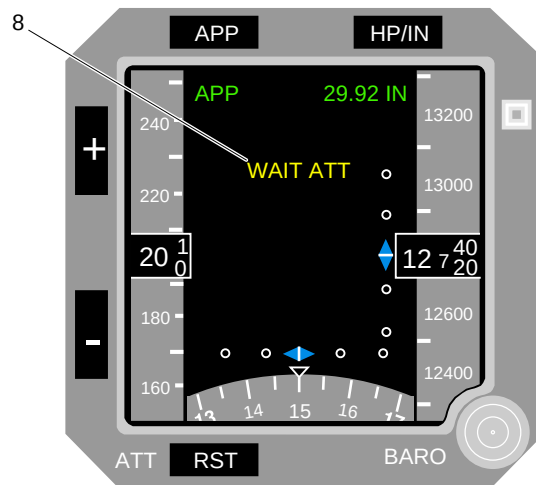
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Figure 16/Graphic - Sheet 03 / 34-22-45-991-016-A01
ERROR Display

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Table 15 : Legend for Figure 16

1:	A red ATT flag is shown in place of the gyro horizon display if the instrument cannot supply the attitude information for more than 10 seconds. In this case, the airplane symbol, pitch, roll and acceleration are not displayed. <u>NOTE:</u>
2:	A red ALT flag is shown in place of the altitude scale if the instrument cannot supply the altitude data.
3:	A red SPD flag is shown in place of the airspeed scale if the instrument cannot supply the airspeed data.
4:	A red HDG flag is shown in place of the heading scale if the magnetic heading label is invalid. In this case, the heading triangle index and scale are not displayed.
5:	A red G/S flag is shown in place of the G/S bar if the G/S label is invalid. G/S flag is displayed only in front course approach.
6:	A red LOC flag is shown in place of the LOC bar if the LOC label is invalid. LOC flag is displayed either in front course or in backcourse approach.
7:	A red PROGRAM PIN ERROR indication is displayed in two lines (PROGRAM on first line) is shown if the mode discrete or the pin programming parity are invalid.
8:	A yellow WAIT ATT indication is shown if one of the sensors is out of capture range. In this case, the roll and pitch scales, the lateral acceleration symbol and the airplane symbol are not displayed. If the sensor is out of the capture range for more than 10 seconds, the WAIT ATT flag is replaced by the ATT:RST yellow flag. In this case, the pilot must fly the aircraft in straight and level flight and at constant speed and then press the RST button to cage the instrument. The ATT:RST flag is then replaced by the indication "ATT 10s" in yellow. Only for C16221QA., a yellow ATT:RST is displayed after power-on limitactivation. This message is intended to prompt the flight crew to operate the "ATT:RST" pushbutton on the bezel. In this case, the roll and pitch scales, the lateral acceleration symbol and the airplane symbol are not displayed.

NOTE: The flags ATT, ALT, SPD, HDG, G/S and LOC are yellow for C16221QA.. indicated by an asterix * ([Refer to Table 18](#)).

(j) Hardware Operation

1 The instrument receives these types of data:

- Digital data on the ARINC 429 buses (one Low Speed --> APP Bus and one High Speed -> HDG Bus and only for C16221QA.. indicated by an asterix * ([Refer to Table 18](#)) two supplementary Low Speed --> Total and static pressures of SADM Bus).
- Analog data (0-5VAC/400Hz bezel dimming and +28VDC).
- Discrete data:
 - Parity check (Connector J1-k)
 - Activation of the HEADING display or Mach display (Connector J1-n)
 - The selection between two airplane symbol shapes (Connector J1-P)

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- Display of the altitude in meters (Connector J1-Z)
- IBIT request (Connector J1-q)
- Configuration of the operating mode discretes of the instrument (Connector J1-X,-h,-a)
- Back course, ILS selection (Connector J1-f,-n).
- Slip selection (Connector J1-s).
- Instrument panel configuration (Connector J1-b,-c,-d,-e).

NOTE: * The OPER mode is forced for these configurations.

Table 16 : Instrument Panel Configuration Discretes for C16221KA01/LA01

CONF_PL_4 (Connector J1-b)	CONF_PL_3 (Connector J1-c)	CONF_PL_2(Con nector J1-d)	CONF_PL_1(Con nector J1-e)	TILT ANGLE	AIR DATA AIDING
OC	OC	OC	OC	0 DEG (Lab.)	NO
OC	OC	OC	GND	18 DEG	YES
OC	OC	GND	OC	18 DEG	YES
OC	OC	GND	GND	22 DEG	YES
OC	GND	OC	OC	18 DEG	YES
OC	GND	OC	GND	16 DEG	YES
OC	GND	GND	OC	16 DEG	YES
OC	GND	GND	GND	0 DEG	YES
GND	OC	OC	OC	0 DEG	YES
GND	OC	OC	GND	0 DEG	YES
GND	OC	GND	OC	15 DEG	NO
GND	OC	GND	GND	10 DEG	NO
GND	GND	OC	OC	5 DEG	NO
GND	GND	OC	GND	5 DEG	NO
GND	GND	GND	OC	16 DEG	NO
GND	GND	GND	GND	0 DEG	YES

Table 17 : Instrument Panel Configuration Discretes for C16221KA02/LA02/LA03/MA01

CONF_PL_4 (Connector J1-b)	CONF_PL_3 (Connector J1-c)	CONF_PL_2(Con nector J1-d)	CONF_PL_1(Con nector J1-e)	TILT ANGLE	AIR DATA AIDING
OC	OC	OC	OC	0 DEG (Lab.)	NO
OC	OC	OC	GND	18 DEG	YES

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CONF_PL_4 (Connector J1-b)	CONF_PL_3 (Connector J1-c)	CONF_PL_2(Con nector J1-d)	CONF_PL_1(Con nector J1-e)	TILT ANGLE	AIR DATA AIDING
OC	OC	GND	OC	18 DEG	YES
OC	OC	GND	GND	22 DEG	YES
OC	GND	OC	OC	18 DEG	YES
OC	GND	OC	GND	16 DEG	YES
OC	GND	GND	OC	16 DEG	YES
OC	GND	GND	GND	0 DEG	YES
GND	OC	OC	OC	16 DEG	YES
GND	OC	OC	GND	16 DEG	YES
GND	OC	GND	OC	15 DEG	NO
GND	OC	GND	GND	10 DEG	NO
GND	GND	OC	OC	5 DEG	NO
GND	GND	OC	GND	5 DEG	NO
GND	GND	GND	OC	16 DEG	NO
GND	GND	GND	GND	0 DEG	YES

Table 18 : Instrument Panel Configuration Discretes for C16221QA01

CONF_PL_4 (Connector J1-b)	CONF_PL_3 (Connector J1-c)	CONF_PL_2(Con nector J1-d)	CONF_PL_1(Con nector J1-e)	TILT ANGLE	AIR DATA AIDING
OC	OC	OC	OC	0 DEG (Lab.)	NO
OC	OC	OC	GND	18 DEG	YES
OC	OC	GND	OC	18 DEG	YES
OC	OC	GND	GND	22 DEG	YES
OC	GND	OC	OC	18 DEG	YES
OC (*)	GND (*)	OC (*)	GND (*)	22 DEG (*)	YES
OC (*)	GND (*)	GND (*)	OC (*)	22 DEG (*)	YES
OC	GND	GND	GND	0 DEG	YES
GND	OC	OC	OC	16 DEG	YES
GND	OC	OC	GND	16 DEG	YES
GND (*)	OC (*)	GND (*)	OC (*)	0 DEG (*)	NO
GND	OC	GND	GND	10 DEG	NO

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CONF_PL_4 (Connector J1-b)	CONF_PL_3 (Connector J1-c)	CONF_PL_2(Con nector J1-d)	CONF_PL_1(Con nector J1-e)	TILT ANGLE	AIR DATA AIDING
GND	GND	OC	OC	5 DEG	NO
GND	GND	OC	GND	5 DEG	NO
GND	GND	GND	OC	16 DEG	NO
GND	GND	GND	GND	0 DEG	YES

Table 19 : Pin Programming Discretes

FUNCTION/STA TE	APP(Connector J1-n)	BCRS(Connect or J1-f)	MACHNUMBER (Connector J1- m)	VBAR(OPTION) (Connector J1- p)	ALTITUDE METER(OPTIO N) (Connector J1- Z)
OC	Enabled (OC)	Enabled (OC)	Enabled (GND)	Standard symbol (double cue) (OC)	No Alt.in meters (OC)

NOTE: * The OPER mode is forced for these configurations.

- 2 After acquisition of the data, the instrument processes them for display on the color LCD screen. The backlighting of the LCD screen is supplied from fluorescent lamps; it is controlled automatically from a light sensor on the front face and/or from the dimming control. The screen is a high-resolution active-matrix LCD of the flat panel type.
- 3 The instrument supplies these output data:
 - Digital data on the ARINC 429 buses (one Low Speed --> CMC bus and one High Speed -> INITIAL)
 - Discrete data for the status of the display.
- 4 The instrument is energized from a 28VDC power supply and also receives a 5VAC supply for bezel brightness control.
- 5 The RS422 line is used to download software and for the test of the instrument.

SUBTASK 34-22-45-870-002-A01

(2) Detailed Operation

(a) Illustrated Block Texts (Refer to Figure 17)

- 1 The detailed operation of the instrument is given in the Illustrated Block Texts (IBTs). Each IBT has two sheets:
 - A "Block Diagram" sheet which shows a schematic representation of the related circuit
 - A text sheet which gives details about the block diagram.

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Table 20 : Revision Reference Sheet for Figure 11

Revision Code	Sheet Number	Revision Description	Mod State

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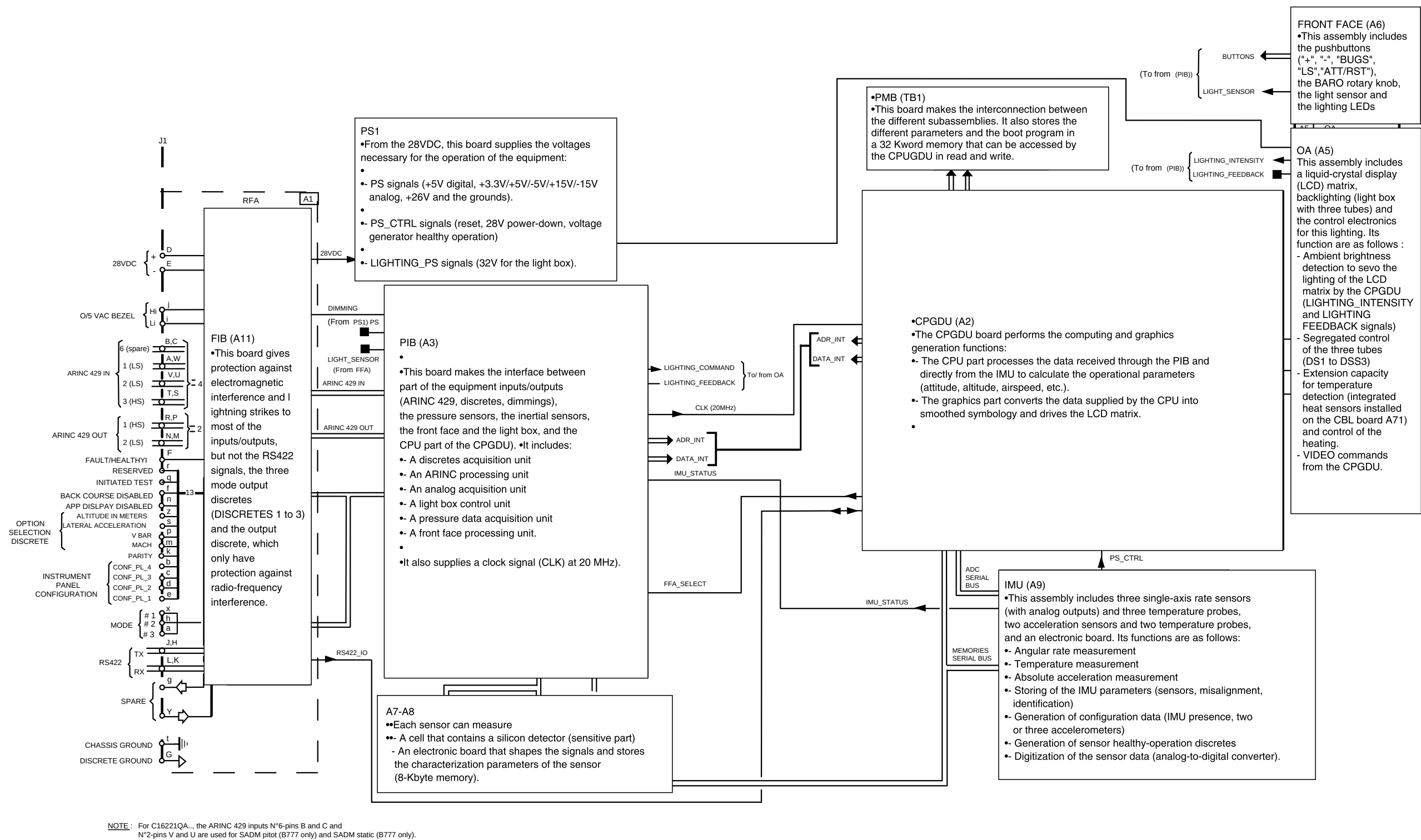


Figure 17/Graphic - Sheet 01 / 34-22-45-991-017-A01
General Block Diagram

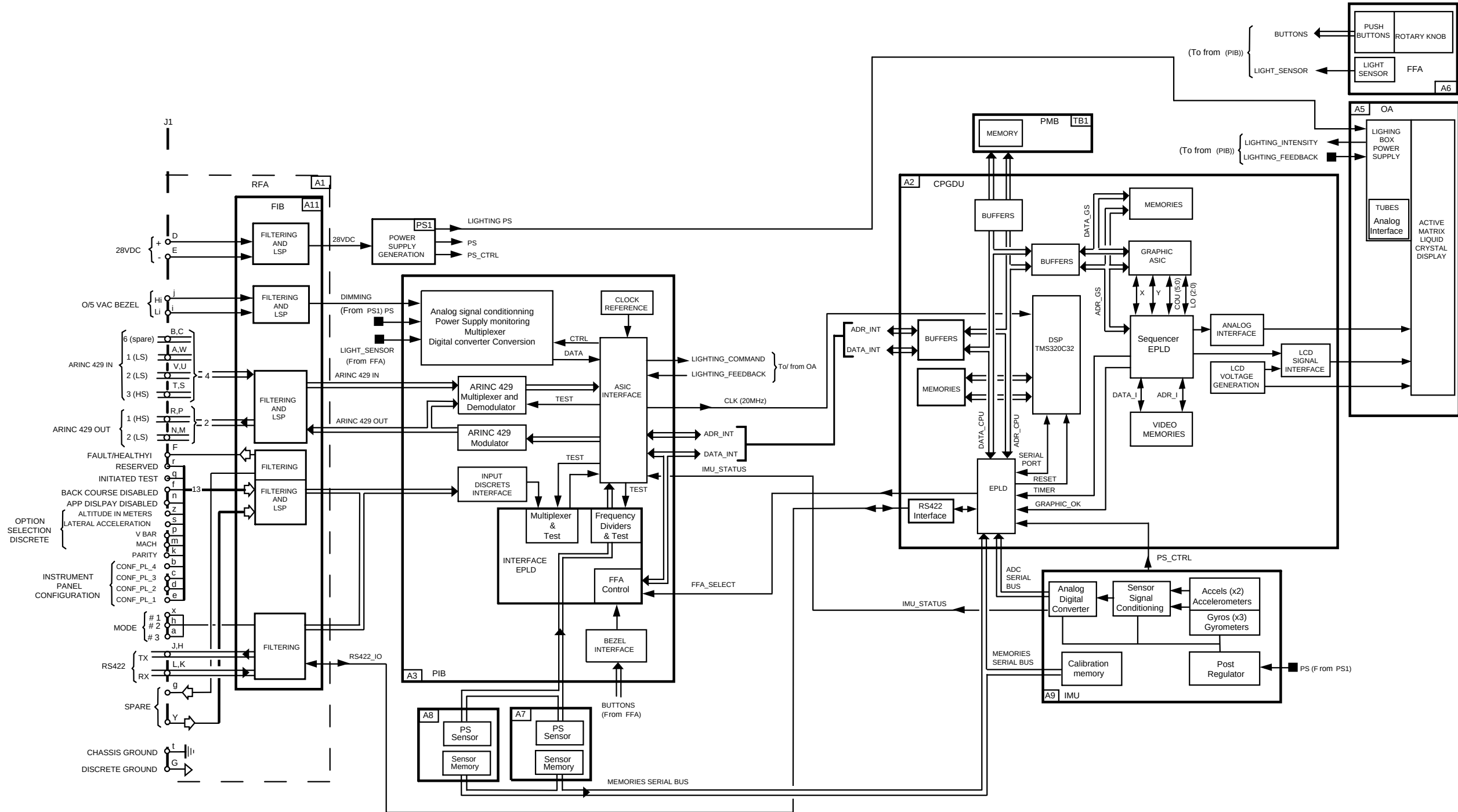
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NOTE: For C16221QA, the ARINC 429 inputs N°6-pins B and C and N°2-pins V and U are used for SADM pitot (B777 only) and SADM static (B777 only).

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Figure 17/Graphic - Sheet 02 / 34-22-45-991-017-A01
General Block Diagram

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TESTING AND FAULT ISOLATION

TASK 34-22-45-750-801-A01

1. Automatic Test requirements

A. Introduction

- (1) This paragraph lists the ATLAS Test Specification for testing the instrument.

B. General

- (1) The purpose of the ATLAS Test Specification is to provide the ATLAS test requirements for the instrument.

C. ATLAS Test Specification

- (1) Refer to CMM 34-22-45A, latest revision (available on our Web site).

D. Media

- (1) For every update of the ATS, the CMM 34-22-45A - Testing and Troubleshooting, will updated and posted on our Web site.
(2) ATLAS Test Specification is provided in PDF format and text format.

E. Ordering

- (1) Access to Thales Website: refer to the back of the Title Page (Page T-2) for website address and front office e-mail address.
(2) The user must have a username and password, to get access to secured technical publications.

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TASK 34-22-45-750-802-A01

2. General

A. Reason for the job

- (1) This section gives the operations to be done to make sure that the hardware operation of the equipment is correct.
- (2) The tests include the bonding and insulation resistance tests and the functional tests.

NOTE: All the tests are done with the equipment in a closed condition.

NOTE: During testing operations, make sure that the environmental conditions are as follows :

- Pressure : No specific environmental level other than the ambient pressure level is necessary.
- Relative humidity: Refer to SOPM 20-02-00, THAV-A-20-02-00-020F0-028A-C - Special instructions in relation to SSD - General procedure.

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

(a) Special Tools, Fixtures and Equipment

1 (Refer to [PGBLK-34-22-45-9000-A01](#))

(b) Standard Tools, Fixtures and Equipment

1 For the test of the repeater function:

- One power supply, 28V $\pm$ 05VDC/5A minimum (instrument power supply)
- One low-frequency generator 5V/400Hz, 10mA (for dimming control)
- Three ARINC 429 generators (one low speed and two high speed) for ILS output(for C16221KA.., C16221LA.. and C16221MA..)
- Four ARINC 429 generators (three low speed and one high speed) for ILS-SADM-ADIRU outputS (for C16221QA..)
- Two ARINC 429 receivers for parameter reception
 - . One low speed: 6 labels at 30 Hz over few minutes
 - . One high speed: 10 labels at 60 Hz over few minutes
- One RS422 transmitter (115200 bauds) (for output of test code)
- One RS422 receiver (115200 bauds) (for parameter reception)
- One personal computer for data collection (with ARINC and RS422 links described above)
- 15 two-position switches (Ground and Open positions) for status of discretes
- One discrete input (Ground and Open positions) for correct operation and test startup
- One DC voltmeter (50VDC scale)
- One brightness and color meter (Minolta model CS-100 or equivalent, refer to the NOTE).
- One light source with an intensity of 5000 Lux (Schott KL 1500 or equivalent)
- One cable compatible with the instrument
- One milliohmeter with a source 10 Amps

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- One ammeter, measurement range 5A, precision 1% (to measure the power consumption).
- One megohmmeter (50VDC - measurement range 0 - 100 Mohms)

NOTE: The performances of the brightness and color meter must be as follows:

- Selenium photocell with filtering to get as near as possible to the CIE relative photopic brightness
- Acceptance angle: 1 degree
- Measurement range: 0.001 to 14,590 fL
- Precision: $\pm 2\%$ ($\pm$ one number on the last digit).

2 For the test of the attitude functions:

a Either:

- One vertical support feature (0-degree tilt angle, precision 0.1 degrees in roll and pitch)
- One two-axis rotating table, remotely controlled, capable of 10 degrees/second in roll and pitch, position precision 0.1 degrees
- One time base (can be included in the personal computer), precision 0.2 seconds
- One artillery level, precision 3 seconds of arc.

b Or:

- One single-axis rotating table (rotation around the Z axis with a minimum value of 90 degrees and a rotation speed between 10 degrees/second and 95 degrees/second)
- One vertical support fixture (0-degree tilt angle, precision 0.1 degrees in roll and pitch)
- One time base (can be included in the personal computer), precision 0.2 seconds
- One support fixture, -15 degree and -16 degree tilt angle in pitch with a precision of 0.1 degrees
- One support fixture, 100 degree tilt in roll with a precision of 0.1 degrees
- One support fixture, +20 degree and -20 degree tilt in roll and pitch with a precision of 0.1 degrees.

3 For the test of the altitude and airspeed functions:

a Either:

- One pressure gauge, range 0hPa to 1600hPa, precision ± 0.2 hPa (for measurement of the pitot pressure)
- One barometer, range 100hPa to 1200hPa, precision ± 0.2 hPa (for measurement of the static pressure)
- Two pressure sources, 100hPa to 1600hPa, precision ± 0.2 hPa (for generation of the pitot and static pressure).

b Or:

- Two calibrated pressure sources, 100hPa to 1600hPa, precision ± 0.2 hPa (for generation and measurement of the pitot and static pressure).

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4 For the test of the lightning strike protection function:

- One pulse generator $\pm 240V \pm 10V$, $60\mu s \pm 10\mu s$ (for the generation of limited voltages)
- One oscilloscope.

(2) Test Conditions

- Unless the instructions tell you differently, the tests of the instrument are done in these conditions:
 - Position: normal operating position.
 - Temperature: $20^{\circ}C \pm 5^{\circ}C$.
 - Relative humidity: Refer to SOPM 20-02-00, THAV-A-20-02-00-020F0-028A-C - Special instructions in relation to SSD - General procedure.
 - Pressure: No specific environmental level other than the ambient pressure level.

C. Job Set-up

(1) Not applicable.

D. Procedure

(1) Not applicable.

E. Close-up

(1) Not applicable.

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TASK 34-22-45-750-803-A01

3. Tests for C16221KA.. and C16221LA..

- A. Reason for the job
 - (1) Not applicable.
- B. Job Set-up Information
 - (1) [\(Refer to TASK-34-22-45-750-802-A01\)](#)
- C. Job Set-up
 - (1) Not applicable.
- D. Procedure

SUBTASK 34-22-45-750-001-A01

- (1) Visual Inspection
 - (a) Examine all the inscriptions on the bezel from a distance of approximately 700 mm along a line that goes through the longitudinal axis of the instrument.
 - (b) Make sure that the unit is complete, that it has an inspection seal, and that protective covers are installed on the power supply connector and pressure unions.
 - (c) Make sure that no defect is visible to the naked eye.
 - (d) Make sure that the seals are in good condition.
 - (e) Make sure that the identification plates are correct [\(Refer to PGBLK-34-22-45-1-A01\)](#).
 - (f) The table that follows gives the cross reference between the Boeing part numbers and the Thales part numbers [\(Refer to Table 1001\)](#).

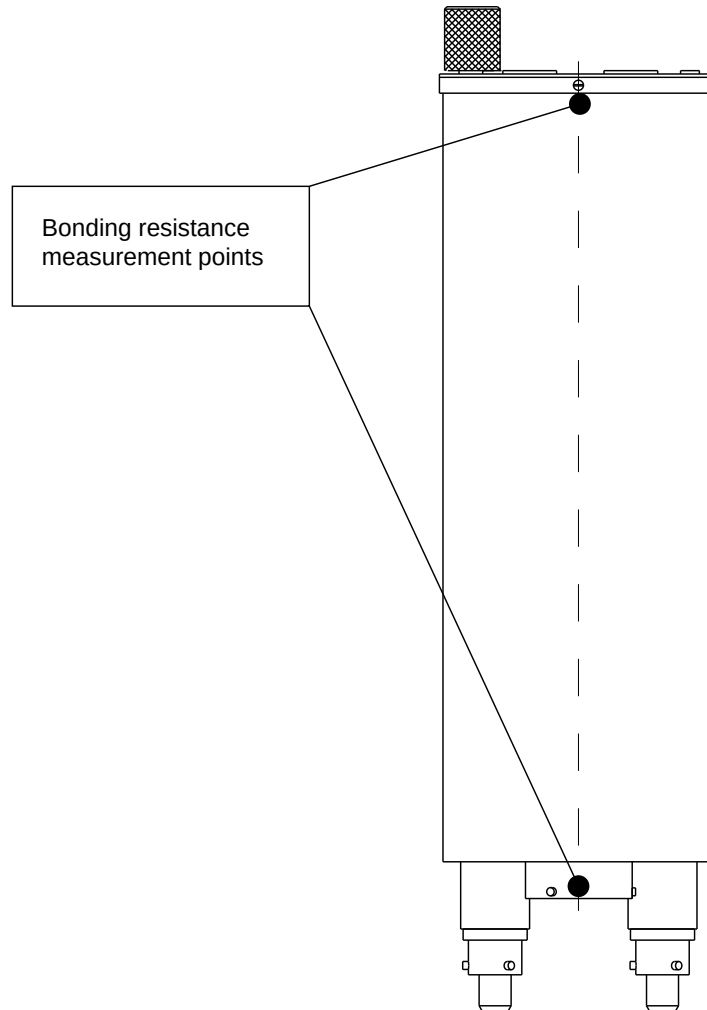
Table 1001 : Part Numbers Cross Table

BOEING PN	THALES PN
S231A120-1	C16221KA01
S231A120-5	C16221KA02
S231A120-2	C16221LA01
S231A120-3	C16221LA02
S231A120-6	C16221LA03

SUBTASK 34-22-45-750-002-A01

- (2) Electrical Bonding Test
 - (a) With a current of 10A applied, measure the bonding resistance between a point on the body of connector J1 and a point on the cover near the bezel [\(Refer to Figure 1001\)](#).
 - (b) Make sure that the bonding resistance is equal to or less than 10 mohms.

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Figure 1001/Graphic 34-22-45-991-101-A01
Bonding Resistance Measurement Points

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SUBTASK 34-22-45-750-003-A01

(3) Insulation Resistance Test

- (a) Set a megohmmeter to a voltage of 40VDC.
- (b) Measure the insulation resistance between pins D and E (40VDC input) and pin t (0VDC input).
- (c) Make sure that the insulation resistance is equal to or higher than 40 Mohms.

SUBTASK 34-22-45-750-004-A01

(4) Test of the Presence of the Lightning Strike Protection Diodes

- (a) Except C16221KA02 with amendment K and C16221LA03 with amendment K.

- 1 Make the test setup shown on the diagram ([Refer to Figure 1002](#)).
- 2 Do the calibration of the pulses at a voltage higher than the limited voltage under test as follows:
 - a Connect the pulse generator to an oscilloscope through a BNC cable.
 - b Set the repetition period to 10ms, the pulsewidth to $60\mu s \pm 10\mu s$ and the amplitude to $+240V \pm 10V$ and $-240V \pm 10V$.
- 3 Generate the pulses as follows:

NOTE: The limiting of the voltage applied is done in the instrument.

 - a Connect the pulse generator to the pin under test ([Refer to Table 1002](#)), with the ground on pin G.
 - b Transmit positive and then negative pulses, and read the limited voltage value for each type of pulse.
 - c ([Refer to Table 1002](#)) for the voltage values.

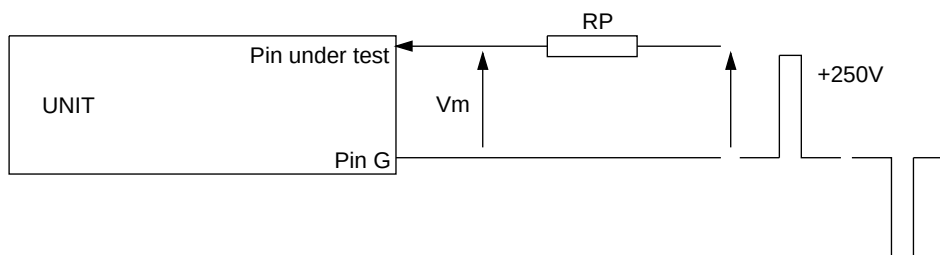
Table 1002 : Table of Test Pins

Signal	Pins	Pins under test	+240V pulse		-240V pulse	
			Rprotection (Ohms)	Vmeasured (V)	Rprotection (Ohms)	Vmeasured (V)
Power supply	D, E	D, E	255	50 ± 10	test not to be performed	
Bezel dimming	i, j	i, j	220K	60 ± 15	220K	-60 ± 15
ARINC In and spare	A,W,U,V,S,T,B,C	A,W,U,V,S,T,B,C	10K	35 ± 5	10K	-35 ± 5
ARINC Out	M,N,R,P	M,N,R,P	1K	15 ± 5	1K	-15 ± 5
IN discrete	b,c,d,e,k,m,p,r,s,q,Z,f	b,c,d,e,k,m,p,r,s	10K	35 ± 5	10K	-35 ± 5
OUT discrete	g	g	10K	60 ± 10	test not to be performed	

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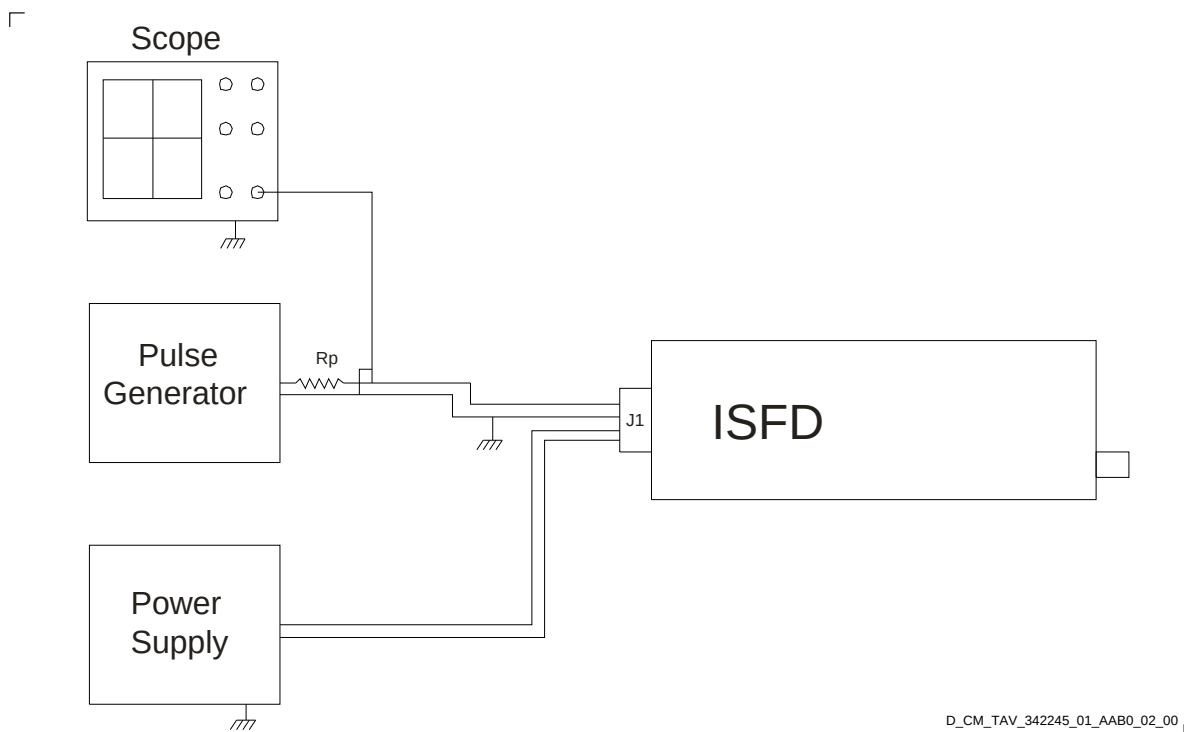
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Figure 1002/Graphic 34-22-45-991-102-A01
Test Setup for Presence of Lightning Strike Diodes

- (b) Only for C16221KA02 with amendment K and C16221LA03 with amendment K.
- 1 Make the test setup shown on the diagram
NOTE: The power supply is connected only for the test of ARINC429 output.
 - 2 Do the calibration of the pulses at a voltage higher than the limited voltage under test as follows:
 - a Connect the pulse generator to an oscilloscope through a BNC cable.
 - b Adjust pulses width at $60\mu s \pm 10\mu s$.
 - c Adjust pulse level at V_{test} on open circuit.
 - 3 ISIS connection
 - a Connect pin t to ground.
 - b If power supply needed, connect 28VDC power supply between pin D and E, and connect E to ground.
 - c Note: during test on pin D, E shall be disconnected.
 - 4 Generate the pulses as follows:
 - a Adjust pulses generator level at V_{test}
 - b Connect the pulse generator to the pin under test (Refer to Table 1003) through resistor R_p , with the ground on pin G.
 - c Apply 5 pulses.
 - d Transmit positive and then negative pulses, and read the limited voltage value for each type of pulse (Refer to Table 1003).

Table 1003 : Table of Test Pins

Signal	Pin	Power Supply	Vtest VDC	Rp	Measurement VDC	
					Upper Limit	Low Limit
28V 3 × SMCJ15A	D	OFF	240	255Ω	59	49
			-240		-1	-7
ARINC 429 Inputs	A, W, U, V, S, T, B,C	OFF	240	10kΩ	55	43
			-240		-43	-55
Discrete Inputs	b, c, d, e, k, m, p, r, s, f, q, Z	OFF	240	10kΩ	45	33
			-240		-33	-45
Discrete Input	n	OFF	240	10kΩ	55	43
			-240		-43	-55
Discrete outputs	g	OFF	80	10kΩ	55	45
			-80		-45	-55
ARINC 429 Outputs	M, N, R, P	ON	180	1kΩ	19.5	17.5
			-180		-17.5	-19.5



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Figure 1003/Graphic 34-22-45-991-102-A02
Test Setup for Presence of Lightning Strike Diodes

(c)

1 Dimming Inputs : Input impedance measurement (Power off)

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- a Connect +40V voltage source successively between pin i, j, and pin t (or pin G).
- b Measure input current i, j : $1.0\text{mA} < I < 1.3\text{mA}$,
- c Or Measure input impedance (V/I) : $30\text{k} < R < 40\text{k}$.

SUBTASK 34-22-45-750-005-A01

(5) Instrument Operating Position and Check of the Fault Memory

- (a) Attach the instrument to a suitable test fixture and connect it in the operational configuration ([Refer to Table 1004](#)).
- (b) Adjust the test fixture to 0 degrees in roll with a precision of ± 1 degree.
- (c) Adjust the test fixture to 0 degrees in pitch with a precision of ± 1 degree.
- (d) Change to maintenance mode as follows:
 - 1 Connect the discrete pins X and a to the common ground discrete (pin G).
- (e) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (f) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (g) Energize the instrument.
- (h) Transmit the word "S" on the RS422 input bus and make sure that the instrument transmits the content of the fault memory on the RS422 output bus. Record the result for subsequent comparison ([Refer to SUBTASK-34-22-45-750-024-A01](#)).

SUBTASK 34-22-45-750-006-A01

(6) Power Consumption and Validity Output Discrete Test

- (a) De-energize the instrument.
- (b) Connect the instrument in its operational configuration ([Refer to Table 1004](#)).
- (c) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (e) Energize the instrument in average ambient lighting conditions (between approximately 300 and 600 lux).
- (f) Let the instrument operate for one minute and then measure the power consumption.
- (g) Make sure that the power consumption is equal to or less than 0.7A.
- (h) Make sure that the validity output discrete F is connected to ground (pin G).

SUBTASK 34-22-45-750-007-A01

(7) Test of the Operating Configuration

- (a) Connect the instrument in the operational configuration ([Refer to Table 1004](#)).

Table 1004 : Operational Configuration for 0-degree Instrument Panel Slant Angle

J1 pin	SIGNAL	I/O	COMMENTS	WIRED
B		I	SPARE	
C		I	SPARE	
t	CHASSIS_GND		Case ground	YES
D	28V	I	+28V DC	YES

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E	0_28V	I	0V DC	YES
A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
W	ARINC_E_1_B	I		
j	0_LUM_SCR	I	Bezel Lighting Control	YES
V	ARINC_E_2_A	I	SPARE	YES
U	ARINC_E_2_B	I		
i	LUM_SCR	I	Bezel Lighting Control	YES
T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
S	ARINC_E_3_B	I		
h	DISCRET_E_1	I	Mode selection n°2	
R	ARINC_S_1_A	O	IR parameters for Cat III option	
P	ARINC_S_1_B	O		
g			SPARE	
N	ARINC_S_2_A	O	ADR parameters for Cat III option	
M	ARINC_S_2_B	O		
f	DISCRET_E_2	I	Backcourse display disabled	
X	DISCRET_E_3	I	Mode selection n°1	
Y			SPARE	
Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
a	DISCRET_E_5	I	Mode selection n°3	
b	DISCRET_E_6	I	A/C configuration 8/0	
c	DISCRET_E_7	I	A/C configuration 4/0	
d	DISCRET_E_8	I	A/C configuration 2/0	
e	DISCRET_E_9	I	A/C configuration 1/0	
k	DISCRET_E_10	I	A/C configuration parity	
m	DISCRET_E_11	I	Mag heading display disabled	
n	DISCRET_E_12	I	APP display disabled	
p	DISCRET_E_13	I	SPARE (for Vbar option)	
q	DISCRET_E_14	I	Initiated test	
r	DISCRET_E_15	I	SPARE	

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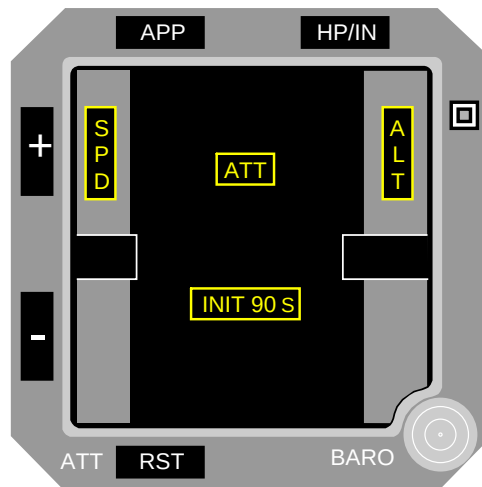
Component Maintenance Manual

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s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
G	ELEC_GND	I	Discrete common/ground	
L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	
K	RS422_E_L	I		
J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	
H	RS422_S_L	O		

- (b) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (c) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (d) Energize the instrument.
- (e) Make sure that the initialization phase 1 display is shown ([Refer to Figure 1004](#)).

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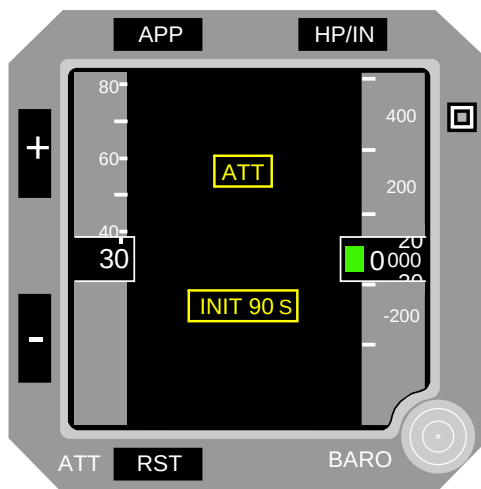
D_CM_TAV_342245_01_AAC0_01_00

Figure 1004/Graphic - Sheet 01 / 34-22-45-991-103-A01
Initialization Phase 1

THALES

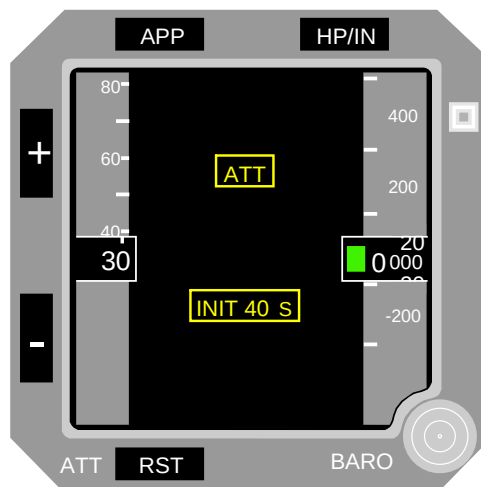
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D_CM_TAV_342245_01_AAC0_02_00

Figure 1004/Graphic - Sheet 02 / 34-22-45-991-103-A01
Initialization Phase 2 (For C16221KA01, C16221LA01 and C16221LA02)



D_CM_TAV_342245_01_AAC0_03_00

Figure 1004/Graphic - Sheet 03 / 34-22-45-991-103-A01
Initialization Phase 2 (For C16221KA02 and C16221LA03)

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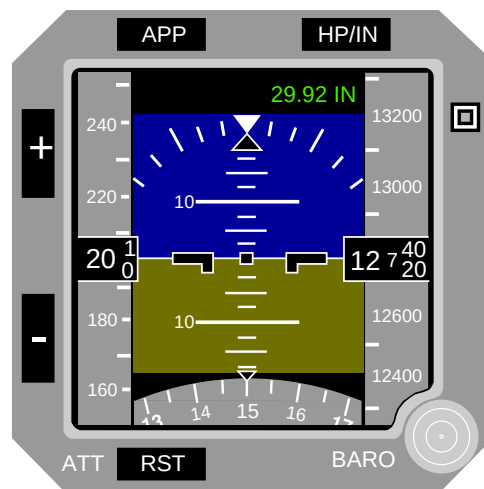
C16221K

- (f) Make sure that the display changes to the initialization phase 2 not more than 15 seconds after power-up (Refer to Figure 1004) or (Refer to Figure 1004).

NOTE: Ignore the altitude, airspeed and initialization time values shown on the display.

- (g) At the end of the initialization phase (duration less than 100 seconds), make sure that the instrument shows the operational display (Refer to Figure 1005).

NOTE: Ignore the barometric pressure unit (which can be HP for hectopascals or IN for inches of mercury) and the numeric values.



D_CM_TAV_342245_01_AAD0_01_00

Figure 1005/Graphic - Sheet 01 / 34-22-45-991-104-A01
Operational Display 1

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D_CM_TAV_342245_01_AAD0_02_00

Figure 1005/Graphic - Sheet 02 / 34-22-45-991-104-A01
Operational Display 2



D_CM_TAV_342245_01_AAD0_03_00

Figure 1005/Graphic - Sheet 03 / 34-22-45-991-104-A01
Operational Display 3

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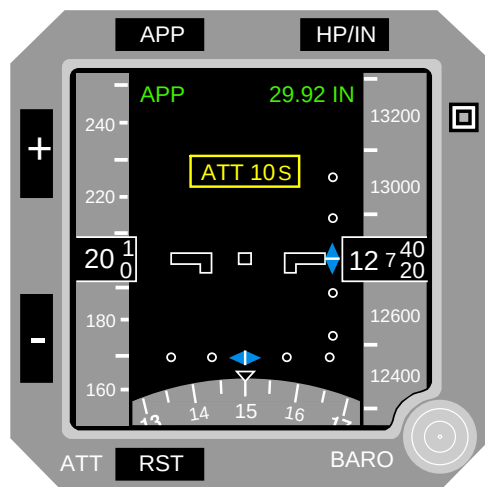
- (h) Make sure that the bezel does not have any defects that can interfere with its use.
- (i) Make sure that the markings and inscriptions are easy to see and read.

SUBTASK 34-22-45-750-008-A01

(8) Test of the Pushbuttons and Knobs

- (a) Press the "+" pushbutton once and make sure that the brightness increases.
- (b) Press the "-" pushbutton once and make sure that the brightness decreases.
- (c) Press the "APP" pushbutton three times and make sure that the operational display changes from no ILS to APP, then to BCRS, and then back to no ILS (Refer to Figure 1005).
- (d) Press the HP/IN pushbutton once and make sure that the barometric pressure unit changes (from HP to IN or from IN to HP).
- (e) Turn the BARO knob clockwise (approximately a quarter turn) and make sure that the barometric pressure value increases.
- (f) Turn the BARO knob counterclockwise (approximately a quarter turn) and make sure that the barometric pressure value decreases.
- (g) Press the BARO knob and make sure that the barometric setting changes from the adjustable barometric pressure to the standard value.
- (h) Press the RST pushbutton for 2 seconds. Make sure that the instrument resets to zero and that the message "ATT10s" is displayed (Refer to Figure 1006).

NOTE: Ignore the values displayed.



D_CM_TAV_342245_01_AAE0_01_00

Figure 1006/Graphic 34-22-45-991-105-A01
ATT 10s Page

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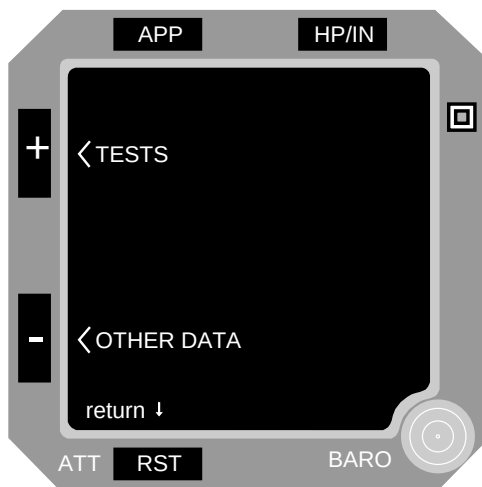
C16221K

SUBTASK 34-22-45-750-009-A01

(9) Test of the TEST Function

- (a) Press the HP/IN and APP buttons together for 2 seconds.
- (b) Make sure that the first test page is displayed ([Refer to Figure 1007](#)).

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D_CM_TAV_342245_01_AAF0_01_00

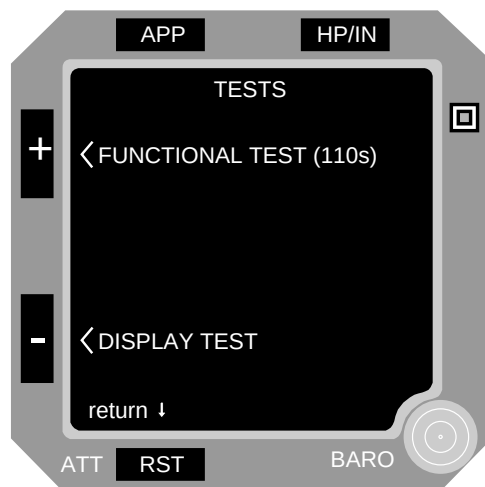
Figure 1007/Graphic 34-22-45-991-106-A01
First Test Page

- (c) Press the "+" button and make sure that the second test page is displayed ([Refer to Figure 1008](#)).

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D_CM_TAV_342245_01_AAG0_01_00

Figure 1008/Graphic 34-22-45-991-107-A01
Second Test Page

- (d) Press the "+" button again and make sure that "Test in progress 110s" is displayed in less than 10 seconds.
- (e) At the end of the test, make sure that "Test OK" is displayed.
- (f) Press the RST button three times and make sure that the screen goes back to the operational display (Refer to Figure 1005).
- (g) Connect the pin q to ground for at least 0.4 seconds and then put it in open circuit in less than 6 seconds.
- (h) Make sure that "Test in progress 110s" is displayed in less than 10 seconds.
- (i) After 110 seconds, make sure that the screen goes back to the operational display (Refer to Figure 1005).

SUBTASK 34-22-45-750-010-A01

(10) Test of the HEADING Function

- (a) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (b) Transmit the label 312 (with its value set to zero) and the label 320 (with its value set to 30) on the HS bus 3 (pins T and S).
- (c) Make sure that the heading value displayed is 30 degrees.
- (d) De-energize the instrument.
- (e) Connect the pins m and k to the discrete ground (pin G) to disable the heading display.
- (f) Energize the instrument.
- (g) At the end of the initialization phase, make sure that no heading value is displayed.
- (h) De-energize the instrument and put the pins m and k in open circuit.

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SUBTASK 34-22-45-750-011-A01

(11) Test of the ILS Function

(a) Test of the ILS Front Course Function

- 1 Transmit the label 173 (with its value set to 0.0775) and the label 174 (with its value set to 0.0875) on the LS bus 1 (pins A and W).
- 2 Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- 3 Energize the instrument.
- 4 At the end of the initialization phase, press the APP button to display the glideslope and localizer data.
- 5 Make sure that the APP indication is displayed in the upper- left corner.
- 6 Make sure that the ILS indications are correct ([Refer to Table 1005](#)).

Table 1005 : ILS Front Course Indications

	GLIDESLOPE		LOCALIZER	
	FLAG	BAR	FLAG	BAR
Position	Not in view	First mark down	Not in view	First mark right

(b) Test of the ILS Back Course Function

- 1 Press the APP button to change to backcourse approach mode.
- 2 Make sure that the BCRS indication is displayed in the upper- left corner.
- 3 Make sure that the ILS indications are correct ([Refer to Table 1006](#)).

Table 1006 : ILS Back Course Indications

	GLIDESLOPE		LOCALIZER	
	FLAG	BAR	FLAG	BAR
Position	Not in view	Not in view	Not in view	First mark left

- 4 Press the APP button and make sure that the localizer and glideslope data go out of view.

SUBTASK 34-22-45-750-012-A01

(12) Bezel Lighting Test

- (a) Put a mask in front of the photoelectric cell to simulate night lighting conditions.
- (b) Set the bezel lighting to the maximum brightness value 5VAC (7.07Vpeak) - 400Hz.
- (c) Make sure that the lighting of the pushbuttons (only) is uniform.
- (d) Make sure that there is no unwanted leakage of light.
- (e) For all PNRs C16221KA02 and C16221LA03.

NOTE: Except amendment K If amendment K is to be tested, it shall be retrofitted with amendment L before test.

- 1 Adjust the bezel lighting control to 1.945VAC (2.75Vpeak) - 400 Hz, minimum brightness before extinction.

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- a Make sure that the lighting of all buttons are lighted.
 - b Make sure that there is no unwanted leakage of light.
- 2 Adjust the bezel lighting control to 1.66VAC (2.35Vpeak) - 400 Hz, minimum brightness before extinction.
 - a Make sure that the lighting of all buttons are not lighted.

SUBTASK 34-22-45-750-013-A01

(13) Tests of the Options (V-bar, Slip and Altitude in Meters)

- (a) De-energize the instrument.
- (b) Connect the discretes for V-bar selection (pin p), slip selection (pin s) and altitude in meters selection (pin Z) to the discrete ground (pin G) and set the discrete on pin m to the open state.
- (c) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (e) Energize the instrument.
- (f) At the end of the initialization phase, make sure that the correct page is displayed ([Refer to Figure 1009](#)).

NOTE: Ignore the values displayed.



D_CM_TAV_342245_01_AA00_01_00

Figure 1009/Graphic 34-22-45-991-108-A01
V-Bars, Slip and Altitude In Meters Display

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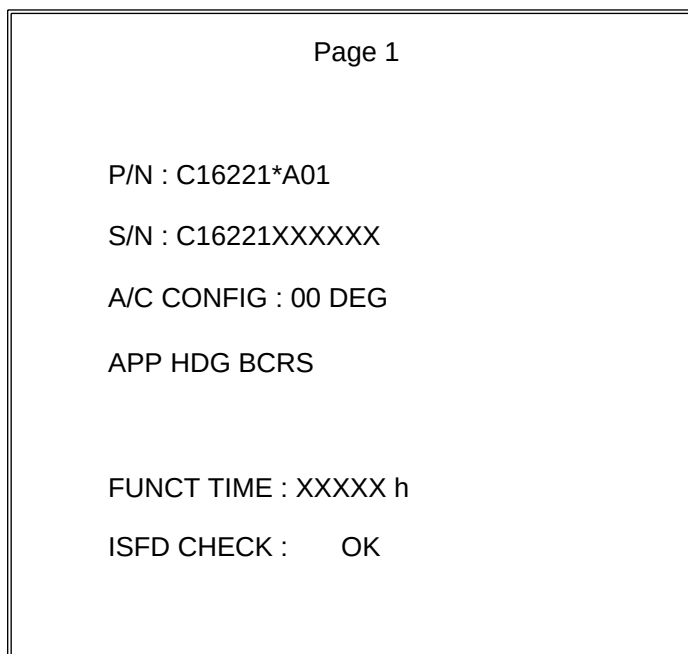
C16221K

SUBTASK 34-22-45-750-014-A01

(14) Display of Self-Test Results

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration ([Refer to Table 1004](#)).
- (c) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (d) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (e) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (f) Energize the instrument.
- (g) Make sure that the self-test page 1 is displayed in less than 10 seconds ([Refer to Figure 1010](#)).
 - If the instrument front face is gray, the instrument P/N is C16221KA.
 - If the instrument front face is brown, the instrument P/N is C16221LA.
- (h) Transmit the word "T" on the RS422 input bus and make sure that the related display data are transmitted on the RS422 output bus.
- (i) After 1 minute, press the RST button (or transmit the words "A" and then "B" on the RS422 bus) to make the display change to page 2 ([Refer to Figure 1010](#)).

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D_CM_TAV_342245_01_AA10_01_00

Figure 1010/Graphic 34-22-45-991-109-A01
Self-Test Display (Page 1)

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Page 2

HOURS	FAILURE CODE
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX

D_CM_TAV_342245_01_AA30_01_00

Figure 1011/Graphic 34-22-45-991-110-A01
Self-Test Display (Page 2)

- (j) Transmit the word "T" on the RS422 input bus and make sure that the related display data are transmitted on the RS422 output bus.

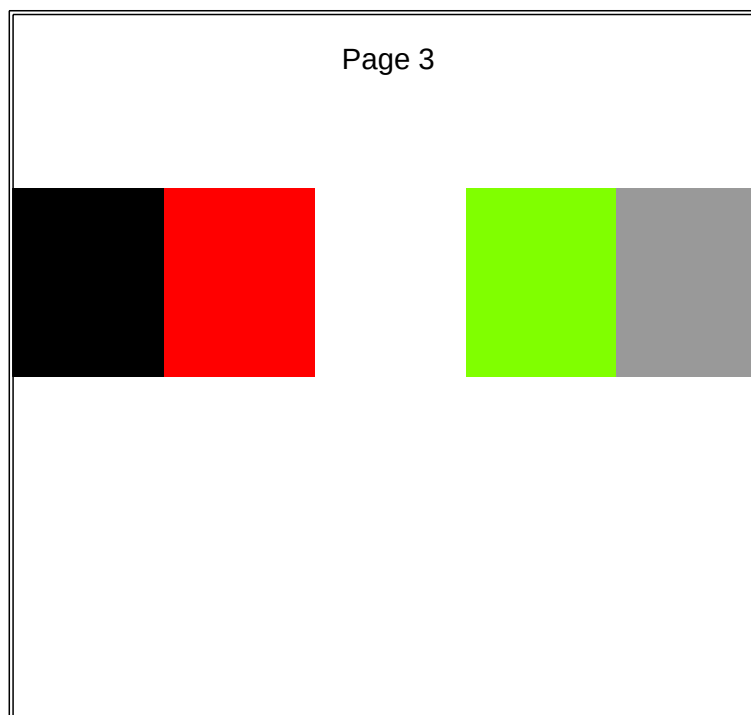
SUBTASK 34-22-45-750-015-A01

(15) Test of the Brightness and Contrast Test Displays

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 3 ([Refer to Figure 1012](#)).

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D_CM_TAV_342245_01_AAK0_01_00

Figure 1012/Graphic 34-22-45-991-111-A01
Self-Test Display (Page 3)

- (b) Press the RST pushbutton again (or transmit the words "A" and then "B" on the RS422 bus) and make sure that the display changes to page 4 ([Refer to Figure 1013](#)).

SUBTASK 34-22-45-750-016-A01

(16) Test of the Checksum Page Display

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 5 ([Refer to Figure 1014](#)).

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Page 4

CONTRAST RATIO

D_CM_TAV_342245_01_AAL0_01_00

Figure 1013/Graphic 34-22-45-991-112-A01
Self-Test Display (Page 4)

Page 5

CHK CPU	2254 BC16
CHK GS	0002 4865
CHK PAR	AAA5 DBBD

D_CM_TAV_342245_01_AAM0_01_00

Figure 1014/Graphic - Sheet 01 / 34-22-45-991-113-A01
Checksum Test Display (Page 5) (For C16221KA01 and C16221LA01)

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Page 5

CHK CPU 2245 BC16

CHK GS 0002 4865

CHK PAR B003 E56C

D_CM_TAV_342245_01_AAM0_02_00

Figure 1014/Graphic - Sheet 02 / 34-22-45-991-113-A01
Checksum Test Display (Page 5) (For C16221LA02)

Page 5

CHK CPU B867 33B7

CHK GS 0002 3270

CHK PAR 1B23 A0EF

D_CM_TAV_342245_01_AAM0_03_00

Figure 1014/Graphic - Sheet 03 / 34-22-45-991-113-A01
Checksum Test Display (Page 5) (For C16221KA02 and C16221LA03)

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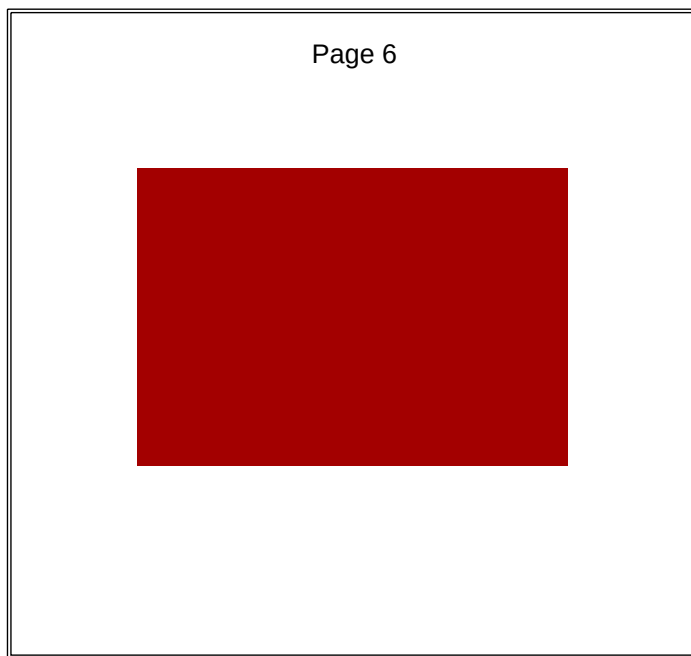
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SUBTASK 34-22-45-750-017-A01

(17) Test of the Color Test Page Display

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 6 [\(Refer to Figure 1015\)](#).

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D_CM_TAV_342245_01_AAN0_01_00

Figure 1015/Graphic 34-22-45-991-114-A01
Color Test Page (Page 6)

- (b) Make sure that the rectangle shows the ground color.
- (c) Press the RST button again and make sure that the rectangle changes to the sky color.
- (d) Press the RST button again and make sure that the rectangle changes to gray.
- (e) Press the RST button again and make sure that the rectangle changes to white.
- (f) Press the RST button again and make sure that the rectangle changes to red.
- (g) Press the RST button again and make sure that the rectangle changes to green.
- (h) Press the RST button again and make sure that the rectangle changes to yellow.
- (i) Press the RST button again and make sure that the rectangle changes to magenta.
- (j) De-energize the instrument.

SUBTASK 34-22-45-750-018-A01

(18) Test of the Display Brightness and Color

NOTE: All the brightness tests are done in the dark (ambient lighting less than 0.3 fc or 3 Lux). The brightness and color are measured perpendicular to the screen.

- (a) Connect the instrument in the operational configuration [\(Refer to Table 1004\)](#).
- (b) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.

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- (c) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (e) Energize the instrument.
- (f) Press the RST button two times (or transmit the words "A" and then "B" two times on the RS422 input bus) to make the display change to page 3 ([Refer to Figure 1012](#)).
- (g) Measure the brightness value at the center of the white square.

SUBTASK 34-22-45-750-019-A01

(19) Test of the Display Brightness in Night Conditions

- (a) Put a mask in front of the photoelectric cell to simulate night conditions.
- (b) Press the "-" button for approximately 6 seconds (or transmit the word "C" and then, after 6 seconds, the word "D" on the RS422 input bus) to set the screen brightness to the minimum value.
- (c) Let the instrument stabilize at this value for 3 minutes.
- (d) Make sure that the brightness value is between 0.02fL and 0.3fL ($0. \pm 9 \text{ Cd/m}^2$ and 1.03 Cd/m^2).

SUBTASK 34-22-45-750-020-A01

(20) Test of the Display Brightness in Full Sunlight Conditions

- (a) Apply a lighting level of 5000 Lux minimum to the photoelectric cell to simulate full sunlight conditions.
- (b) Press the "+" button for approximately 6 seconds (or transmit the word "E" and then, after 6 seconds, the word "F" on the RS422 input bus) to set the screen brightness to the maximum value.
- (c) Let the instrument stabilize at this value for 3 minutes.
- (d) Make sure that the brightness value is between 180fL and 205fL (617 Cd/m^2 and 702 Cd/m^2).

SUBTASK 34-22-45-750-021-A01

(21) Test of the Ground and Sky Color Coordinates

- (a) Press the RST button (or transmit the word "A" and then "B" on the RS422 input bus three times) to change to page 6a ([Refer to Figure 1015](#)).
- (b) Measure the color coordinates in the center of the colored rectangle.
- (c) Make sure that the 1976 CIE color coordinates measures are as follows:
- (d) Make sure that the brightness value is between 180fL and 205fL (617 Cd/m^2 and 702 Cd/m^2).
 - Ground: $u' = 0.22 \pm 0.02$, $v' = 0.54 \pm 0.02$
- (e) Press the RST button (or transmit the word "A" and then "B" on the RS422 input bus one time) to change to page 6b ([Refer to Figure 1015](#)).
- (f) Make sure that the brightness value is between 180fL and 205fL (617 Cd/m^2 and 702 Cd/m^2).
 - Sky: $u' = 0.14 \pm 0.03$, $v' = 0.360 \pm 0.03$.

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SUBTASK 34-22-45-750-022-A01

(22) Tests of the Attitude Functions

NOTE: For this test, it is possible to use two procedures, Procedure 1 or Procedure 2.

(a) Procedure 1

- 1 Measure the ambient room temperature and the temperature at a point on the instrument case near the bezel (Refer to Figure 1001) for the measurement point).
- 2 Make sure that the difference between the two temperatures is less than 3°C.
- 3 Install the instrument on the test support designed for an instrument panel slant angle of 0 degrees, with a precision of 0.1 degrees in roll and pitch.
- 4 Install the assembly on a two-axis table.
- 5 Change to operational mode with RS422 (Refer to Table 1007) .

Table 1007 : Operational Configuration for 0-degree Instrument Panel Slant Angle with RS422

J1 No.	J1 pin	SIGNAL	I/O	COMMENTS	WIRED
1	B		I	SPARE	
2	C		I	SPARE	
3	t	CHASSIS_GND		Case ground	YES
4	D	28V	I	+28V DC	YES
5	E	0_28V	I	0V DC	YES
6	A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
7	W	ARINC_E_1_B	I		
8	j	0_LUM_SCR	I	Bezel Lighting Control	YES
9	V	ARINC_E_2_A	I	SPARE	YES
10	U	ARINC_E_2_B	I		
11	i	LUM_SCR	I	Bezel Lighting Control	YES
12	T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
13	S	ARINC_E_3_B	I		
14	h	DISCRET_E_1	I	Mode selection n°2	
15	R	ARINC_S_1_A	O	IR parameters for Cat III option	
16	P	ARINC_S_1_B	O		
17	g			SPARE	
18	N	ARINC_S_2_A	O	ADR parameters for Cat III option	
19	M	ARINC_S_2_B	O		

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20	f	DISCRET_E_2	I	Backcourse display disabled	
21	X	DISCRET_E_3	I	Mode selection n°1	
22	Y			SPARE	
23	Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
24	a	DISCRET_E_5	I	Mode selection n°3	
25	b	DISCRET_E_6	I	A/C configuration 8/0	
26	c	DISCRET_E_7	I	A/C configuration 4/0	
27	d	DISCRET_E_8	I	A/C configuration 2/0	
28	e	DISCRET_E_9	I	A/C configuration 1/0	
29	k	DISCRET_E_10	I	A/C configuration parity	
30	m	DISCRET_E_11	I	Mag heading display disabled	
31	n	DISCRET_E_12	I	APP display disabled	
32	p	DISCRET_E_13	I	SPARE (for Vbar option)	
33	q	DISCRET_E_14	I	Initiated test	
34	r	DISCRET_E_15	I	SPARE	
35	s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
36	F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
37	G	ELEC_GND	I	Discrete common/ground	
38	L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
39	K	RS422_E_L	I		
40	J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
41	H	RS422_S_H	O		

- 6 Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- 7 Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- 8 Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:
 - a For C16221KA01, C16221LA01 and C16221LA02
 - Pitch: label 324
 - Roll: label 325
 - p: label 327

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- q: label 326
- r: label 330
- Xacc: label 331
- Yacc: label 332
- $\epsilon\theta$: label 353
- $\epsilon\Phi$: label 352
- $\epsilon\psi$: label 155
- dpi: label 157
- dqj: label 160
- dri: label 161
- Sw_NAV: label 351

NOTE: For the labels 331 and 332, the resolution is improved (LSB = 11 and not 17 as before). It is necessary to take this modification into consideration for the tests done on these labels.

b For C16221KA02 and C16221LA03

- Pitch: label 324
- Roll: label 325
- p: label 327
- q: label 326
- r: label 330
- Xacc: label 331
- Yacc: label 332
- dpi: label 157
- dqj: label 160
- dri: label 161
- Sw_NAV1: label 351
- Sw_NAV2: label 352

9 Initialization Flag

- a Energize the instrument.
- b Make sure that the labels are present on the ARINC reader.
- c For versions C16221KA01, C16221LA01 and C16221LA02, read the discrete B_INI_NAV on label 351 (bit 14 changes from 0 to 1) to make sure that the attitude-function initialization time of the instrument is equal to or less than 91 seconds.
- d For versions C16221KA02 and C16221LA03, read the discrete S_O_B_NAV_MODE on label 351 (bit 12 changes from 0 to 1) to make sure that the attitude-function initialization time of the instrument is equal to or less than 91 seconds.
- e For versions C16221KA02 and C16221LA03, read the discrete S_I_B_3ACC on label 352 (bit 18 = 0) to make sure that there are only two accelerometers.

NOTE: The numbers of the 32 bits go from 1 to 32.

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10 Horizon Line Movements

- a Record the values of the labels 324 (pitch) and 325 (roll) for one minute (for C16221KA01, C16221LA01 and C16221LA02) or three minutes (for C16221KA02 and C16221LA03) at a frequency of 60Hz.
- b For the pitch and roll parameters, make sure that:
 - The average value calculated on the recorded samples is equal to 0 ± 0.5 degrees.
 - The noise value calculated on the recorded samples is less than 0.15 degrees for 1 sigma (for C16221KA01, C16221LA01 and C16221LA02) or less than 0.17 degrees for 2 sigma (for C16221KA02 and C16221LA03).
- c For versions C16221KA01, C16221LA01 and C16221LA02, read the values of the labels 157 (dpi), 160 (dqi) and 161 (dri).
- d Make sure that the average value of the dpi, dqi and dri parameters calculated on the recorded samples is less than 2 degrees/second.
- e Adjust the table to 20 degrees nose-up in pitch at a rate of 5 deg/sec.
- f After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- g Make sure that the average pitch value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- h Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- i Adjust the table to 20 degrees nose-down in pitch at a rate of 5 deg/sec.
- j Record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- k Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- l Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- m Reset the table to 0 degrees in pitch at the rate of 5 deg/sec.
- n After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- o Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- p Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- q Adjust the table to 20 degrees in clockwise roll at a rate of 5 deg/sec.
- r After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- s Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- t Make sure that the average roll value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- u Adjust the table to 20 degrees in counterclockwise roll at a rate of 5 deg/sec.

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- v After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- w Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- x Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- y Reset the table to 0 degrees in roll at the rate of 5 deg/sec.
- z After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- aa Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- ab Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- ac Adjust the table to 5 degrees in clockwise roll and, at the same time, to 5 degrees in nose-up pitch at the rate of 20 deg/sec.
- ad After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- ae Make sure that the average pitch value calculated on the recorded sample is equal to 5 ± 0.5 degrees.
- af Make sure that the average roll value calculated on the recorded sample is equal to 5 ± 0.5 degrees.
- ag Reset the table to 0 degrees in roll and pitch at a rate of 5 deg/sec.
- ah After one minute, record the values of labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- ai Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- aj Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

11 Instrument Panel Slant Angle Programming Pins

- a De-energize the instrument.
- b Install the instrument on a support fixture set for an instrument panel slant angle of 16 degrees.
- c [\(Refer to Table 1008\)](#) for the test setup.

Table 1008 : Operational Configuration for 16-degree Instrument Panel Slant Angle

J1 n°	J1 pin	SIGNAL	I/O	COMMENTS	WIRED
1	B		I	SPARE	
2	C		I	SPARE	
3	t	CHASSIS_GND		Case ground	YES

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4	D	28V	I	+28V DC	YES
5	E	0_28V	I	0V DC	YES
6	A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
7	W	ARINC_E_1_B	I		
8	j	0_LUM_SCR	I	Bezel Lighting Control	YES
9	V	ARINC_E_2_A	I	SPARE	YES
10	U	ARINC_E_2_B	I		
11	i	LUM_SCR	I	Bezel Lighting Control	YES
12	T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
13	S	ARINC_E_3_B	I		
14	h	DISCRET_E_1	I	Mode selection n°2	
15	R	ARINC_S_1_A	O	IR parameters for Cat III option	
16	P	ARINC_S_1_B	O		
17	g			SPARE	
18	N	ARINC_S_2_A	O	ADR parameters for Cat III option	
19	M	ARINC_S_2_B	O		
20	f	DISCRET_E_2	I	Backcourse display disabled	
21	X	DISCRET_E_3	I	Mode selection n°1	
22	Y			SPARE	
23	Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
24	a	DISCRET_E_5	I	Mode selection n°3	
25	b	DISCRET_E_6	I	A/C configuration 8/0	
26	c	DISCRET_E_7	I	A/C configuration 4/0	ground
27	d	DISCRET_E_8	I	A/C configuration 2/0	
28	e	DISCRET_E_9	I	A/C configuration 1/0	ground
29	k	DISCRET_E_10	I	A/C configuration parity	
30	m	DISCRET_E_11	I	Mag heading display disabled	
31	n	DISCRET_E_12	I	APP display disabled	
32	p	DISCRET_E_13	I	SPARE (for Vbar option)	
33	q	DISCRET_E_14	I	Initiated test	

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34	r	DISCRET_E_15	I	SPARE	
35	s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
36	F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
37	G	ELEC_GND	I	Discrete common/ground	
38	L	RS422_E_H	I	Downloading	YES
39	K	RS422_E_L	I		
40	J	RS422_S_H	O	Downloading	YES
41	H	RS422_S_H	O		

- d Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- e Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- f Energize the instrument.
- g At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for a period of one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- h Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- i Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- j De-energize the instrument.
- k Adjust the table to an instrument panel slant angle of 15 degrees.
- l Connect the pins b and d to ground (pin G) and put the pins c and e in open circuit.
- m Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- n Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- o Energize the instrument.
- p At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- q Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- r Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

12 Instrument Steady State Test

- a This test can be done after the instrument panel slant angle programming pins test or during the test of the altitude and airspeed functions ([Refer to SUBTASK-34-22-45-750-023-A01](#)).

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- b Install the instrument on a vertical support fixture set to an instrument panel slant angle of 0 degrees in roll and pitch with a precision of 0.1 degrees.
- c Attach the instrument and support to a two-axis table set to 0 degrees in roll and pitch, or to a horizontal surface table with a precision of 0.1 degrees.
- d Change to operational mode ([Refer to Table 1007](#)) .
- e Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- f Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).

NOTE: Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:

- Pitch: label 324
- Roll: label 325.

- g Energize the instrument.
- h At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- i Make sure that the average values calculated on the recorded samples are equal to 0 ± 0.5 degrees.
- j Let the instrument operate for 1 hour 30 minutes.
- k At the end of this time, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- l Make sure that the average values calculated on the recorded samples are equal to 0 ± 0.5 degrees.

(b) Procedure 2

- 1 Measure the ambient room temperature and the temperature at a point on the instrument case near the bezel ([Refer to Figure 1001](#) for the measurement point).
- 2 Make sure that the difference between the two temperatures is less than 3°C.
- 3 Install the instrument on the test fixture designed for an instrument panel slant angle of 0 degrees, with a precision of 0.1 degrees in roll and pitch.
- 4 Attach the assembly to a single-axis table.
- 5 Change to operational mode with RS422 ([Refer to Table 1007](#)) .
- 6 Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- 7 Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- 8 Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:
 - a For C16221KA01, C16221LA01 and C16221LA02
 - Pitch: label 324
 - Roll: label 325

- p: label 327
- q: label 326
- r: label 330
- Xacc: label 331
- Yacc: label 332
- $\epsilon\Phi$: label 352
- $\epsilon\theta$: label 353
- $\epsilon\psi$: label 155
- dpi: label 157
- dqi: label 160
- dri: label 161
- Sw_NAV: label 351.

b For C16221KA02 and C16221LA03

- Pitch: label 324
- Roll: label 325
- p: label 327
- q: label 326
- r: label 330
- Xacc: label 331
- Yacc: label 332
- dpi: label 157
- dqi: label 160
- dri: label 161
- Sw_NAV1: label 351.
- Sw_NAV2: label 352.

9 Initialization Flag

- a Energize the instrument.
- b Make sure that the labels are present on the ARINC reader.
- c Make sure that the attitude function initialization time is equal to or less than 91 seconds as follows:
 - Monitor the B_INI_NAV discrete on label 351 (bit 14 changes from 0 to 1).

NOTE: The numbers of the 32 bits go from 1 to 32.

10 Horizon Line Movements

- a Record the values of the labels 324 (pitch) and 325 (roll) for one minute (for C16221KA01, C16221LA01 and C16221LA02) or three minutes (for C16221KA02 and C16221LA03) at a frequency of 60Hz.
- b For the pitch and roll parameters, make sure that:
 - The average value calculated on the recorded samples is equal to 0 ± 0.5 degrees.
 - The noise value calculated on the recorded samples is less than 0.15 degrees for 1 sigma (for C16221KA01, C16221LA01 and C16221LA02) or less than

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0.17 degrees for 2 sigma (for C16221KA02 and C16221LA03).

- c For versions C16221KA01, C16221LA01 and C16221LA02, read the values of the labels 157 (dpi), 160 (dqi) and 161 (dri) at a frequency of 60Hz.
- d Make sure that the average value of the dpi, dqi and dri parameters calculated on the recorded samples is less than 2 degrees/second.
- e Adjust the table to 20 degrees nose-up in pitch at a rate of less than 95 deg/sec.
- f After three minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- g Make sure that the average pitch value calculated on the recorded sample is 20 ± 0.5 degrees.
- h Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.
- i Adjust the table to 20 degrees nose-down in pitch at a rate of less than 95 deg/sec.
- j After three minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- k Make sure that the average pitch value calculated on the recorded sample, is -20 ± 0.5 degrees.
- l Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.
- m Keep the same pitch angle and turn the instrument through 20 degrees in clockwise roll at a rate of less than 95 deg/sec.
- n After three minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- o Make sure that the average pitch value calculated on the recorded sample is -20 ± 0.5 degrees.
- p Make sure that the average roll value calculated on the recorded sample is 20 ± 0.5 degrees.
- q Keep the same pitch angle and turn the instrument through 40 degrees in counterclockwise roll at a rate of less than 95 deg/sec (to a roll angle of -20 degrees).
- r After three minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- s Make sure that the average pitch value calculated on the recorded sample is -20 ± 0.5 degrees.
- t Make sure that the average roll value calculated on the recorded sample is -20 ± 0.5 degrees.
- u Turn the single-axis table through an angle between 90 and 180 degrees around the Z axis at a rate between 10 and 95 deg/sec.
- v During the rotation, make sure that the labels 324 and 325 stay at a value of -20 ± 1 degrees.

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- w After three minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- x Make sure that the average pitch value calculated on the recorded sample is -20 ± 0.5 degrees.
- y Make sure that the average roll value calculated on the recorded sample is -20 ± 0.5 degrees.
- z Keep the same pitch angle and turn the instrument through 20 degrees in clockwise roll to reset the roll angle to 0 degrees at a rate of less than 95 deg/sec.
- aa After three minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- ab Make sure that the average pitch value calculated on the recorded sample is -20 ± 0.5 degrees.
- ac Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.
- ad Reset the instrument pitch angle to 0 degrees at a rate of less than 95 deg/sec.
- ae After 1 hour 30 minutes, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- af Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- ag Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.

11 Instrument Panel Slant Angle Programming Pins

- a De-energize the instrument.
- b Set the instrument support fixture to an instrument panel slant angle of 16 degrees.
- c Connect the instrument in the configuration for a 16-degree instrument panel slant angle ([Refer to Table 1008](#)).
- d Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- e Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- f Energize the instrument.
- g At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- h Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- i Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.
- j De-energize the instrument.
- k Set the instrument support fixture to an instrument panel slant angle of 15 degrees.

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- l Connect the pins b and d to ground (pin G) and put the pins c and e in open circuit.
- m Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- n Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- o Energize the instrument.
- p At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- q Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- r Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.

12 Instrument Steady State Test

- a This test can be done after the test of the instrument panel slant angle programming pins or during the test of the altitude and airspeed functions ([Refer to SUBTASK-34-22-45-750-023-A01](#)).
- b Install the instrument on a vertical support fixture set to an instrument panel slant angle of 0 degrees with a precision of 0.1 degrees in roll and pitch.
- c Install the instrument and support fixture assembly on the single-axis table set to a angle of 0 degrees in roll and pitch, or on a horizontal surface (with a precision of 0.1 degrees).
- d Connect the instrument in the operational configuration with RS422 ([Refer to Table 1007](#)).
- e Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- f Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).

NOTE: Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:

- Pitch: label 324
- Roll: label 325.

- g Energize the instrument.
- h At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- i Make sure that the average pitch and roll values calculated on the recorded samples are 0 ± 0.5 degrees.
- j Let the instrument operate for 1 hour 30 minutes.
- k At the end of this time, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz (for C16221KA01, C16221LA01 and C16221LA02) or three minutes at a frequency of 60Hz (for C16221KA02 and C16221LA03).
- l Make sure that the average pitch and roll values calculated on the recorded samples are 0 ± 0.5 degrees.

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(23) Test of Altitude and Airspeed Functions

(a) Test Position

- 1 Attach the instrument to the test fixture designed for an instrument panel slant angle of 0 degrees.
- 2 Connect the instrument in the operational configuration (with the heading function disabled) ([Refer to Table 1007](#)) .

NOTE: Connect the ARINC 429 Low Speed output line (pins N and M of connector J1) to the ARINC 429 reader to get the data that follow:

- Indicated airspeed (IAS): label 206
 - Corrected baro altitude (Zbaro): label 204
 - Standard altitude (Zstd): label 203
 - Barometric pressure in hPa (Pbaro): label 234
 - Barometric pressure in in.Hg (PbaroUS): label 235
 - Mach: label 205.
- 3 Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
 - 4 Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
 - 5 Connect the pressure ports as follows:
 - The static pressure port to a pressure source and a barometer, or to a calibrated pressure source
 - The pitot pressure port to a pressure source and a pressure gauge, or to a calibrated pressure source.

(b) Test of Leakage on the Static and Pitot Ports and Mach Number Display

- 1 Do the test of the leakage on the static and pitot ports as follows:

NOTE: These measurements must be done with the instrument in a power-off condition and in normal test conditions.

 - a Decrease the pressure to the values that follow at a maximum rate of change of 20 hPa/sec (1200 hPa/min):
 - Static pressure (Ps): 500 hPa (absolute value)
 - Pitot pressure (Pt): between 525 and 560 hPa.
 - b Isolate the instrument-barometer assembly from the pressure sources. The unwanted volume isolated (tubing and barometer) must be less than 0.05 dm3 for each of the sensors.
 - c Let the instrument stabilize for one minute, then take an initial reading on the two barometers
 - d After another minute, take a second reading on the two barometers.
 - e Make sure that the difference between the two readings is not more than 1.7 hPa for each of the pitot and static pressure ports.

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- f Increase the static and pitot pressure values back to normal atmospheric pressure at a maximum rate of change of 20 hPa/sec (1200 hPa/min)
- 2 Do the test of the Mach number display as follows:
 - NOTE: To select the Mach number display, connect the discretes m and k to the discrete ground (pin G).
 - a Energize the instrument.
 - b At the end of the initialization phase, set the pressure values as follows at a maximum rate of change of 20 hPa/sec (1200 hPa/min):
 - Static pressure (Ps): 1013.25 hPa (absolute value)
 - Pitot pressure (Pt): 1250 hPa.
 - c Make sure that the Mach number is shown in the bottom left corner of the display.
 - d Set the static and pitot pressure values to normal atmospheric pressure at a maximum rate of change of 20 hPa/sec (1200 hPa/min).
- (c) Test of the Altitude and Airspeed Functions
 - 1 Test of Altitude Accuracy
 - a De-energize the instrument.
 - b Disconnect the discretes m and k from the ground (pin G).
 - c Energize the instrument.
 - d At the end of the initialization phase, press the BARO pushbutton to set the barometric pressure to a value equal to the standard pressure.
 - NOTE: During this test, keep the pitot pressure adjusted to show an airspeed of 200±5 kts.
 - e Increase the pressure-altitude from -2000 feet to +50,000 feet, at a rate of approximately 5000 feet/min. Decrease this rate gradually before each test point ([Refer to Table 1009](#)) to prevent overshoot of the test point.

Table 1009 : Altitude Accuracy Test Points

Pressure-altitude Za in feet (1)	Static pressure (Ps) in hPa (2)	Tolerance ± in feet (3)
-2000	1088.66	20
0*	1013.25	20
1000	977.17	20
2000*	942.13	30
6000	812.00	40
10,000*	696.81	50
20,000	465.63	75
30,000*	300.89	120

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Pressure-altitude Za in feet (1)	Static pressure (Ps) in hPa (2)	Tolerance ± in feet (3)
40,000	187.54	150
50,000	115.97	180

f For each static pressure value equivalent to a typical Za value (Refer to Table 1009), keep the pressure stable for more than 20 seconds and then record the value Zm shown by the altimeter (label 203).

g Make sure that the difference Zm-Za is not more than the tolerance (Refer to Table 1009).

h Stop for two minutes at the value of +50,000 feet.

i Decrease the pressure-altitude from +50,000 to 0 feet at a rate of approximately 5000 ft/min. Decrease this rate gradually before each test point to prevent overshoot of the test point.

j For each static pressure value equivalent to a typical Za value (Refer to Table 1009), keep the pressure stable for more than 20 seconds and then record the value Zstd shown by the instrument (label 203).

NOTE: It is possible to do this test only at the altitude values shown with an asterisk (Refer to Table 1009).

k Make sure that the difference Zstd-Za is not more than the tolerance (Refer to Table 1009).

l Reset the pitot pressure Pt to ambient room pressure.

2 Test of Airspeed Accuracy

a Connect the pitot and static pressure ports to an adjustable pressure source and a barometer, or to a calibrated pressure source

NOTE: During this test, keep the static pressure Ps adjusted to show an altitude of 10,000 ±50 feet.

Table 1010 : Airspeed Accuracy Test Points

Airspeed (A) in Knots (1)	Differential pressure (Pt-Ps) in hPa (2)	Tolerance ± in Knots (3)
40*	2.60	4
70	7.97	4
100	16.30	4
140*	32.13	3
180	53.50	3
200	66.34	3

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Airspeed (A) in Knots (1)	Differential pressure (Pt-Ps) in hPa (2)	Tolerance $\pm$ in Knots (3)
300*	153.55	3
350	212.86	3
400	283.94	3
520	510.28	3

- b Increase the airspeed from 0 to 520 Knots at a rate of approximately 240 Kts/min.
- c For each differential pressure value (Pt-Ps) (Refer to Table 1010) equivalent to the airspeed (A) in column (1), keep the pressure stable for more than 20 seconds and then record the airspeed indication (IAS) shown by the instrument (label 206).
- d Make sure that the difference between the airspeed indication and the value A in column 1 (Refer to Table 1010) is not more than the tolerance in column 3 (Refer to Table 1010).
- e Stop at the value of 520 Knots for 2 minutes.
- f Decrease the airspeed from 520 to 0 Knots at a rate of approximately 240 Kts/min.
- g For each differential pressure value (Pt-Ps) (Refer to Table 1010) equivalent to the airspeed values (A) shown with an asterisk in column (1), keep the pressure stable for more than 20 seconds and then record the airspeed indication (IAS) shown by the instrument (label 206).
- h Make sure that the difference between the airspeed indication and the value A in column 1 (Refer to Table 1010) is not more than the tolerance in column 3 (Refer to Table 1010) .
- i Reset the static and pitot pressures to ambient room pressure.

SUBTASK 34-22-45-750-024-A01

(24) Final Operational Check

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration (Refer to Table 1004) .
- (c) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (d) Energize the instrument.
- (e) Make sure that the self-test page 1 is displayed (Refer to Figure 1010) (or that the related data is output on the RS422 line after the word "T" is transmitted on the RS422 input bus).
- (f) Make sure that all the results of the self-tests are correct.
- (g) Record the value shown on the time counter.
- (h) Transmit the word "S" on the RS422 input bus.

NOTE: The instrument transmits the content of the failure memory on the RS422 output line.

- (i) (Refer to SUBTASK-34-22-45-750-005-A01) and make sure that the content of the failure memory is the same.

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- (j) If the content is not the same, make sure that only the fault codes related to the internal photocell monitoring (T21, code 0020 0000 0000 subcode internal) are shown and that a maximum of ten errors for a maximum of three power-ups are recorded.
- (k) If the maximum number of errors is too high, do the additional check as follows:
- 1 For a manual additional test:
 - Install the instrument on the vertical support fixture.
 - Connect the instrument in the operational configuration for a 0-degree panel slant angle ([Refer to Table 1013](#)).
 - Energize the instrument.
 - Do not transmit the labels 312 or 320 on the HS bus 3 (T-S).
 - Do not transmit the labels 173 or 174 on the LS bus 1 (A-W).
 - At the end of the initialization phase, push the APP and HP/IN buttons for more than two seconds to go to the test display.
 - Push the "-" button to go to the next display.
 - Push the "-" button again to go to the Engineering Data display.
 - Make sure that the fault code shown on page 1/5 is 0000 0200 8020 (related to ILS, Heading and Groundspeed error T47, T37 and T9), and that no additional photocell monitoring is recorded.
 - 2 For an automatic additional test:
 - Energize the instrument.
 - Do not transmit the labels 312 or 320 on the HS bus 3 (T-S).
 - After 20 seconds, transmit the word "S" on the RS422 input bus to get the instrument to send the content of the failure memory on the RS422 output bus.
 - Compare the content of the failure memory with the content recorded before.
 - Make sure that the new fault code recorded is 0000 0200 8020 (related to ILS, Heading and Groundspeed error T47, T37 and T9), and that no additional photocell monitoring is recorded.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-750-804-A01

4. Tests for C16221MA..

- A. Reason for the job
 - (1) Not applicable.
- B. Job Set-up Information
 - (1) [\(Refer to TASK-34-22-45-750-802-A01\)](#)
- C. Job Set-up
 - (1) Not applicable.
- D. Procedure

SUBTASK 34-22-45-750-025-A01

- (1) Visual Inspection
 - (a) Examine all the inscriptions on the bezel from a distance of approximately 700 mm along a line that goes through the longitudinal axis of the instrument.
 - (b) Make sure that the unit is complete, that it has an inspection seal, and that protective covers are installed on the power supply connector and pressure unions.
 - (c) Make sure that no defect is visible to the naked eye.
 - (d) Make sure that the seals are in good condition.
 - (e) Make sure that the identification plates are correct [\(Refer to PGBLK-34-22-45-1-A01\)](#).
 - (f) The table that follows gives the cross reference between the Boeing part numbers and the Thales part numbers [\(Refer to Table 1011\)](#).

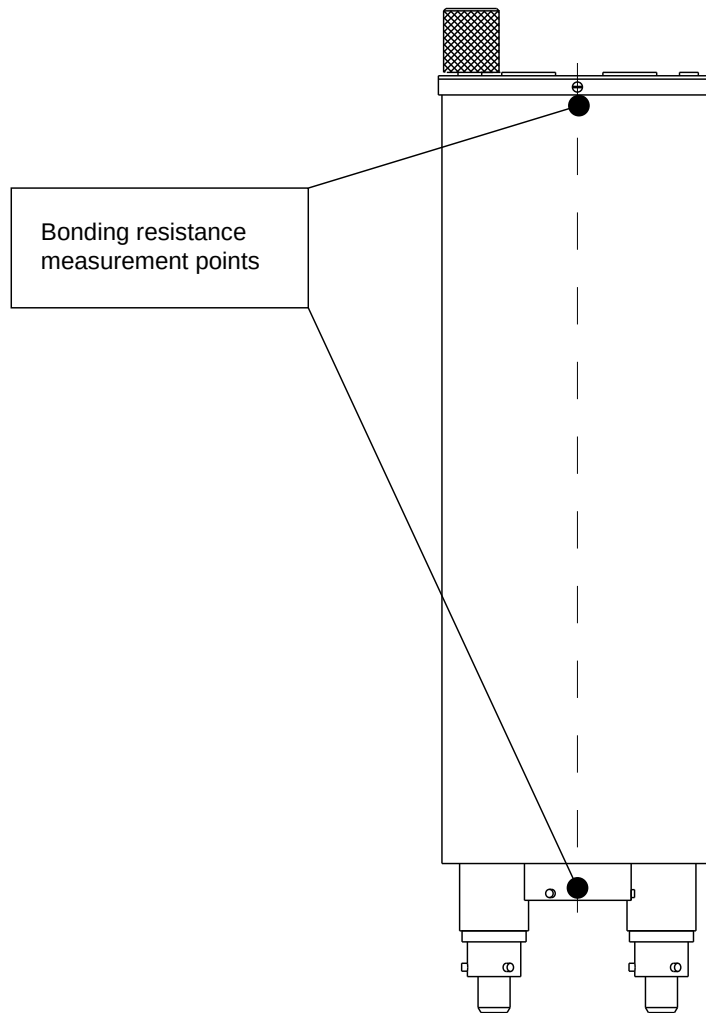
Table 1011 : Part Number Cross Reference

BOEING PN	THALES PN
S231A120-4	C16221MA01

SUBTASK 34-22-45-750-026-A01

- (2) Electrical Bonding Test
 - (a) With a current of 10A applied, measure the bonding resistance between a point on the body of connector J1 and a point on the cover near the bezel [\(Refer to Figure 1016\)](#).
 - (b) Make sure that the bonding resistance is equal to or less than 10 mohms.

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D_CM_TAV_342245_01_AA00_01_00_

Figure 1016/Graphic 34-22-45-991-115-A01
Bonding Resistance Measurement Points

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SUBTASK 34-22-45-750-027-A01

(3) Insulation Resistance Test

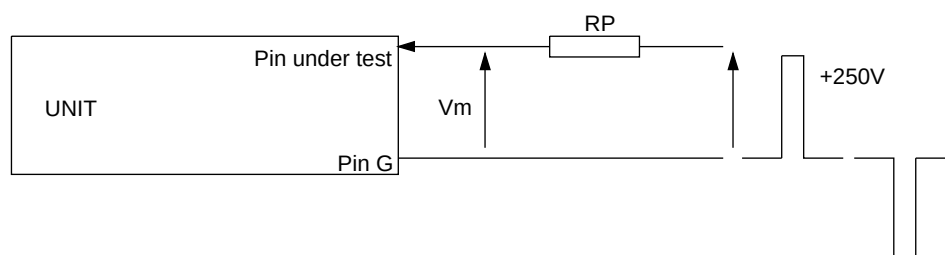
- (a) Set a megohmmeter to a voltage of 40VDC.
- (b) Measure the insulation resistance between pins D and E (40VDC input) and pin t (0VDC input).
- (c) Make sure that the insulation resistance is equal to or higher than 40 Mohms.

SUBTASK 34-22-45-750-028-A01

(4) Test of the Presence of the Lightning Strike Protection Diodes

- (a) Make the test setup shown on the diagram ([Refer to Figure 1017](#)).

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D_CM_TAV_342245_01_AAP0_01_00

Figure 1017/Graphic 34-22-45-991-116-A01
Test Setup for Presence of Lightning Strike Diodes

- (b) Do the calibration of the pulses at a voltage higher than the limited voltage under test as follows:
 - 1 Connect the pulse generator to an oscilloscope through a BNC cable.
 - 2 Set the repetition period to 10ms, the pulsewidth to $60\mu\text{s} \pm 10\mu\text{s}$ and the amplitude to $+240\text{V} \pm 10\text{V}$ and $-240\text{V} \pm 10\text{V}$.
- (c) Generate the pulses as follows:

NOTE: The limiting of the voltage applied is done in the instrument.

 - 1 Connect the pulse generator to the pin under test ([Refer to Table 1012](#)), with the ground on pin G.
 - 2 Transmit positive and then negative pulses, and read the limited voltage value for each type of pulse.
 - 3 ([Refer to Table 1012](#)) for the voltage values.

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Table 1012 : Table of Test Pins

			+240V pulse		-240V pulse	
Signal	Pins	Pins under test	Rprotection (Ohms)	Vmeasured (V)	Rprotection (Ohms)	Vmeasured (V)
Power supply	D, E	D, E	255	50±10	test not to be performed	
Bezel dimming	i, j	i, j	220K	60±15	220K	-60±15
ARINC In and spare	A,W,U,V,S,T,B,C	A,W,U,V,S,T,B,C	10K	35±5	10K	-35±5
ARINC Out	M,N,R,P	M,N,R,P	1K	15±5	1K	-15±5
IN discrete	b,c,d,e,k,m,p,r,s,q, Z,f	b,c,d,e,k,m,p,r,s	10K	35±5	10K	-35±5
OUT discrete	g	g	10K	60±10	test not to be performed	

SUBTASK 34-22-45-750-029-A01

(5) Instrument Operating Position and Check of the Fault Memory

- (a) Attach the instrument to a suitable test fixture and connect it in the operational configuration ([Refer to Table 1013](#)).
- (b) Adjust the test fixture to 0 degrees in roll with a precision of ±1 degree.
- (c) Adjust the test fixture to 0 degrees in pitch with a precision of ±1 degree.
- (d) Change to maintenance mode as follows:
 - 1 Connect the discrete pins X and a to the common ground discrete (pin G).
- (e) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (f) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (g) Energize the instrument.
- (h) Transmit the word "S" on the RS422 input bus and make sure that the instrument transmits the content of the fault memory on the RS422 output bus. Record the result for subsequent comparison ([Refer to SUBTASK-34-22-45-750-048-A01](#)).

SUBTASK 34-22-45-750-030-A01

(6) Power Consumption and Validity Output Discrete Test

- (a) De-energize the instrument.
- (b) Connect the instrument in its operational configuration ([Refer to Table 1013](#)).
- (c) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (e) Energize the instrument in average ambient lighting conditions (between approximately 300 and 600 lux).
- (f) Let the instrument operate for one minute and then measure the power consumption.

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(g) Make sure that the power consumption is equal to or less than 0.7A.

(h) Make sure that the validity output discrete F is connected to ground (pin G).

SUBTASK 34-22-45-750-031-A01

(7) Test of the Operating Configuration

(a) Connect the instrument in the operational configuration [\(Refer to Table 1013\)](#).

Table 1013 : Operational Configuration for 0-degree Instrument Panel Slant Angle

J1 pin	SIGNAL	I/O	COMMENTS	WIRED
B		I	SPARE	
C		I	SPARE	
t	CHASSIS_GND		Case ground	YES
D	28V	I	+28V DC	YES
E	0_28V	I	0V DC	YES
A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
W	ARINC_E_1_B	I		
j	0_LUM_SCR	I	Bezel Lighting Control	YES
V	ARINC_E_2_A	I	SPARE	YES
U	ARINC_E_2_B	I		
i	LUM_SCR	I	Bezel Lighting Control	YES
T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
S	ARINC_E_3_B	I		
h	DISCRET_E_1	I	Mode selection n°2	
R	ARINC_S_1_A	O	IR parameters for Cat III option	
P	ARINC_S_1_B	O		
g			SPARE	
N	ARINC_S_2_A	O	ADR parameters for Cat III option	
M	ARINC_S_2_B	O		
f	DISCRET_E_2	I	Backcourse display disabled	
X	DISCRET_E_3	I	Mode selection n°1	
Y			SPARE	
Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	

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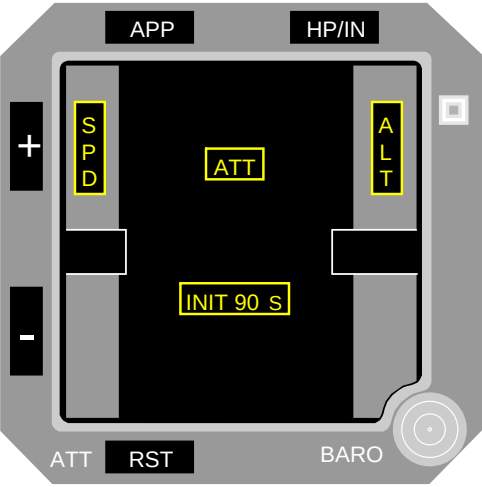
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a	DISCRET_E_5	I	Mode selection n°3	
b	DISCRET_E_6	I	A/C configuration 8/0	
c	DISCRET_E_7	I	A/C configuration 4/0	
d	DISCRET_E_8	I	A/C configuration 2/0	
e	DISCRET_E_9	I	A/C configuration 1/0	
k	DISCRET_E_10	I	A/C configuration parity	
m	DISCRET_E_11	I	Mag heading display disabled	
n	DISCRET_E_12	I	APP display disabled	
p	DISCRET_E_13	I	SPARE (for Vbar option)	
q	DISCRET_E_14	I	Initiated test	
r	DISCRET_E_15	I	SPARE	
s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
G	ELEC_GND	I	Discrete common/ground	
L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	
K	RS422_E_L	I		
J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	
H	RS422_S_L	O		

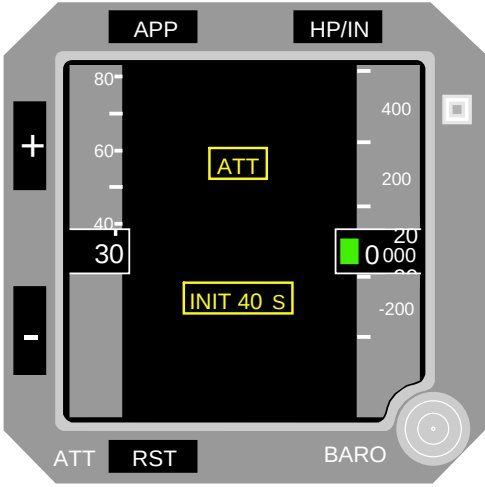
- (b) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (c) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (d) Energize the instrument.
- (e) Make sure that the initialization phase 1 display is shown ([Refer to Figure 1018](#)).

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D_CM_TAV_342245_01_AAQ0_01_00

Figure 1018/Graphic - Sheet 01 / 34-22-45-991-117-A01
Initialization Phase 1



D_CM_TAV_342245_01_AAQ0_02_00

Figure 1018/Graphic - Sheet 02 / 34-22-45-991-117-A01
Initialization Phase 2

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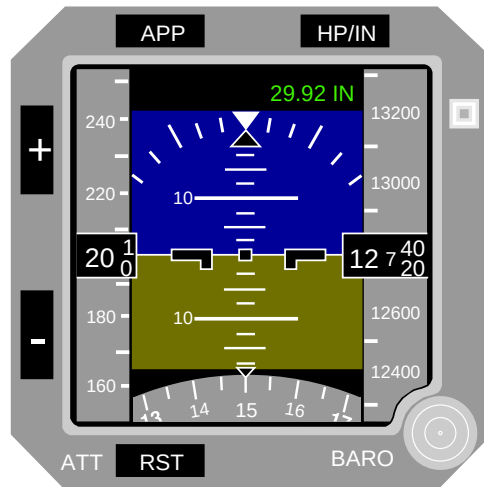
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- (f) Make sure that the display changes to the initialization phase 2 not more than 15 seconds after power-up (Refer to Figure 1018).

NOTE: Ignore the altitude and airspeed values shown on the display.

- (g) At the end of the initialization phase, make sure that the instrument shows the operational display (Refer to Figure 1019).

NOTE: Ignore the barometric pressure unit (which can be HP for hectopascals or IN for inches of mercury) and the numeric values.



D_CM_TAV_342245_01_AAR0_01_00

Figure 1019/Graphic - Sheet 01 / 34-22-45-991-118-A01
Operational Display 1

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D_CM_TAV_342245_01_AAR0_02_00

Figure 1019/Graphic - Sheet 02 / 34-22-45-991-118-A01
Operational Display 2



D_CM_TAV_342245_01_AAR0_03_00

Figure 1019/Graphic - Sheet 03 / 34-22-45-991-118-A01
Operational Display 3

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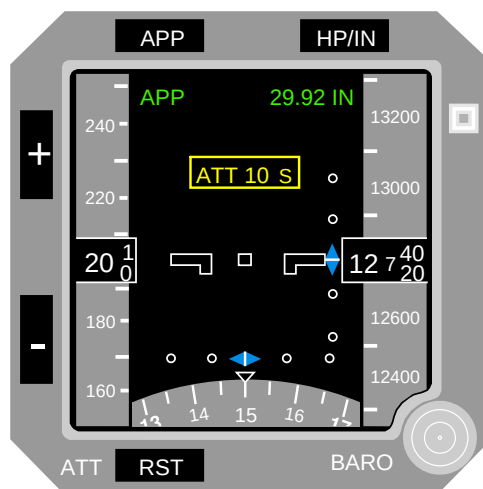
- (h) Make sure that the bezel does not have any defects that can interfere with its use.
- (i) Make sure that the markings and inscriptions are easy to see and read.

SUBTASK 34-22-45-750-032-A01

(8) Test of the Pushbuttons and Knobs

- (a) Press the "+" pushbutton once and make sure that the brightness increases.
- (b) Press the "-" pushbutton once and make sure that the brightness decreases.
- (c) Press the "APP" pushbutton three times and make sure that the operational display changes from no ILS to ATT, then to BCRS, and then back to no ILS (Refer to Figure 1019).
- (d) Press the HP/IN pushbutton once and make sure that the barometric pressure unit changes (from HP to IN or from IN to HP).
- (e) Turn the BARO knob clockwise (approximately a quarter turn) and make sure that the barometric pressure value increases.
- (f) Turn the BARO knob counterclockwise (approximately a quarter turn) and make sure that the barometric pressure value decreases.
- (g) Press the BARO knob and make sure that the barometric setting changes from the adjustable barometric pressure to the standard value.
- (h) Press the RST pushbutton for 2 seconds. Make sure that the instrument resets to zero and that the message "ATT10s" is displayed (Refer to Figure 1020).

NOTE: Ignore the values displayed.



D_CM_TAV_342245_01_AAS0_01_00

Figure 1020/Graphic 34-22-45-991-119-A01
ATT 10s Page

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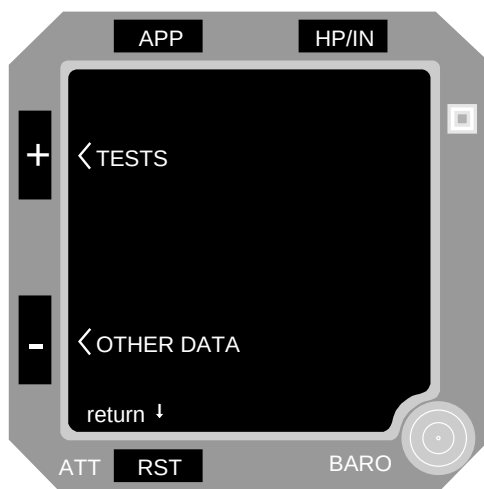
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SUBTASK 34-22-45-750-033-A01

(9) Test of the TEST Function

- (a) Press the HP/IN and APP buttons together for 2 seconds.
- (b) Make sure that the first test page is displayed ([Refer to Figure 1021](#)).

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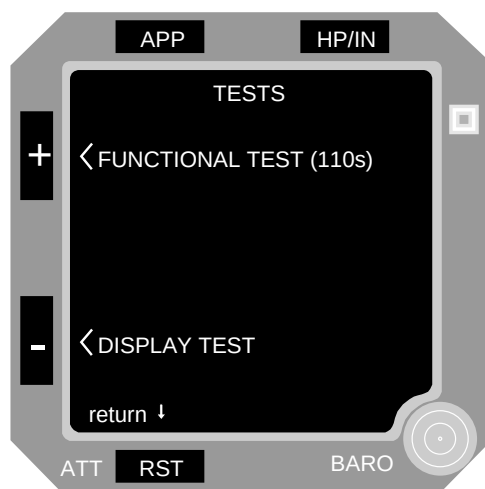
Figure 1021/Graphic 34-22-45-991-120-A01
First Test Page

- (c) Press the "+" button and make sure that the second test page is displayed ([Refer to Figure 1022](#)).

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D_CM_TAV_342245_01_AA00_01_00

Figure 1022/Graphic 34-22-45-991-121-A01
Second Test Page

- (d) Press the "+" button again and make sure that "Test in progress 110s" is displayed in less than 10 seconds.
- (e) At the end of the test, make sure that "Test OK" is displayed.
- (f) Press the RST button three times and make sure that the screen goes back to the operational display ([Refer to Figure 1019](#)).
- (g) Connect the pin q to ground for at least 0.4 seconds and then put it in open circuit in less than 6 seconds.
- (h) Make sure that "Test in progress 110s" is displayed in less than 10 seconds.
- (i) After 110 seconds, make sure that the screen goes back to the operational display ([Refer to Figure 1019](#)).

SUBTASK 34-22-45-750-034-A01

(10) Test of the HEADING Function

- (a) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (b) Transmit the label 312 (with its value set to zero) and the label 320 (with its value set to 30) on the HS bus 3 (pins T and S).
- (c) Make sure that the heading value displayed is 30 degrees.
- (d) De-energize the instrument.
- (e) Connect the pins m and k to the discrete ground (pin G) to disable the heading display.
- (f) Energize the instrument.
- (g) At the end of the initialization phase, make sure that no heading value is displayed.
- (h) De-energize the instrument and put the pins m and k in open circuit.

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SUBTASK 34-22-45-750-035-A01

(11) Test of the ILS Function

(a) Test of the ILS Front Course Function

- 1 Transmit the label 173 (with its value set to 0.0775) and the label 174 (with its value set to 0.0875) on the LS bus 1 (pins A and W).
- 2 Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- 3 Energize the instrument.
- 4 At the end of the initialization phase, press the APP button to display the glideslope and localizer data.
- 5 Make sure that the indication APP is displayed in the upper-left corner.
- 6 Make sure that the ILS indications are correct ([Refer to Table 1014](#)).

Table 1014 : ILS Front Course Indications

	GLIDESLOPE		LOCALIZER	
	FLAG	BAR	FLAG	BAR
Position	Not in view	First mark down	Not in view	First mark right

(b) Test of the ILS Back Course Function

- 1 Press the APP button to change to backcourse approach mode.
- 2 Make sure that the indication BCRS is displayed in the upper-left corner.
- 3 Make sure that the ILS indications are correct ([Refer to Table 1015](#)).

Table 1015 : ILS Back Course Indications

	GLIDESLOPE		LOCALIZER	
	FLAG	BAR	FLAG	BAR
Position	Not in view	Not in view	Not in view	First mark left

- 4 Press the APP button and make sure that the localizer and glideslope data go out of view.

SUBTASK 34-22-45-750-036-A01

(12) Bezel Lighting Test

- (a) Put a mask in front of the photoelectric cell to simulate night lighting conditions.
- (b) Set the bezel lighting to the maximum brightness value (5VAC/400Hz).
- (c) Make sure that the lighting of the pushbuttons (only) is uniform.
- (d) Make sure that there is no unwanted leakage of light.

SUBTASK 34-22-45-750-037-A01

(13) Tests of the Options (V-bar, Slip and Altitude in Meters)

- (a) De-energize the instrument.

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- (b) Connect the discretes for V-bar selection (pin p), slip selection (pin s) and altitude in meters selection (pin Z) to the discrete ground (pin G) and set the discrete on pin m to the open state.
- (c) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (e) Energize the instrument.
- (f) At the end of the initialization phase, make sure that the correct page is displayed ([Refer to Figure 1023](#)).

NOTE: Ignore the values displayed.



D_CM_TAV_342245_01_AAV0_01_00

Figure 1023/Graphic 34-22-45-991-122-A01
V-Bars, Slip and Altitude In Meters Display

SUBTASK 34-22-45-750-038-A01

(14) Display of Self-Test Results

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration ([Refer to Table 1013](#)).
- (c) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (d) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (e) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (f) Energize the instrument.

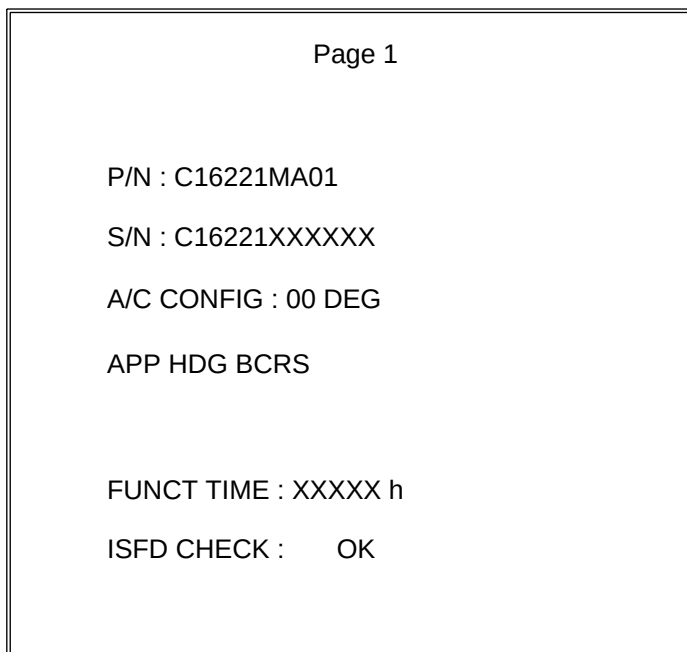
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- (g) Make sure that the self-test page 1 is displayed in less than 10 seconds (Refer to Figure 1024).
- (h) Transmit the word "T" on the RS422 input bus and make sure that the related display data are transmitted on the RS422 output bus.
- (i) After 1 minute, press the RST button (or transmit the words "A" and then "B" on the RS422 bus) to make the display change to page 2 (Refer to Figure 1025).

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Figure 1024/Graphic 34-22-45-991-123-A01
Self-Test Display (Page 1)

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Page 2

HOURS	FAILURE CODE
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX

D_CM_TAV_342245_01_AAX0_01_00

Figure 1025/Graphic 34-22-45-991-124-A01
Self-Test Display (Page 2)

- (j) Transmit the word "T" on the RS422 input bus and make sure that the related display data are transmitted on the RS422 output bus.

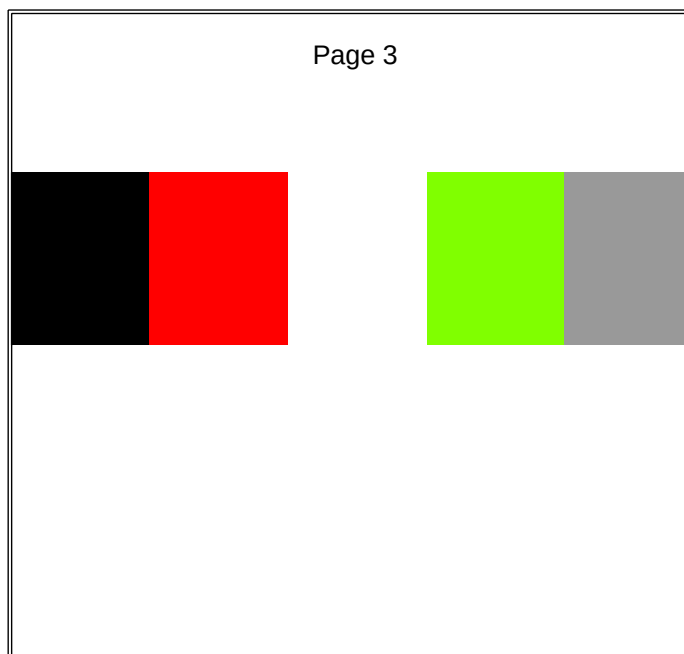
SUBTASK 34-22-45-750-039-A01

(15) Test of the Brightness and Contrast Test Displays

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 3 ([Refer to Figure 1026](#)).

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D_CM_TAV_342245_01_AAY0_01_00

Figure 1026/Graphic 34-22-45-991-125-A01
Self-Test Display (Page 3)

- (b) Press the RST pushbutton again (or transmit the words "A" and then "B" on the RS422 bus) and make sure that the display changes to page 4 ([Refer to Figure 1027](#)).

SUBTASK 34-22-45-750-040-A01

(16) Test of the Checksum Page Display

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 5 ([Refer to Figure 1028](#)).

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Page 4

CONTRAST RATIO

D_CM_TAV_342245_01_AAZ0_01_00

Figure 1027/Graphic 34-22-45-991-126-A01
Self-Test Display (Page 4)

Page 5

CHK CPU B867 33B7

CHK GS 0002 3270

CHK PAR 1B23 A0EF

D_CM_TAV_342245_01_ABA0_01_00

Figure 1028/Graphic 34-22-45-991-127-A01
Checksum Test Display (Page 5)

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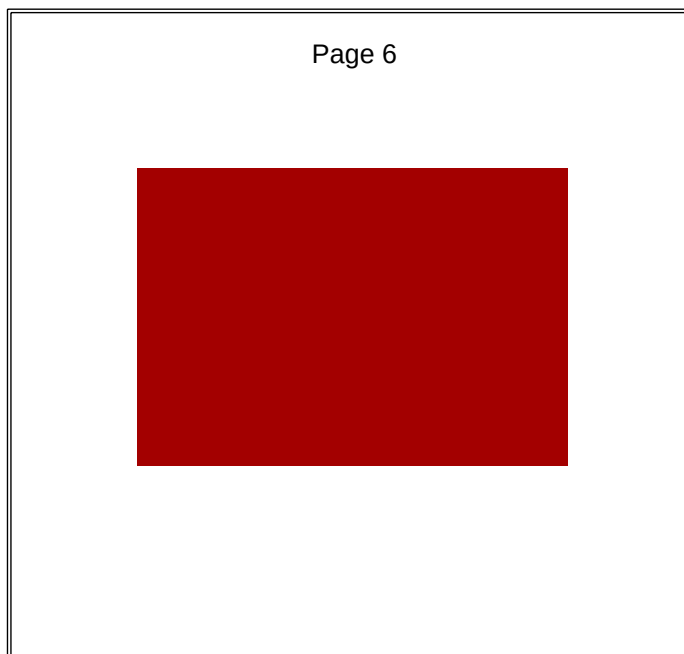
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SUBTASK 34-22-45-750-041-A01

(17) Test of the Color Test Page Display

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 6 ([Refer to Figure 1029](#)).

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D_CM_TAV_342245_01_ABB0_01_00

Figure 1029/Graphic 34-22-45-991-128-A01
Color Test Page (Page 6)

- (b) Make sure that the rectangle shows the ground color.
- (c) Press the RST button again and make sure that the rectangle changes to the sky color.
- (d) Press the RST button again and make sure that the rectangle changes to gray.
- (e) Press the RST button again and make sure that the rectangle changes to white.
- (f) Press the RST button again and make sure that the rectangle changes to red.
- (g) Press the RST button again and make sure that the rectangle changes to green.
- (h) Press the RST button again and make sure that the rectangle changes to yellow.
- (i) Press the RST button again and make sure that the rectangle changes to magenta.
- (j) De-energize the instrument.

SUBTASK 34-22-45-750-042-A01

(18) Test of the Display Brightness and Color

NOTE: All the brightness tests are done in the dark (ambient lighting less than 0.3 fc or 3 Lux). The brightness and color are measured perpendicular to the screen.

- (a) Connect the instrument in the operational configuration ([Refer to Table 1013](#)).
- (b) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.

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- (c) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (e) Energize the instrument.
- (f) Press the RST button two times (or transmit the words "A" and then "B" two times on the RS422 input bus) to make the display change to page 3 ([Refer to Figure 1026](#)).
- (g) Measure the brightness value at the center of the white square.

SUBTASK 34-22-45-750-043-A01

(19) Test of the Display Brightness in Night Conditions

- (a) Put a mask in front of the photoelectric cell to simulate night conditions.
- (b) Press the "-" button for approximately 6 seconds (or transmit the word "C" and then, after 6 seconds, the word "D" on the RS422 input bus) to set the screen brightness to the minimum value.
- (c) Let the instrument stabilize at this value for 3 minutes.
- (d) Make sure that the brightness value is between 0.02fL and 0.3fL (0.069 Cd/m² and 1.03Cd/m²).

SUBTASK 34-22-45-750-044-A01

(20) Test of the Display Brightness in Full Sunlight Conditions

- (a) Apply a lighting level of 5000 Lux minimum to the photoelectric cell to simulate full sunlight conditions.
- (b) Press the "+" button for approximately 6 seconds (or transmit the word "E" and then, after 6 seconds, the word "F" on the RS422 input bus) to set the screen brightness to the maximum value.
- (c) Let the instrument stabilize at this value for 3 minutes.
- (d) Make sure that the brightness value is between 180fL and 205fL (617Cd/m² and 702Cd/m²).

SUBTASK 34-22-45-750-045-A01

(21) Test of the Ground and Sky Color Coordinates

- (a) Press the RST button (or transmit the word "A" and then "B" on the RS422 input bus three times) to change to page 6a ([Refer to Figure 1029](#)).
- (b) Measure the color coordinates in the center of the colored rectangle.
- (c) Make sure that the 1976 CIE color coordinates measures are as follows:
- (d) Make sure that the brightness value is between 180fL and 205fL (617Cd/m² and 702Cd/m²).
 - Ground: $u' = 0.22 \pm 0.02$, $v' = 0.54 \pm 0.02$
- (e) Press the RST button (or transmit the word "A" and then "B" on the RS422 input bus one time) to change to page 6b ([Refer to Figure 1029](#)).
- (f) Make sure that the brightness value is between 180fL and 205fL (617Cd/m² and 702Cd/m²).
 - Sky: $u' = 0.14 \pm 0.03$, $v' = 0.360 \pm 0.03$.

SUBTASK 34-22-45-750-046-A01

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(22) Tests of the Attitude Functions

NOTE: For this test, it is possible to use two procedures, Procedure 1 or Procedure 2, but it is preferable to use Procedure 2.

(a) Procedure 1

- 1 Measure the ambient room temperature and the temperature at a point on the instrument case near the bezel (Refer to Figure 1016) for the measurement point).
- 2 Make sure that the difference between the two temperatures is less than 3°C.
- 3 Install the instrument on the test support designed for an instrument panel slant angle of 0 degrees, with a precision of 0.1 degrees in roll and pitch.
- 4 Install the assembly on a two-axis table.
- 5 Change to operational mode with RS422 (Refer to Table 1016).

Table 1016 : Operational Configuration for 0-degree Instrument Panel Slant Angle with RS422

J1 No.	J1 pin	SIGNAL	I/O	COMMENTS	WIRED
1	B		I	SPARE	
2	C		I	SPARE	
3	t	CHASSIS_GND		Case ground	YES
4	D	28V	I	+28V DC	YES
5	E	0_28V	I	0V DC	YES
6	A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
7	W	ARINC_E_1_B	I		
8	j	0_LUM_SCR	I	Bezel Lighting Control	YES
9	V	ARINC_E_2_A	I	SPARE	YES
10	U	ARINC_E_2_B	I		
11	i	LUM_SCR	I	Bezel Lighting Control	YES
12	T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
13	S	ARINC_E_3_B	I		
14	h	DISCRET_E_1	I	Mode selection n°2	
15	R	ARINC_S_1_A	O	IR parameters for Cat III option	
16	P	ARINC_S_1_B	O		
17	g			SPARE	
18	N	ARINC_S_2_A	O	ADR parameters for Cat III option	
19	M	ARINC_S_2_B	O		

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20	f	DISCRET_E_2	I	Backcourse display disabled	
21	X	DISCRET_E_3	I	Mode selection n°1	
22	Y			SPARE	
23	Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
24	a	DISCRET_E_5	I	Mode selection n°3	
25	b	DISCRET_E_6	I	A/C configuration 8/0	
26	c	DISCRET_E_7	I	A/C configuration 4/0	
27	d	DISCRET_E_8	I	A/C configuration 2/0	
28	e	DISCRET_E_9	I	A/C configuration 1/0	
29	k	DISCRET_E_10	I	A/C configuration parity	
30	m	DISCRET_E_11	I	Mag heading display disabled	
31	n	DISCRET_E_12	I	APP display disabled	
32	p	DISCRET_E_13	I	SPARE (for Vbar option)	
33	q	DISCRET_E_14	I	Initiated test	
34	r	DISCRET_E_15	I	SPARE	
35	s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
36	F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
37	G	ELEC_GND	I	Discrete common/ground	
38	L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
39	K	RS422_E_L	I		
40	J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
41	H	RS422_S_H	O		

- 6 Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- 7 Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- 8 Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:
- Pitch: label 324
 - Roll: label 325
 - p: label 327
 - q: label 326

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- r: label 330
 - Xacc: label 331
 - Yacc: label 332
 - dpi: label 157
 - dqj: label 160
 - dri: label 161
 - IMU_Sw_NAV1: label 351
 - IMU_Sw_NAV2: label 352
- 9 For the labels 331, 332 and 333, the resolution has been improved (LSB = 11 and not 17). This modification must be taken into consideration for the following test done on these labels.

10 Initialization Flag

- a Energize the instrument.
 - b Make sure that the labels are present on the ARINC reader.
 - c Read the discrete S_O_B_NAV_MODE on label 351 (bit 12 changes from 0 to 1) to make sure that the attitude-function initialization time of the instrument is equal to or less than 91 seconds.
 - d Read the discrete S_I_B_3ACC on label 352 (bit 18 = 1) to make sure that the third accelerometer is present.
- NOTE: The numbers of the 32 bits go from 1 to 32.

11 Horizon Line Movements

- a Record the values of the labels 324 (pitch) and 325 (roll) for a period of three minutes at a frequency of 60Hz.
- b Record the values of the labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for a period of three minutes at a frequency of 60Hz.
- c Record the values of the labels 327 (p), 326 (q) and 330 (r) for a period of three minutes at a frequency of 60Hz.
- d For the pitch and roll parameters, make sure that:
 - The SSM of labels 324 and 325 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to 0 ± 0.5 degrees.
 - The noise value calculated on the recorded samples is less than 0.17 degrees for 2 sigma.
- e For the Xacc, Yacc and Zacc parameters, make sure that:
 - The SSM of labels 331, 332 and 333 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to $0g \pm 0.01g$.
 - The noise value calculated on the recorded samples is less than 0.002g for 2 sigma.
- f For the p, q and r parameters, make sure that:
 - The SSM of labels 327, 326 and 330 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.

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- The noise value calculated on the recorded samples is less than 0.08 deg/sec for 2 sigma.
- g Adjust the table to 20 degrees nose-up in pitch at a rate of 5 deg/sec.
- h After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- i Record the values of labels 331 (Zacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- j Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- k Make sure that the average pitch value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- l Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- m Make sure that the average Xacc value calculated on the recorded sample is equal to $0.342g \pm 0.01g$.
- n Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- o Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- p Adjust the table to -20 degrees in pitch at a rate of 5 deg/sec.
- q After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- r Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- s Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- t Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- u Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- v Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- w Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- x Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- y Reset the table to 0 degrees in pitch at the rate of 5 deg/sec.
- z After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aa Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ab Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- ac Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

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- ad Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- ae Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- af Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ag Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ah Adjust the table to 20 degrees in clockwise roll at a rate of 5 deg/sec.
- ai After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aj Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ak Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- al Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- am Make sure that the average roll value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- an Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ao Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- ap Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- aq Adjust the table to 20 degrees in counterclockwise roll at a rate of 5 deg/sec.
- ar After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- as Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- at Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- au Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- av Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- aw Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ax Make sure that the average Yacc value calculated on the recorded sample is equal to $0.342g \pm 0.01g$.
- ay Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- az Reset the table to 0 degrees in roll at the rate of 5 deg/sec.
- ba After one minute, record the values of labels 324 (pitch) and 325 (roll) three minutes at a frequency of 60Hz.

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- bb Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bc Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bd Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- be Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bf Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bg Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bh Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bi Adjust the table to 5 degrees in clockwise roll and, at the same time, to 5 degrees in nose-up pitch at the rate of 20 deg/sec.
- bj After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bk Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bl Make sure that the average pitch value calculated on the recorded sample is equal to 5 ± 0.5 degrees.
- bm Make sure that the average roll value calculated on the recorded sample is equal to 5 ± 0.5 degrees.
- bn Make sure that the average Xacc value calculated on the recorded sample is equal to $0.087g \pm 0.01g$.
- bo Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.087g \pm 0.01g$.
- bp Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.01g \pm 0.01g$.
- bq Reset the table to 0 degrees in roll and pitch at a rate of 5 deg/sec.
- br After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bs Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bt Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bu Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bv Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bw Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bx Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- by Set the table to 100 degrees in roll at a rate of 5 deg/sec.

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- bz After one minute, record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ca Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- cb Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- cc Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.985g \pm 0.01g$.
- cd Make sure that the average Zacc value calculated on the recorded sample is equal to $-1.174g \pm 0.01g$.
- ce Reset the table to 0 degrees in roll at a rate of 5 deg/sec.
- cf After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- cg Record the values of labels 331 (Xacc), 332 Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ch Record the values of labels 327 (p), 326 (q) and 330 (r) for three minutes at a frequency of 60Hz.
- ci Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- cj Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- ck Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- cl Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- cm Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- cn Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- co Make sure that the average value of p calculated on the recorded sample is equal to $0 \text{ deg. sec} \pm 0.5 \text{ deg/sec}$.
- cp Make sure that the average value of q calculated on the recorded sample is equal to $0 \text{ deg. sec} \pm 0.5 \text{ deg/sec}$.
- cq Make sure that the average value of r calculated on the recorded sample is equal to $0 \text{ deg. sec} \pm 0.5 \text{ deg/sec}$.

12 Instrument Panel Slant Angle Programming Pins

- a De-energize the instrument.
- b Install the instrument on a support fixture set for an instrument panel slant angle of 16 degrees.
- c [\(Refer to Table 1017\)](#) for the test setup.

Table 1017 : Operational Configuration for 16-degree Instrument Panel Slant Angle

J1 n°	J1 pin	SIGNAL	I/O	COMMENTS	WIRED
1	B		I	SPARE	

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2	C		I	SPARE	
3	t	CHASSIS_GND		Case ground	YES
4	D	28V	I	+28V DC	YES
5	E	0_28V	I	0V DC	YES
6	A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
7	W	ARINC_E_1_B	I		
8	j	0_LUM_SCR	I	Bezel Lighting Control	YES
9	V	ARINC_E_2_A	I	SPARE	YES
10	U	ARINC_E_2_B	I		
11	i	LUM_SCR	I	Bezel Lighting Control	YES
12	T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
13	S	ARINC_E_3_B	I		
14	h	DISCRET_E_1	I	Mode selection n°2	
15	R	ARINC_S_1_A	O	IR parameters for Cat III option	
16	P	ARINC_S_1_B	O		
17	g			SPARE	
18	N	ARINC_S_2_A	O	ADR parameters for Cat III option	
19	M	ARINC_S_2_B	O		
20	f	DISCRET_E_2	I	Backcourse display disabled	
21	X	DISCRET_E_3	I	Mode selection n°1	
22	Y			SPARE	
23	Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
24	a	DISCRET_E_5	I	Mode selection n°3	
25	b	DISCRET_E_6	I	A/C configuration 8/0	
26	c	DISCRET_E_7	I	A/C configuration 4/0	ground
27	d	DISCRET_E_8	I	A/C configuration 2/0	
28	e	DISCRET_E_9	I	A/C configuration 1/0	ground
29	k	DISCRET_E_10	I	A/C configuration parity	
30	m	DISCRET_E_11	I	Mag heading display disabled	
31	n	DISCRET_E_12	I	APP display disabled	

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32	p	DISCRET_E_13	I	SPARE (for Vbar option)	
33	q	DISCRET_E_14	I	Initiated test	
34	r	DISCRET_E_15	I	SPARE	
35	s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
36	F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
37	G	ELEC_GND	I	Discrete common/ground	
38	L	RS422_E_H	I	Downloading	YES
39	K	RS422_E_L	I		
40	J	RS422_S_H	O	Downloading	YES
41	H	RS422_S_H	O		

- d Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- e Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- f Energize the instrument.
- g At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for a period of one minute at a frequency of 60Hz.
- h Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- i Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- j De-energize the instrument.
- k Adjust the table to an instrument panel slant angle of 15 degrees.
- l Connect the pins b and d to ground (pin G) and put the pins c and e in open circuit.
- m Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- n Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- o Energize the instrument.
- p At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- q Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- r Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

13 Instrument Steady State Test

- a This test can be done after the instrument panel slant angle programming pins test or during the test of the altitude and airspeed functions ([Refer to SUBTASK-34-22-45-750-047-A01](#)).

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- b Install the instrument on a vertical support fixture set to an instrument panel slant angle of 0 degrees in roll and pitch with a precision of 0.1 degrees.
- c Attach the instrument and support to a two-axis table set to 0 degrees in roll and pitch, or to a horizontal surface table with a precision of 0.1 degrees.
- d Change to operational mode ([Refer to Table 1016](#)).
- e Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- f Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).

NOTE: Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:

- Pitch: label 324
- Roll: label 325.

- g Energize the instrument.
- h At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- i Make sure that the average values calculated on the recorded samples are equal to 0 ± 0.5 degrees.
- j Let the instrument operate for 1 hour 30 minutes.
- k At the end of this time, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- l Make sure that the average values calculated on the recorded samples are equal to 0 ± 0.5 degrees.

(b) Procedure 2

- 1 Measure the ambient room temperature and the temperature at a point on the instrument case near the bezel ([Refer to Figure 1016](#)) for the measurement point.
- 2 Make sure that the difference between the two temperatures is less than 3°C.
- 3 Install the instrument on the test fixture designed for an instrument panel slant angle of 0 degrees, with a precision of 0.1 degrees in roll and pitch.
- 4 Attach the assembly to a Z-axis table.
- 5 Change to operational mode with RS422 ([Refer to Table 1016](#)).
- 6 Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- 7 Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- 8 Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:
 - Pitch: label 324
 - Roll: label 325
 - p: label 327
 - q: label 326
 - r: label 330
 - Xacc: label 331

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- Yacc: label 332
- $\epsilon\Phi$: label 352
- $\epsilon\theta$: label 353
- $\epsilon\psi$: label 155
- dpi: label 157
- dqj: label 160
- dri: label 161
- IMU_Sw_NAV1: label 351.
- IMU_Sw_NAV2: label 352.

NOTE: For the labels 331, 332 and 333, the resolution is improved (LSB = 11 and not 17 as before). It is necessary to take this modification into consideration for the tests done on these labels.

9 Initialization Flag

- a Energize the instrument.
- b Make sure that the labels are present on the ARINC reader.
- c Read the discrete S_O_B_NAV_MODE on label 351 (bit 12 changes from 0 to 1) to make sure that the attitude-function initialization time of the instrument is equal to or less than 91 seconds.
- d Read the discrete S_I_B_3ACC on label 352 (bit 1_ = 1) to make sure that the third accelerometer is present.

NOTE: The numbers of the 32 bits go from 1 to 32.

10 Horizon Line Movements

- a Record the values of the labels 324 (pitch) and 325 (roll) for a period of three minutes at a frequency of 60Hz.
- b Record the values of the labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for a period of three minutes at a frequency of 60Hz.
- c Record the values of the labels 327 (p), 326 (q) and 330 (r) for a period of three minutes at a frequency of 60Hz.
- d For the pitch and roll parameters, make sure that:
 - The SSM of labels 324 and 325 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to 0 ± 0.5 degrees.
 - The noise value calculated on the recorded samples is less than 0.17 degrees for 2 sigma.
- e For the Xacc, Yacc and Zacc parameters, make sure that:
 - The SSM of labels 331, 332 and 333 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to $0g \pm 0.01g$.
 - The noise value calculated on the recorded samples is less than 0.002g for 2 sigma.
- f For the p, q and r parameters, make sure that:
 - The SSM of labels 327, 326 and 330 is NO (bits 30 and 31 set to 1).

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- The average value calculated on the recorded samples is equal to 0 deg/sec ± 0.05 deg/sec.
- The noise value calculated on the recorded samples is less than 0.08 deg/sec for 2 sigma.
- g Adjust the table to 20 degrees nose-up in pitch at a rate of less than 95 deg/sec.
- h After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- i Record the values of labels 331 (Zacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- j Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- k Make sure that the average pitch value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- l Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- m Make sure that the average Xacc value calculated on the recorded sample is equal to $0.342g \pm 0.01g$.
- n Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- o Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- p Adjust the table to -20 degrees in pitch at a rate of less than 95 deg/sec.
- q After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- r Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- s Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- t Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- u Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- v Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- w Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- x Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- y At the same pitch angle, turn the instrument through 20 degrees in roll in the clockwise direction at a rate of less than 95 deg/sec.
- z After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aa Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ab Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).

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- ac Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- ad Make sure that the average roll value calculated on the recorded sample is equal to 20 ± 0.5 degrees.
- ae Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- af Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.321g \pm 0.01g$.
- ag Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.117g \pm 0.01g$.
- ah At the same pitch angle, turn the instrument through 40 degrees in roll in the counterclockwise direction (to set its roll to -20 degrees) at a rate of less than 95 deg/sec.
- ai After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aj Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ak Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- al Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- am Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- an Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- ao Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.321g \pm 0.01g$.
- ap Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.117g \pm 0.01g$.
- aq Turn the table around its Z axis through an angle between 90 and 180 degrees at an angular rate between 10 deg/sec and 90 deg/sec.
- ar During the movement, record the values of labels 324 (pitch) and 325 (roll) and make sure that they stay at -20 ± 0.5 degrees.
- as Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- at After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- au Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- av Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- aw Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- ax Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.

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- ay Make sure that the average Yacc value calculated on the recorded sample is equal to $0.321g \pm 0.01g$.
- az Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.117g \pm 0.01g$.
- ba At the same pitch angle, turn the table through 20 degrees in the clockwise direction at a rate of less than 95 deg/sec to set the roll angle to 0 degrees.
- bb After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bc Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bd Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- be Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- bf Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bg Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- bh Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bi Make sure that the average Zacc value calculated on the recorded sample is equal to $0.06g \pm 0.01g$.
- bj Set the instrument pitch angle to 0 degrees (20 degrees nose-up) at a rate of less than 95 deg/sec.
- bk After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bl Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bm Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bn Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bo Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bp Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bq Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- br Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bs At the same pitch angle, turn the instrument counterclockwise through 100 degrees at a rate of less than 95 deg/sec.
- bt After three minutes, record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bu Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).

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- bv Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - bw Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.985g \pm 0.01g$.
 - bx Make sure that the average Zacc value calculated on the recorded sample is equal to $-1.174g \pm 0.01g$.
 - by At the same pitch angle, turn the instrument clockwise through 100 degrees to set it to 0 degrees at a rate of less than 95 deg/sec.
 - bz After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
 - ca Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
 - cb Record the values of labels 327 (p), 326 (q) and 330 (r) for three minutes at a frequency of 60Hz.
 - cc Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
 - cd Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.01 degrees.
 - ce Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.01 degrees.
 - cf Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - cg Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - ch Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - ci Make sure that the average p value calculated on the recorded sample is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
 - cj Make sure that the average qvalue calculated on the recorded sample is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
 - ck Make sure that the average r value calculated on the recorded sample is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
- 11 Instrument Panel Slant Angle Programming Pins
- a De-energize the instrument.
 - b Set the instrument support fixture to an instrument panel slant angle of 16 degrees.
 - c Connect the instrument in the configuration for a 16-degree instrument panel slant angle ([Refer to Table 1017](#)).
 - d Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
 - e Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
 - f Energize the instrument.
 - g At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.

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- h Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- i Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.
- j De-energize the instrument.
- k Set the instrument support fixture to an instrument panel slant angle of 15 degrees.
- l Connect the pins b and d to ground (pin G) and put the pins c and e in open circuit.
- m Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- n Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- o Energize the instrument.
- p At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- q Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- r Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.

12 Instrument Steady State Test

- a This test can be done after the test of the instrument panel slant angle programming pins or during the test of the altitude and airspeed functions ([Refer to SUBTASK-34-22-45-750-047-A01](#)).
- b Install the instrument on a vertical support fixture set to an instrument panel slant angle of 0 degrees with a precision of 0.1 degrees in roll and pitch.
- c Install the instrument and support fixture assembly on the two-axis table set to a angle of 0 degrees in roll and pitch, or on a horizontal surface (with a precision of 0.1 degrees).
- d Connect the instrument in the operational configuration with RS422 ([Refer to Table 1016](#)).
- e Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- f Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).

NOTE: Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:

- Pitch: label 324
- Roll: label 325.

- g Energize the instrument.
- h At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- i Make sure that the average pitch and roll values calculated on the recorded samples are 0 ± 0.5 degrees.
- j Let the instrument operate for 1 hour 30 minutes.

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- k At the end of this time, record the labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- l Make sure that the average pitch and roll values calculated on the recorded samples are 0 ± 0.5 degrees.

SUBTASK 34-22-45-750-047-A01

(23) Test of Altitude and Airspeed Functions

(a) Test Position

- Attach the instrument to the test fixture designed for an instrument panel slant angle of 0 degrees.
- Connect the instrument in the operational configuration (with the heading function disabled) ([Refer to Table 1016](#)).

NOTE:

- Connect the ARINC 429 Low Speed output line (pins N and M of connector J1) to the ARINC 429 reader to get the data that follow:
- Indicated airspeed (IAS): label 206
- Corrected baro altitude (Zbaro): label 204
- Standard altitude (Zstd): label 203
- Barometric pressure in hPa (Pbaro): label 234
- Barometric pressure in in.Hg (PbaroUS): label 235
- Mach: label 205.

- 1 Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- 2 Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).

(b) Test of Leakage on the Static and Pitot Ports and Mach Number Display

- 1 Do the test of the leakage on the static and pitot ports as follows:

NOTE: These measurements must be done with the instrument in a power-off condition and in normal test conditions.

- a Connect the static pressure port to a pressure source and barometer, or to a calibrated pressure source.
- b Connect the pitot pressure port to a pressure source and pressure gauge, or to a calibrated pressure source.
- c Decrease the pressure to the values that follow at a maximum rate of change of 20 hPa/sec (1200 hPa/min):
 - Static pressure (Ps): 500 hPa (absolute value)
 - Pitot pressure (Pt): between 525 and 560 hPa.
- d Isolate the instrument-barometer assembly from the pressure sources. The unwanted volume isolated (tubing and barometer) must be less than 0.05 dm³ for each of the sensors.
- e Let the instrument stabilize for one minute, then take an initial reading on the two barometers
- f After another minute, take a second reading on the two barometers.
- g Make sure that the difference between the two readings is not more than 1.7 hPa for each of the pitot and static pressure ports.

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- h Bring the static and pitot pressures back to the normal atmospheric pressure at a maximum rate of change of 20 hPa/sec (1200 hPa/min).
- 2 Do the test of the Mach number display as follows:
 - NOTE: To select the Mach number display, connect the discretes m and k to the discrete ground (pin G).
 - a Energize the instrument.
 - b At the end of the initialization phase, set the pressure values as follows at a maximum rate of change of 20 hPa/sec (1200 hPa/min):
 - Static pressure (Ps): 1013.25 hPa (absolute value)
 - Pitot pressure (Pt): 1250 hPa.
 - c Make sure that the Mach number is shown in the bottom left corner of the display.
 - d Bring the static and pitot pressures back to the normal atmospheric pressure at a maximum rate of change of 20 hPa/sec (1200 hPa/min).
- (c) Test of the Altitude and Airspeed Functions
 - 1 Test of Altitude Accuracy
 - a De-energize the instrument.
 - b Disconnect the discretes m and k from the ground (pin G).
 - c Energize the instrument.
 - d At the end of the initialization phase, press the BARO pushbutton to set the barometric pressure to a value equal to the standard pressure.
 - NOTE: During this test, keep the pitot pressure adjusted to show an airspeed of 200±5 kts.
 - e Increase the pressure-altitude from -2000 feet to +50,000 feet, at a rate of approximately 5000 feet/min. Decrease this rate gradually before each test point ([Refer to Table 1018](#)) to prevent overshoot of the test point.

Table 1018 : Altitude Accuracy Test Points

Pressure-altitude Za in feet (1)	Static pressure (Ps) in hPa (2)	Tolerance ± in feet (3)
-2000	1088.66	20
0*	1013.25	20
1000	977.17	20
2000*	942.13	30
6000	812.00	40
10,000*	696.81	50
20,000	465.63	75
30,000*	300.89	120

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Pressure-altitude Za in feet (1)	Static pressure (Ps) in hPa (2)	Tolerance ± in feet (3)
40,000	187.54	150
50,000	115.97	180

- f For each static pressure value equivalent to a typical Za value (Refer to Table 1018), keep the pressure stable for more than 20 seconds and then record the value Zm shown by the altimeter (label 203).
 - g Make sure that the difference Zm-Za is not more than the tolerance (Refer to Table 1018).
 - h Stop for two minutes at the value of +50,000 feet.
 - i Decrease the pressure-altitude from +50,000 to 0 feet at a rate of approximately 5000 ft/min. Decrease this rate gradually before each test point to prevent overshoot of the test point.
 - j For each static pressure value equivalent to a typical Za value (Refer to Table 1018), keep the pressure stable for more than 20 seconds and then record the value Zstd shown by the altimeter (label 203).
- NOTE: It is possible to do this test only at the altitude values shown with an asterisk (Refer to Table 1018).
- k Make sure that the difference Zstd-Za is not more than the tolerance (Refer to Table 1018).
 - l Reset the pitot pressure Pt to ambient room pressure.

2 Test of Airspeed Accuracy

- a Connect the pitot and static pressure ports to an adjustable pressure source and a barometer, or to a calibrated pressure source

NOTE: During this test, keep the static pressure Ps adjusted to show an altitude of 10,000 ±50 feet.

Table 1019 : Airspeed Accuracy Test Points

Airspeed (A) in Knots (1)	Differential pressure (Pt-Ps) in hPa (2)	Tolerance ± in Knots (3)
40*	2.60	4
70	7.97	4
100	16.30	4
140*	32.13	3
180	53.50	3
200	66.34	3

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Airspeed (A) in Knots (1)	Differential pressure (Pt-Ps) in hPa (2)	Tolerance $\pm$ in Knots (3)
300*	153.55	3
350	212.86	3
400	283.94	3
520	510.28	3

- b Increase the airspeed from 0 to 520 Knots at a rate of approximately 240 Kts/min.
- c For each differential pressure value (Pt-Ps) (Refer to Table 1019) equivalent to the airspeed (A) in column (1), keep the pressure stable for more than 20 seconds and then record the airspeed indication (IAS) shown by the instrument (label 206).
- d Make sure that the difference between the airspeed indication and the value A in column 1 (Refer to Table 1019) is not more than the tolerance in column 3 (Refer to Table 1019).
- e Stop at the value of 520 Knots for 2 minutes.
- f Decrease the airspeed from 520 to 0 Knots at a rate of approximately 240 Kts/min.
- g For each differential pressure value (Pt-Ps) (Refer to Table 1019) equivalent to the airspeed values (A) shown with an asterisk in column (1), keep the pressure stable for more than 20 seconds and then record the airspeed indication (IAS) shown by the instrument (label 206).
- h Make sure that the difference between the airspeed indication and the value A in column 1 (Refer to Table 1019) is not more than the tolerance in column 3 (Refer to Table 1019).
- i Reset the static and pitot pressures to ambient room pressure.

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(24) Final Operational Check

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration (Refer to Table 1013).
- (c) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (d) Energize the instrument.
- (e) Make sure that the self-test page 1 is displayed (Refer to Figure 1024) (or that the related data is output on the RS422 line after the word "T" is transmitted on the RS422 input bus).
- (f) Make sure that all the results of the self-tests are correct.
- (g) Record the value shown on the time counter.
- (h) Transmit the word "S" on the RS422 input bus.

NOTE: The instrument transmits the content of the failure memory on the RS422 output line.

- (i) (Refer to SUBTASK-34-22-45-750-029-A01) and make sure that the content of the failure memory is the same.

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E. Close-up

(1) Not applicable.

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TASK 34-22-45-750-805-A01

5. Tests for C16221QA..

- A. Reason for the job
 - (1) Not applicable.
- B. Job Set-up Information
 - (1) [\(Refer to TASK-34-22-45-750-802-A01\)](#)
- C. Job Set-up
 - (1) Not applicable.
- D. Procedure

SUBTASK 34-22-45-750-049-A01

- (1) Visual Inspection
 - (a) Examine all the inscriptions on the bezel from a distance of approximately 700 mm along a line that goes through the longitudinal axis of the instrument.
 - (b) Make sure that the unit is complete, that it has an inspection seal, and that protective covers are installed on the power supply connector and pressure unions.
 - (c) Make sure that no defect is visible to the naked eye.
 - (d) Make sure that the seals are in good condition.
 - (e) Make sure that the identification plates are correct [\(Refer to PGBLK-34-22-45-1-A01\)](#).
 - (f) The table that follows gives the cross reference between the Boeing part numbers and the Thales part numbers [\(Refer to Table 1020\)](#).

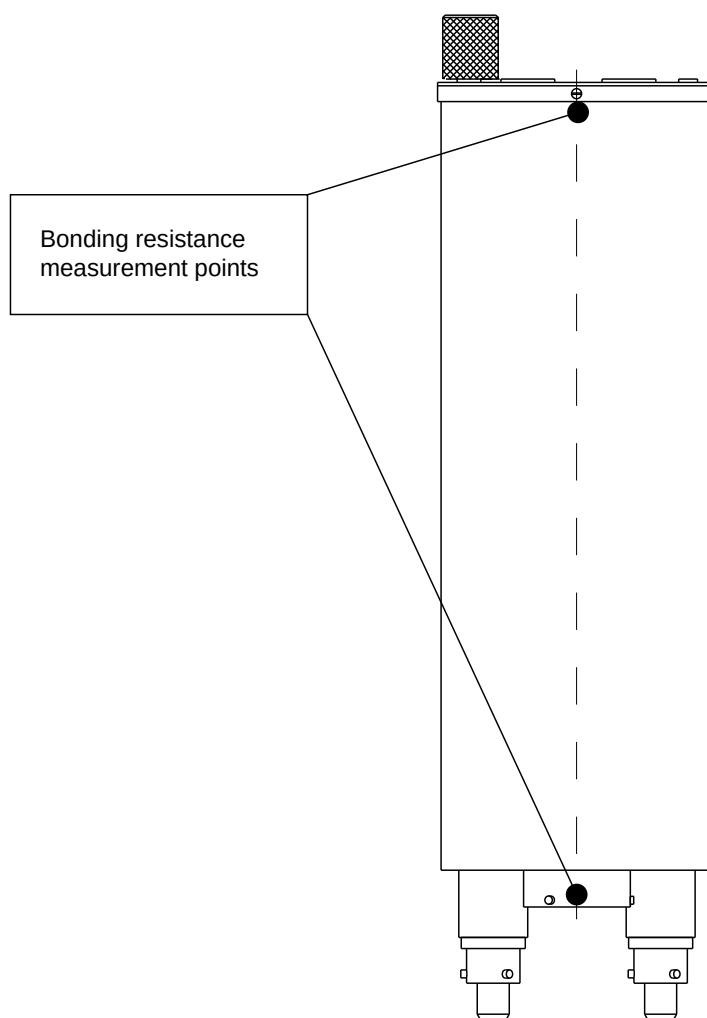
Table 1020 : Part Number Cross Reference

BOEING PN	THALES PN
S231A120-7	C16221QA01

SUBTASK 34-22-45-750-050-A01

- (2) Electrical Bonding Test
 - (a) With a current of 10A applied, measure the bonding resistance between a point on the body of connector J1 and a point on the cover near the bezel [\(Refer to Figure 1030\)](#).
 - (b) Make sure that the bonding resistance is equal to or less than 10 mohms.

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D_CM_TAV_342245_01_ABC0_01_00

Figure 1030/Graphic 34-22-45-991-129-A01
Bonding Resistance Measurement Points

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SUBTASK 34-22-45-750-051-A01

(3) Insulation Resistance Test

- (a) Set a megohmmeter to a voltage of 40VDC.
- (b) Measure the insulation resistance between pins D and E (40VDC input) and pin t (0VDC input).
- (c) Make sure that the insulation resistance is equal to or higher than 40 Mohms.

SUBTASK 34-22-45-750-052-A01

(4) Test of the Presence of the Lightning Strike Protection Diodes

(a) Without amendment H

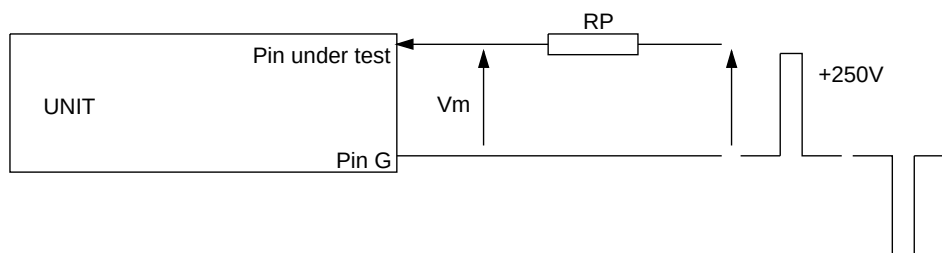
- 1 Make the test setup shown on the diagram ([Refer to Figure 1031](#)).
- 2 Do the calibration of the pulses at a voltage higher than the limited voltage under test as follows:
 - a Connect the pulse generator to an oscilloscope through a BNC cable.
 - b Set the repetition period to 10ms, the pulsewidth to $60\mu s \pm 10\mu s$ and the amplitude to $+240V \pm 10V$ and $-240V \pm 10V$.
- 3 Generate the pulses as follows:

NOTE: The limiting of the voltage applied is done in the instrument.

 - a Connect the pulse generator to the pin under test ([Refer to Table 1021](#)), with the ground on pin G.
 - b Transmit positive and then negative pulses, and read the limited voltage value for each type of pulse.
 - c ([Refer to Table 1021](#)) for the voltage values.

Table 1021 : Table of Test Pins

Signal	Pins	Pins under test	+250V pulse		-250V pulse	
			Rprotection (Ohms)	Vmeasured (V)	Rprotection (Ohms)	Vmeasured (V)
Power supply	D, E	D, E	255	50 ± 10	test not to be performed	
Bezel dimming	i, j	i, j	10K	180 ± 18	10K	-180 ± 18
ARINC In and spare	A,W,U,V,S,T,B,C	A,W,U,V,S,T,B,C	10K	35 ± 5	10K	-35 ± 5
ARINC Out	M,N,R,P	M,N,R,P	1K	15 ± 5	1K	-15 ± 5
IN discrete	b,c,d,e,k,m,p,r,s,q,Z,f	b,c,d,e,k,m,p,r,s	10K	35 ± 5	10K	-35 ± 5
OUT discrete	g	g	10K	60 ± 10	test not to be performed	



D_CM_TAV_342245_01_ABD0_01_00

Figure 1031/Graphic 34-22-45-991-130-A01
Test Setup for Presence of Lightning Strike Diodes

(b) With amendment H

- 1 Make the test setup shown on the diagram
NOTE: The power supply is connected only for the test of ARINC429 output.
- 2 Do the calibration of the pulses at a voltage higher than the limited voltage under test as follows:
 - a Connect the pulse generator to an oscilloscope through a BNC cable.
 - b Adjust pulses width at $60\mu s \pm 10\mu s$.
 - c Adjust pulse level at V_{test} on open circuit.
- 3 ISIS connection
 - a Connect pin t to ground.
 - b If power supply needed, connect 28VDC power supply between pin D and E, and connect E to ground.
 - c Note: during test on pin D, E shall be disconnected.
- 4 Generate the pulses as follows:
 - a Adjust pulses generator level at V_{test}
 - b Connect the pulse generator to the pin under test (Refer to Table 1022) through resistor R_p , with the ground on pin G.
 - c Apply 5 pulses.
 - d Transmit positive and then negative pulses, and read the limited voltage value for each type of pulse (Refer to Table 1022).

Table 1022 : Table of Test Pins

Signal	Pin	Power Supply	Vtest VDC	Rp	Measurement VDC	
					Upper Limit	Low Limit
28V	D	OFF	240	255Ω	59	49
			-240		-1	-7
ARINC 429 Inputs	A, W, U, V, S, T, B,C	OFF	240	10kΩ	55	43
			-240		-43	-55
Discrete Inputs	b, c, d, e, k, m, p, r, s, f, q, Z	OFF	240	10kΩ	45	33
			-240		-33	-45
Discrete Input	n	OFF	240	10kΩ	55	43
			-240		-43	-55
Discrete outputs	g	OFF	80	10kΩ	55	45
			-80		-45	-55
ARINC 429 Outputs	M, N, R, P	ON	180	1kΩ	19.5	17.5
			-180		-17.5	-19.5

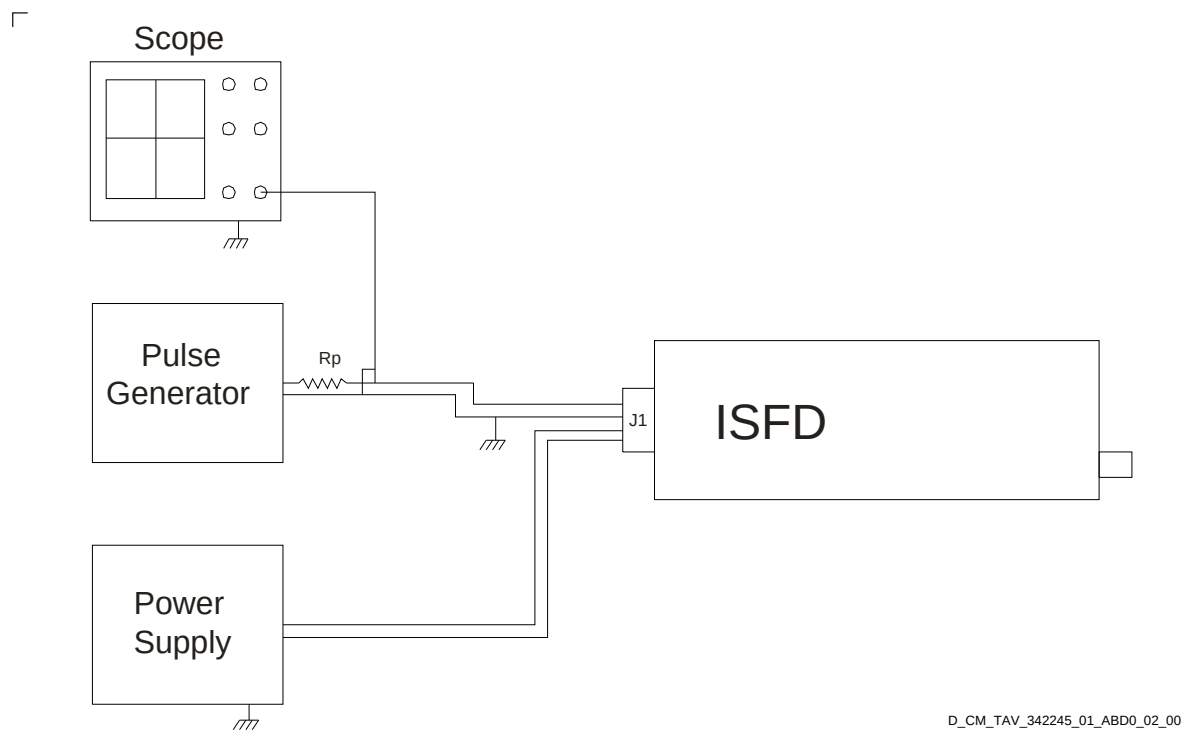


Figure 1032/Graphic 34-22-45-991-130-A02
Test Setup for Presence of Lightning Strike Diodes

(c)

1 Dimming Inputs : Input impedance measurement (Power off)

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- a Connect +40V voltage source successively between pin i, j, and pin t (or pin G).
- b Measure input current i, j : $1.0\text{mA} < I < 1.3\text{mA}$,
- c Or Measure input impedance (V/I) : $30\text{k} < R < 40\text{k}$.

SUBTASK 34-22-45-750-053-A01

(5) Instrument Operating Position and Check of the Fault Memory

- (a) Attach the instrument to a suitable test fixture and connect it in the operational configuration ([Refer to Table 1023](#))S.
- (b) Adjust the test fixture to 0 degrees in roll with a precision of ± 1 degree.
- (c) Adjust the test fixture to 0 degrees in pitch with a precision of ± 1 degree.
- (d) Change to maintenance mode as follows:
 - 1 Connect the discrete pins X and a to the common ground discrete (pin G).
- (e) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (f) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (g) Energize the instrument.
- (h) Transmit the word "S" on the RS422 input bus and make sure that the instrument transmits the content of the fault memory on the RS422 output bus. Record the result for subsequent comparison ([Refer to SUBTASK-34-22-45-750-074-A01](#)).

Table 1023 : Operational Configuration for 0-degree Instrument Panel Slant Angle

J1 pin	SIGNAL	I/O	COMMENTS	WIRED
B	ARINC_E_6_A	I	Arinc link for static pressure SADM on B777	YES
C	ARINC_E_6_B	I		
t	CHASSIS_GND		Case ground	YES
D	28V	I	+28V DC	YES
E	0_28V	I	0V DC	YES
A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
W	ARINC_E_1_B	I		
j	0_LUM_SCR	I	Bezel Lighting Control	YES
V	ARINC_E_2_A	I	Arinc link for pitot pressure SADM on B777	YES
U	ARINC_E_2_B	I		
i	LUM_SCR	I	Bezel Lighting Control	YES
T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
S	ARINC_E_3_B	I		
h	DISCRET_E_1	I	Mode selection n°2	
R	ARINC_S_1_A	O	IR parameters	YES

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P	ARINC_S_1_B	O		
g			SPARE	
N	ARINC_S_2_A	O	ADR parameters	YES
M	ARINC_S_2_B	O		
f	DISCRET_E_2	I	Backcourse display disabled	
X	DISCRET_E_3	I	Mode selection n°1	
Y			SPARE	
Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
a	DISCRET_E_5	I	Mode selection n°3	
b	DISCRET_E_6	I	A/C configuration 8/0	
c	DISCRET_E_7	I	A/C configuration 4/0	
d	DISCRET_E_8	I	A/C configuration 2/0	
e	DISCRET_E_9	I	A/C configuration 1/0	
k	DISCRET_E_10	I	A/C configuration parity	
m	DISCRET_E_11	I	Mag heading display disabled	
n	DISCRET_E_12	I	APP display disabled	
p	DISCRET_E_13	I	SPARE (for Vbar option)	
q	DISCRET_E_14	I	Initiated test	
r	DISCRET_E_15	I	SPARE	
s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
G	ELEC_GND	I	Discrete common/ground	
L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
K	RS422_E_L	I		
J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
H	RS422_S_H	O		

SUBTASK 34-22-45-750-054-A01

(6) Power Consumption and Validity Output Discrete Test

- (a) De-energize the instrument.
- (b) Connect the instrument in its operational configuration ([Refer to Table 1023](#)).
- (c) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).

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- (d) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (e) Energize the instrument in average ambient lighting conditions (between approximately 300 and 600 lux).
- (f) Let the instrument operate for one minute and then measure the power consumption.
- (g) Make sure that the power consumption is equal to or less than 0.7A.
- (h) Make sure that the validity output discrete F is connected to ground (pin G).

SUBTASK 34-22-45-750-055-A01

(7) Test of the Operating Configuration

- (a) Connect the instrument in the operational configuration ([Refer to Table 1024](#)).

Table 1024 : Operational Configuration for 0-degree Instrument Panel Slant Angle
(B777)

J1 pin	SIGNAL	I/O	COMMENTS	WIRED
B	ARINC_E_6_A	I	Arinc link for static pressure SADM on B777	YES
C	ARINC_E_6_B	I		
t	CHASSIS_GND		Case ground	YES
D	28V	I	+28V DC	YES
E	0_28V	I	0V DC	YES
A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
W	ARINC_E_1_B	I		
j	0_LUM_SCR	I	Bezel Lighting Control	YES
V	ARINC_E_2_A	I	Arinc link for pitot pressure SADM on B777	YES
U	ARINC_E_2_B	I		
i	LUM_SCR	I	Bezel Lighting Control	YES
T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
S	ARINC_E_3_B	I		
h	DISCRET_E_1	I	Mode selection n°2	
R	ARINC_S_1_A	O	IR parameters	YES
P	ARINC_S_1_B	O		
g			SPARE	
N	ARINC_S_2_A	O	ADR parameters	YES
M	ARINC_S_2_B	O		
f	DISCRET_E_2	I	Backcourse display disabled	

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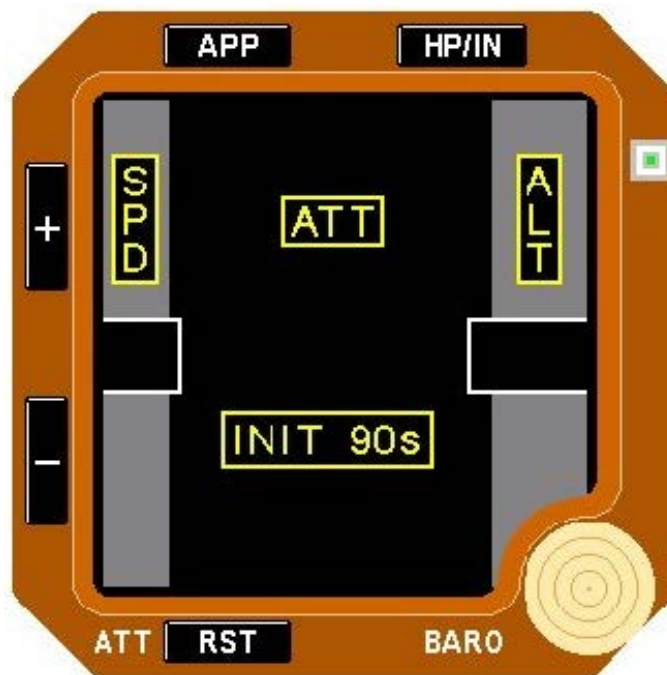
X	DISCRET_E_3	I	Mode selection n°1	
Y			SPARE	
Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
a	DISCRET_E_5	I	Mode selection n°3	
b	DISCRET_E_6	I	A/C configuration 8/0	GND
c	DISCRET_E_7	I	A/C configuration 4/0	
d	DISCRET_E_8	I	A/C configuration 2/0	GND
e	DISCRET_E_9	I	A/C configuration 1/0	
k	DISCRET_E_10	I	A/C configuration parity	
m	DISCRET_E_11	I	Mag heading display disabled	
n	DISCRET_E_12	I	APP display disabled	
p	DISCRET_E_13	I	SPARE (for Vbar option)	
q	DISCRET_E_14	I	Initiated test	
r	DISCRET_E_15	I	SPARE	
s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
G	ELEC_GND	I	Discrete common/ground	
L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
K	RS422_E_L	I		
J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
H	RS422_S_H	O		

- (b) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (c) Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (d) Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- (e) Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- (f) Connect the ARINC 429 LS output bus (bus 5, pins N and M) to the ARINC 429 receiver to acquire the following data:
 - ISFD status and fault: Label 350
- (g) Energize the instrument.
- (h) Make sure that the initialization phase 1 display is shown ([Refer to Figure 1033](#)).

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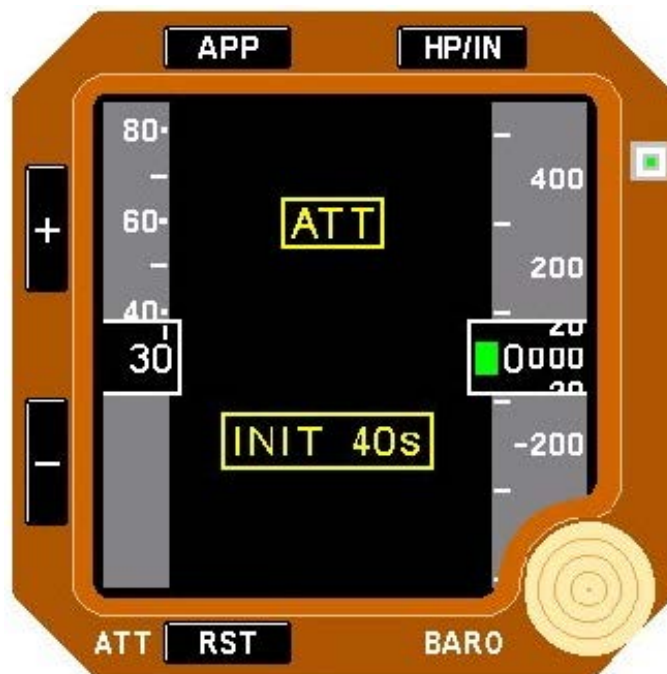
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D_CM_TAV_342245_01_ABE0_01_00

Figure 1033/Graphic - Sheet 01 / 34-22-45-991-131-A01
Initialization Phase 1



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Figure 1033/Graphic - Sheet 02 / 34-22-45-991-131-A01
Initialization Phase 2

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- (i) Make sure that the display changes to the initialization phase 2 not more than 15 seconds after power-up (Refer to Figure 1033).

NOTE: Ignore the altitude and airspeed values shown on the display.

- (j) Make sure that all other bits are reset.

- (k) Make sure the SSM = 00 and SDI = 00.

- (l) At the end of the initialization phase, make sure that the instrument shows the operational display (Refer to Figure 1034).

NOTE: Ignore the barometric pressure unit (which can be HP for hectopascals or IN for inches of mercury) and the numeric values.



D_CM_TAV_342245_01_ABF0_01_00

Figure 1034/Graphic - Sheet 01 / 34-22-45-991-132-A01
Operational Display 1

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D_CM_TAV_342245_01_ABFO_02_00

Figure 1034/Graphic - Sheet 02 / 34-22-45-991-132-A01
Operational Display 2



D_CM_TAV_342245_01_ABFO_03_00

Figure 1034/Graphic - Sheet 03 / 34-22-45-991-132-A01
Operational Display 3

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- (m) Make sure that the bezel does not have any defects that can interfere with its use.
- (n) Make sure that the markings and inscriptions are easy to see and read.
- (o) Make sure that only bit 26 of label 350 is set to 1.
- (p) Make sure that the display does not present any defects (line failure) representing a hindrance to its use.

SUBTASK 34-22-45-750-056-A01

(8) Test of the Pushbuttons and Knobs

- (a) Press the HP/IN pushbutton once and make sure that the barometric pressure unit changes (from HP to IN or from IN to HP).
- (b) Turn the BARO knob clockwise (approximately a quarter turn) and make sure that the barometric pressure value increases.
- (c) Turn the BARO knob counterclockwise (approximately a quarter turn) and make sure that the barometric pressure value decreases.
- (d) Press the BARO knob and make sure that the barometric setting changes from the adjustable barometric pressure to the standard value.
- (e) Press the RST pushbutton for 2 seconds. Make sure that the instrument resets to zero and that the message "ATT10s" is displayed (Refer to Figure 1035).

NOTE: Ignore the values displayed.



D_CM_TAV_342245_01_ABG0_01_00

Figure 1035/Graphic 34-22-45-991-133-A01
ATT 10s Page

SUBTASK 34-22-45-750-057-A01

(9) Test of the TEST Function

- (a) Press the HP/IN and APP buttons together for 2 seconds.

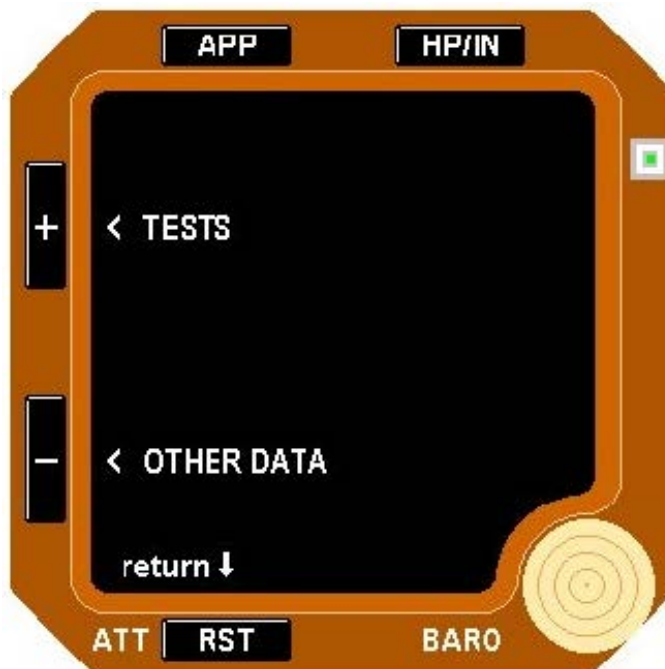
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- (b) Make sure that the first test page is displayed ([Refer to Figure 1036](#)).

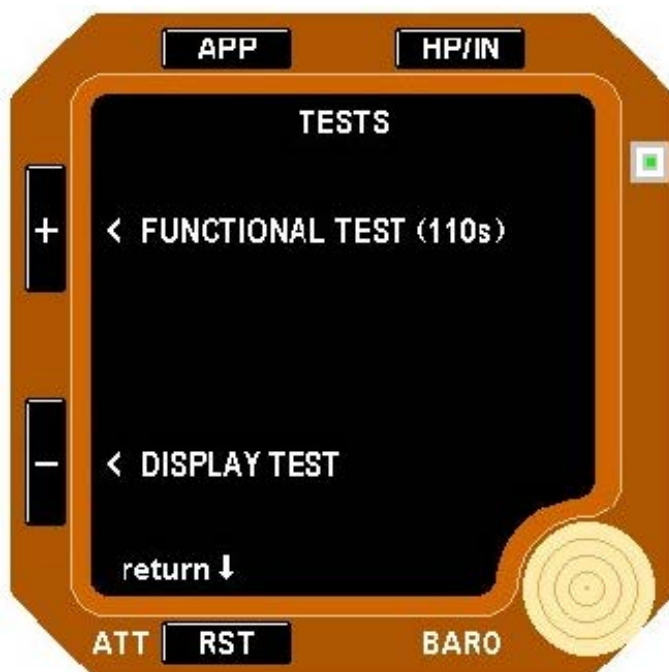
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D_CM_TAV_342245_01_ABH0_01_00

Figure 1036/Graphic 34-22-45-991-134-A01
First Test Page

- (c) Press the "+" button and make sure that the second test page is displayed ([Refer to Figure 1037](#)).



D_CM_TAV_342245_01_ABI0_01_00

Figure 1037/Graphic 34-22-45-991-135-A01
Second Test Page

- (d) Press the "+" button again and make sure that "Test in progress 110s" is displayed in less than 10 seconds.
- (e) At the end of the test, make sure that "Test OK" is displayed.
- (f) Press the RST button three times and make sure that the screen goes back to the operational display (Refer to Figure 1034).
- (g) Connect the pin q to ground for at least 0.4 seconds and then put it in open circuit in less than 6 seconds.
- (h) Make sure that "Test in progress 110s" is displayed in less than 10 seconds.
- (i) After 110 seconds, make sure that the screen goes back to the operational display (Refer to Figure 1034).

SUBTASK 34-22-45-750-058-A01

(10) Test of the HEADING Function

- (a) Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- (b) Transmit the label 312 (with its value set to zero) and the label 320 (with its value set to 30) on the HS bus 3 (pins T and S).
- (c) Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- (d) Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- (e) Make sure that the heading value displayed is 30 degrees.
- (f) De-energize the instrument.
- (g) Connect the pins m and k to the discrete ground (pin G) to disable the heading display.
- (h) Energize the instrument.

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- (i) At the end of the initialization phase, make sure that no heading value is displayed.
 - (j) De-energize the instrument and put the pins m and k in open circuit.
- SUBTASK 34-22-45-750-059-A01

(11) Test of the ILS Function

(a) Test of the ILS Front Course Function

- 1 Transmit the label 173 (with its value set to 0.0775) and the label 174 (with its value set to 0.0875) on the LS bus 1 (pins A and W).
- 2 Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- 3 Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- 4 Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- 5 Energize the instrument.
- 6 At the end of the initialization phase, press the APP button to display the glideslope and localizer data.
- 7 Make sure that the indication APP is displayed in the upper-left corner.
- 8 Make sure that the ILS indications are correct ([Refer to Table 1025](#)).

Table 1025 : ILS Front Course Indications

	GLIDESLOPE		LOCALIZER	
	FLAG	BAR	FLAG	BAR
Position	Not in view	First mark down	Not in view	First mark right

(b) Test of the ILS Back Course Function

- 1 Press the APP button to change to backcourse approach mode.
- 2 Make sure that the indication BCRS is displayed in the upper-left corner.
- 3 Make sure that the ILS indications are correct ([Refer to Table 1026](#)).

Table 1026 : ILS Back Course Indications

	GLIDESLOPE		LOCALIZER	
	FLAG	BAR	FLAG	BAR
Position	Not in view	Not in view	Not in view	First mark left

- 4 Press the APP button and make sure that the localizer and glideslope data go out of view.

SUBTASK 34-22-45-750-060-A01

(12) ILS Discretes

- (a) De-energize the instrument.
- (b) Set the discrete signals n and k to discrete ground (pin G) in order to disable APP display.
- (c) Energize the instrument.
- (d) At the end of the initialization phase, press the APP button four times.
- (e) Make sure that only BCRS page is displayed.

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- (f) De-energize the instrument.
- (g) Set the discrete signals n to open state and f and k to the discrete ground (pin G) in order to disable BCRS display.
- (h) Energize the instrument.
- (i) At the end of the initialization phase, press the APP button four times.
- (j) Make sure that only APP page is displayed.

SUBTASK 34-22-45-750-061-A01

(13) Bezel Lighting Test

- (a) Put a mask in front of the photoelectric cell to simulate night lighting conditions.
- (b) Set the bezel lighting to the maximum brightness value (5VAC/400Hz).
- (c) Make sure that the lighting of the pushbuttons (only) is uniform.
- (d) Make sure that there is no unwanted leakage of light.

SUBTASK 34-22-45-750-062-A01

(14) Tests of the Options (V-bar, Slip and Altitude in Meters)

- (a) De-energize the instrument.
- (b) Connect the discretes for V-bar selection (pin p), slip selection (pin s) and altitude in meters selection (pin Z) to the discrete ground (pin G) and set the discrete on pin m to the open state.
- (c) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (e) Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- (f) Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- (g) Energize the instrument.
- (h) At the end of the initialization phase, make sure that the correct page is displayed ([Refer to Figure 1038](#)).

NOTE: Ignore the values displayed.

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Figure 1038/Graphic 34-22-45-991-136-A01
V-Bars, Slip and Altitude In Meters Display

SUBTASK 34-22-45-750-063-A01

(15) Display of Self-Test Results

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration ([Refer to Table 1024](#)).
- (c) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (d) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (e) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (f) Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- (g) Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- (h) Energize the instrument.
- (i) Make sure that the self-test page 1 is displayed in less than 10 seconds ([Refer to Figure 1039](#)).
- (j) Transmit the word "T" on the RS422 input bus and make sure that the related display data are transmitted on the RS422 output bus.
- (k) Press the RST button (or transmit the words "A" and then "B" on the RS422 bus) to make the display change to page 2 ([Refer to Figure 1040](#)).

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Page 1

P/N : C16221QA01

S/N : C16221XXXXXX

A/C CONFIG : 00 DEG

APP HDG BCRS

FUNCT TIME : XXXXX h

ISFD CHECK : OK

D_CM_TAV_342245_01_ABK0_01_00

Figure 1039/Graphic 34-22-45-991-137-A01
Self-Test Display (Page 1)

Page 2

HOURS	FAILURE CODE
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX
XXXXX	XXXX XXXX XXXX

D_CM_TAV_342245_01_ABL0_01_00

Figure 1040/Graphic 34-22-45-991-138-A01
Self-Test Display (Page 2)

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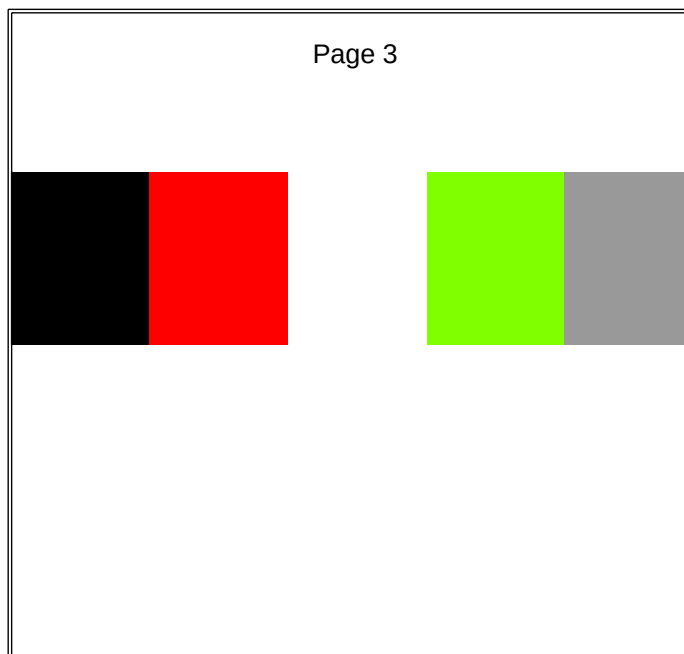
- (l) Transmit the word "T" on the RS422 input bus and make sure that the related display data are transmitted on the RS422 output bus.

SUBTASK 34-22-45-750-064-A01

(16) Test of the Brightness and Contrast Test Displays

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 3 ([Refer to Figure 1041](#)).

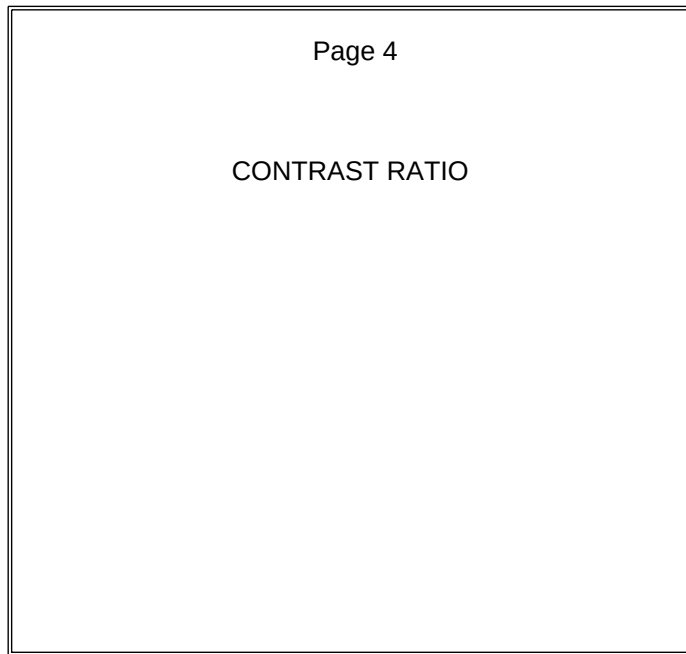
┌



D_CM_TAV_342245_01_ABM0_01_00

Figure 1041/Graphic 34-22-45-991-139-A01
Self-Test Display (Page 3)

- (b) Press the RST pushbutton again (or transmit the words "A" and then "B" on the RS422 bus) and make sure that the display changes to page 4 ([Refer to Figure 1042](#)).



D_CM_TAV_342245_01_ABNO_01_00

Figure 1042/Graphic 34-22-45-991-140-A01
Self-Test Display (Page 4)

SUBTASK 34-22-45-750-065-A01

(17) Test of the Checksum Page Display

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 5 ([Refer to Figure 1043](#)).

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Page 5	
CHK CPU	70EE 3675
CHK GS	0003 66EE
CHK PAR	56A9 7E86

D_CM_TAV_342245_01_ABO0_01_00

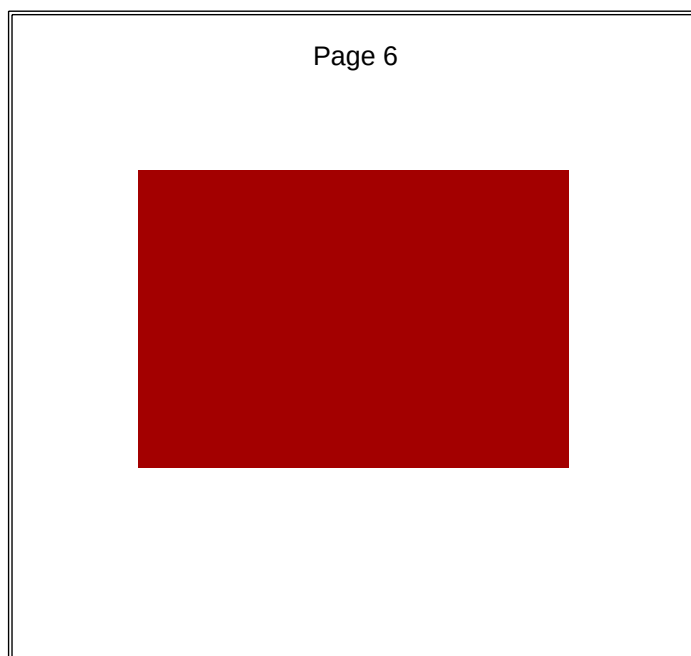
Figure 1043/Graphic 34-22-45-991-141-A01
Checksum Test Display (Page 5)

SUBTASK 34-22-45-750-066-A01

(18) Test of the Color Test Page Display

- (a) Press the RST button (or transmit the words "A" and then "B" on the RS422 input bus) and make sure that the display changes to page 6 ([Refer to Figure 1044](#)).

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D_CM_TAV_342245_01_ABP0_01_00

Figure 1044/Graphic 34-22-45-991-142-A01
Color Test Page (Page 6)

- (b) Make sure that the rectangle shows the ground color.
- (c) Press the RST button again and make sure that the rectangle changes to the sky color.
- (d) Press the RST button again and make sure that the rectangle changes to the ground color.
- (e) Press the RST button again and make sure that the rectangle changes to gray.
- (f) Press the RST button again and make sure that the rectangle changes to white.
- (g) Press the RST button again and make sure that the rectangle changes to black.
- (h) Press the RST button again and make sure that the rectangle changes to black.
- (i) Press the RST button again and make sure that the rectangle changes to green.
- (j) Press the RST button again and make sure that the rectangle changes to red.
- (k) Press the RST button again and make sure that the rectangle changes to yellow.
- (l) Press the RST button again and make sure that the rectangle changes to white.
- (m) Press the RST button again and make sure that the rectangle changes to black.
- (n) Press the RST button again and make sure that the rectangle changes to magenta.
- (o) Press the RST button again and make sure that the rectangle changes to yellow.
- (p) Press the RST button again and make sure that the rectangle changes to green + white.
- (q) Press the RST button again and make sure that the rectangle changes to white.
- (r) De-energize the instrument.

SUBTASK 34-22-45-750-067-A01

(19) Test of SADM Pressure via ARINC input test in B777 configuration

NOTE: For all these measurements, the acquisition window for average of altitude (label

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204) and IAS ARINC (label 206) is 3 seconds.

The acquisition is started 10 seconds after Ps, Pt labels update.

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration ([Refer to Table 1024](#)).
- (c) Connect the pins m and k to discrete ground (pin G) to change to select Mach number display.
- (d) Transmit the labels 173 (with the value set to 0.0775) and 174 (with the value set to 0.0875) on the LS bus 1 (pins A and W).
- (e) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (f) Transmit the labels 242 (Pt- with the value set to 1013.25) on the LS bus 2 (pins V and U).
- (g) Transmit the labels 245 (Ps- with the value set to 1013.25) on the LS bus 6 (pins B and C).
- (h) Connect the ARINC 429 LS output bus (bus 5- pins N and M on connector J1) to the ARINC 429 receiver to acquire the following data:
 - Baro Corrected Altitude: label 204,
 - Computed Airspeed: label 206,
- (i) Energize the instrument.
- (j) At the end of the initialization phase, press the BARO pushbutton in order to have barometric pressure = standard pressure. make sure that the instrument shows the operational display.
- (k) Make sure that altitude and speed tapes are displayed.
- (l) Make sure that altitude value is 0 ft $\pm$ 1 ft (output LSB).
- (m) Set the Ps and Pt pressure values to 942.13 hPa (absolute value), at a maximum variation rate of 20 hPa/s.
- (n) Make sure that altitude value is 2000 ft $\pm$ 1 ft (output LSB).
- (o) Set the Ps and Pt pressure values to 1013.25 hPa (absolute value), at a maximum variation rate of 20 hPa/s.
- (p) Make sure that altitude value is 0 ft $\pm$ 1 ft (output LSB).
- (q) Set the Ps and Pt pressure values to 1013.25 $\pm$ 32.13 hPa (absolute value), at a maximum variation rate of 20 hPa/s.
- (r) Make sure that altitude value is 140 kts $\pm$ 0.0625 kts (output LSB).
- (s) Set the Pt pressure value to 1250 hPa, at a maximum variation rate of 20 hPa/s (1200 hPa/mn).
- (t) Make sure that Mach number is displayed at the lower left corner.
- (u) Set the Ps and Pt pressure values to 1013.25 (absolute value), at a maximum variation rate of 20 hPa/s.
- (v) Make sure that altitude value is 0 ft $\pm$ 1 ft (output LSB).

SUBTASK 34-22-45-750-068-A01

(20) Test of the Display Brightness and Color

NOTE: All the brightness tests are done in the dark (ambient lighting less than 0.3 fc or 3 Lux). The brightness and color are measured perpendicular to the screen.

- (a) Connect the instrument in the operational configuration ([Refer to Table 1024](#)).

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NOTE: In this configuration, brightness is controlled by dimmer input signal.
"+" and "-" pushbuttons shall not be used and "Button correction coefficient" on brightness is assumed to be "1" (power-on default value) during the tests.

- (b) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (c) Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- (d) Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- (e) Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- (f) Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- (g) Connect the DC voltage generator to pin i, j (dimmer input) and adjust DC voltage to 1VDC.
- (h) Energize the instrument.
- (i) Press the RST button two times (or transmit the words "A" and then "B" two times on the RS422 input bus) to make the display change to page 3 ([Refer to Figure 1041](#)).
- (j) Measure the brightness value at the center of the white square.

SUBTASK 34-22-45-750-069-A01

(21) Test of the Display Brightness in Night Conditions

- (a) Put a mask in front of the photoelectric cell to simulate night conditions.
- (b) Let the instrument stabilize at this value for 3 minutes.
- (c) Make sure that the brightness value is between 0.1fL and 0.3fL (0.34Cd/m2 and 1.03Cd/m2).
- (d) Increase the dimmer DC voltage until 4.8 ± 0.02 VDC.
- (e) Let the instrument stabilize at this value for 1 minutes.
- (f) Make sure that the brightness value is between 0.75fL and 1.05fL (2.55Cd/m2 and 3.57Cd/m2).

SUBTASK 34-22-45-750-070-A01

(22) Test of the Display Brightness in Full Sunlight Conditions

- (a) Apply a lighting level of 5000 Lux minimum to the photoelectric cell to simulate full sunlight conditions.
- (b) Increase the dimmer DC voltage until 7 VDC and wait for 10 seconds.
- (c) Let the instrument stabilize at this value for 3 minutes.
- (d) Make sure that the brightness value is between 180fL and 205fL (617Cd/m2 and 702Cd/m2).

SUBTASK 34-22-45-750-071-A01

(23) Test of the Ground and Sky Color Coordinates

- (a) Increase the dimmer DC voltage until 7VDC and wait for 10 seconds.
- (b) Press the RST button (or transmit the word "A" and then "B" on the RS422 input bus four times) to change to page 6b (Custom Shade Sky).
- (c) Measure the color coordinates in the center of the colored rectangle.
- (d) Make sure that the 1976 CIE color coordinates measures are as follows:

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- (e) Make sure that the brightness value is between 180fL and 205fL (617Cd/m2 and 702Cd/m2).
 - Sky: $u' = 0.14 \pm 0.03$, $v' = 0.360 \pm 0.03$
- (f) Press the RST button (or transmit the word "A" and then "B" on the RS422 input bus one time) to change to page 6c (Custom Shade Ground).
- (g) Make sure that the brightness value is between 180fL and 205fL (617Cd/m2 and 702Cd/m2).
 - Ground: $u' = 0.22 \pm 0.02$, $v' = 0.53 \pm 0.02$.

SUBTASK 34-22-45-750-072-A01

(24) Tests of the Attitude Functions

NOTE: For this test, it is possible to use two procedures, Procedure 1 or Procedure 2, but it is preferable to use Procedure 2.

(a) Procedure 1

- 1 Measure the ambient room temperature and the temperature at a point on the instrument case near the bezel ([Refer to Figure 1030](#)) for the measurement point.
- 2 Make sure that the difference between the two temperatures is less than 3°C.
- 3 Install the instrument on the test support designed for an instrument panel slant angle of 0 degrees, with a precision of 0.1 degrees in roll and pitch.
- 4 Install the assembly on a two-axis table.
- 5 Change to operational mode with RS422 ([Refer to Table 1027](#)).

Table 1027 : Operational Configuration for 0-degree Instrument Panel Slant Angle with RS422

J1 No.	J1 pin	SIGNAL	I/O	COMMENTS	WIRED
1	B	ARINC_E_6_A	I	ARINC link for static pressure SADM on B777	
2	C	ARINC_E_6_B	I		
3	t	CHASSIS_GND		Case ground	YES
4	D	28V	I	+28V DC	YES
5	E	0_28V	I	0V DC	YES
6	A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
7	W	ARINC_E_1_B	I		
8	j	0_LUM_SCR	I	Bezel Lighting Control	YES
9	V	ARINC_E_2_A	I	ARINC link for pitot pressure SADM on B777	YES
10	U	ARINC_E_2_B	I		
11	i	LUM_SCR	I	Bezel Lighting Control	YES
12	T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
13	S	ARINC_E_3_B	I		

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14	h	DISCRET_E_1	I	Mode selection n°2	
15	R	ARINC_S_1_A	O	IR parameters for Cat III option	
16	P	ARINC_S_1_B	O		
17	g			SPARE	
18	N	ARINC_S_2_A	O	ADR parameters for Cat III option	
19	M	ARINC_S_2_B	O		
20	f	DISCRET_E_2	I	Backcourse display disabled	
21	X	DISCRET_E_3	I	Mode selection n°1	
22	Y			SPARE	
23	Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
24	a	DISCRET_E_5	I	Mode selection n°3	
25	b	DISCRET_E_6	I	A/C configuration 8/0	
26	c	DISCRET_E_7	I	A/C configuration 4/0	
27	d	DISCRET_E_8	I	A/C configuration 2/0	
28	e	DISCRET_E_9	I	A/C configuration 1/0	
29	k	DISCRET_E_10	I	A/C configuration parity	
30	m	DISCRET_E_11	I	Mag heading display disabled	
31	n	DISCRET_E_12	I	APP display disabled	
32	p	DISCRET_E_13	I	SPARE (for Vbar option)	
33	q	DISCRET_E_14	I	Initiated test	
34	r	DISCRET_E_15	I	SPARE	
35	s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
36	F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
37	G	ELEC_GND	I	Discrete common/ground	
38	L	RS422_E_H	I	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
39	K	RS422_E_L	I		
40	J	RS422_S_H	O	Downloading : 115200 Bds, 8 data bits, 1 stop bit, not parity	YES
41	H	RS422_S_H	O		

6 Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).

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- 7 Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- 8 Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:
 - Pitch: label 324
 - Roll: label 325
 - p: label 327
 - q: label 326
 - r: label 330
 - Xacc: label 331
 - Yacc: label 332
 - dpi: label 157
 - dqj: label 160
 - dri: label 161
 - Sw_NAV1: label 351
 - Sw_NAV2: label 352
- 9 For the labels 331, 332 and 333, the resolution has been improved (LSB = 11 and not 17). This modification must be taken into consideration for the following test done on these labels.
- 10 Initialization Flag
 - a Energize the instrument.
 - b Make sure that the labels are present on the ARINC reader.
 - c Read the discrete S_O_B_NAV_MODE on label 351 (bit 12 changes from 0 to 1) to make sure that the attitude-function initialization time of the instrument is equal to or less than 91 seconds.
 - d Read the discrete S_I_B_3ACC on label 352 (bit 18 = 1) to make sure that the two accelerometer is present.

NOTE: The numbers of the 32 bits go from 1 to 32.
- 11 Horizon Line Movements
 - a Record the values of the labels 324 (pitch) and 325 (roll) for a period of three minutes at a frequency of 60Hz.
 - b Record the values of the labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for a period of three minutes at a frequency of 60Hz.
 - c Record the values of the labels 327 (p), 326 (q) and 330 (r) for a period of three minutes at a frequency of 60Hz.
 - d For the pitch and roll parameters, make sure that:
 - The SSM of labels 324 and 325 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to 0 ± 0.5 degrees.
 - The noise value calculated on the recorded samples is less than 0.17 degrees for 2 sigma.
 - e For the Xacc, Yacc and Zacc parameters, make sure that:
 - The SSM of labels 331, 332 and 333 is NO (bits 30 and 31 set to 1).

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- The average value calculated on the recorded samples is equal to $0g \pm 0.01g$.
- The noise value calculated on the recorded samples is less than $0.002g$ for 2 sigma.
- f For the p, q and r parameters, make sure that:
 - The SSM of labels 327, 326 and 330 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
 - The noise value calculated on the recorded samples is less than 0.08 deg/sec for 2 sigma.
- g Adjust the table to 20 degrees nose-up in pitch at a rate of 5 deg/sec .
- h After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- i Record the values of labels 331 (Zacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- j Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- k Make sure that the average pitch value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- l Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- m Make sure that the average Xacc value calculated on the recorded sample is equal to $0.342g \pm 0.01g$.
- n Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- o Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- p Adjust the table to -20 degrees in pitch at a rate of 5 deg/sec .
- q After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- r Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- s Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- t Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- u Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- v Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- w Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- x Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- y Reset the table to 0 degrees in pitch at the rate of 5 deg/sec .

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- z After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aa Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ab Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- ac Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- ad Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- ae Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- af Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ag Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ah Adjust the table to 20 degrees in clockwise roll at a rate of 5 deg/sec.
- ai After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aj Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ak Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- al Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- am Make sure that the average roll value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- an Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ao Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- ap Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- aq Adjust the table to 20 degrees in counterclockwise roll at a rate of 5 deg/sec.
- ar After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- as Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- at Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- au Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- av Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- aw Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.

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- ax Make sure that the average Yacc value calculated on the recorded sample is equal to $0.342g \pm 0.01g$.
- ay Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- az Reset the table to 0 degrees in roll at the rate of 5 deg/sec.
- ba After one minute, record the values of labels 324 (pitch) and 325 (roll) three minutes at a frequency of 60Hz.
- bb Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bc Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bd Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- be Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bf Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bg Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bh Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bi Adjust the table to 5 degrees in clockwise roll and, at the same time, to 5 degrees in nose-up pitch at the rate of 20 deg/sec.
- bj After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bk Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bl Make sure that the average pitch value calculated on the recorded sample is equal to 5 ± 0.5 degrees.
- bm Make sure that the average roll value calculated on the recorded sample is equal to 5 ± 0.5 degrees.
- bn Make sure that the average Xacc value calculated on the recorded sample is equal to $0.087g \pm 0.01g$.
- bo Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.087g \pm 0.01g$.
- bp Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.01g \pm 0.01g$.
- bq Reset the table to 0 degrees in roll and pitch at a rate of 5 deg/sec.
- br After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bs Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bt Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bu Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

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- bv Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - bw Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - bx Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - by Set the table to 100 degrees in roll at a rate of 5 deg/sec.
 - bz After one minute, record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
 - ca Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
 - cb Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - cc Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.985g \pm 0.01g$.
 - cd Make sure that the average Zacc value calculated on the recorded sample is equal to $-1.174g \pm 0.01g$.
 - ce Reset the table to 0 degrees in roll at a rate of 5 deg/sec.
 - cf After one minute, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
 - cg Record the values of labels 331 (Xacc), 332 Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
 - ch Record the values of labels 327 (p), 326 (q) and 330 (r) for three minutes at a frequency of 60Hz.
 - ci Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
 - cj Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
 - ck Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
 - cl Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - cm Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - cn Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
 - co Make sure that the average value of p calculated on the recorded sample is equal to $0 \text{ deg.sec} \pm 0.5 \text{ deg/sec}$.
 - cp Make sure that the average value of q calculated on the recorded sample is equal to $0 \text{ deg.sec} \pm 0.5 \text{ deg/sec}$.
 - cq Make sure that the average value of r calculated on the recorded sample is equal to $0 \text{ deg.sec} \pm 0.5 \text{ deg/sec}$.
- 12 Instrument Panel Slant Angle Programming Pins
- a De-energize the instrument.
 - b Install the instrument on a support fixture set for an instrument panel slant angle of 16 degrees.

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c (Refer to Table 1028) for the test setup.

Table 1028 : Operational Configuration for 16-degree Instrument Panel Slant Angle

J1 n°	J1 pin	SIGNAL	I/O	COMMENTS	WIRED
1	B	ARINC_E_6_A	I	ARINC link for static pressure SADM on B777	YES
2	C	ARINC_E_6_B	I		
3	t	CHASSIS_GND		Case ground	YES
4	D	28V	I	+28V DC	YES
5	E	0_28V	I	0V DC	YES
6	A	ARINC_E_1_A	I	APP for display and ADR bus (for simulation)	YES
7	W	ARINC_E_1_B	I		
8	j	0_LUM_SCR	I	Bezel Lighting Control	YES
9	V	ARINC_E_2_A	I	ARINC link for pitot pressure SADM on B777	YES
10	U	ARINC_E_2_B	I		
11	i	LUM_SCR	I	Bezel Lighting Control	YES
12	T	ARINC_E_3_A	I	Heading for displayed and IR bus (for simulation)	YES
13	S	ARINC_E_3_B	I		
14	h	DISCRET_E_1	I	Mode selection n°2	
15	R	ARINC_S_1_A	O	IR parameters for Cat III option	
16	P	ARINC_S_1_B	O		
17	g			SPARE	
18	N	ARINC_S_2_A	O	ADR parameters for Cat III option	
19	M	ARINC_S_2_B	O		
20	f	DISCRET_E_2	I	Backcourse display disabled	
21	X	DISCRET_E_3	I	Mode selection n°1	
22	Y			SPARE	
23	Z	DISCRET_E_4	I	SPARE (for altitude in meters option)	
24	a	DISCRET_E_5	I	Mode selection n°3	
25	b	DISCRET_E_6	I	A/C configuration 8/0	ground
26	c	DISCRET_E_7	I	A/C configuration 4/0	
27	d	DISCRET_E_8	I	A/C configuration 2/0	

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28	e	DISCRET_E_9	I	A/C configuration 1/0	
29	k	DISCRET_E_10	I	A/C configuration parity	ground
30	m	DISCRET_E_11	I	Mag heading display disabled	
31	n	DISCRET_E_12	I	APP display disabled	
32	p	DISCRET_E_13	I	SPARE (for Vbar option)	
33	q	DISCRET_E_14	I	Initiated test	
34	r	DISCRET_E_15	I	SPARE	
35	s	DISCRET_E_16	I	SPARE (for lateral accel. option)	
36	F	DISCRET_S_1	O	ISFD validity : fault/healthy	YES
37	G	ELEC_GND	I	Discrete common/ground	
38	L	RS422_E_H	I	Downloading	YES
39	K	RS422_E_L	I		
40	J	RS422_S_H	O	Downloading	YES
41	H	RS422_S_H	O		

- d Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- e Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- f Energize the instrument.
- g At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for a period of one minute at a frequency of 60Hz.
- h Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- i Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- j De-energize the instrument.
- k Adjust the table to an instrument panel slant angle of 15 degrees.
- l Connect the pins c, d and e to ground (pin G) and put the pin b in open circuit.
- m Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- n Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- o Transmit the labels 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- p Transmit the labels 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- q Energize the instrument.

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- r - At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- s - Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- t - Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

13 Instrument Steady State Test

- a - This test can be done after the instrument panel slant angle programming pins test or during the test of the altitude and airspeed functions ([Refer to SUBTASK-34-22-45-750-073-A01](#)).
- b - Install the instrument on a vertical support fixture set to an instrument panel slant angle of 0 degrees in roll and pitch with a precision of 0.1 degrees.
- c - Attach the instrument and support to a two-axis table set to 0 degrees in roll and pitch, or to a horizontal surface table with a precision of 0.1 degrees.
- d - Change to operational mode ([Refer to Table 1027](#)).
- e - Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- f - Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).

NOTE: Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:

- Pitch: label 324
- Roll: label 325.
- Temperature Gyro: label 202.
- IMU Status word NAV2: label 352.

- g - Energize the instrument.
- h - At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- i - Make sure that the average values calculated on the recorded samples are equal to 0 ± 0.5 degrees.
- j - Let the instrument operate for 1 hour 30 minutes.
- k - At the end of this time, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- l - Make sure that the average values calculated on the recorded samples are equal to 0 ± 0.5 degrees.
- m - Make sure that the temperature drift (label 202) is less than 0.1Degree Celsius/mn.
- n - Make sure that the booleans S_O_B_LOCK_P, S_O_B_LOCK_Q and S_O_B_LOCK_R of label 352 remain set during the three minutes ([Refer to Table 1029](#)).

Table 1029 : Label 352 bit definition

Bit number		DESCRIPTION
1-8	Label 352	Label number

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Bit number		DESCRIPTION
9	S_O_B_ACC_FUNCTION_OK	Linear accelerations consistency flag
10	S_O_B_GYRO_FUNCTION_OK	Angular rates consistency flag
11	S_O_B_AZ_VALIDITY	Z axis linear acceleration consistency flag
12	S_O_B_BANK_AND_TURN	Bank detected
13	S_O_B_GYRO_MOVE	Excessive angular rate
14	S_O_B_AX_MOVE	Excessive motion detected by X accelerometer
15	S_O_B_LOCK_P	X gyrometer saturation flag
16	S_O_B_LOCK_Q	Y gyrometer saturation flag
17	S_O_B_LOCK_R	Z gyrometer saturation flag
18	S_I_B_3ACC	When true, IMU with 3 accelerometers
19	S_I_B_PARAM_ACC_OK	Accelerometers model parameters healthy flag
20	S_I_B_PARAM_GYR_OK	Gyrometers model parameters healthy flag
21	S_I_B_PARAM_TEMP_OK	Temperature model parameters healthy flag
22	S_I_B_ACCELEROS_OK	Accelerometers sensors healthy flag
23	S_I_B_GYROS_OK	Gyrometers acquisition healthy flag
24	S_I_B_TEMPERATURES_OK	Temperature sensors healthy flag
25	S_I_B_ACC_Z_OK	Third accelerometer measurement validity flag
26	S_O_B_OUT_TEMP_OK	Sensor temperature healthy flag
27	S_O_B_S_O_B_RECONFIG_TEMP	Failure detected on temperature probes
28	S_O_B_HEATING_APPLICATION	Memorized temperature in use
29	S_O_B_MEMO_E2PROM	Gyrometers drift storage authorized
30-31	SSM	SSM
32	Parity bit	ARINC parity

(b) Procedure 2

- 1 Measure the ambient room temperature and the temperature at a point on the instrument case near the bezel ([Refer to Figure 1030](#)) for the measurement point).
- 2 Make sure that the difference between the two temperatures is less than 3°C.
- 3 Install the instrument on the test fixture designed for an instrument panel slant angle of 0 degrees, with a precision of 0.1 degrees in roll and pitch.
- 4 Attach the assembly to a Z-axis table.
- 5 Change to operational mode with RS422 ([Refer to Table 1027](#)).

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- 6 Transmit the labels 173 and 174 (with their values set to zero) on the LS bus 1 (pins A and W).
- 7 Transmit the labels 312 and 320 (with their values set to zero) on the HS bus 3 (pins T and S).
- 8 Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:
 - Pitch: label 324
 - Roll: label 325
 - p: label 327
 - q: label 326
 - r: label 330
 - Xacc: label 331
 - Yacc: label 332
 - $\epsilon\Phi$: label 352
 - $\epsilon\theta$: label 353
 - $\epsilon\psi$: label 155
 - dpi: label 157
 - dqj: label 160
 - dri: label 161
 - IMU_Sw_NAV1: label 351.
 - IMU_Sw_NAV2: label 352.

NOTE: For the labels 331, 332 and 333, the resolution is improved (LSB = 11 and not 17 as before). It is necessary to take this modification into consideration for the tests done on these labels.

- 9 Initialization Flag
 - a Energize the instrument.
 - b Make sure that the labels are present on the ARINC reader.
 - c Read the discrete S_O_B_NAV_MODE on label 351 (bit 12 changes from 0 to 1) to make sure that the attitude-function initialization time of the instrument is equal to or less than 91 seconds.
 - d Read the discrete S_I_B_3ACC on label 352 (bit 1_ = 1) to make sure that the third accelerometer is present.

NOTE: The numbers of the 32 bits go from 1 to 32.

- 10 Horizon Line Movements
 - a Record the values of the labels 324 (pitch) and 325 (roll) for a period of three minutes at a frequency of 60Hz.
 - b Record the values of the labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for a period of three minutes at a frequency of 60Hz.
 - c Record the values of the labels 327 (p), 326 (q) and 330 (r) for a period of three minutes at a frequency of 60Hz.
 - d For the pitch and roll parameters, make sure that:
 - The SSM of labels 324 and 325 is NO (bits 30 and 31 set to 1).

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- The average value calculated on the recorded samples is equal to 0 ± 0.5 degrees.
- The noise value calculated on the recorded samples is less than 0.17 degrees for 2 sigma.
- e For the Xacc, Yacc and Zacc parameters, make sure that:
 - The SSM of labels 331, 332 and 333 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to $0g \pm 0.01g$.
 - The noise value calculated on the recorded samples is less than 0.002g for 2 sigma.
- f For the p, q and r parameters, make sure that:
 - The SSM of labels 327, 326 and 330 is NO (bits 30 and 31 set to 1).
 - The average value calculated on the recorded samples is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
 - The noise value calculated on the recorded samples is less than 0.08 deg/sec for 2 sigma.
- g Adjust the table to 20 degrees nose-up in pitch at a rate of less than 95 deg/sec.
- h After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- i Record the values of labels 331 (Zacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- j Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- k Make sure that the average pitch value calculated on the recorded sample is equal to $+20 \pm 0.5$ degrees.
- l Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- m Make sure that the average Xacc value calculated on the recorded sample is equal to $0.342g \pm 0.01g$.
- n Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- o Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- p Adjust the table to -20 degrees in pitch at a rate of less than 95 deg/sec.
- q After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- r Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- s Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- t Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- u Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- v Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.

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- w Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- x Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.06g \pm 0.01g$.
- y At the same pitch angle, turn the instrument through 20 degrees in roll in the clockwise direction at a rate of less than 95 deg/sec.
- z After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aa Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ab Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- ac Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- ad Make sure that the average roll value calculated on the recorded sample is equal to 20 ± 0.5 degrees.
- ae Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- af Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.321g \pm 0.01g$.
- ag Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.117g \pm 0.01g$.
- ah At the same pitch angle, turn the instrument through 40 degrees in roll in the counterclockwise direction (to set its roll to -20 degrees) at a rate of less than 95 deg/sec.
- ai After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- aj Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- ak Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- al Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- am Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- an Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- ao Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.321g \pm 0.01g$.
- ap Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.117g \pm 0.01g$.
- aq Turn the table around its Z axis through an angle between 90 and 180 degrees at an angular rate between 10 deg/sec and 90 deg/sec.
- ar During the movement, record the values of labels 324 (pitch) and 325 (roll) and make sure that they stay at -20 ± 0.5 degrees.

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- as Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- at After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- au Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- av Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- aw Make sure that the average roll value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- ax Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- ay Make sure that the average Yacc value calculated on the recorded sample is equal to $0.321g \pm 0.01g$.
- az Make sure that the average Zacc value calculated on the recorded sample is equal to $-0.117g \pm 0.01g$.
- ba At the same pitch angle, turn the table through 20 degrees in the clockwise direction at a rate of less than 95 deg/sec to set the roll angle to 0 degrees.
- bb After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bc Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bd Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- be Make sure that the average pitch value calculated on the recorded sample is equal to -20 ± 0.5 degrees.
- bf Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bg Make sure that the average Xacc value calculated on the recorded sample is equal to $-0.342g \pm 0.01g$.
- bh Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bi Make sure that the average Zacc value calculated on the recorded sample is equal to $0.06g \pm 0.01g$.
- bj Set the instrument pitch angle to 0 degrees (20 degrees nose-up) at a rate of less than 95 deg/sec.
- bk After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- bl Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bm Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bn Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.5 degrees.
- bo Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.5 degrees.

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- bp Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bq Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- br Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bs At the same pitch angle, turn the instrument counterclockwise through 100 degrees at a rate of less than 95 deg/sec.
- bt After three minutes, record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- bu Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- bv Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- bw Make sure that the average Yacc value calculated on the recorded sample is equal to $-0.985g \pm 0.01g$.
- bx Make sure that the average Zacc value calculated on the recorded sample is equal to $-1.174g \pm 0.01g$.
- by At the same pitch angle, turn the instrument clockwise through 100 degrees to set it to 0 degrees at a rate of less than 95 deg/sec.
- bz After three minutes, record the values of labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- ca Record the values of labels 331 (Xacc), 332 (Yacc) and 333 (Zacc) for three minutes at a frequency of 60Hz.
- cb Record the values of labels 327 (p), 326 (q) and 330 (r) for three minutes at a frequency of 60Hz.
- cc Make sure that the SSM of labels 324, 325, 331, 332, 333, 327, 326 and 330 is NO (bits 30 and 31 set to 1).
- cd Make sure that the average pitch value calculated on the recorded sample is equal to 0 ± 0.01 degrees.
- ce Make sure that the average roll value calculated on the recorded sample is equal to 0 ± 0.01 degrees.
- cf Make sure that the average Xacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- cg Make sure that the average Yacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ch Make sure that the average Zacc value calculated on the recorded sample is equal to $0g \pm 0.01g$.
- ci Make sure that the average p value calculated on the recorded sample is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
- cj Make sure that the average qvalue calculated on the recorded sample is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.
- ck Make sure that the average r value calculated on the recorded sample is equal to $0 \text{ deg/sec} \pm 0.05 \text{ deg/sec}$.

11 Instrument Panel Slant Angle Programming Pins

- a De-energize the instrument.

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- b Set the instrument support fixture to an instrument panel slant angle of 16 degrees.
- c Connect the instrument in the configuration for a 16-degree instrument panel slant angle ([Refer to Table 1028](#)).
- d Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- e Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- f Energize the instrument.
- g At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- h Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- i Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.
- j De-energize the instrument.
- k Set the instrument support fixture to an instrument panel slant angle of 15 degrees.
- l Connect the pins c, d and e to ground (pin G) and put the pin b in open circuit.
- m Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- n Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- o Transmit the label 242 (with the value set to 1013.25) on the LS bus 2 (pins V and U).
- p Transmit the label 245 (with the value set to 1013.25) on the LS bus 6 (pins B and C).
- q Energize the instrument.
- r At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for one minute at a frequency of 60Hz.
- s Make sure that the average pitch value calculated on the recorded sample is 0 ± 0.5 degrees.
- t Make sure that the average roll value calculated on the recorded sample is 0 ± 0.5 degrees.

12 Instrument Steady State Test

- a This test can be done after the test of the instrument panel slant angle programming pins or during the test of the altitude and airspeed functions ([Refer to SUBTASK-34-22-45-750-073-A01](#)).
- b Install the instrument on a vertical support fixture set to an instrument panel slant angle of 0 degrees with a precision of 0.1 degrees in roll and pitch.
- c Install the instrument and support fixture assembly on the two-axis table set to a angle of 0 degrees in roll and pitch, or on a horizontal surface (with a precision of 0.1 degrees).
- d Connect the instrument in the operational configuration with RS422 ([Refer to Table 1027](#)).

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- e Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
- f Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).

NOTE: Connect the ARINC 429 High Speed output line (pins R and P of connector J1) to the ARINC 429 reader to get the data that follow:

- Pitch: label 324
- Roll: label 325.
- Temperature Gyro: label 202.
- IMU Status word NAV2: label 352.

- g Energize the instrument.
- h At the end of the initialization phase, record the labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- i Make sure that the average pitch and roll values calculated on the recorded samples are 0 ± 0.5 degrees.
- j Let the instrument operate for 1 hour 30 minutes.
- k At the end of this time, record the labels 324 (pitch) and 325 (roll) for three minutes at a frequency of 60Hz.
- l Make sure that the average pitch and roll values calculated on the recorded samples are 0 ± 0.5 degrees.
- m Make sure that the temperature drift (label 202) is less than 0.1Degree Celsius/mn.
- n Make sure that the booleans S_O_B_LOCK_P, S_O_B_LOCK_Q and S_O_B_LOCK_R of label 352 remain set during the three minutes ([Refer to Table 1029](#)).

SUBTASK 34-22-45-750-073-A01

(25) Test of Altitude and Airspeed Functions

(a) Test Position

- Attach the instrument to the test fixture designed for an instrument panel slant angle of 0 degrees.
- Connect the instrument in the operational configuration (with the heading function disabled) ([Refer to Table 1027](#)).

NOTE: Connect the ARINC 429 Low Speed output line (pins N and M of connector J1) to the ARINC 429 reader to get the data that follow:

- Indicated airspeed (IAS): label 206
- Corrected baro altitude (Zbaro): label 204
- Standard altitude (Zstd): label 203
- Barometric pressure in hPa (Pbaro): label 234
- Barometric pressure in in.Hg (PbaroUS): label 235
- Mach: label 205.
- Pitot pressure: label 242.
- Static source pressure: label 245.

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- 1 Transmit the labels 173 and 174 (with their values set to 0) on the LS bus 1 (pins A and W).
 - 2 Transmit the labels 312 and 320 (with their values set to 0) on the HS bus 3 (pins T and S).
- (b) Test of Leakage on the Static and Pitot Ports and Mach Number Display
- 1 Do the test of the leakage on the static and pitot ports as follows:
NOTE: These measurements must be done with the instrument in a power-off condition and in normal test conditions.
 - a Connect the static pressure port to a pressure source and barometer, or to a calibrated pressure source.
 - b Connect the pitot pressure port to a pressure source and pressure gauge, or to a calibrated pressure source.
 - c Decrease the pressure to the values that follow at a maximum rate of change of 20 hPa/sec (1200 hPa/min):
 - Static pressure (Ps): 500 hPa (absolute value)
 - Pitot pressure (Pt): between 525 and 560 hPa.
 - d Isolate the instrument-barometer assembly from the pressure sources. The unwanted volume isolated (tubing and barometer) must be less than 0.05 dm³ for each of the sensors.
 - e Let the instrument stabilize for one minute, then take an initial reading on the two barometers
 - f After another minute, take a second reading on the two barometers.
 - g Make sure that the difference between the two readings is not more than 1.7 hPa for each of the pitot and static pressure ports.
 - h Bring the static and pitot pressures back to the normal atmospheric pressure at a maximum rate of change of 20 hPa/sec (1200 hPa/min).
 - 2 Do the test of the Mach number display as follows:
NOTE: To select the Mach number display, connect the discretes m and k to the discrete ground (pin G).
 - a Energize the instrument.
 - b At the end of the initialization phase, set the pressure values as follows at a maximum rate of change of 20 hPa/sec (1200 hPa/min):
 - Static pressure (Ps): 1013.25 hPa (absolute value)
 - Pitot pressure (Pt): 1250 hPa.
 - c Make sure that the Mach number is shown in the bottom left corner of the display.
 - d Bring the static and pitot pressures back to the normal atmospheric pressure at a maximum rate of change of 20 hPa/sec (1200 hPa/min).
- (c) Test of the Altitude and Airspeed Functions
- 1 Test of Altitude Accuracy
 - a De-energize the instrument.
 - b Disconnect the discretes m and k from the ground (pin G).
 - c Energize the instrument.

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- d At the end of the initialization phase, press the BARO pushbutton to set the barometric pressure to a value equal to the standard pressure.
NOTE: During this test, keep the pitot pressure adjusted to show an airspeed of 200±5 kts.
- e Increase the pressure-altitude from -2000 feet to +50,000 feet, at a rate of approximately 5000 feet/min. Decrease this rate gradually before each test point ([Refer to Table 1030](#)) to prevent overshoot of the test point.

Table 1030 : Altitude Accuracy Test Points

Pressure-altitude Za in feet (1)	Static pressure (Ps) in hPa (2)	Tolerance ± in feet (3)
-2000	1088.66	20
0*	1013.25	20
1000	977.17	20
2000*	942.13	30
6000	812.00	40
10,000*	696.81	50
20,000	465.63	75
30,000*	300.89	120
40,000	187.54	150
50,000	115.97	180

- f For each static pressure value equivalent to a typical Za value ([Refer to Table 1030](#)), keep the pressure stable for more than 20 seconds and then record the value Zm shown by the altimeter (label 203).
- g Make sure that the difference Zm-Za is not more than the tolerance ([Refer to Table 1030](#)).
- h Stop for two minutes at the value of +50,000 feet.
- i Decrease the pressure-altitude from +50,000 to 0 feet at a rate of approximately 5000 ft/min. Decrease this rate gradually before each test point to prevent overshoot of the test point.
- j For each static pressure value equivalent to a typical Za value ([Refer to Table 1030](#)), keep the pressure stable for more than 20 seconds and then record the value Zstd shown by the altimeter (label 203).
NOTE: It is possible to do this test only at the altitude values shown with an asterisk ([Refer to Table 1030](#)).
- k Make sure that the difference Zstd-Za is not more than the tolerance ([Refer to Table 1030](#)).

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- l At a specific Zstd value (30,000 feet), make sure that the noise level on the pressure sensors with the following procedure:
 - 1 During at least one minute, record and process the Ps and Pt ARINC label values:
 - (max - average) value of Pt, Ps is less than 0.2 hPa
 - (average - min) value of Pt, Ps is less than 0.2 hPa
 - root mean square deviation on Pt, Ps is less than 0.1 hPa
 - m Reset the static and pitot pressures to ambient room pressure.
- 2 Test of Airspeed Accuracy
 - a Connect the pitot and static pressure ports to an adjustable pressure source and a barometer, or to a calibrated pressure source

NOTE: During this test, keep the static pressure Ps adjusted to show an altitude of 10,000 ±50 feet.

Table 1031 : Airspeed Accuracy Test Points

Airspeed (A) in Knots (1)	Differential pressure (Pt-Ps) in hPa (2)	Tolerance ± in Knots (3)
40*	2.60	4
70	7.97	4
100	16.30	4
140*	32.13	3
180	53.50	3
200	66.34	3
300*	153.55	3
350	212.86	3
400	283.94	3
520	510.28	3

- b Increase the airspeed from 0 to 520 Knots at a rate of approximately 240 Kts/min.
- c For each differential pressure value (Pt-Ps) (Refer to Table 1031) equivalent to the airspeed (A) in column (1), keep the pressure stable for more than 20 seconds and then record the airspeed indication (IAS) shown by the instrument (label 206) during 3 seconds.
- d Make sure that the difference between the airspeed indication and the value A in column 1 (Refer to Table 1031) is not more than the tolerance in column 3 (Refer to Table 1031).
- e Stop at the value of 520 Knots for 2 minutes.
- f Decrease the airspeed from 520 to 0 Knots at a rate of approximately 240 Kts/min.

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- g For each differential pressure value (Pt-Ps) [\(Refer to Table 1031\)](#) equivalent to the airspeed values (A) shown with an asterisk in column (1), keep the pressure stable for more than 20 seconds and then record the airspeed indication (IAS) shown by the instrument (label 206).
- h Make sure that the difference between the airspeed indication and the value A in column 1 [\(Refer to Table 1031\)](#) is not more than the tolerance in column 3 [\(Refer to Table 1031\)](#).
- i Reset the static and pitot pressures to ambient room pressure.

SUBTASK 34-22-45-750-074-A01

(26) Final Operational Check

- (a) De-energize the instrument.
- (b) Connect the instrument in the operational configuration [\(Refer to Table 1024\)](#).
- (c) Connect the pins X and a (discretes 1 and 3) to ground (pin G) to change to maintenance mode.
- (d) Energize the instrument.
- (e) Make sure that the self-test page 1 is displayed [\(Refer to Figure 1039\)](#) (or that the related data is output on the RS422 line after the word "T" is transmitted on the RS422 input bus).
- (f) Make sure that all the results of the self-tests are correct.
- (g) Record the value shown on the time counter.
- (h) Transmit the word "S" on the RS422 input bus.

NOTE: The instrument transmits the content of the failure memory on the RS422 output line.

- (i) [\(Refer to SUBTASK-34-22-45-750-053-A01\)](#) and make sure that the content of the failure memory is the same.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-750-807-A01

6. Fault Isolation

A. Acceptance test procedure diagnostic

SUBTASK 34-22-45-750-075-A01

(1) General

- (a) The following tables clarify functional blocks straight associated to each failure step of the Acceptance Test Procedure (ATP). The failure has be completed with the failure memory analysis (Refer to User's Guide ref. Z1385698).

Table 1032 : Operator tests (Not operating)

Test	SRU	Mnemonic	Function	Failure mode	Comment
Appearance	No specific diagnosis				
Mechanical checks	No specific diagnosis				
Continuity	Verify the assembly of the A6 (Equipped Front Face) and the A1 (Rear Interface Module) on the ISFD				
Insulation	A11	PALI	EMI and lightning strike protection	Loss of power supply	No display
	PS1	ALIM	Equipment power supply	Loss of power supply	No display

Table 1033 : Operator tests (Lightning protection test)

Tested pins	SRU	Mnemonic	Function	Failure mode	Comment
D,E	A11	PALI	EMI and lightning strike protection	Loss of power supply	No display
i, j		PDIM		Loss of A6 lighting dimming	Loss of instrument A6 dimming function
A, W, U, V, S, T, B, C		PARI		Loss of ARINC reception	Loss of LS, loss of heading
M, N, R, P		PARI		Loss of ARINC emission	
b, c, d, e, k, m, p, r, s, q, Z, f		PDIS		Detected false configuration	
g		PDIS		Detected false configuration	

Table 1034 : Operator tests (Failure memory control)

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Test	SRU	Mnemonic	Function	Failure mode	Comment
Failure memory control					and if display is not active, then change SRU A2 (take care to download in the correct configuration)
	A2	CEPM	Sequencer	Loss of EPLD	Loss of all functions
	TB1	MOMEM	EEPROM boot/errors	Loss of boot and error memory	
		MOCON	Interconnections between electronic boards of the equipment	Loss of interconnection signals	
	A2	M422	EMI and lightning strike protection	Loss of RS422 reception / transmission	
	A11	P422			

Table 1035 : Operator tests (Power consumption and validity output discrete)

Test	SRU	Mnemonic	Function	Failure mode	Comment
Power consumption	PS1	ERACLE	Equipment power supply		
		ALIM			
	A2	CCPU	Parameter calculation and equipment management		
		CEPM	Sequencer		
		GSEQ	Graphics sequencer		
	A3	IPLD	Discretes interface		
Validity output discrete	For this step, functional blocks localization is more explicit with previous or next ATP steps, and with the failure memory analysis				

Table 1036 : Operator tests (Tests of secondary functions of bezel)

Test	SRU	Mnemonic	Function	Failure mode	Comment
Operational configuration					
Init phase 1	A5	OCBL	High voltage for tubes	Loss of high voltage	No display

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Test	SRU	Mnemonic	Function	Failure mode	Comment
	PS1	ERACLE	Equipment power supply	Loss of equipment power supply	
	A2	GCONV	Analog interface	Loss of dot matrix brightness commands	Loss of graphics functions
	PS1	ALIM	Equipment power supply	Loss of matrix voltages	
	A2	GGSM	Graphics processor	Loss of graphics processor	Blank screen
	A5	OAMLCD	Color LCD matrix	Degradation of matrix operation	Degraded display operation
		TUBES	Tubes failed	Loss of 1 or 2 tubes	
	A2	GMEMI	Image Memory	Loss of image data	Blank screen
		GSEQ	Graphics sequencer	Loss of sequencer	
	A3	IOM	Signal acquisition and transfer to processor	Loss of pressure informations, ARINC lines	
	A2	GVOL	AMLCD voltage generation	Loss of image	Loss of all graphics functions
		GSLCD	AMLCD interface		
	A3	IQTZ	Quartz 40MHz	Loss of Quartz	Blank screen
	A7	PMEMS	Static pressure sensor memory	Loss of static pressure memory	Flags ALT, SPD, M
	A8	PMEMT	Total pressure sensor memory	Loss of total pressure memory	Flags SPD, M
	A3	IPLD	Discretes interface	Frequency divider	Flags ALT, SPD, M
Init phase 2	A7	PS	Static pressure sensor	Sensor saturated	
	A8	PT	Total pressure sensor	Sensor saturated	Flags SPD, M
Conformity of operational page	A9	GYROS	Supplies velocity data	Electronical failure	Flag ATT
		ICAN	Analog to digital conversion	Loss sensor informations	
		ACCELS	Supplies acceleration data	Accelerometer saturated	

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Test	SRU	Mnemonic	Function	Failure mode	Comment
	A2	MINT	Dialog buffers between INT and CPU	Loss of dimming reference	
	A9	ISEN	Signal conditioning	Gyrometers conditioning	Flag ATT
		ICAN	Analog to digital conversion	Loss of temperature informations	
	A3	IARE	Analog multiplexing	Loss of ARINC monitoring	Flag LS, LOC, G/S, HDG
	A9	ISEN	Signal conditioning	Accelerometers conditioning	Flag ATT
		IREG	Gyro power supply regulation	Loss of regulation	
		INUM	Calibration memory and gyros logic	Failure on calibration memory and gyros discrete logic	
	A3	IARE	Analog multiplexing	Loss of ARINC monitoring and ARINC input	Flag LS, LOC, G/S, HDG
	A9	GYROS	Supplies velocity data	Gyrometer quartz saturated	Flag ATT
Push-button/knob tests					
'+' and '-' button	A3	IOM	Signal acquisition and transfer to processor	Loss of dimming reference	
	A3	IEFF	Interface for front face data	Front face interface fail	Ineffective push buttons
	A6	PBUT	Front face assy	Loss of '+' button	
		PBUT	Front face assy	Loss of '-' button	
	A5	OCBL	High voltage for tubes	Loss of dimming	
	A3	IPLD	Discretes interface	Dimming fail PWM	Ineffective A6 diming
'APP' button	A6	PBUT	Front face assy	Loss of 'APP' button	Ineffective push button
'HP/IN' button	A6	PBUT	Front face assy	Loss of 'HP/IN' button	
'RST' button	A6	PBUT	Front face assy	Loss of 'RST' button	
'BARO' knob	A6	BROT	Front face assy	Loss of STD/BARO rotary knob	Ineffective rotary knob

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Test	SRU	Mnemonic	Function	Failure mode	Comment
TEST function tests	A11	PDIS	EMI and lightning strike protection	Test discrete non functional	
	A3	IDIS		Loss of test command	Test command ineffective
		IPLD	Discretes interface	Discrete test fail	
		IARE	Analog multiplexing	Loss of ARINC inputs	Flags LS, LOC, GS, HDG
HEADING function tests	A11	PARI	EMI and lightning strike protection	Loss of ARINC 429 reception	Flags LOC, GS, HDG
ILS function tests	A3	IARE	Analog multiplexing	Loss of ARINC inputs	Flags LS, LOC, GS, HDG
	A11	PARI	EMI and lightning strike protection	Loss of ARINC 429 reception	Flags LOC, GS, HDG
Bezel lighting check	A6	PBUT	Equipped front face	Degradation of buttons	Ineffective push buttons
	A3	IADC	Analog interface for dimming and power supply monitoring	Loss of EFF dimming reference	
	A6	LIGHT	Front face assy	Loss of lighting leds	Loss of buttons lighting
Option tests : V-bar, slip and altitude in meter	A11	PDIS	EMI and lightning strike protection	Test discrete non functional	
	A3	IDIS		Loss of test command	Test command ineffective

Table 1037 : Optical Tests

Test	SRU	Mnemonic	Function	Failure mode	Comment
Display of auto-test results	No specific diagnosis				
Display of maintenance 1	No specific diagnosis				
Maintenance page 1 sent on RS422 output bus	No specific diagnosis				
Display of maintenance 2	No specific diagnosis				

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Test	SRU	Mnemonic	Function	Failure mode	Comment
Maintenance page 2 sent on RS422 output bus	No specific diagnosis				
Display of luminance measurement maintenance page	A2	GCONV	Analog interface	Loss of dot matrix brightness commands	Loss of graphics functions
	A5	OAMLCD	Color LCD matrix	Degradation of matrix operation	
	A2	GVOL	AMLCD voltage generation	Loss of image	Loss of all graphics functions
		GSLCD	AMLCD interface	Loss of image	
Display of contrast measurement maintenance page	A5	OAMLCD	Color LCD matrix	Degradation of matrix operation	
Display of checksum maintenance page	A2	CCPU	Param calculation and equipment management	CPU failure	Loss of computing processor
		CGCON	A2 connector	Fail on connector	Loss of computing functions
Display of colors maintenance page	A5	OAMLCD	Color LCD matrix	Loss of drivers	
SADM Pressure via ARINC input	A3	IARE	Analog multiplexing	Loss of Arinc inputs	Flag LS, G/S, LOC, HDG
	A11	PARI	EMI and lightning strike protection	Loss of Arinc reception	Flag LS, G/S, LOC, HDG
Lighting function and color test					
Brightness in night conditions Full sunlight brightness	A5	OAMLCD	Color LCD matrix	Degradation of matrix operation	Degraded display operation
		TUBES	Tubes failed	Loss of tubes	
	A2	MINT	Dialog buffers between INT and CPU	Loss of dimming reference	
	A6	ALS	Equipped front face	Loss of ambient light sensor	

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Test	SRU	Mnemonic	Function	Failure mode	Comment
	A5	OCBL	High voltage for tubes	Loss of calibration	
Ground and sky color coordinates	A5	OAMLCD	Color LCD matrix	Loss of drivers	

Table 1038 : Attitude and Altitude Tests

Test	SRU	Mnemonic	Function	Failure mode	Comment
Attitude function test					
Difference Temperature	No specific diagnosis				
Emission ARINC	A3	IOM	Signal acquisition and transfer processor	Loss of ARINC lines	
		IARS	ARINC 429 modulator	Loss of ARINC outputs	
	A11	PARI	EMI and lightning strike protection	Loss of ARINC 429 transmission	
Initialisation of attitude function					
Horizon line movements	A9	ACCELS	Supplies acceleration data	Accelerometer locked	Flag ATT
				Accelerometer drift	
		IREG	Gyro power supply regulation	Regulation degraded	
		GYROS	Supplies velocity data	Gyrometer quartz locked	
				Gyrometer quartz drift	
Pin-prog for tilt angle	A11	PDIS	EMI and lightning strike protection	Detected false configuration	
	A3	IDIS	EMI and lightning strike protection		
Horizon line checking when stabilized	A9	ACCELS	Supplies acceleration data	Accelerometer locked	Flag ATT
				Accelerometer drift	
		IREG	Gyro power supply regulation	Regulation degraded	

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Test	SRU	Mnemonic	Function	Failure mode	Comment
		GYROS	Supplies velocity data	Gyrometer quartz locked	
				Gyrometer quartz drift	
Altitude and airspeed function test					
Test of altitude and airspeed functions	A7	PS	Static pressure sensor	Sensor locked	
	A8	PT	Total pressure sensor		
	A7	PS	Static pressure sensor	Sensor drift	
	A8	PT	Total pressure sensor		
	A3	IQTZ	Quartz 40MHz	Quartz drift	
Final operation verification					
Display of 1st maintenance page	No specific diagnosis				
Auto tests	No specific diagnosis				
Failure memory	No specific diagnosis				
Comp memory fault	Refer to Bite User's Guide in taking account the minute counter				
Attitude Discrepancies	(Refer to TASK-34-22-45-750-808-A01)				

Table 1039 : Cross reference Mnemonic and functional block

Mnemonic	Functional block
A11	Filter Interconnection board
PALI	Power supply lightning protection and EMI/EMC filter
PDIM	Front Face Assy lighting dimming lightning protection and EMI/EMC filter
PARI	ARINC429 input/output lightning protection and EMI/EMC filter
PDIS	I/O discrete lightning protection and EMI/EMC filter
P422	RS 422 I/O lightning protection and EMI/EMC filter

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Mnemonic	Functional block
PS1	Power supply Unit
ALIM	Power supply generation
ERACLE	Power supply generation
A3	Programmed Interface Board
IADC	Analog signal conditioning, multiplexer and digital converter.
IQTZ	40 MHz quartz Clock reference for IOM frequency measurements
IOM	IOM ASIC Frequency acquisition (for PS1 data acquisition) Pulse counting acquisition (for A5 lighting control) Discrete I/O ARINC I/O
IARE	ARINC multiplexer : input selection from A11 (operational)/IARS (test). ARINC demodulator (DAR333) : ARINC level to logic level conversion
IARS	ARINC modulator (MODAR V15)
IDIS	Input discrete Open/Ground to logic level conversion Output discrete logic level to Open/Ground level conversion
IPLD	Interface EPLD (7128S) A11 Discrete signals multiplexing to IOM, IOM discrete acquisition test. PS1 signal frequency division to IOM, IOM frequency acquisition test. Front Face Assy buttons acquisition and Front Face Assy lighting control
IEFF	Front Face Assy push buttons and rotating knob interface
ICON	A3 to TB1 connector
IALI	Power supply
A8	Static Pressure Sensor Assy
PS	Pressure Sensor
PMEMS	Sensor memory : contains calibration data
A7	Total Pressure Sensor Assy
PT	Pressure Sensor

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Mnemonic	Functional block
PMEMT	Sensor memory : contains calibration data
A9	Inertial Measurement Unit
Accels	2 accelerometers
Gyros	3 gyrometers
ISEN	Sensor Signal Conditioning
ICAN	High accuracy Analog to Digital Converter
INUM	Calibration memory and Gyros logic
IREG	Power supply post regulator
ACCEL Z	1 accelerometer on Z axis
A2	Control Processor and Graphic Display Unit
MINT	Address, Data and Control signal buffer from/to A3
CMEM	CPU memories (SRAM and Flash PROM)
CCPU	Control Processing Unit Digital Signal Processing
CEPM	EPLD (7256S) Watchdog, Reset, power interruption management Address decoding Serial bus multiplexing (RS422 , IMU CAN SERIAL BUS, MEMORIES SERIAL BUS) IMU acquisition management Internal discrete signal acquisition (Timer, ALIM_OK, Graphic_OK)
M422	RS422 to logic level conversion
GBUFF	CPU/Graphic communication buffers
GMENT	Graphic memories (SRAM and Flash PROM)
GGSM	ASIC GSM2 : Graphic Symbology generation
GSEQ	EPLD FLEX10K - Graphic sequencer Anti-aliasing treatment OAMLCD digital RGB signals and clock generation
GMEMI	SRAM Video memories
GCONV	OAMLCD RGB signals digital to analog conversion

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Mnemonic	Functional block
GVOL	OAMLCD DC voltages generation
GSLCD	OAMLCD synchronization clocks signals level adaptation
GCON	CPGDM-MOB connector
A5	Optical Assy
OCBL	Lighting box power supply
	Lighting box intensity sensor acquisition and conditioning
TUBES	Light Emitting Tubes (3 units)
OAMLCD	Active Matrix Liquid Crystal Display
A6	Front Face Assy
PBUT	Push Buttons
BROT	Rotary knob
ALS	Light sensor
LIGHT	Leds
TB1	Programmed Motherboard
MOMEM	BOOT, error and parameters memory.
MOCON	Connectors
PROT	Power supply protection
PROT	Power supply protection

- B. Reason for the job
 - (1) Not applicable.
- C. Job Set-up Information
 - (1) [\(Refer to TASK-34-22-45-750-802-A01\)](#)
- D. Job Set-up
 - (1) Not applicable.
- E. Procedure
 - (1) Not applicable.
- F. Close-up
 - (1) Not applicable.

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TASK 34-22-45-750-808-A01

7. Fault Isolation for INERTIAL MEASUREMENT UNIT AND PRESSURE SENSOR

A. General

- (1) The fault isolation procedures for the Inertial Measurement Unit and the pressure sensors.

B. Reason for the Job

- (1) Not applicable.

C. Job Set-up information

- (1) [\(Refer to TASK-34-22-45-650-804-A01\)](#)

D. Job Set-up

- (1) Three hours test

- (a) This test can be done if:

- 1 The reason for removal is not easy to find.
- 2 The reason for removal is related to the attitude, altitude and/or airspeed functions.

- (b) [\(Refer to TASK-34-22-45-650-804-A01\)](#)

- (2) Fault isolation for the Three hours test

- (a) The table that follows shows the subassemblies directly related to each failure step of the three hours test.

Table 1040 : Operator test

TEST	SUB-ASSEMBLIES
ALIGNING PHASE TEST	INERTIAL MEASUREMENT UNIT (IMU)
THREE HOURS TEST UMI	INERTIAL MEASUREMENT UNIT (IMU)
THREE HOURS TEST PRESSURES	EQUIPPED PRESSURE SENSORS PROGRAMMED INTERFACE PCB

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-750-901-A01

8. Procedure for the Adjustment of the Optical Assembly Brightness

A. Reason for the job

- (1) Do this procedure for the adjustment of the Optical Assembly Brightness when the display brightness test for ATLAS Specification is faulty.

B. Job Set-up Information

NOTE: For the detailed information of the special tools called below. ([Refer to PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#))

NOTE: Equivalent alternatives may be used for the items listed.

(1) Tools, Fixtures and Equipment

Table 1041 : Tools, Fixtures and Equipment

CMM CODE	PART NUMBER	QTY	NAME
ST26	D16845AB01	1	Optical Assembly Test bench
	Not specific	1	34410A USB digital multimeter, AGILENT
	Not specific	1	Power supply 28VDC, 3A
	Not specific	1	Industrial laptop, Windows XP, with RS422 I/F and RS232 I/F
	Not specific	1	Photometer CS200 MINOLTA

(2) Referenced Information

Table 1042 : Referenced Information

PART NUMBER	NAME
SOPM 20-02-00	Standard Overhaul Practices Manual

C. Job Set-up

SUBTASK 34-22-45-210-050-A01

(1) Preliminary Operations

- (a) Refer to Standard Overhaul Practices Manual reference SOPM 20-02-00 for the equipment contains electrostatic sensitive devices.
- (b) Remove the Covers ([Refer to TASK-34-22-45-000-802-A01](#)).
- (c) Remove the Front Face A6 ([Refer to TASK-34-22-45-000-803-A01](#)).
- (d) Removal of the Optical Assembly A5 ([Refer to TASK-34-22-45-000-804-A01](#)).

D. Procedure

SUBTASK 34-22-45-210-051-A01

(1) Hardware setup

([Refer to Figure 1045](#))

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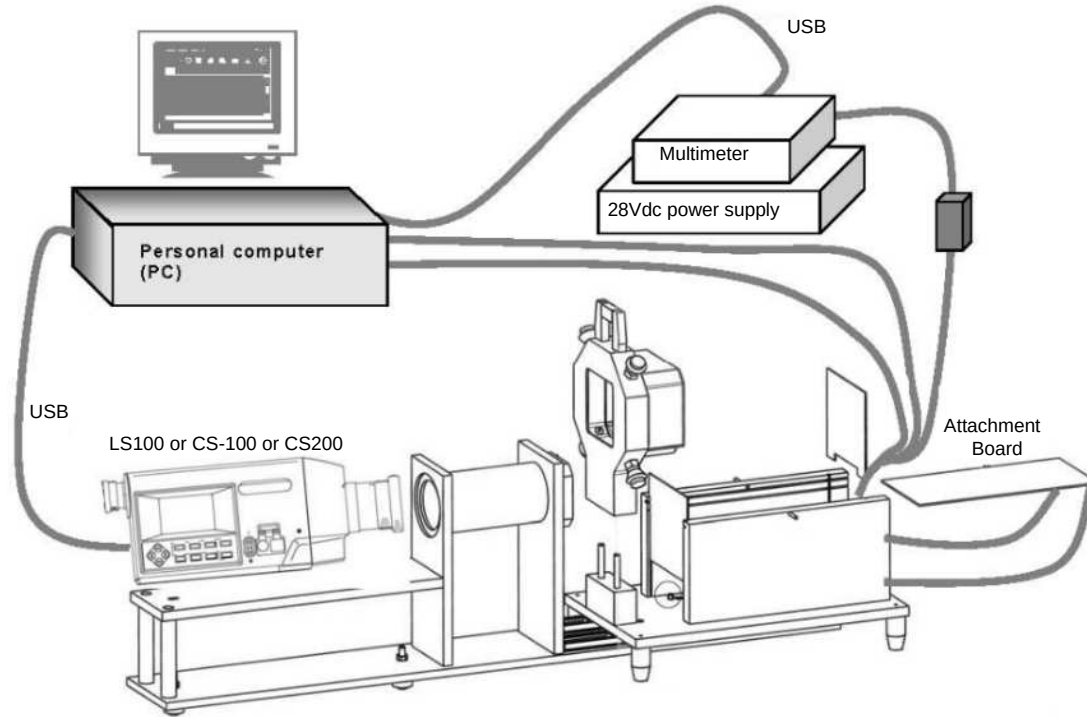
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- (a) Install the optical assembly on the 3ATI tool.
- (b) Install the 3ATI tool on the OPM test bench.
- (c) Connect the attachment board to the CBL PCB of the optical assembly.



D_CM_TAV_342245_01_ACLO_01_00

Figure 1045/Graphic 34-22-45-991-919-A01
OPM test bench

SUBTASK 34-22-45-210-053-A01

- (2) Test description
 - (a) Click on the shortcut icon to start the program.

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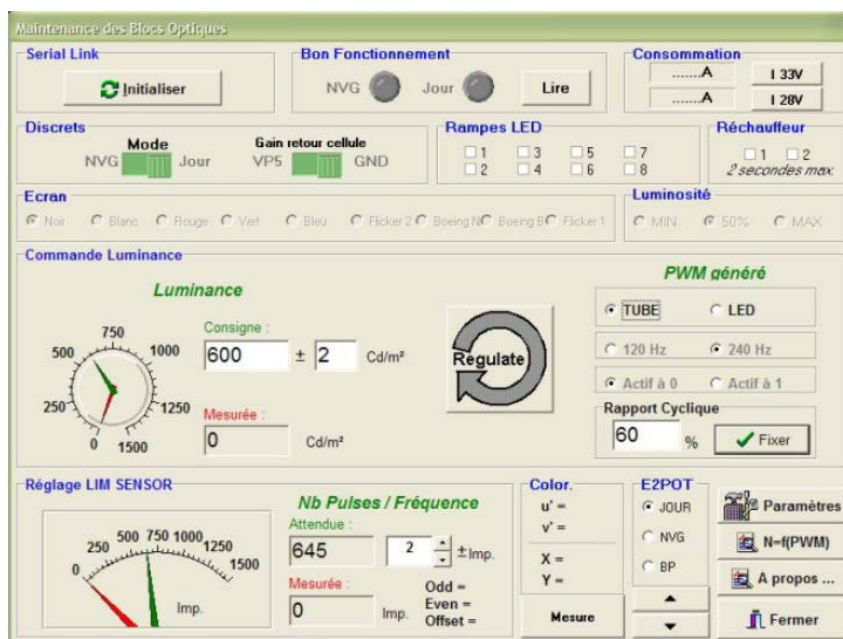
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D_CM_TAV_342245_01_ACMO_01_00

Figure 1046/Graphic 34-22-45-991-920-A01
Short cut icon on Windows desktop

(b) The following window displays:



D_CM_TAV_342245_01_ACMO_01_00

Figure 1047/Graphic 34-22-45-991-921-A01

(c) Adjust the output voltage of the power supply to $28 \pm 0.5\text{VDC}$.

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- (d) Click on the button "Initialiser"

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D_CM_TAV_342245_01_ACPO_01_00

Figure 1048/Graphic 34-22-45-991-922-A01
"Initialiser" button

- (e) Tick the box "TUBE"

NOTE: By default the box "240HZ" and the box "Actif à 0" are ticked.



D_CM_TAV_342245_01_ACQO_01_00

Figure 1049/Graphic 34-22-45-991-923-A01

- (f) Select the “Discret Mode” in “jour” position and the “Discret Gain retour cellule” in “GND” position.



D_CM_TAV_342245_01_ACRO_01_00

Figure 1050/Graphic 34-22-45-991-924-A01

- (g) Tick the box “Blanc”

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D_CM_TAV_342245_01_ACSO_01_00

Figure 1051/Graphic 34-22-45-991-925-A01

(h) Click on the button "Paramètres"



D_CM_TAV_342245_01_ACTO_01_00

Figure 1052/Graphic 34-22-45-991-926-A01

(i) The following window displays:

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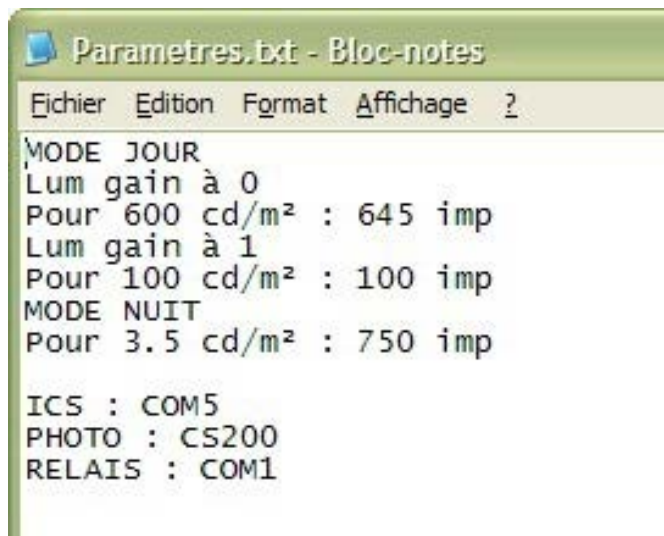
34-22-45

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D_CM_TAV_342245_01_ACUO_01_00

Figure 1053/Graphic 34-22-45-991-927-A01

- (j) Do a check of the order (consigne) given the window above.
NOTE: By default the value must be for 600 cd/m² : 645 imp, if not, write the good value.
- (k) Put the photometer against the optical assembly.
- (l) Click on the button “Regulate” to activate it and wait 15 minutes.

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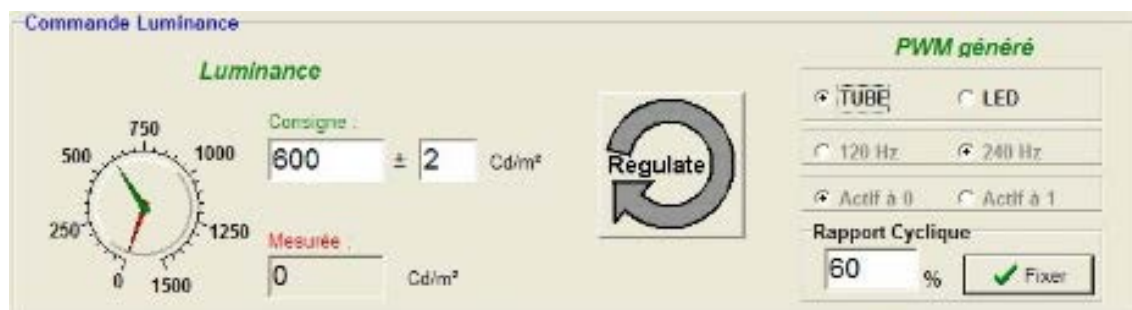
C16221K



D_CM_TAV_342245_01_ACVO_01_00

Figure 1054/Graphic 34-22-45-991-928-A01

- (m) Wait that the measured value which displays in the box “Mesurée” reaches the order value (Consigne).



D_CM_TAV_342245_01_ACWO_01_00

Figure 1055/Graphic 34-22-45-991-929-A01

- (n) Set the LIM sensor as follows:

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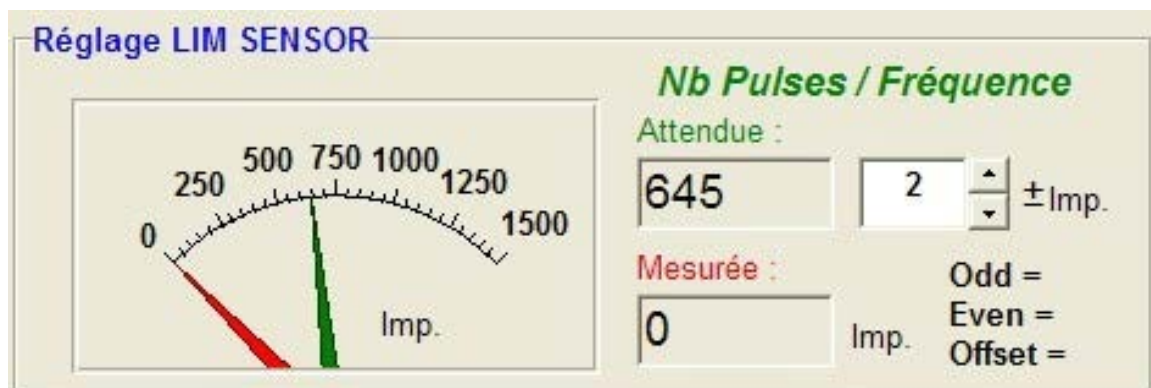
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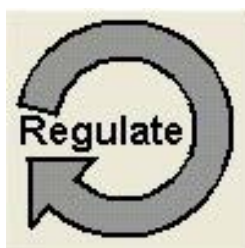
- 1 Adjust the potentiometer R40 of the CBL PCB to obtain 645 ± 2 pulses in the box "Mesurée" (Refer to IPL 8-270)



D_CM_TAV_342245_01_ACXO_01_00

Figure 1056/Graphic 34-22-45-991-930-A01

- (o) Click on the button "Regulate" to deactivate it.



D_CM_TAV_342245_01_ACYO_01_00

Figure 1057/Graphic 34-22-45-991-931-A01

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- (p) Move away the photometer from the optical assembly.
- (q) Set the flicker as follows:
 - 1 Tick the box "Flicker 1".
 - 2 Adjust the potentiometer R36 of the CBL PCB to dim the flicker to its minimum value (Refer to IPL 8-270).

└

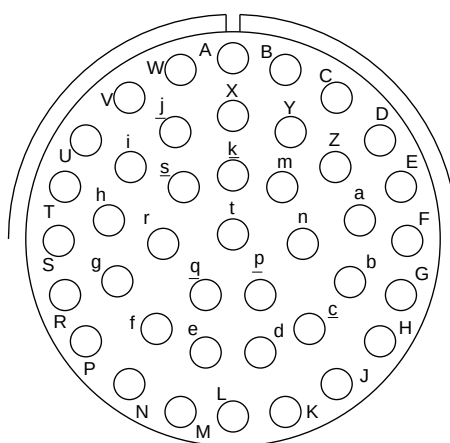


D_CM_TAV_342245_01_ACZO_01_00

Figure 1058/Graphic 34-22-45-991-932-A01

- (r) De-energize the 28 V power supply.
 - (s) Disconnect the attachment board from the CBL PCB of the optical assembly.
 - (t) Remove the optical assembly from the 3ATI tool.
 - (u) Put a drop of red paint on the two potentiometers screws
 - (v) Remove the 3ATI tool from the OPM test bench.
- E. Close up
- SUBTASK 34-22-45-210-052-A01
- (1) Final Operations
 - (a) Install the Optical Assembly A5 (Refer to TASK-34-22-45-400-811-A01).
 - (b) Install the Front Face A6 (Refer to TASK-34-22-45-400-812-A01).
 - (c) Install the Covers (Refer to TASK-34-22-45-400-813-A01).

- (1) The interconnection of the equipment with the external environment is made through one 41-pin connector J1.



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TASK 34-22-45-995-802-A01

2. IN/OUT Pin Assignment

A. Lists

- (1) The table that follows gives the list of all the pins of the external connector, the generic signal names and types (input and output) and the real signal names.

Table 2001 : Connector

PIN NUMBER	GENERIC SIGNAL NAME	REAL SIGNAL NAME	I/O
A	ARINC_E_1_A	ARINC 429 BUS 1 (HI) LS (APP for display and APP+ADR for simulation/simulator mode)	I
B		SPARE	I
C		SPARE	I
D	28V	28 VDC	I
E	0_28V	0V DC	I
F	DISCRETE_S_1	FAULT/HEALTHY	O
G	DISCRETE	COMMON/GROUND	I
H	RS422_OUT_L	RS422_TX_L	O
J	RS422_OUT_H	RS422_TX_H	O
K	RS422_IN_L	RS422_RX_L	I
L	RS422_IN_H	RS422_RX_H	I
M	ARINC_S_2_B	ARINC 429 BUS 5 (LO) LS (ADR parameters for Cat III option)	O
N	ARINC_S_2_B	ARINC 429 BUS 5 (HI) LS (ADR parameters for Cat III option)	O
P	ARINC_S_1_B	ARINC 429 BUS 4 (LO) HS (IR parameters for Cat III option)	O
R	ARINC_S_1_A	ARINC 429 BUS 4 (HI) HS (IR parameters for Cat III option)	O
S	ARINC_E_3_B	ARINC 429 BUS 3 (LO) HS (HDG for displayed and IR bus for simulation)	I
T	ARINC_E_3_A	ARINC 429 BUS 3 (HI) HS (HDG for displayed and IR bus for simulation)	I
U	ARINC_E_2_B	SPARE (Reserved BOEING)	I
V	ARINC_E_2_A	SPARE (Reserved BOEING)	I

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PIN NUMBER	GENERIC SIGNAL NAME	REAL SIGNAL NAME	I/O
W	ARINC_E_1_B	ARINC 429 BUS 1 (LO) LS (APP for display and APP+ADR for simulation/simulator mode)	I
X	DISCRETE_E_3	MODE SELECTION =1	I
Y		SPARE	
Z	DISCRETE_E_4	SPARE (ALT in meter option)	I
a	DISCRETE_E_5	MODE SELECTION =3	I
b	DISCRETE_E_6	CONF_PL_4 (AC configuration 8/0)	I
c	DISCRETE_E_7	CONF_PL_3 (AC configuration 4/0)	I
d	DISCRETE_E_8	CONF_PL_2 (AC configuration 2/0)	I
e	DISCRETE_E_9	CONF_PL_1 (AC configuration 1/0)	I
f	DISCRETE_E_2	BACKCOURSE DISPLAY DISABLED	I
g		SPARE	
h	DISCRETE_E_1	MODE SELECTION =2	I
i	EFF_DIM_L	0/5V AC LO BEZEL	I
j	EFF_DIM_H	0/5V AC HI BEZEL	I
k	DISCRETE_E_10	PARITY DISCRETE	I
m	DISCRETE_E_11	RESERVED FOR THALES (MACH NUMBER OPTION)	I
n	DISCRETE_E_12	APP DISPLAY DISABLED	I
p	DISCRETE_E_13	RESERVED FOR THALES (V-BAR OPTION)	I
q	DISCRETE_E_14	RESERVED (INITIATED TEST)	I
r	DISCRETE_E_15	SPARE	I
s	DISCRETE_E_16	RESERVED FOR THALES (LATERAL ACCELERATION OPTION)	I
t	CHASSIS_GND	CASE GROUND	I

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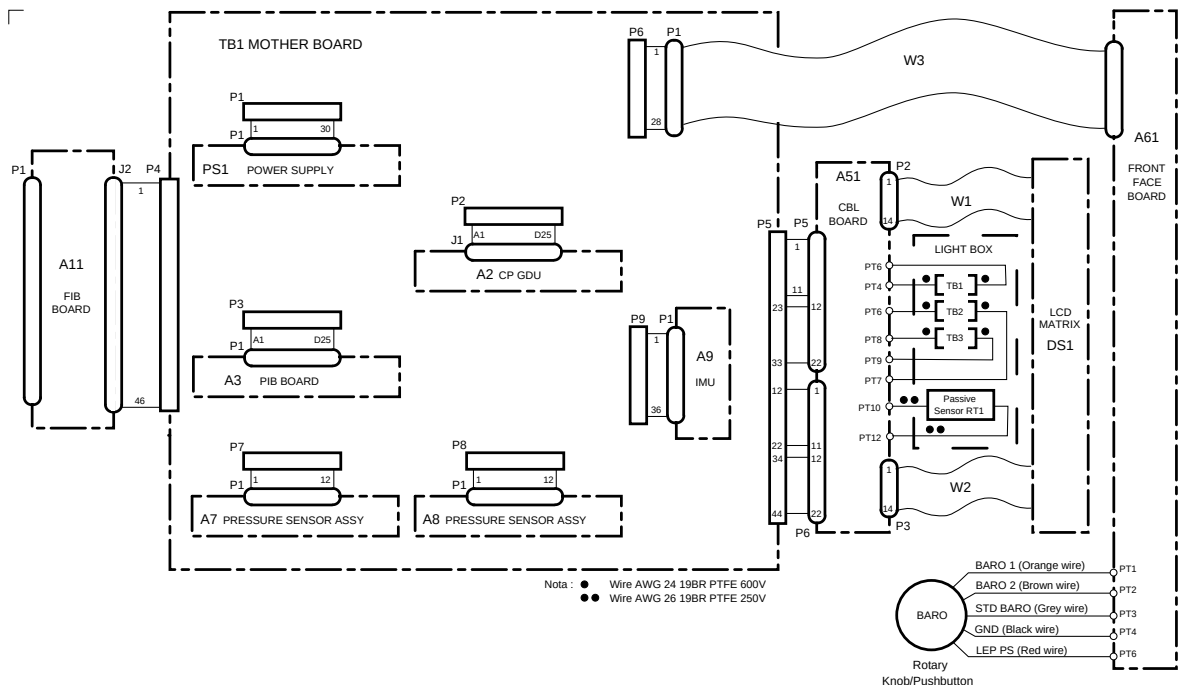
C16221K

TASK 34-22-45-995-803-A01

3. Internal Interconnections

A. Wiring List

- (1) The internal interconnection of the instrument is made through a programmed interconnection TB1 (Refer to Figure 2002).



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Figure 2002/Graphic - Sheet 01 / 34-22-45-991-202-A01
Internal Interconnections (ISIS with CBL Board E24100BB)

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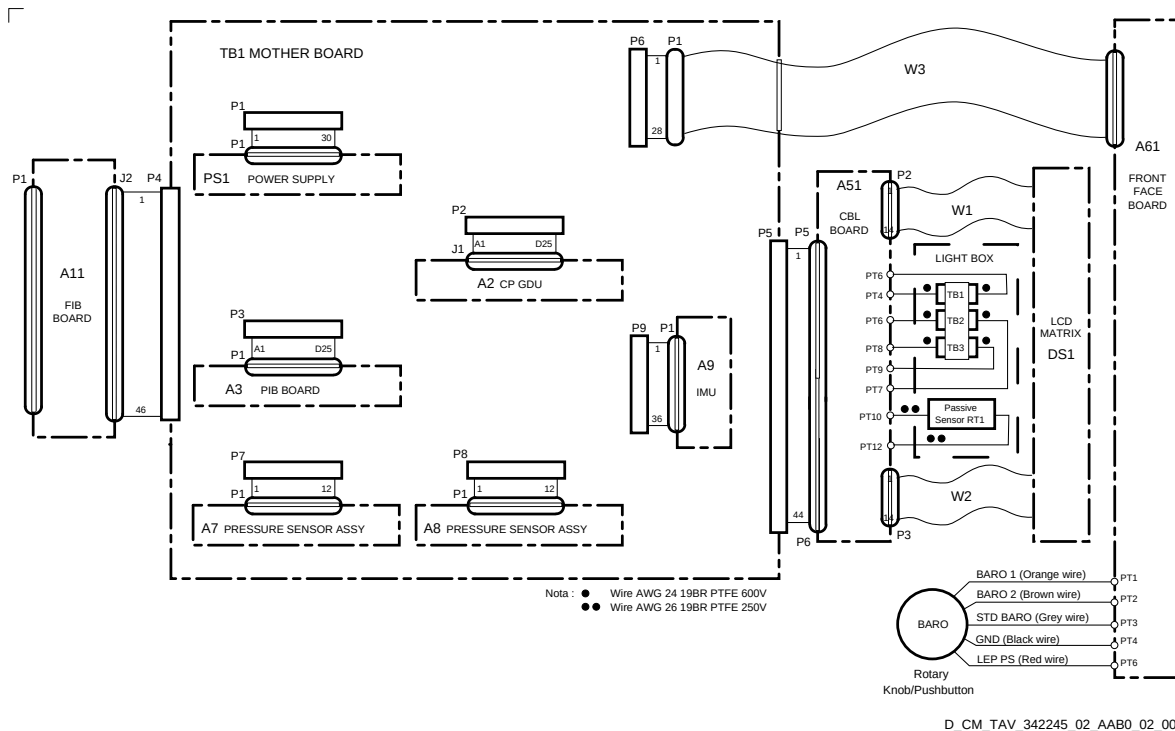


Figure 2002/Graphic - Sheet 02 / 34-22-45-991-202-A01
Internal Interconnections (ISIS with CBL Board E24100EA)

(2) Cross-reference Tables

- (a) Table 2002 gives the list of connectors and the related Figures and Tables which contain the pin number and the signal name.
- (b) Table 2003 gives the interconnections and signal names between the connector P5 of the motherboard TB1 and the connectors P5 and P6 of the CBL board A51 (E24100BB) or the connector P5 of the CBL board A51 (E24100EA).
- (c) Table 2004 gives the pin assignment and signal names of the connector P2 of the CBL board, which is connected to the flat cable W1.
- (d) Table 2005 gives the pin assignment and signal names of the connector P3 of the CBL board, which is connected to the flat cable W2.

Table 2002 : Connector Cross-reference Table

Subassemblies and Connectors		Refer to Figures and Tables
External connector P1		(Refer to Figure 2001) and (Refer to Table 2001)
Motherboard TB1	Connectors P1 with P1 (Power supply PS1)	(Refer to Figure 2010) and (Refer to Figure 2011)
	Connectors P2 with J1 (CPGDU A2)	
	Connectors P3 with P1 (PIB Board A3)	

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Subassemblies and Connectors		Refer to Figures and Tables
	Connectors P4 with J2 (FIB Board A11)	(Refer to Figure 2004) (Refer to Figure 2010) and (Refer to Figure 2011)
	Connectors P5 with P5 and P6 (CBL Board A51)	(Refer to Table 2003) (Refer to Figure 2006) (Refer to Figure 2010) and (Refer to Figure 2011)
	Connectors P5 with P5 (CBL Board A51)	(Refer to Table 2003) (Refer to Figure 2006) (Refer to Figure 2010) and (Refer to Figure 2011)
	Connectors P6 with P1 (Flat cable W3)	(Refer to Figure 2008) (Refer to Figure 2009) (Refer to Figure 2010) and (Refer to Figure 2011)
	Connectors P7 with P1 (Pressure sensor assy A7)	(Refer to Figure 2010) and (Refer to Figure 2011)
	Connectors P8 with P1 (Pressure sensor assy A8)	
	Connectors P9 with P1 (IMU A9)	
LCD Matric (DS1)	Flat cable W1 with P2 (CBL Board A51)	(Refer to Table 2004)
	Flat cable W2 with P3 (CBL Board A51)	(Refer to Table 2005)

Table 2003 : Signal Names and Pin Numbers of Connector P5 (Motherboard TB1) with P5 and P6 (CBL Board A51) or P5 (CBL Board A51)

ISIS with CBL Board A51 (E24100BB)			ISIS with CBL Board A51 (E24100EA)		
Connector P5 (Motherboard TB1)	Connector P5 (CBL Board A51)	Connector P6 (CBL Board A51)	Connector P5 (Motherboard TB1)	Connector P5 (CBL Board A51)	Signal Name
1	1		1	1	LIM_SENSOR_GAIN
2	2		2	2	
3	3		3	3	
4	4		4	4	GND
5	5		5	5	VR1
6	6		6	6	VB1
7	7		7	7	OE
8	8		8	8	CPH1
9	9		9	9	VR2

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ISIS with CBL Board A51 (E24100BB)			ISIS with CBL Board A51 (E24100EA)		Signal Name
Connector P5 (Motherboard TB1)	Connector P5 (CBL Board A51)	Connector P6 (CBL Board A51)	Connector P5 (Motherboard TB1)	Connector P5 (CBL Board A51)	
10	10		10	10	VP5A
11	11		11	11	R_LAST
12		1	12	12	CLOCK2
13		2	13	13	C_COM
14		3	14	14	VG_OFF
15		4	15	15	VEE
16		5	16	16	GND
17		6	17	17	GND
18		7	18	18	VN15
19		8	19	19	
20		9	20	20	GND_32V
21		10	21	21	VP28
22		11	22	22	VP28
23	12		23	23	LIM_SENSOR
24	13		24	24	PWM_DAY
25	14		25	25	
26	15		26	26	
27	16		27	27	VG1
28	17		28	28	VB
29	18		29	29	CPH3
30	19		30	30	CPH2
31	20		31	31	STH
32	21		32	32	VG2
33	22		33	33	VB2
34		12	34	34	INPUT
35		13	35	35	CLOCK1

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ISIS with CBL Board A51 (E24100BB)			ISIS with CBL Board A51 (E24100EA)		
Connector P5 (Motherboard TB1)	Connector P5 (CBL Board A51)	Connector P6 (CBL Board A51)	Connector P5 (Motherboard TB1)	Connector P5 (CBL Board A51)	Signal Name
36		14	36	36	VP13
37		15	37	37	GND
38		16	38	38	VP5N
39		17	39	39	GND
40		18	40	40	VP15
41		19	41	41	
42		20	42	42	VP32
43		21	43	43	GND_28V
44		22	44	44	GND_28V

Table 2004 : Pin Assignment of Flat cable W1

Connector	Pin Number	Signal Name
P2 (CBL Board A51)	1	
	2	VB2
	3	VG2
	4	VR2
	5	STH
	6	CPH1
	7	CPH2
	8	CPH3
	9	OE
	10	VB
	11	VB1
	12	VG1
	13	VR1
	14	GND

Table 2005 : Pin Assignment of Flat Cable W2

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Connector	Pin Number	Signal Name
P3 (CBL Board A51 TB1)	1	GND
	2	VDD
	3	VEE
	4	VG_OFF
	5	GND
	6	GND
	7	C_COM
	8	CLOCK1
	9	CLOCK2
	10	INPUT
	11	R_LAST
	12	GP
	13	BP
	14	GND

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TASK 34-22-45-995-804-A01

4. Electricals Diagrams

A. General

SUBTASK 34-22-45-995-001-A01

(1) Introduction

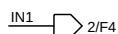
- (a) The electrical diagram has a number of sheets. On these sheets, some specials symbols are used and their definitions are given below:

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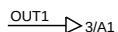


Link between sheets (or supply voltage)

The related inscriptions have these meanings :

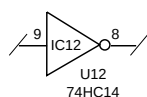
IN1 : CAD system designation

2/F4 : Sheet number (2) and geographical location
(F4 : F = horizontal and 4 = vertical) of the follow-on
sheet of the electrical diagram.



Internal sheet link

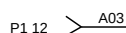
The inscriptions have the same meaning as the link between
sheets.



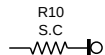
Integrated circuit with manufacturer P/N 74HC14 and equipment
designator IC12, terminals 9 and 8. It is installed on the board
but not connected.

CLK_N

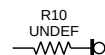
CLK/ = CLK (Complemented CLK signal)



Input signal, the CAD designation of which is A03 and which is
applied to pin 12 of connector P1.



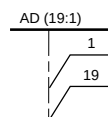
The resistor with the equipment designator R10 is short-circuited
(S.C).



The value of the resistor with the equipment designator R10 is
undefined (UNDEF.).



Resistor network between terminals 1 and 8 made by four 47 kilohm
resistors. the equipment designator is RN7.



Connection by bus, the signals of which are called bit1,
bit2, ... bit19.

The value of the components are shown by a number followed by a letter.
The definitions of the letters are as follows :

K : Kilo

N : Nano

U : Unit (Ohm, Farad...)

M : Mega, Micro

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Figure 2003/Graphic 34-22-45-991-203-A01
Special Symbols

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B. Electrical Diagram for FIB Board A11
SUBTASK 34-22-45-995-002-A01

(1) Revision Reference Sheet

Table 2006 : Revision Reference Sheet for Figure 2004

Revision Code	Sheet Number	Revision Description	Mod State

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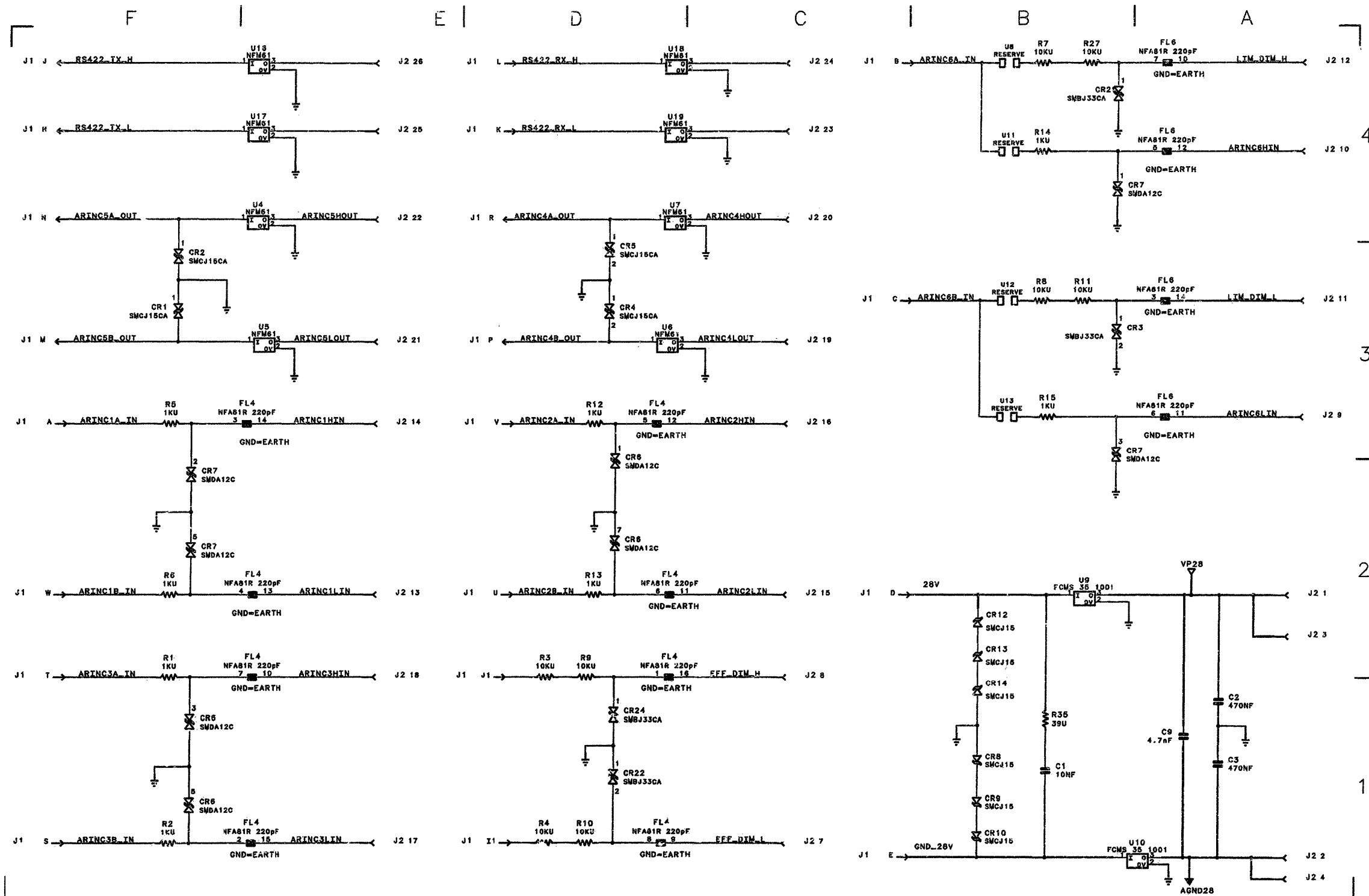


Figure 2004/Graphic - Sheet 01 / 34-22-45-991-204-A01
Electrical Diagram for FIB Board A11 (E16368CA)

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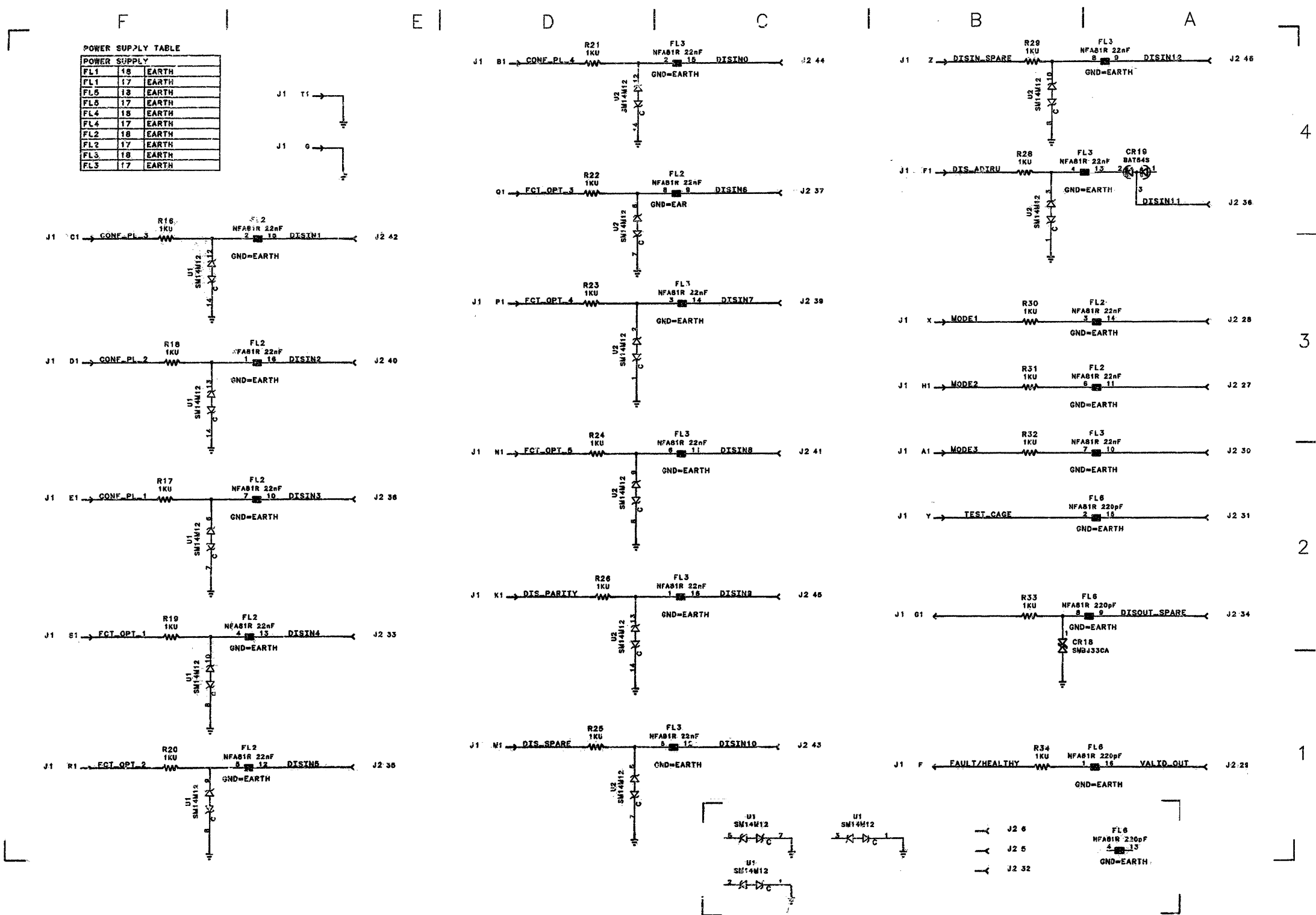
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SUBTASK 34-22-45-995-009-A01

(2) Revision Reference Sheet

Table 2007 : Revision Reference Sheet for Figure 2005

Revision Code	Sheet Number	Revision Description	Mod State

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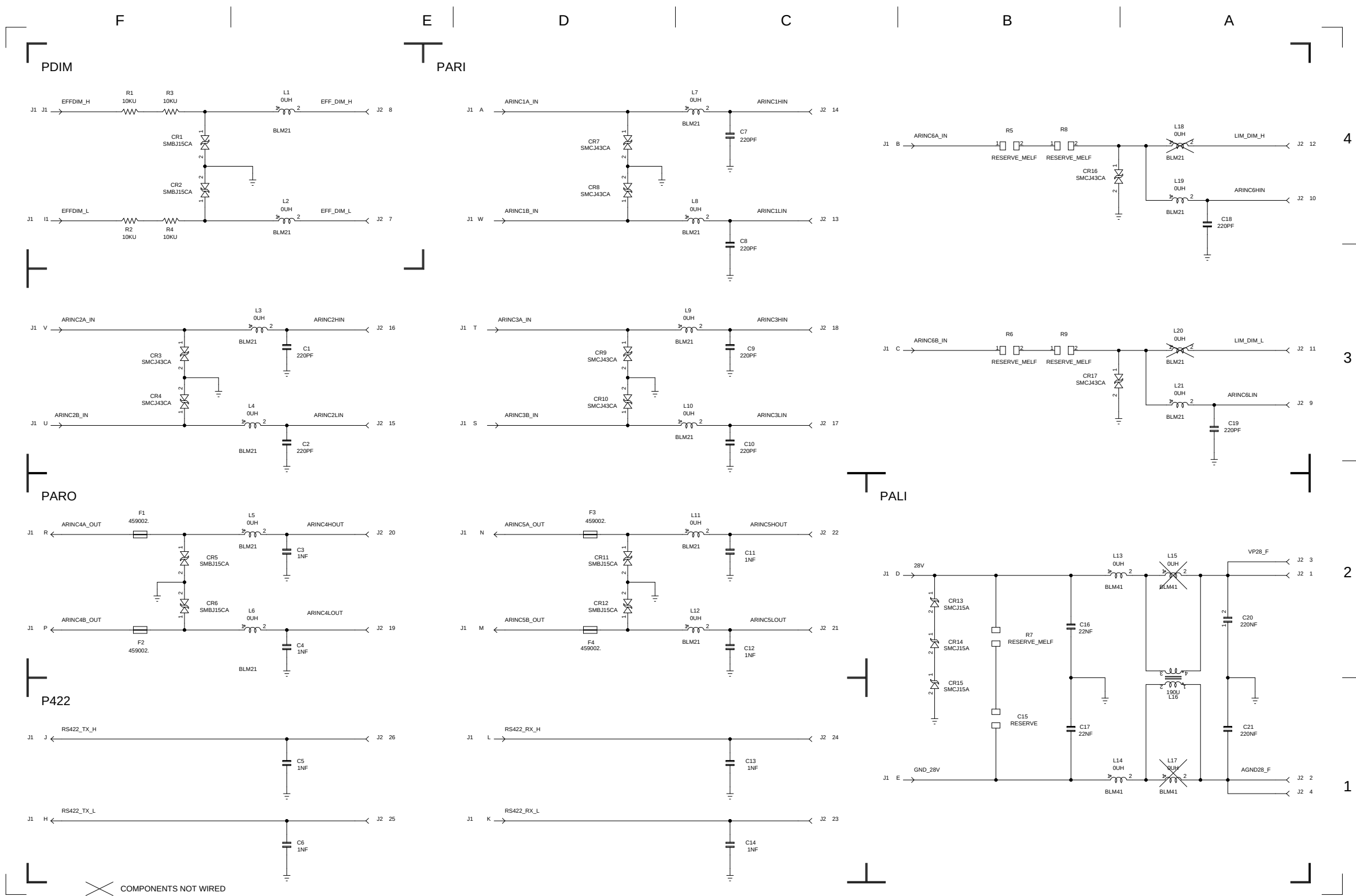


Figure 2005/Graphic - Sheet 01 / 34-22-45-991-205-A01
Electrical Diagram for FIB Board A11 (E16368FA)

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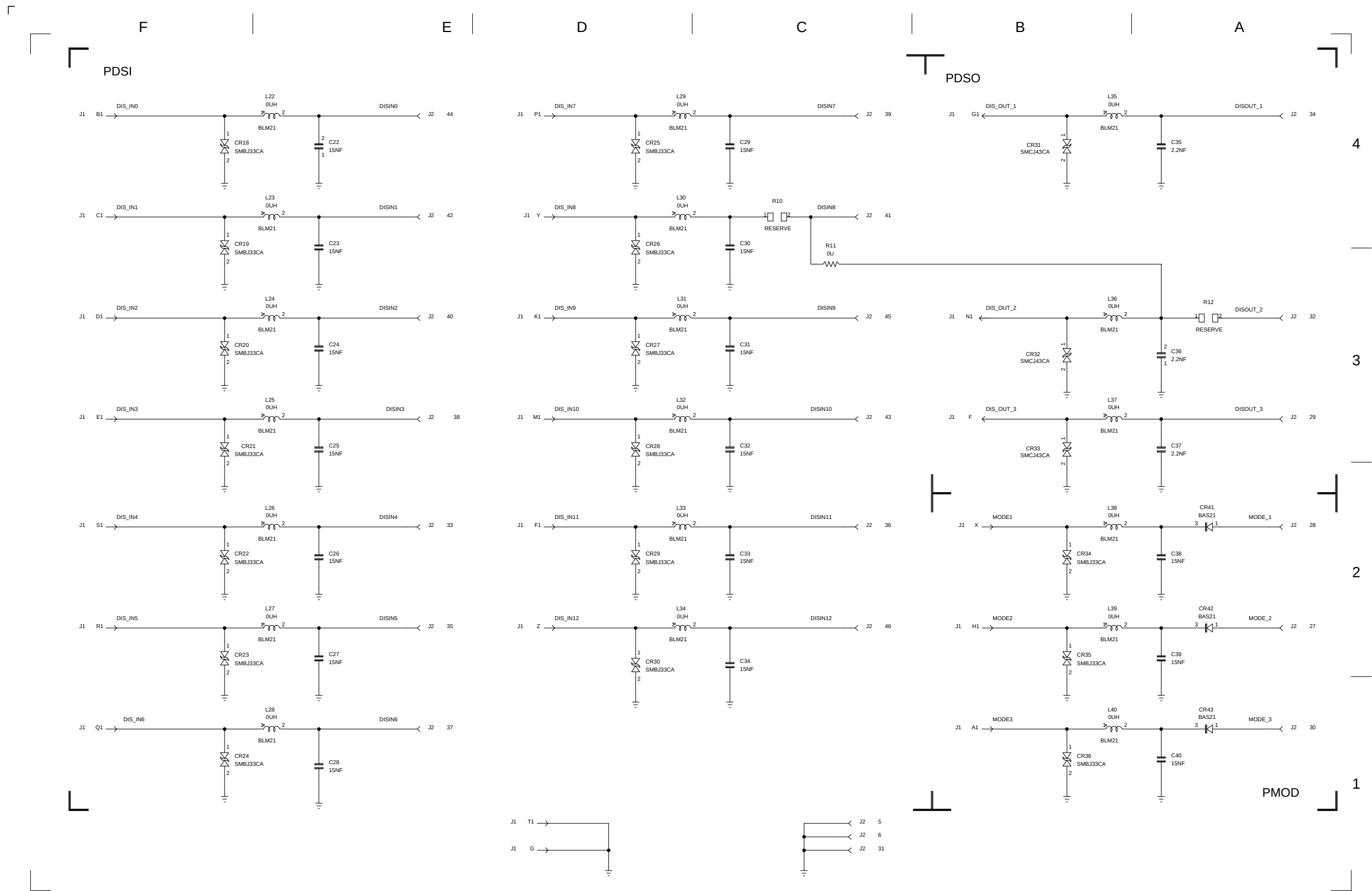


Figure 2005/Graphic - Sheet 02 / 34-22-45-991-205-A01
Electrical Diagram for FIB Board A11 (E16368FA)

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C. Electrical Diagram for CBL Board A51
SUBTASK 34-22-45-995-003-A01

(1) Revision Reference Sheet

Table 2008 : Revision Reference Sheet for Figure 2006

Revision Code	Sheet Number	Revision Description	Mod State

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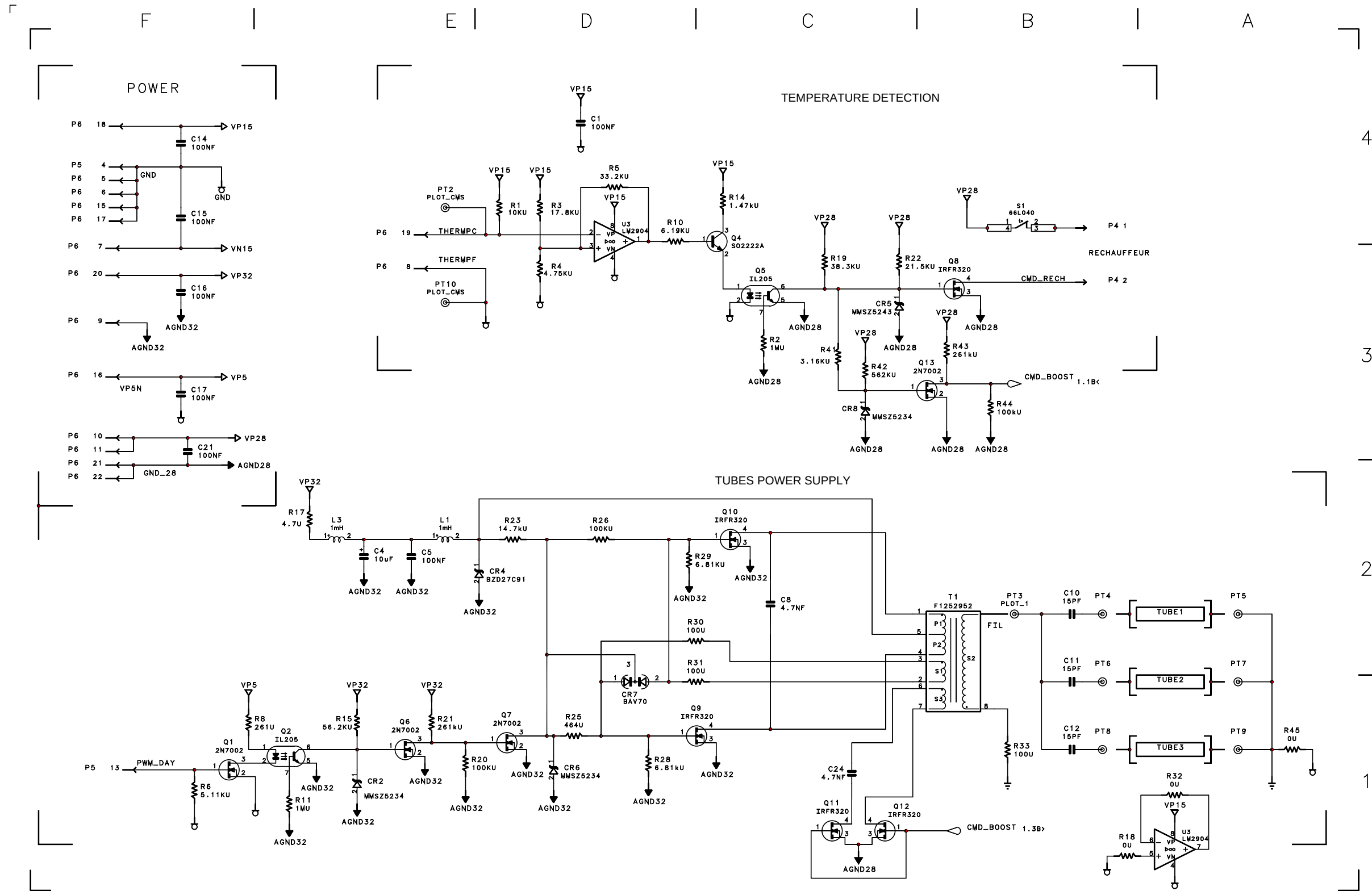


Figure 2006/Graphic - Sheet 01 / 34-22-45-991-206-A01
Electrical Diagram for CBL Board A51 (E24100BB)

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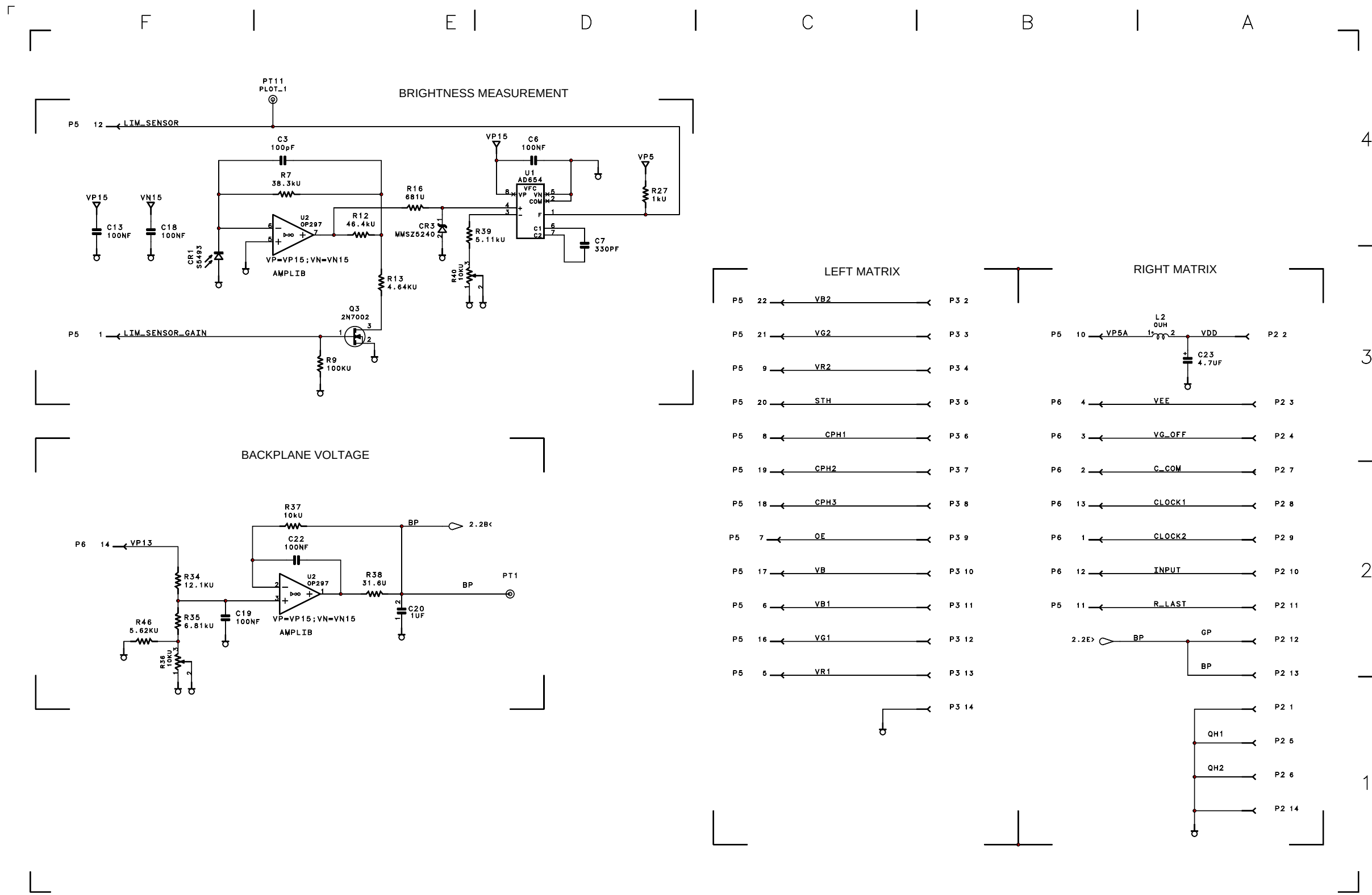


Figure 2006/Graphic - Sheet 02 / 34-22-45-991-206-A01
Electrical Diagram for CBL Board A51 (E24100BB)

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SUBTASK 34-22-45-995-004-A01

(2) Revision Reference Sheet

Table 2009 : Revision Reference Sheet for Figure 2007

Revision Code	Sheet Number	Revision Description	Mod State

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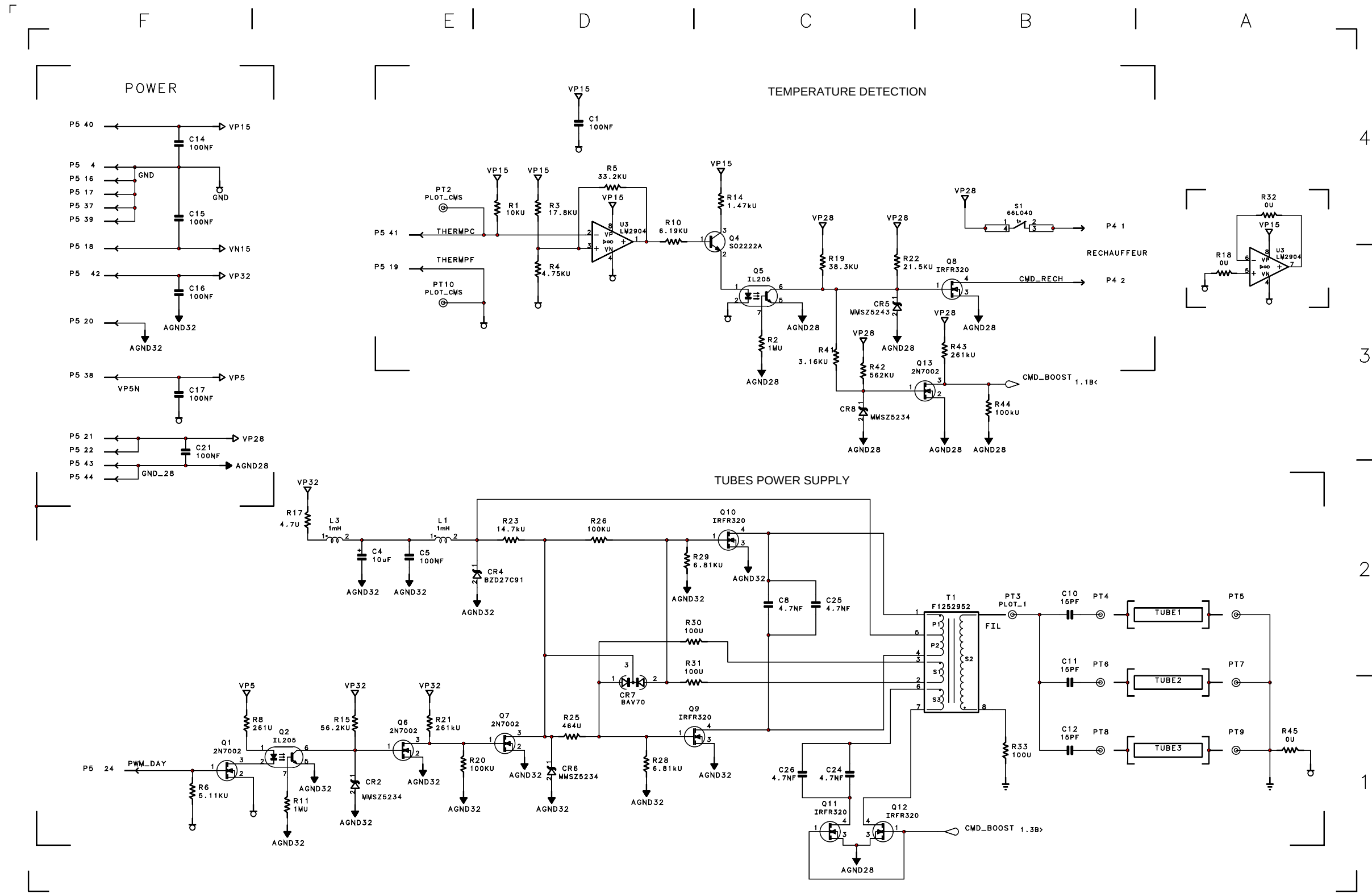


Figure 2007/Graphic - Sheet 01 / 34-22-45-991-207-A01
Electrical Diagram for CBL Board A51 (E24100EA)

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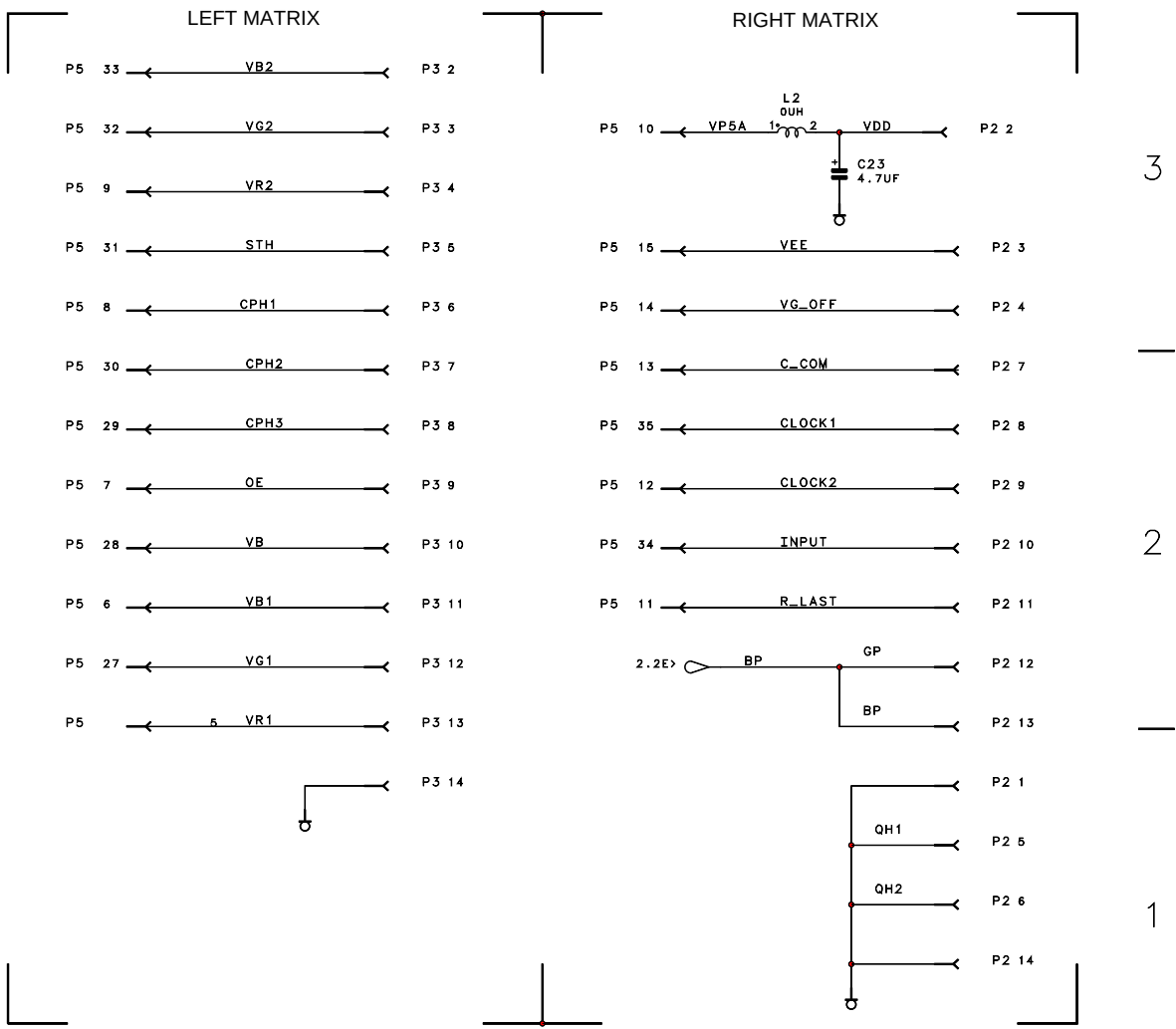
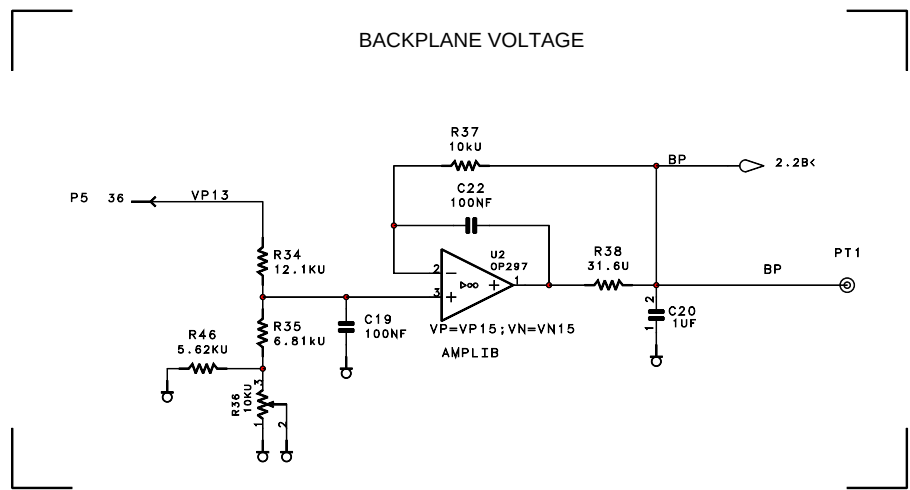
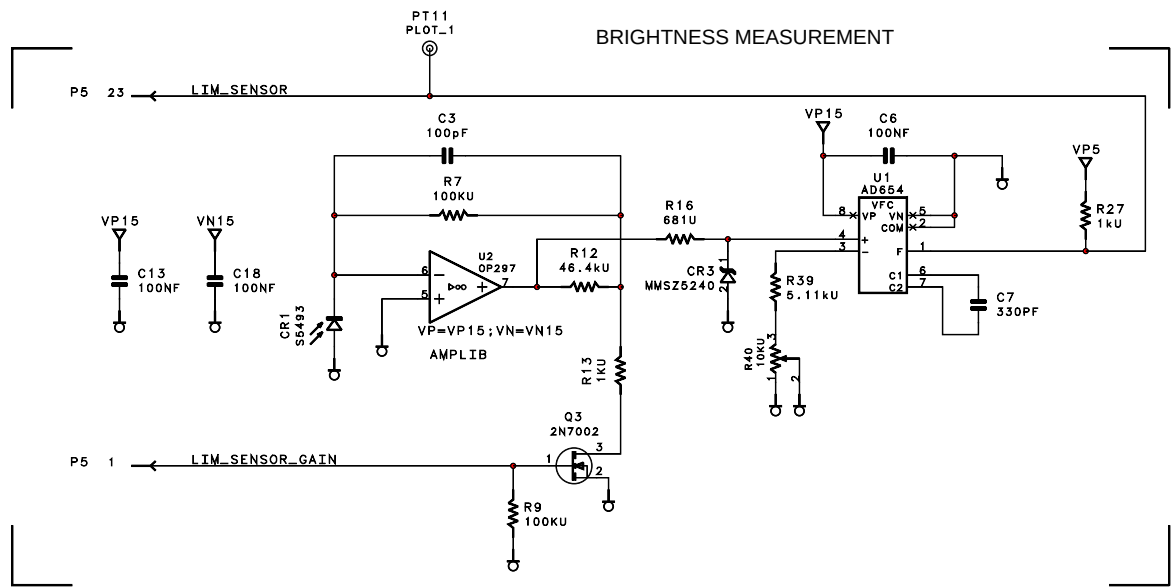


Figure 2007/Graphic - Sheet 02 / 34-22-45-991-207-A01
Electrical Diagram for CBL Board A51 (E24100EA)

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D. Electrical Diagram for Front Face Board A61
SUBTASK 34-22-45-995-005-A01

(1) Revision Reference Sheet

Table 2010 : Revision Reference Sheet for Figure 2008

Revision Code	Sheet Number	Revision Description	Mod State

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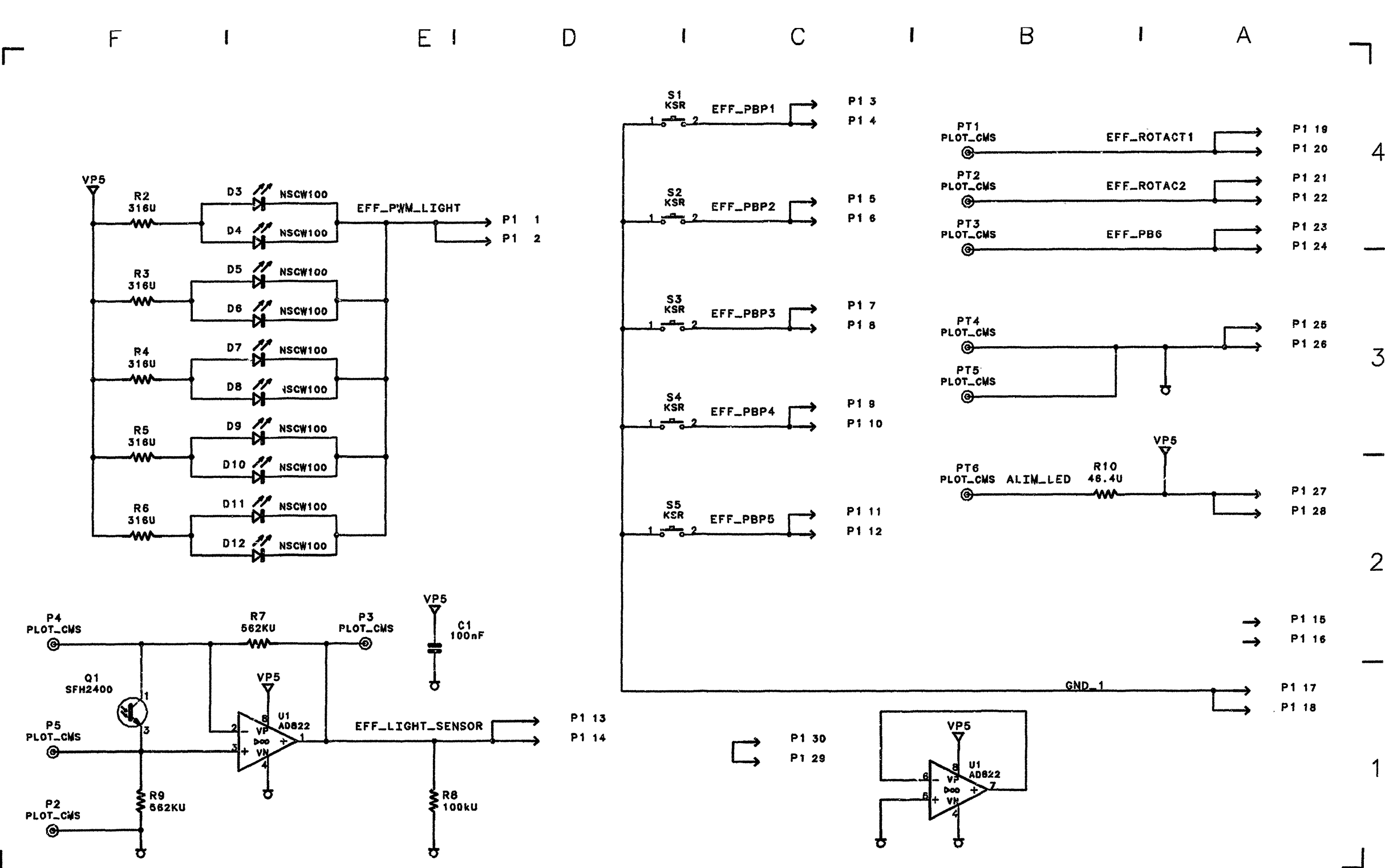


Figure 2008/Graphic 34-22-45-991-208-A01
Electrical Diagram for Front Face Board A61(E16389JA)

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SUBTASK 34-22-45-995-006-A01

(2) Revision Reference Sheet

Table 2011 : Revision Reference Sheet for Figure 2009

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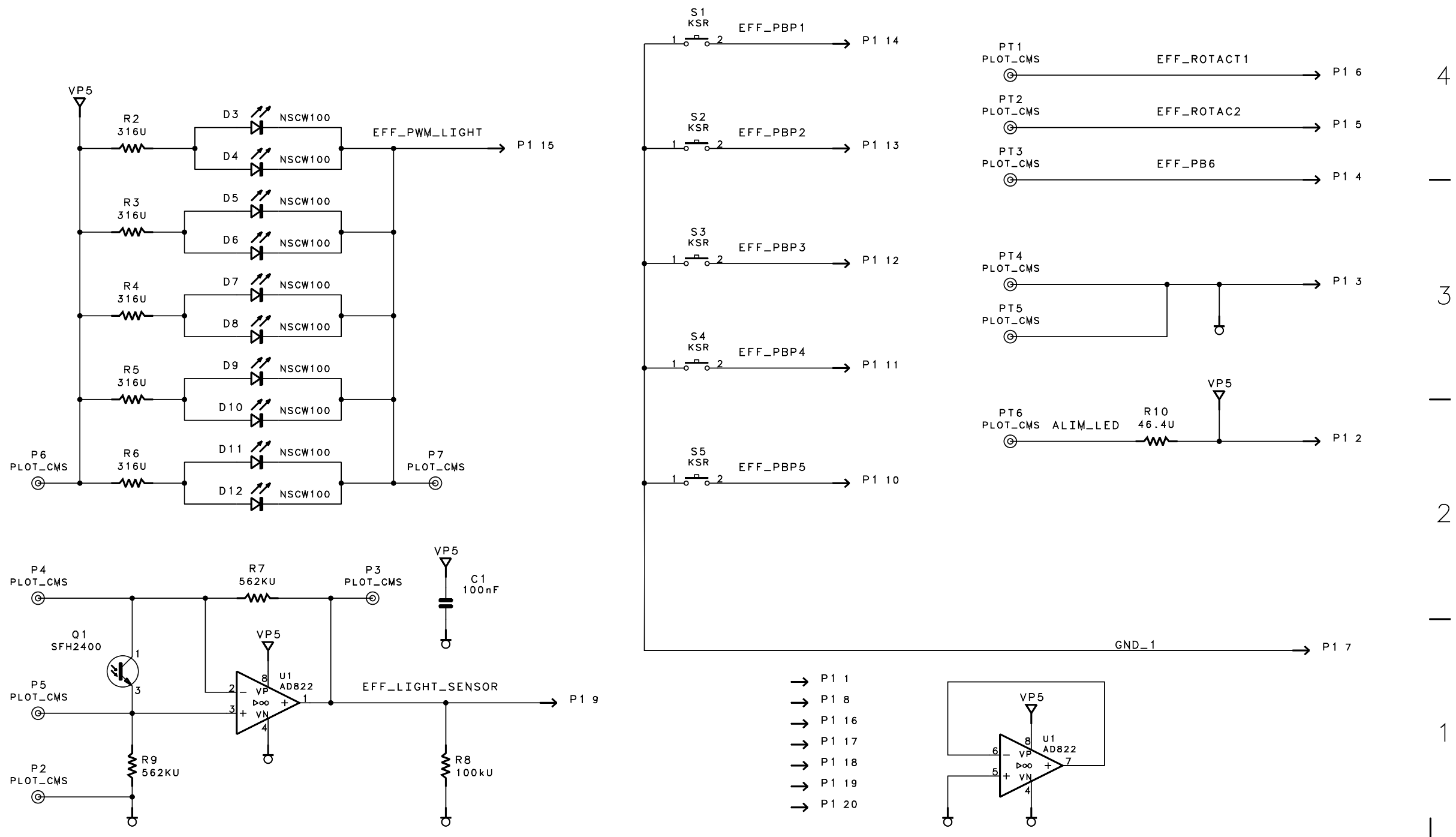


Figure 2009/Graphic 34-22-45-991-209-A01
Electrical Diagram for Front Face Board A61(E16389JB)

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E. Electrical Diagram for Programmed MotherBoard TB1
SUBTASK 34-22-45-995-007-A01

(1) Revision Reference Sheet

Table 2012 : Revision Reference Sheet for Figure 2010

Revision Code	Sheet Number	Revision Description	Mod State

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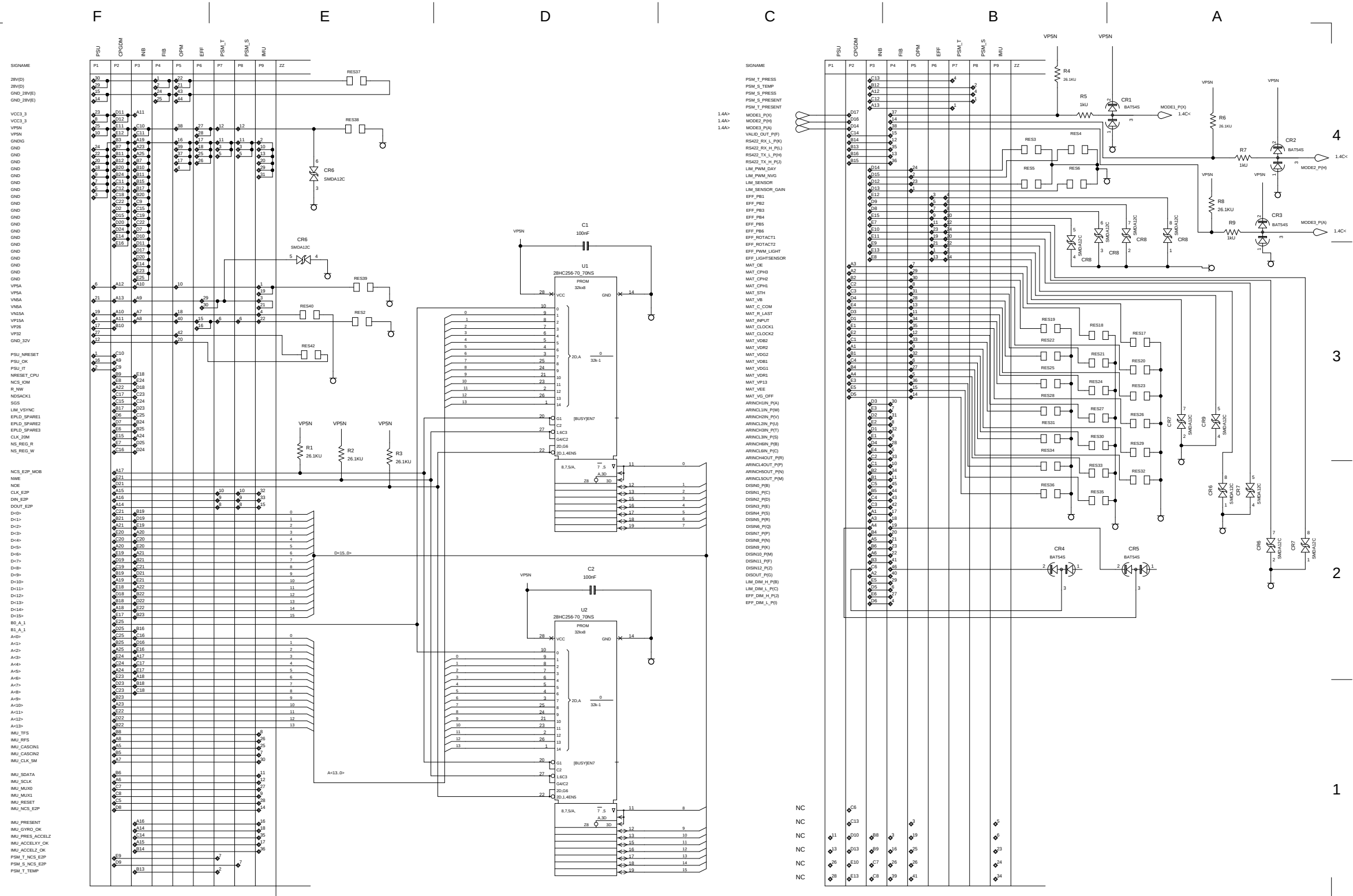


Figure 2010/Graphic 34-22-45-991-210-A01
Electrical Diagram for Programmed MotherBoard TB1(E16486BA.)

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SUBTASK 34-22-45-995-008-A01

(2) Revision Reference Sheet

Table 2013 : Revision Reference Sheet for Figure 2011

Revision Code	Sheet Number	Revision Description	Mod State

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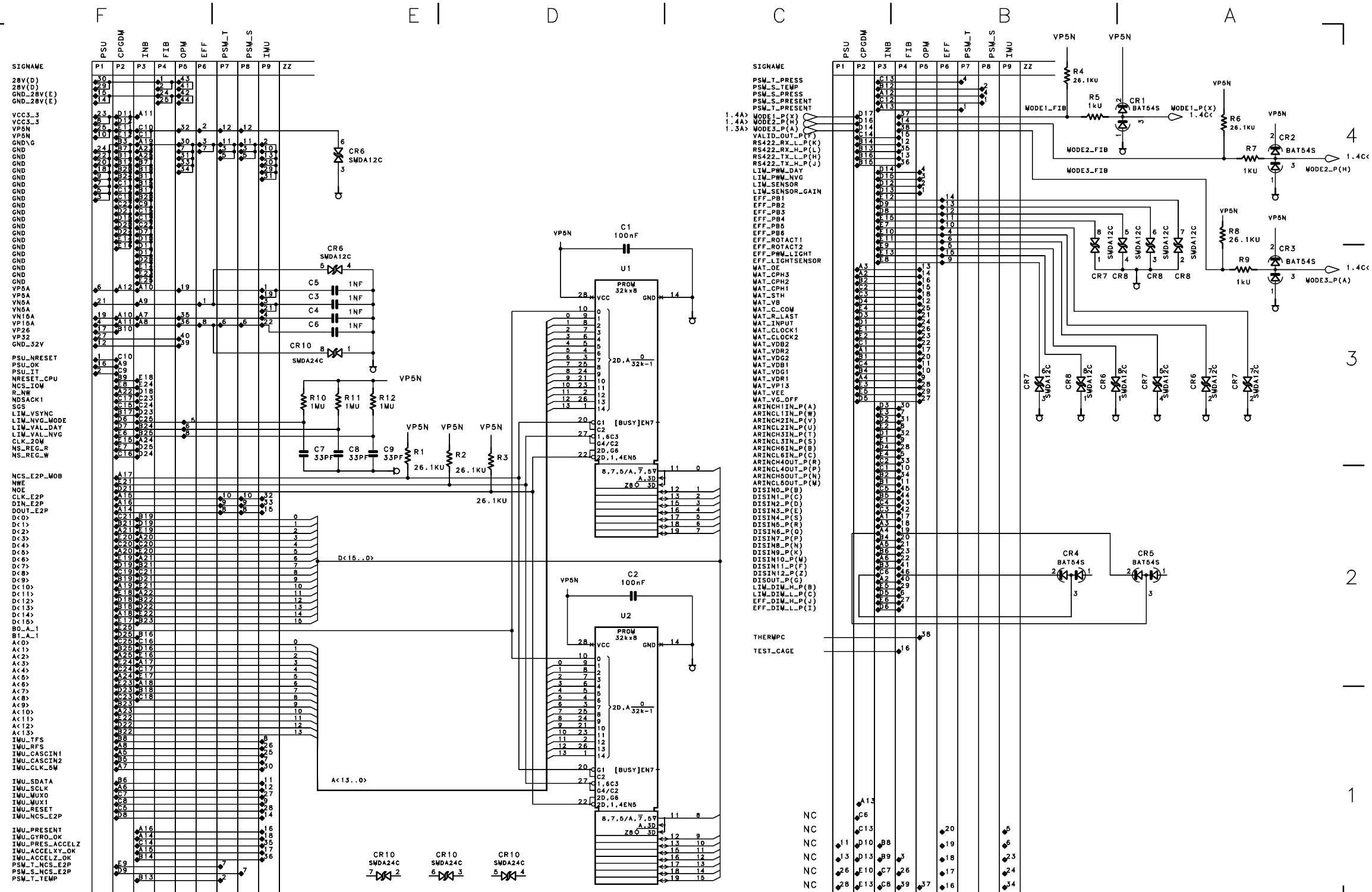


Figure 2011/Graphic 34-22-45-991-211-A01
Electrical Diagram for Programmed MotherBoard TB1(E16486BB. and E16486JA.)

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DISASSEMBLY

TASK 34-22-45-000-801-A01

1. General

A. Introduction

- (1) Obey the CAUTION and WARNING information when you do work on the boards and assemblies.

B. Reason for the Job

- (1) The procedures that follow tell the user how to disassemble the unit to the extent necessary for its repair.

C. Job Set-up Information

(1) Standard Tools, Fixtures and Equipment

- Standard electronics technician's toolkit
- Set of Torx screwdrivers (8x50mm)

(2) Special Tools, Fixtures and Equipment

NOTE: The special tools are identified in the text by an alphanumeric code.

Example: ST01

For more details, (Refer to [PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)) pageblock.

- Extraction tool A5 (ST04)
- Extraction tool A2 (ST05)
- Extraction tool A3 (ST06)

(3) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

For more details, (Refer to [PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)) pageblock.

- Acetone (Ol-so-013)
- Gloves

(4) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)), ([Refer to FIGURE-34-22-45-950-030-A01](#)), ([Refer to FIGURE-34-22-45-950-031-A01](#)), ([Refer to FIGURE-34-22-45-950-040-A01](#)), ([Refer to FIGURE-34-22-45-950-050-A01](#)), ([Refer to FIGURE-34-22-45-950-990-A01](#)) to identify the parts.

NOTE: The first number in brackets is the figure number and the second is the number of the item on the figure.

SUBTASK 34-22-45-000-007-A01

(1) Job Set-up

- (a) ([Refer to PAGEBLOCK TESTING AND FAULT ISOLATION](#)) to find the cause of a malfunction. Disassemble the unit only to the extent necessary for fault isolation.

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- (b) Read the history or record card, if there is one, to find possible defective areas which you must look at.

SUBTASK 34-22-45-000-008-A01

(1) General Instructions and Special Precautions

(a) General Instructions

- 1 Disassemble the equipment only as much as is necessary to replace the defective parts.
- 2 The screws are safetied with red paint (Pa-pa-010) (Refer to [PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)). This paint can be removed with a acetone (Ol-so-013) (Refer to [PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)).
- 3 Replace all the sealing rings removed with new sealing rings.
- 4 Do the disassembly operations in a clean environment (no dust).
- 5 The references to the left and right, front and rear of the equipment are given in relation to its normal test position.

(b) Special Precautions

- 1 Disconnect the equipment from all power sources before disassembly.
- 2 Use gloves to touch the optical part.
- 3 When the equipment contains electrostatic sensitive devices, refer to the Standard Overhaul Practices Manual SOPM 20-02-00 for the precautions to use.

D. Close-up

- (1) Not applicable.

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TASK 34-22-45-000-802-A01

2. Removal of the covers

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and equipment

- Standard electronics technician's toolkit

- (2) Consumables

- None

- (3) Referenced information

- (a) (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01) to identify the parts.

C. Job Set-up

- (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.

D. Procedure

- Remove the screws (2 or 2A or 2B or 2C/120).
- Remove the screws (2 or 2A or 2B or 2C/130).
- Remove the covers (2 or 2A or 2B or 2C/110).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-000-803-A01

3. Removal of the Front Face A6

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and equipment

- Standard electronics technician's toolkit

- (2) Consumables

- None

- (3) Referenced information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-000-001-A01

- (1) Removal

CAUTION: BE CAREFUL NOT TO CAUSE DAMAGE TO THE FLEX CABLE DURING THESE STEPS.

- Remove the screws (2 or 2A or 2B or 2C/150).
- Disconnect the flex cable from the board TB1.
- Tag the wires of the BARO encoder (3 or 3A/130) on the front face board A61 and unsolder them.
- Remove the screws (3 or 3A/70).
- Remove the front face A6 , together with the BARO encoder (3 or 3A/130).

SUBTASK 34-22-45-000-002-A01

- (2) Disassembly

- Remove the keys (3 or 3A/80 to 3 or 3A/120).
- Remove the screws (3 or 3A/140).
- Remove the BARO encoder (3 or 3A/130).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-000-804-A01

4. Removal of the Optical Assembly A5

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and equipment

- Standard electronics technician's toolkit
- Extraction tool A5 (ST04)

(2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- (a) Gloves (mi-mi-084)

(3) Referenced information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-000-003-A01

(1) Removal

- Remove the screws (2 or 2A or 2B or 2C/180).
- Remove the optical assembly A5 with the special tool ST04.

SUBTASK 34-22-45-000-004-A01

(2) Disassembly

- Disconnect the two flex connectors from the board A51.
- Remove the three screws (4/60).
- Remove the LCD matrix (DS1), the polarizer (4/70), the diffusers (4/80 and 4/90) and the assembly that includes the board (A51), the lighting assy unit and the heater (4/100).
- Cut the tie that attaches the two wires of the passive sensor RT1.
- Tag the two wires that connect the passive sensor RT1 to the board A51 and unsolder them.
- Tag the six wires that connect the tubes (DS2 to DS4) to the board A51 and unsolder them.
- Disconnect the connector J4 from the heater (3/100) to the board A51.
- Remove the screws (4/230) and washers (4/240).
- Remove the board A51.
- Remove the nuts (4/210).
- Remove the caps (4/190) and the tubes DS2 to DS4.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-000-805-A01

5. Removal of the Inertial Measurement Unit A9

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and equipment

- Standard electronics technician's toolkit

- (2) Consumables

- None

- (3) Referenced information

- (a) (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01) to identify the parts.

C. Job Set-up

- (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-000-009-A01

- (1) Removal of the Inertial Measurement Unit A9

- Remove the six screws (2 or 2A or 2B or 2C/200) and the two screws (2 or 2A or 2B or 2C/210).
- Remove the inertial measurement unit A9.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-000-806-A01

6. Removal of the Control Processor Graphic Display Unit A2

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and equipment
 - Standard electronics technician's toolkit
 - Extraction tool A2 (ST05)
 - (2) Consumables
 - (a) None
 - (3) Referenced information
 - (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.
- C. Job Set-up
 - (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.
- D. Procedure

SUBTASK 34-22-45-000-010-A01

 - (1) Removal of the Control Processor Graphic Display Unit A2
 - Release the slides.
 - Install the two screws of the special tool ST05 on the board A2.
 - Install the two screws of the special tool ST05 on the board A2 and use the extraction tool to remove the board.
- E. Close-up
 - (1) Not applicable.

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TASK 34-22-45-000-807-A01

7. Removal of the Programming Interface Board A3

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and Equipment
 - Standard electronics technician's toolkit
 - Extraction tool A3 (ST06)
 - (2) Consumables
 - (a) None
 - (3) Referenced Information
 - (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.
- C. Job Set-up
 - (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.
- D. Procedure

SUBTASK 34-22-45-000-011-A01

 - (1) Removal of the Programming Interface Board A3
 - With the extraction tool ST06, put two pins in the holes of A3.
 - Pull the structure (2 or 2A/450 or 2B/460) to remove the board A3.
- E. Close-up
 - (1) Not applicable.

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TASK 34-22-45-000-808-A01

8. Removal of the Power Supply PS1

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and Equipment
 - Standard electronics technician's toolkit
 - Extraction tool A3 (ST06)
 - (2) Consumables
 - (a) None
 - (3) Referenced information
 - (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.
- C. Job Set-up
 - (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.
- D. Procedure

SUBTASK 34-22-45-000-012-A01

 - (1) Removal of the Power Supply PS1
 - Remove the four screws (2 or 2A or 2B or 2C/250).
 - Remove the power supply PS1 with the extraction tool ST06.
- E. Close-up
 - (1) Not applicable.

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TASK 34-22-45-000-809-A01

9. Removal of the Rear Module

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Standard electronics technician's toolkit

- (2) Consumables

- (a) None

- (3) Referenced information

- (a) (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01), (Refer to FIGURE-34-22-45-950-050-A01), (Refer to FIGURE-34-22-45-950-990-A01) to identify the parts.

C. Job Set-up

- (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-000-005-A01

- (1) Removal

- Remove the screws (2 or 2A/340 or 2B or 2C/360).
- Disconnect the hoses from the pitot and static pressure unions.
- Disconnect the rear module A1 from the board TB1.
- Remove the protective caps (99/10 and 99/20) (only for C16221KA../LA../MA..).
- Remove the pressure caps (2C/365 and 2C/385) (only for C16221QA..).
- Remove the quick unions (2 or 2A/350, 2B or 2C/370 and 2 or 2A/370, 2B or 2C/390) and the sealing rings (2 or 2A/360, 2B or 2C/380 and 2 or 2A/380, 2B or 2C/400).

SUBTASK 34-22-45-000-006-A01

- (2) Disassembly

- Remove the four screws (5/70) and washers (5/80).
- Remove the seven screws (5/90) and washers (5/100).
- Remove the backplate (5/60).
- Remove the varnish locally from the pins of connector J1 on the FIB Board A11(5/130).
- Desolder the pins of connector J1(5/110).
- Remove the connector J1(5/110) and the flange (5/120).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-000-810-A01

10. Removal of the Programmed Motherboard TB1

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and Equipment
 - Standard electronics technician's toolkit
 - (2) Consumables
 - None
 - (3) Referenced information
 - (a) (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01) to identify the parts.
- C. Job Set-up
 - (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.
- D. Procedure

SUBTASK 34-22-45-000-013-A01

 - (1) Removal of the Programmed Motherboard TB1
 - Remove the four screws (2 or 2A/430) and washers (2 or 2A/440) or the four screws (2 or 2A/430A) (only for versions C16221KA.. and C16221LA..).
 - Remove the four screws (2B or 2C/450) (only for version C16221MA../QA..).
 - Remove the programmed motherboard TB1.
- E. Close-up
 - (1) Not applicable.

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TASK 34-22-45-000-811-A01

11. Removal of the Pressure Sensor Assemblies A7 and A8

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and Equipment
 - Standard electronics technician's toolkit
 - (2) Consumables
 - None
 - (3) Referenced information
 - (a) [\(Refer to Figure 10002\)](#), [\(Refer to Figure 10003\)](#), [\(Refer to Figure 10004\)](#), [\(Refer to Figure 10005\)](#), [\(Refer to Figure 10017\)](#) to identify the parts.
- C. Job Set-up
 - (1) [\(Refer to TASK-34-22-45-400-801-A01\)](#), Safety Precautions.
- D. Procedure

SUBTASK 34-22-45-000-014-A01

- (1) Removal of the Pressure Sensor Assemblies A7 and A8 except PNRs F1407830 and F1407831
 - Cut the two ties at the level of the pressure union (one tie for each pressure union).
 - Cut the three ties to release the hose of the pitot pressure union P from the board TB1.
 - Disconnect the hoses from the pitot pressure union P and static pressure union S.
 - Remove the screws (2 or 2A or 2B or 2C/300 and 307).
 - Remove the pressure sensor assemblies A7 and A8.
 - Remove the eight spacers (2 or 2A/310, 2B or 2C/330) and screws (2 or 2A/320, 2B or 2C/340).
 - Remove the pressure sensor assemblies A7 and A8.

SUBTASK 34-22-45-000-017-A01

- (2) Removal of the Pressure Sensor Assemblies A7 and A8 only for PNRs F1407830 and F1407831
 - Cut the two ties at the level of the pressure union (one tie for each pressure union).
 - Cut the three ties to release the hose of the pitot pressure union P from the board TB1.
 - Disconnect the hoses from the pitot pressure union P and static pressure union S.
 - Remove the screws (2 or 2A or 2B or 2C/300 and 307).
 - Remove the pressure sensor equipped A7 and A8.
 - Remove the eight screws (10/20).
 - Remove the shielding (10/10).
 - Remove the eight screws (10/40) and the eight pillars (10/30).
 - Remove the pressure sensor assemblies A7 and A8.

- E. Close-up
 - (1) Not applicable.

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TASK 34-22-45-000-812-A01

12. Removal of the Sensor Shielding Assembly

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and equipment
 - Standard electronics technician's toolkit
 - (2) Consumables
 - None
 - (3) Referenced Information
 - (a) (Refer to Figure 10002), (Refer to Figure 10003), (Refer to Figure 10004), (Refer to Figure 10005), (Refer to Figure 10017) to identify the parts.

- C. Job Set-up
 - (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-000-015-A01

- (1) Removal of the Sensor Shielding Assembly except PNRs F1407830 and F1407831
 - Remove the screws (2 or 2A or 2B or 2C/270) and washers (2 or 2A or 2B or 2C/280).
 - Remove the sensor shielding assembly (2 or 2A or 2B or 2C/260).

SUBTASK 34-22-45-000-018-A01

- (2) Removal of the Sensor Shielding Assembly only for PNRs F1407830 and F1407831
 - Remove the pressure sensor assemblies A7 and A8 (Refer to SUBTASK-34-22-45-000-017-A01).

E. Close-up

- (1) Not applicable.

THALES

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TASK 34-22-45-000-813-A01

13. Removal of the Structure Assembly

- A. Reason for the job
 - (1) Self explanatory
- B. Job Set-up Information
 - (1) Tools, Fixtures and equipment
 - Standard electronics technician's toolkit
 - (2) Consumables
 - None
 - (3) Referenced Information
 - (a) (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01) to identify the parts.
- C. Job Set-up
 - (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.
- D. Procedure

SUBTASK 34-22-45-000-016-A01

 - (1) Removal of the Structure Assembly
 - Remove the screws (2 or 2A/400, 2B or 2C/420) and the nuts (2 or 2A/410, 2B or 2C/430).
 - Remove the four board guides (2 or 2A/390, 2B or 2C/410).
- E. Close-up
 - (1) Not applicable.

CLEANING

TASK 34-22-45-100-801-A01

1. Cleaning of the LRU

A. General

- (1) The procedures in this section contain instructions to fully clean the component.
- (2) It is necessary to clean the component after assembly of the component and before storage or use.
- (3) You must do this removing all traces of finger marks, oil, grease, dust and dirt.
- (4) Do the cleaning procedure in a clean room

B. Reason for the Job

- (1) Self explanatory.

C. Job Set-Up Information

- (1) Tools, Fixtures and Equipment

- None.

- (2) Consumables

(Refer to TASK-34-22-45-940-803-A01)

NOTE: Equivalent substitutes can be used for the items that follow.

- Precision wipe.
- Silk paint brush.
- Isopropyl alcohol (OI-cl-011).
- Substrate cleaner (OI-cl-017).

- (3) Referenced Information

- (Refer to TASK-34-22-45-940-803-A01), Consumables.

D. Job Set-Up

SUBTASK 34-22-45-160-001-A01

- (1) Job Set-Up

- (a) Make sure that the indicator is fully assembled.
- (b) Disconnect the equipment from all electrical power sources.
- (c) Wash your hands before the start of the cleaning procedure.

E. Procedure

SUBTASK 34-22-45-160-002-A01

- (a) Optical Parts

CAUTION: DO NOT USE MATERIAL SUCH AS TISSUE PAPER WHEN YOU CLEAN THE OPTICAL PARTS OF THE LRU. SUCH MATERIAL CAN CAUSE DAMAGE TO THE LRU.

- For a finger print or recent mark without grease or oil:
 - Breathe on the stained area and clean with a soft precision wipe.

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- For other marks:
 - Soak a precision wipe in Substrate cleaner (OI-cl-017).
 - Carefully clean the optical part.
 - Use a new precision wipe each time to prevent scratches to the optical parts with the collected dust.
 - Do the same step again until the optical parts are fully clean.

SUBTASK 34-22-45-160-003-A01

(a) Painted Parts

- Soak a precision wipe in isopropyl alcohol (OI-cl-011).
- Carefully rub the painted part with a precision wipe (IC04).

SUBTASK 34-22-45-160-004-A01

(a) Mechanical Parts

- For traces of dust:
 - Clean the structure with a clean precision wipe.
 - If you cannot get access to the parts easily, use the silk paint brush.
- For traces of grease, finger prints, etc:
 - Clean the parts with a clean precision wipe soaked in isopropyl alcohol (OI-cl-011).

SUBTASK 34-22-45-160-005-A01

(a) Electrical Parts

CAUTION: DO NOT TOUCH THE PINS OF THE ELECTRICAL CONNECTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE LRU.

- Clean the electrical connectors with a soft silk paint brush soaked in isopropyl alcohol (OI-cl-011).

F. Close-up

- (1) Not applicable.

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CHECK

TASK 34-22-45-200-801-A01

1. Visual inspection

A. General

- (1) The procedures in this section give the instructions for a full check of the LRU.
- (2) The check is necessary after repair of the LRU and during assembly. You must do this procedure to make sure that the parts of the LRU and its subassemblies are physically correct and show no signs of wear, damage or risk of failure.
- (3) The Standard Overhaul Practices Manual SOPM 20-02-00 gives standard procedures.

B. Reason for the job

- (1) The procedures that follow tell the user how to do the inspection/check of the parts in the unit to find if they are serviceable.

C. Job Set-Up Information

- (1) Tools, Fixtures and Equipment

(Refer to TASK-34-22-45-940-802-A01)

NOTE: Equivalent substitutes can be used for the items that follow.

- Jeweller's magnifying glass .
- (2) Consumables
 - None
 - (3) Referenced Information:
 - (Refer to TASK-34-22-45-940-803-A01), Tools, Fixtures and Equipment
 - (Refer to TASK-34-22-45-100-801-A01), Cleaning of the LRU.

D. Job Set-up

SUBTASK 34-22-45-220-001-A01

- (1) Job Set-up

- (a) Read the history or record card, if there is one, to find the possible defective areas which you must look at.

E. Procedure

NOTE: During assembly of SRUs after a repair, you must do one or all of the checks that follow (i.e. inspection of mechanical and optical parts).

SUBTASK 34-22-45-220-002-A01

- (a) Optical Parts

- Do a check of the optical parts for scratches, spots and rings.
- Make sure that no part of the optical light sensor is damaged and darkened.
- Make sure there are no fingerprints, grease spots and other marks.

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SUBTASK 34-22-45-220-003-A01

(a) Painted Parts

- Make sure that there are no scratches or blistering of the paint.

SUBTASK 34-22-45-220-004-A01

(a) Mechanical Parts

- Use a magnifying glass to examine all the mechanical parts (heatsinks, screws, etc.)
- Do a check of all these parts for signs of shocks or burrs.
- Do a check of the condition of the captive screw threads.
- Do a check of the screw torques.
- Make sure that the spring tension washers and the seals do not show signs of cracking or unusual hardness or softness.

SUBTASK 34-22-45-220-005-A01

(a) Electrical Parts

- Use a magnifying glass to examine the connector parts (pins, terminal strips, etc.).
- Make sure that the connectors are correctly connected during SRU installation.
- Do a check of the binding for condition and safe attachment.
- Make sure that the wires and the harnesses do not show signs of marks or damage.
- Do a check of the ribbon cables for broken wires near the soldered connections.
- Make sure that the sensors are in the correct positions.
- Do a check of the solder joints.

F. Close-up

- (1) Not applicable.

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REPAIR

TASK 34-22-45-300-801-A01

1. General

A. Introduction

NOTE: All operations not described in this document must be done by manufacturer, on his premises and in accordance with the approved industrial file.

(1) The standard procedures are included in the Manual ([REFER TO SOPM 20-02-00.](#))

B. Reason for the Job

(1) Self explanatory.

C. Job Set-up Information

(1) Tools, Fixtures and Equipment

([Refer to TASK-34-22-45-940-802-A01](#))

NOTE: Equivalent substitutes can be used for the items that follow.

- ZIF Package (ST08).
- Press.

(2) Consumables

([Refer to TASK-34-22-45-940-803-A01](#))

NOTE: Equivalent substitutes can be used for the items that follow.

- Tiflex XF 160 black ink (Pa-in-004)
- Isopropyl alcohol (OI-cl-011)

(3) Referenced Information

(a) Refer to the Manual ([REFER TO SOPM 20-02-00.](#))

SUBTASK 34-22-45-300-005-A01

(1) Job Set-up

(a) Repair Conditions

NOTE: During repair operations, make sure that the environmental conditions are as follows:

- Pressure: No specific environmental level other than ambient pressure level is necessary.
- Relative humidity: Refer to SPOM 20-02-00, THAV-A-20-02-00-020F0-028A-C - Special instructions in relation to SSD - General procedure.

D. Procedure

(1) Not applicable.

E. Close-up

(1) Not applicable.

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TASK 34-22-45-300-802-A01

2. Replacement (or Modification) of the Labels

A. General

- (1) This procedure is for the replacement or modification of the identification and/or amendment labels.

B. Reason for the Job

- (1) Self explanatory.

C. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- None.

- (2) Consumables

(Refer to TASK-34-22-45-940-803-A01)

NOTE: Equivalent substitutes can be used for the items that follow.

- Tiflex XF 160 black ink (Pa-in-004).
- Isopropyl alcohol (OI-cl-011).

- (3) Referenced Information:

- (Refer to TASK-34-22-45-940-803-A01), Tools, Fixtures and Equipment
- (Refer to TASK-34-22-45-100-801-A01), Cleaning of the LRU.

D. Job Set-up

- (1) Not applicable.

E. Procedure

SUBTASK 34-22-45-300-001-A01

- (1) Replacement (or Modification) of Amendment label

NOTE: It is not usually necessary to replace the amendment label each time that the amendment label changes (unless the label shows signs of deterioration). The Modifiable Bar Code Label is replaced when the Part Number of the instrument changes.

- (a) Remove the transparent cover from the label.

- (b) Clean the related amendment letter or its position carefully with a lintless cloth soaked in isopropyl alcohol (OI-cl-011).

NOTE: Make sure that there is no grease or dust and that all the adhesive compound from the old label is removed.

- (c) Modify the labels as follows:

- On the labels which have letters, use black ink (Pa-in-004) to put a circle round the related amendment letter.
- On new labels, make sure that the same amendment letters are shown as on the old label.

- (d) Put a transparent cover on the amendment label.

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SUBTASK 34-22-45-300-002-A01

(2) Replacement of the Identification Label

NOTE: It is necessary to replace these labels if they show signs of deterioration. The Modifiable Bar Code Label is also replaced when the Part Number of the instrument changes.

- (a) Make sure that the new label has all the correct indications
- (b) With tweezers or a scalpel, remove the old label. Be careful not to damage the support protection.
- (c) With a clean, lint-free cloth soaked in isopropyl alcohol (OI-cl-011), clean the area where the labels is to be attached to eliminate all traces of grease or dust and to remove the remaining traces of adhesive from the position of the old label.
- (d) Put the new label in the correct position.
- (e) Make sure that it is in position before you apply pressure.
- (f) Press lightly to bond the label.

F. Close-up

- (1) Not applicable.

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TASK 34-22-45-300-803-A01

3. Replacement of the Connectors on the inside of the Programmed Motherboard TB1

A. General

- (1) This procedure is for the replacement of the plug-in connectors.

B. Reason for the Job

- (1) Self explanatory.

C. Job Set-up Information

- (1) Tools, Fixtures and Equipment

(Refer to TASK-34-22-45-940-802-A01)

NOTE: Equivalent substitutes can be used for the items that follow.

- ZIF Package (ST08).
- Press.

- (2) Consumables

(Refer to TASK-34-22-45-940-803-A01)

NOTE: Equivalent substitutes may be used for the items that follow.

- None.

- (3) Referenced Information:

- (Refer to TASK-34-22-45-940-802-A01), Tools, Fixtures and Equipment
- (Refer to TASK-34-22-45-100-801-A01), Cleaning of the LRU.

D. Job Set-up

- (1) Not applicable.

E. Procedure

SUBTASK 34-22-45-300-003-A01

- (1) Replacement of connectors on the inside of the programmed motherboard TB1

- (a) Install the anvil No. 2 (ST08) on the support marked MOB (ST08) opposite the connector to be removed.
- (b) Install the programmed motherboard TB1 on the support marked MOB (ST08).
- (c) Put the pusher No. 6 (ST08) in position on the connector to be removed.
- (d) With the press, push out the connector to be removed.
- (e) Remove the pusher No. 6 (ST08).
- (f) Remove the programmed motherboard from the support marked MOB (ST08).
- (g) Remove the anvil No. 2 (ST08).
- (h) Install the anvil No. 3 (ST08) on the support marked MOB (ST08).
- (i) Install the programmed motherboard TB1 on the support marked MOB (ST08).
- (j) Put the pusher No. 6 (ST08) in position on the connector to be removed.
- (k) With the press, push out the connector to be removed.
- (l) Install the anvil No. 1 (ST08) on the support marked MOB (ST08) opposite the connector to be removed.

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- (m) Install the programmed motherboard TB1 on the support marked MOB (ST08).
 - (n) Put the related connector in position.
 - (o) Put the pusher No. 4 (ST08) in position on the connector.
 - (p) With the press, install the connector.
- F. Close-up
- (1) Not applicable.

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TASK 34-22-45-300-804-A01

4. Inspection of the Connectors P2 and P3 on the Programmed Motherboard TB1

A. General

- (1) This procedure is for the inspection of the connectors P2 and P3 on the programmed motherboard TB1 and for the soldering of the pins of these connectors made by the manufacturer FCI (Refer to SB C16221-34-004).

B. Reason for the Job

- (1) Self explanatory.

C. Job Set-up Information

- (1) Tools, Fixtures and Equipment

(Refer to TASK-34-22-45-940-802-A01)

NOTE: Equivalent substitutes can be used for the items that follow:

- Standard electronics technician's toolkit
- Extraction tool A2 (ST05)
- Extraction tool A3 (ST06).

- (2) Consumables

(Refer to TASK-34-22-45-940-803-A01)

NOTE: Equivalent substitutes can be used for the items that follow:

- Adhesive tape (Mi-ta-003)
- Isopropyl alcohol (OI-cl-011).
- Solder (So-so-010).

- (3) Referenced Information:

- (Refer to TASK-34-22-45-000-802-A01), Remove the covers
- (Refer to TASK-34-22-45-000-806-A01), Remove the control processor graphic display unit A2
- (Refer to TASK-34-22-45-000-807-A01), Remove the programming interface board A3
- (Refer to TASK-34-22-45-400-808-A01), Install the programming interface board A3
- (Refer to TASK-34-22-45-000-809-A01), Install the control processor graphic display unit A2
- (Refer to TASK-34-22-45-000-813-A01), Install the covers.

D. Job Set-up

- (1) Not applicable.

E. Procedure

SUBTASK 34-22-45-300-004-A01

- (1) Inspection of connectors P2 and P3 on the programmed motherboard TB1.

CAUTION: DO NOT DAMAGE THE GUIDE AND THE WEDGE.

PINS ARE CONNECTED TO THE GROUND AND PADS ARE VERY FRAGILE.
BE CAREFUL WITH THE SOLDER.

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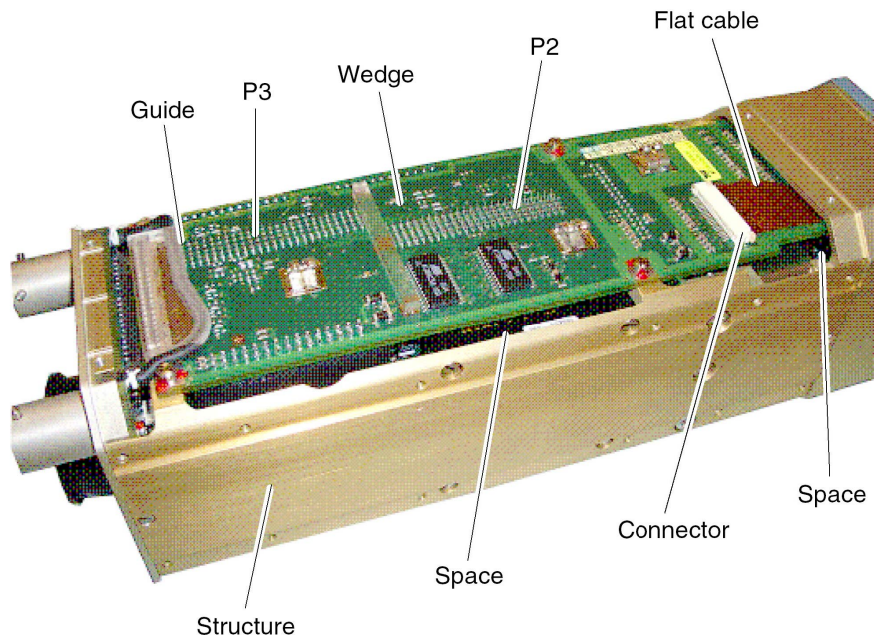
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- (a) Remove the upper cover (Refer to TASK-34-22-45-000-802-A01).
- (b) Remove the control processor graphic display unit A2 (Refer to TASK-34-22-45-000-806-A01).
- (c) Remove the programming interface board A3 (Refer to TASK-34-22-45-000-807-A01).
- (d) If the connector is not white and made by AMP manufacturer (written on the connector), this connector is the good one, therefore do the following operations:
 - 1 Install the programming interface board A3 (Refer to TASK-34-22-45-400-808-A01)
 - 2 Install the control processor graphic display unit A2 (Refer to TASK-34-22-45-000-809-A01).
 - 3 Install the upper cover (Refer to TASK-34-22-45-000-813-A01).
 - 4 Install the inspection label (Refer to Figure 6002).
- (e) If the connector is white and made by FCI manufacturer (written on the connector), do the following operations:
 - 1 Remove the lower cover (Refer to TASK-34-22-45-000-802-A01).
 - 2 Install the adhesive tape (Mi-ta-003) on the guide, the shim, the flat cable and its connector and all the spaces between the programmed motherboard and the structure of the equipment (Refer to Figure 6001).
 - 3 Make sure that the pins of the connectors P2 and P3 on the programmed motherboard TB1 are not varnished. If the pins of the connectors P2 and P3 are varnished, remove the varnish.
 - 4 Make solders (So-so-010) on all the pins of the connectors P2 and P3.
 - 5 Clean the solder joints with isopropyl alcohol (OI-cl-011).
 - 6 Remove the protections.
 - 7 Install the programming interface board A3 (Refer to TASK-34-22-45-400-808-A01).
 - 8 Install the control processor graphic display unit A2 (Refer to TASK-34-22-45-000-809-A01).
 - 9 Install the covers (Refer to TASK-34-22-45-000-813-A01).
 - 10 Install the inspection label (Refer to Figure 6002).

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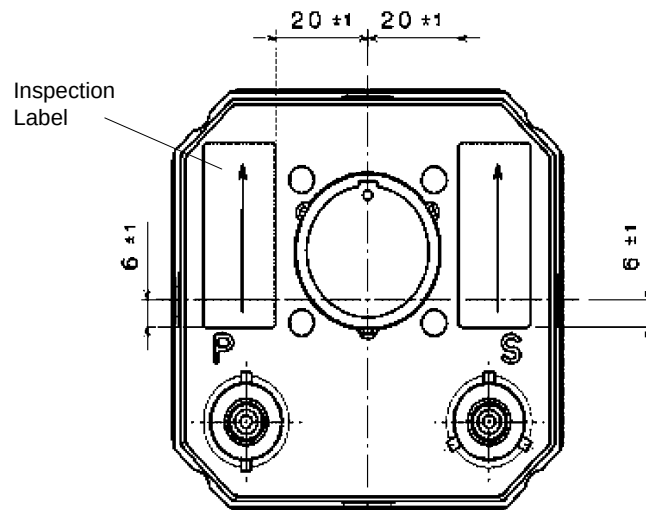
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D_CM_TAV_342245_06_AAA0_01_00

Figure 6001/Graphic 34-22-45-991-601-A01
ISFD (Underside View)



D_CM_TAV_342245_06_AAB0_01_00

Figure 6002/Graphic 34-22-45-991-602-A01
Inspection Label Position

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F. Close-up

(1) Not applicable.

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TASK 34-22-45- 300-805-A01

5. Replacement of a Component of the Boards

A. Reason for the Job

- (1) Self explanatory.

B. Job Set-Up Information

NOTE: All operations not described in this document must be done by the manufacturer, on his premises and in accordance with the approved industrial file.

- (1) Tools, fixtures and Equipment

- (a) None

- (2) Consumable materials

(Refer to TASK-34-22-45-940-803-A01)

NOTE: Equivalent substitutes can be used for the items that follow:

- Solvent SND01L (OI-so-014)
- Araldite 2011 (Ad-ep-001).

- (3) Referenced Information

- Refer to the Manual (REFER TO SOPM 20-02-00.)
- (Refer to PAGEBLOCK DISASSEMBLY), Disassembly
- (Refer to PAGEBLOCK ASSEMBLY), Assembly.

C. Job Set-Up

SUBTASK 34-22-45-350-001-A01

- (1) Removal of the Defective Boards

- (a) (Refer to PAGEBLOCK DISASSEMBLY)

D. Procedure

WARNING: USE THE APPLICABLE ELECTROSTATIC-DISCHARGE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR DEVICES SENSITIVE TO ELECTROSTATIC DISCHARGE. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

SUBTASK 34-22-45-350-002-A01

- (1) Boards

- (a) Refer to the Standard Overhaul Practices Manual SOPM 20-02-00.

SUBTASK 34-22-45-350-003-A01

- (2) Replacement of the switch S1, S2, S3, S4 or S5

- (a) Remove the switch (7-45) (Refer to the Standard Practices Manual SOPM 20-02-00).
 - (b) Clean the printed circuit pads of the switch with solvent SND01L (OI-so-014).
 - (c) Solder the new switch (7-45) to the front face PCB (Refer to the Standard Overhaul Practices Manual SOPM 20-02-00).
 - (d) Clean the solder joints of the switch with solvent SND01L (OI-so-014).

NOTE: The cleaning agent must not migrate inside the switch.

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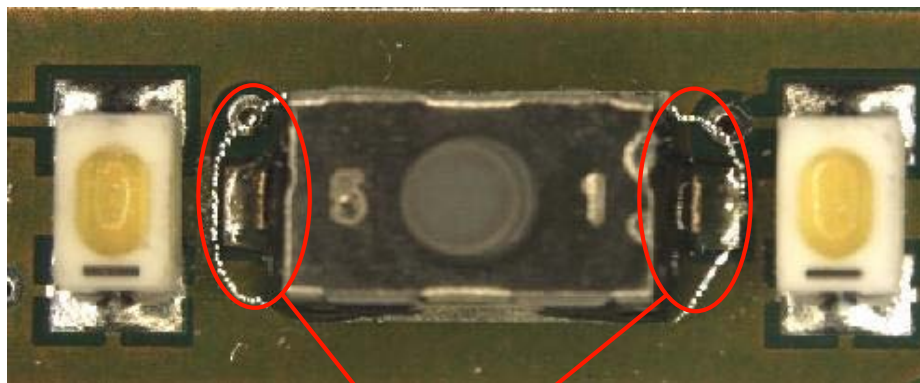
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- (e) Apply Araldite 2011 (Ad-ep-001) with an appropriate applicator on the solder of the switch of the front face PCB (Refer to Figure 6003).

NOTE: The Araldite 2011 (Ad-ep-001) must cover the solder joints.

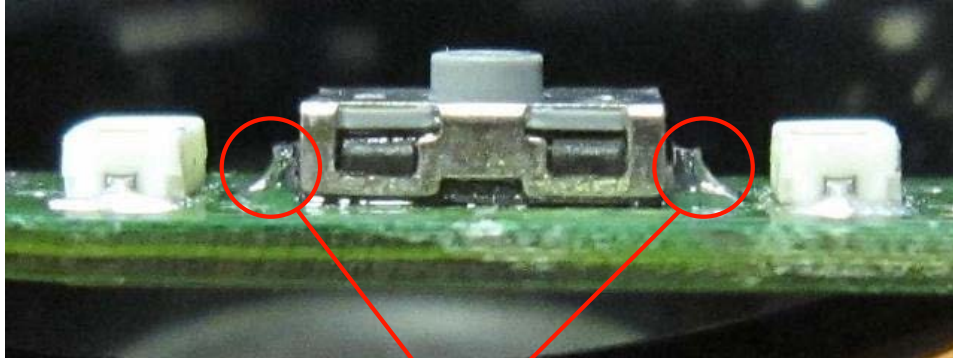
- (f) Let the front face PCB dry for 30 min at the ambient air temperature and for 45 min at 60°C (140 °F).



Bond area

D_CM_TAV_342245_06_AAC0_01_00 _]

Figure 6003/Graphic - Sheet 01 / 34-22-45-991-603-A01
Bonding of the Pushbutton Set



Bond area

D_CM_TAV_342245_06_AAD0_01_00 _]

Figure 6003/Graphic - Sheet 02 / 34-22-45-991-603-A01
Bonding of the Pushbutton Set

- E. Close-up
SUBTASK 34-22-45-350-004-A01
 - (1) Installation of the Repaired Board
 - (a) [\(Refer to PAGEBLOCK ASSEMBLY\)](#).

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ASSEMBLY

TASK 34-22-45-400-801-A01

1. General

A. Reason for the Job

CAUTION: USE THE APPLICABLE ELECTROSTATIC-DISCHARGE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR DEVICES SENSITIVE TO ELECTROSTATIC DISCHARGE. IF YOU DO NOT DO THIS, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

- (1) (Refer to [FIGURE-34-22-45-950-020-A01](#)), (Refer to [FIGURE-34-22-45-950-021-A01](#)), (Refer to [FIGURE-34-22-45-950-022-A01](#)), (Refer to [FIGURE-34-22-45-950-023-A01](#)), (Refer to [FIGURE-34-22-45-950-030-A01](#)), (Refer to [FIGURE-34-22-45-950-031-A01](#)), (Refer to [FIGURE-34-22-45-950-040-A01](#)), (Refer to [FIGURE-34-22-45-950-050-A01](#)), (Refer to [FIGURE-34-22-45-950-990-A01](#)) to identify the parts.

NOTE: The first number in brackets is the figure number and the second is the number of the item on the figure.

B. Job Setup Information

(1) Standard Tools, Fixtures and Equipment:

- Standard electronics technician's toolkit
- Torque screwdriver (Torque: 0 to 1.6 N.m)
- Torque wrench (Torque: 0 to 15 N.m)
- Laminar flow hood Class 100 STD U209E (iso 5 STD 14644-3)
- Decade resistance box (0 to 10 KOhms)

(2) Special Tools, Fixtures and Equipment:

NOTE: The tools are identified in the text by an alphanumeric code.

Example: ST01, SP01

For more detail, ([Refer to PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)) pageblock.

- Shim (ST02)
- Adjustment fixture (ST07)
- Motherboard Hold Tool (ST12)
- End-fitting (ST13)

(3) Consumables:

NOTE: The products are identified in the text by an alphanumeric code.

Example: Ad-si-003

For more detail, ([Refer to PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)) pageblock.

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)

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- Varnish (Ad-si-003)
- Adhesive (Ad-si-006)
- Mastic (Ad-si-011)
- Grease (Lu-gr-012)
- Nylon thread (Mi-mi-025)
- Adhesive foam (Mi-mi-062)
- Adhesive tape (Mi-ta-003)
- Wire (Mi-wi-027)
- Cleaner product (Ol-cl-016)
- Red paint (Pa-pa-024)
- Gloves (Mi-mi-084)

(4) Referenced Information:

- (Refer to TASK-34-22-45-400-801-A01), Safety Precautions
- (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01), (Refer to FIGURE-34-22-45-950-030-A01), (Refer to FIGURE-34-22-45-950-031-A01), (Refer to FIGURE-34-22-45-950-040-A01), (Refer to FIGURE-34-22-45-950-050-A01), (Refer to FIGURE-34-22-45-950-990-A01) to identify the parts.

SUBTASK 34-22-45-400-009-A01

(1) Job Set-up

- (a) Assemble the equipment from the step at which it was disassembled in DISASSEMBLY.
- (b) Read the history or record card, if there is one, to find possible defective areas which you must look at.

C. Safety Precautions

(1) During this maintenance procedure, use the precautions that follow:

- Use the standard precautions for electrostatic discharges sensitive devices (ESDS).
- Disconnect the equipment from all power sources.
- Do the assembly operations in a clean room.
- Use gloves to touch the optical part.

D. Procedure

(1) Not applicable.

E. Close-up

(1) Not applicable.

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TASK 34-22-45-400-802-A01

2. Assembly of the Structure Assembly

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Torque screwdriver

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Red paint (Pa-pa-024)

- (3) Referenced Information

- (a) (Refer to [FIGURE-34-22-45-950-020-A01](#)), (Refer to [FIGURE-34-22-45-950-021-A01](#)), (Refer to [FIGURE-34-22-45-950-022-A01](#)), (Refer to [FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) (Refer to [TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-010-A01

- (1) Assembly of the Structure Assembly

- (a) Attach the four board guides (2 or 2A/390, 2B or 2C/410) with the eight screws (2 or 2A/400, 2B or 2C/420) and nuts (2 or 2A/410, 2B or 2C/430).

NOTE: For the board guides of the pressure sensor assemblies A7 and A8, the slides must be adjacent to the sensor shielding assembly (2 or 2A or 2B/260).

- (b) Torque the nuts (2 or 2A/410, 2B or 2C/430) to 0.2 N.m.

- (c) Safety the heads of the screws (2 or 2A/390, 2B or 2C/410) with red paint (Pa-pa-024).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-803-A01

3. Assembly of the Sensor Shielding Assembly

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Torque screwdriver

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Red paint (Pa-pa-024)

- Varnish (Ad-si-003)

- (3) Referenced Information

- (a) ([Refer to Figure 10002](#)), ([Refer to Figure 10003](#)), ([Refer to Figure 10004](#)), ([Refer to Figure 10005](#)), ([Refer to Figure 10017](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-011-A01

- (1) Assembly of the Sensor Shielding Assembly except PNRs F1407830 and F1407831

- (a) Attach the sensor shielding assembly (2 or 2A or 2B or 2C/260) to the structure (2 or 2A/450, 2B or 2C/460) with the two screws (2 or 2A or 2B or 2C/270) and washers (2 or 2A or 2B or 2C/280).

- (b) Torque the screws (2 or 2A or 2B or 2C/270) to 0.2 N.m.

- (c) Safety the screw heads with red paint (Pa-pa-024).

- (d) Apply a drop of varnish (Ad-si-003) to the attachment points of the shielding assembly (2 or 2A or 2B/260) to prevent vibrations.

SUBTASK 34-22-45-400-020-A01

- (2) Assembly of the Sensor Shielding Assembly only for PNRs F1407830 and F1407831

- (a) Assembly the pressure sensor assemblies A7 and A8 ([Refer to SUBTASK-34-22-45-400-019-A01](#)).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-804-A01

4. Assembly of the Pressure Sensor Assemblies A7 and A8

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Torque screwdriver

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Red paint (Pa-pa-024)

- Loctite (Ad-cr-008)

- (3) Referenced Information

- (a) ([Refer to Figure 10002](#)), ([Refer to Figure 10003](#)), ([Refer to Figure 10004](#)), ([Refer to Figure 10005](#)), ([Refer to Figure 10017](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-012-A01

- (1) Assembly of the Pressure Sensor Assemblies A7 and A8 except PNRs F1407830 and F1407831

CAUTION: ([Refer to Figure 10002](#)) AND THE ASSOCIATED PARTS LIST FOR THE PART NUMBERS OF A7 AND A8.

- (a) Attach the eight spacers (2 or 2A/310, 2B or 2C/330) with the eight screws (2 or 2A/320, 2B or 2C/340) safetied with loctite (Ad-cr-008).
- (b) Torque the screws (2 or 2A/320, 2B or 2C/340) to 0.4 N.m.
- (c) Engage the pressure sensor assemblies A7 and A8 in their housings.
- (d) Attach the pressure sensor assemblies A7 and A8 to the structure with the eight screws (2 or 2A or 2B or 2C/300 and 307) safetied with loctite (Ad-cr-008).
- (e) Torque the screws (2 or 2A or 2B or 2C/300 and 307) to 0.4 N.m.

SUBTASK 34-22-45-400-019-A01

- (2) Assembly of the Pressure Sensor Assemblies A7 and A8 only for PNRs F1407830 and F1407831

CAUTION: ([Refer to Figure 10017](#)) AND THE ASSOCIATED PARTS LIST FOR THE PART NUMBERS OF A7 AND A8.

- (a) Install eight pillars (10/30) on the pitot and static pressure sensors assemblies (50) A7 and A8 and attach them with eight screws (10/40) safetied with locking compound (Ad-cr-008).
- (b) Torque the eight screws (10/40) to 0.4 N.m.
- (c) Engage the pitot and static pressure sensors assemblies (50) A7 and A8 in the shielding (10/10) on and attach them with eight screws (10/20) .

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(d) Torque the eight screws (10/20) to 0.08 N.m.

E. Close-up

(1) Not applicable.

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TASK 34-22-45-400-805-A01

5. Assembly of the Programmed Motherboard TB1

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Screwdriver
- Motherboard Hold Tool (ST12)

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Nylon thread (Mi-mi-025)

- (3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-013-A01

- (1) Assembly of the Programmed Motherboard TB1

- (a) Cut three ties (Mi-mi-025) to a length of 70mm.

- (b) Install the three ties (Mi-mi-025) in the related holes on the board TB1 to attach the pressure sensor assemblies A7 and A8.

NOTE: The loops of the ties must be parallel to the length of the board TB1. The knots will be made on the solder side of the board.

- (c) Install the board TB1 and attach it with the screws (2 or 2A/430) and the washers (2 or 2A/440) or the screws (2 or 2A/430A) (only for C16221KA.. and C16221LA..). Do not tighten the screws fully.

- (d) Install the board TB1 and attach it with the screws (2B or 2C/450) (only for C16221MA.. and C16221QA..). Do not tighten the screws fully.

- (e) Install the Motherboard hold tool (ST12) on the programmed motherboard TB1 to attach the pressure sensor assembly A7.

NOTE: The Motherboard hold tool (ST12) is used to make the assembly easier.

- (f) Do the inspection of the connectors P2 and P3 on the programmed motherboard TB1 ([Refer to TASK-34-22-45-300-804-A01](#)).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-806-A01

6. Assembly of the Rear Module A1

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Torque screwdriver

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)
- Grease (Lu-gr-012)
- Nylon thread (Mi-mi-025)
- Red paint (Pa-pa-024)

- (3) Referenced Information

- (a) (Refer to [FIGURE-34-22-45-950-020-A01](#)), (Refer to [FIGURE-34-22-45-950-021-A01](#)), (Refer to [FIGURE-34-22-45-950-022-A01](#)), (Refer to [FIGURE-34-22-45-950-023-A01](#)), (Refer to [FIGURE-34-22-45-950-050-A01](#)) to identify the parts.

C. Job Set-up

- (1) (Refer to [TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-001-A01

- (1) Assembly

- (a) Install the flange (5/120) and the connector J1(5/110) on the backplate (5/60) with seven screws (5/90) and washers (5/100). Do not tighten the screws fully.

NOTE: Make sure that the internal notch of the connector points to the side opposite to the two pressure unions.

- (b) Put the flange (5/120) and the connector J1(5/110) in the center of the backplate (5/60).

- (c) Torque the screws (5/90) to 0.7 N.m.

- (d) Safety the screw heads (5/90) with red paint (Pa-pa-024).

- (e) Install the FIB Board A11(5/130) on the backplate (5/60) with four screws (5/70) and washers (5/80). Do not tighten the screws fully.

NOTE: Make sure that the board is in the correct position on the pins of connector J1 (5/110).

- (f) Torque the screws (5/70) to 0.4 N.m.

- (g) Safety the screw heads (5/70) with red paint (Pa-pa-024).

- (h) Solder the 41 pins of connector J1(5/110).

- (i) (Refer to [PAGEBLOCK CLEANING](#)) for the cleaning of the electrical parts.

SUBTASK 34-22-45-400-002-A01

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(2) Install

- (a) Attach the rear module A1 with the two screws (2 or 2A/340 or 2B/360), safetied with loctite (Ad-cr-008).
- (b) Torque the screws (2 or 2A/340 or 2B or 2C/360) to 0.4 N.m.
- (c) Torque the screws (2 or 2A/430 or 2B or 2C/450) to 0.4 N.m.
- (d) Attach the hoses to the Programmed Motherboard TB1 with the three ties (Mi-mi-025) installed before.
- (e) Connect the hoses to their related pressure unions.
- (f) Use the ties to attach each hose to the related pressure union.
- (g) Attach the two hoses that are above the screw (2 or 2A/730 or 2B or 2C/450) to prevent interference during tightening.
- (h) Do a visual check to make sure that the hoses are not blocked.
- (i) Apply a layer of grease (Lu-gr-012) to the sealing rings (2 or 2A/360 or 2B or 2C/380 and 2 or 2A/380 or 2B or 2C/400).
- (j) Install the quick unions (2 or 2A/350 or 2B or 2C/370 and 2 or 2A/370 or 2B or 2C/390) and the sealing rings (2 or 2A/360 or 2B or 2C/380 and 2 or 2A/380 or 2B or 2C/400).
- (k) Torque the quick unions (2 or 2A/350 or 2B or 2C/370 and 2 or 2A/370 or 2B or 2C/390) to 15 N.m $\pm$ 1 N.m.
- (l) Install the pressure caps (2C/365 or 2C/385) (only for C16221QA..).
- (m) Install the protective caps (99/10 and 99/20) (only for C16221LA../KA../MA..).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-807-A01

7. Assembly of the Power Supply PS1

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

- Torque screwdriver
- Motherboard Hold Tool (ST12)

(2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)

(3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-014-A01

(1) Assembly of the Power Supply PS1

- (a) Install the board PS1 on the Programmed Motherboard TB1.
- (b) Attach the board PS1 with the four screws (2 or 2A or 2B or 2C/250) safetied with loctite (Ad-cr-008).
- (c) Push down the board PS1 to bring the screws to the bottom of the countersunk holes.
- (d) Torque the screws (2 or 2A or 2B or 2C/250) to 0.7 N.m.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-808-A01

8. Assembly of the Programming Interface Board A3

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Motherboard Hold Tool (ST12)

- (2) Consumables

- None

- (3) Referenced Information

- (a) (Refer to FIGURE-34-22-45-950-020-A01), (Refer to FIGURE-34-22-45-950-021-A01), (Refer to FIGURE-34-22-45-950-022-A01), (Refer to FIGURE-34-22-45-950-023-A01) to identify the parts.

C. Job Set-up

- (1) (Refer to TASK-34-22-45-400-801-A01), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-015-A01

- (1) Assembly of the Programming Interface Board A3

CAUTION: DO NOT APPLY TOO MUCH PRESSURE ON THE BOARD CONNECTORS.

- (a) Engage the board A3 in the board TB1.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-809-A01

9. Assembly of the Control Processor Graphic Display Unit A2

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

- Torque screwdriver
- End-fitting (ST13)
- Motherboard Hold Tool (ST12)

(2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)
- Red paint (Pa-pa-024).

(3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-016-A01

(1) Assembly of the Control Processor Graphic Display Unit A2

- (a) Make sure that the fasteners of the unit A2 are in the correct positions.
- (b) Install the unit A2 in the thermal slides.
- (c) Torque the fasteners to 0.2 N.m with the torque screwdriver and the end-fitting (ST13) to attach the unit A2.
- (d) Safety the board guides (1/520) with red paint (Pa-pa-024).

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-810-A01

10. Assembly of the Initial Measurement Unit A9

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Torque screwdriver

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)

- (3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-017-A01

- (1) Assembly of the Initial Measurement Unit A9

- (a) Engage the unit A9 in the board TB1.

- (b) Attach the unit A9 to the structure (2, 2A/450 or 2B/460) with the six screws (2, 2A, 2B/200) and the two screws (2, 2A, 2B or 2C/210) safetied with Loctite (Ad-cr-008).

- (c) Torque the screws (2, 2A, 2B or 2C/210) to 0.4 N.m.

- (d) Torque the screws (2, 2A, 2B or 2C/200) to 1.6 N.m.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-811-A01

11. Assembly of the Optical Module A5

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

- Torque screwdriver
- Shim (ST02)
- Laminar flow hood Class 100 STD U209E (iso 5 STD 14644-3)

(2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Adhesive (Ad-si-006)
- Loctite (Ad-cr-008)
- Wire (Mi-wi-027)
- Adhesive tape (Mi-ta-003)
- Cleaner product (OI-cl-016)

(3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)), ([Refer to FIGURE-34-22-45-950-040-A01](#)) to identify the parts.

C. Job Set-up

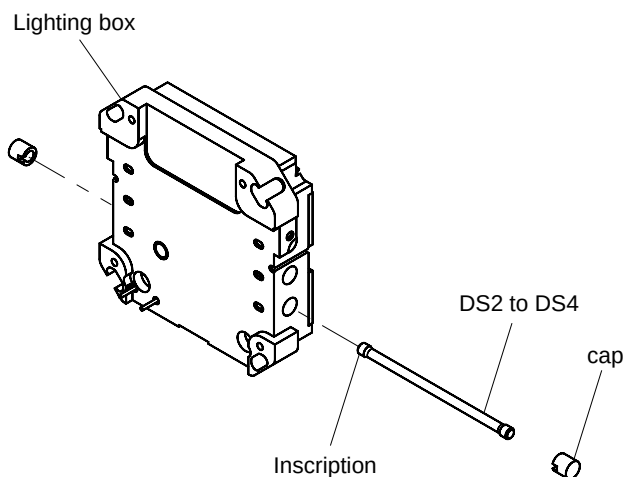
- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-003-A01

(1) Installation of the Tubes DS2 to DS4

- (a) Cut six wires (Mi-wi-027) to a length of 30mm.
- (b) Remove the insulation from the ends of the wires to a length of 3mm and apply tin to them.
- (c) Solder the wires to the bases of the tubes DS2 to DS4. Make sure that the conductors are perpendicular to the axis of the tubes DS2 to DS4.
- (d) Bend the wires to make it easier to install the tubes.
- (e) Do a visual inspection of the solder joints (no dry solder, lack of tin, degassing, etc.).
- (f) Do a visual inspection to make sure that the ends of the tubes are correctly attached and that there are no projections of tin on the bodies of the tubes.
- (g) Install the tubes DS2 to DS4 with the caps (4/130) in their housings and make sure that the inscriptions are in the correct position ([Refer to Figure 7001](#)).
- (h) Put the wires of the tubes DS2 to DS4 through the holes in the lighting box (4/120).
- (i) Lock the tubes DS2 to DS4 in position with the caps (4/130).



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Figure 7001/Graphic 34-22-45-991-701-A01
Installation of the Tubes DS2 to DS4

SUBTASK 34-22-45-400-004-A01

(2) Installation of the Display Assembly

NOTE: Do the operations that follow under a laminar flow hood and with gloves.

- (a) Remove the protective film from the inner side of the LCD matrix.
- (b) Remove the two protective films from the polarizers (3/70).
- (c) Install the polarizer (4/70) in the LCD matrix DS1 with the rough side adjacent to the LCD matrix DS1.
- (d) Examine the polarizer (4/70) and the LCD matrix DS1 to make sure that they are correctly assembled as follows ([Refer to Figure 7002](#)):
 - Light goes through: the assembly is correct.
 - Light does not go through: the polarizer (3/70) is not correctly installed.
- (e) Install the diffuser (4/80) in the LCD matrix DS1 with the matt side against the LCD matrix and the glossy side on the outside.
- (f) Make sure that the LCD matrix DS1, the polarizer (4/70) and the diffuser (4/80) are correctly assembled as follows:
 - Glossy white color: the assembly is correct.
 - Matt gray color: the diffuser (4/70) is not correctly installed.
- (g) Install the anti-sticking diffuser (4/90) on the first diffuser (4/80) with the matt side against the diffuser (4/80) and the glossy side on the outside.
- (h) Install two strips of adhesive tape (Mi-ta-003) under the flexes of the matrix to isolate the heater.

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- (i) Install the heater (4/100) and the lighting box (4/180) in the cell assembly. Make sure that the two wires of the heater (4/100) go in the groove of the lighting box.
- (j) Bond the two wires of the heater (4/100) in the groove with adhesive (Ad-si-006).
- (k) Apply a layer of adhesive (Ad-cr-008) on the threads of the screws (4/60).
- (l) Install the screws (4/60) and torque them to 0.4 N.m.
- (m) Install the four special nuts (4/150) in their housings on the lighting box (4/120).
- (n) Attach the board A51 to the lighting box (4/120) with the four screws (4/170) and washers (4/180).
- (o) Torque the screws (4/170) to 0.4 N.m.
- (p) Safety the screw heads (4/170) with red paint (Pa-pa-024).
- (q) Solder the two wires of the passive sensor RT1 to the pads on the board A51 (Refer to [Figure 7003](#)).

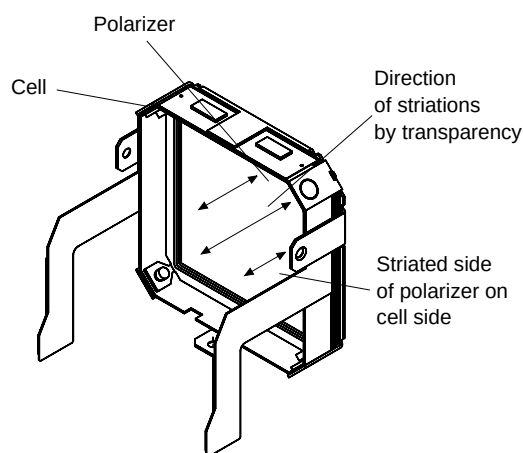
NOTE: Make sure that the wires are not too tight.

- (r) Solder the six wires of the tubes DS2 to DS4 to the pads on the board A51 (Refer to [Figure 7003](#)).

NOTE: Make sure that the wires are not too tight.

- (s) Clean the solder joints with the cleaner product (OI-cl-016).
- (t) Remove the insulation from the two wires of the heater to a length of 2mm.
- (u) Engage the two contacts in the connector.
- (v) Connect the connector P4 to the board A61.

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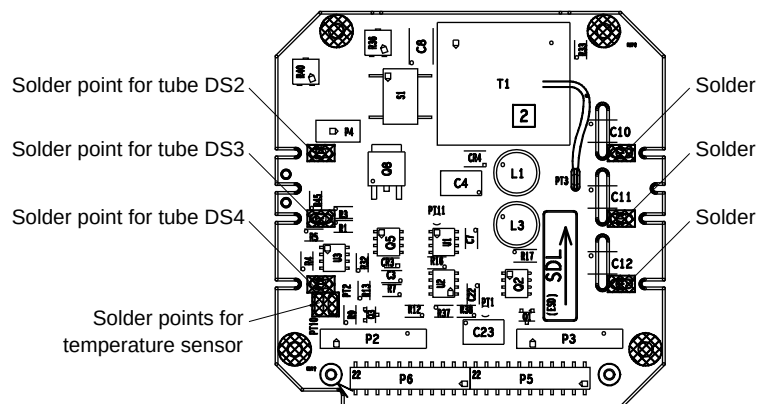
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Figure 7002/Graphic 34-22-45-991-702-A01
Position of Polarizer and LCD Matrix DS1

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Figure 7003/Graphic 34-22-45-991-703-A01
Wire Pads on Board A51

SUBTASK 34-22-45-400-005-A01

(3) Final Assembly

- (a) Apply a strip of adhesive tape (Mi-ta-003) to the edge of the frame of the LCD matrix.
- (b) Install the assembly made by the lighting box (4/120) and the board A51 in the LCD matrix DS1.
- (c) Connect the flex cables of the the cell to the connectors of the board A51.
- (d) Safety the screw heads (4/60) with red paint (Pa-pa-024).

SUBTASK 34-22-45-400-006-A01

(4) Installation of the Optical Module A5

- (a) Put a piece of cardboard (of the same thickness and length as the flex cable) on the front face A61 to make it possible to hold the flexible cable subsequently.
- (b) Bend the flex cables of the matrix to make them straighter.
- (c) Put the cradle in the vertical position, with the front of the instrument at the top.
- (d) Push on the four corners of the matrix frame at the same time to engage the optical assembly A5.
- (e) Apply a drop of bonding compound (Ad-cr-008) to the threads of the four screws (2, 2A, 2B or 2C/180).
- (f) Install the four screws (2, 2A, 2B or 2C/180) in the floating nuts (4/150) but do not tighten them fully.
- (g) Turn the instrument to put the cradle at the top.

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- (h) Put a shim (ST02) under the matrix and push down on the instrument to make sure that the optical assembly A5 is installed at the correct depth.
 - (i) Torque the screws (2, 2A, 2B or 2C/180) to 0.4 N.m.
- E. Close-up
- (1) Not applicable.

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TASK 34-22-45-400-812-A01

12. Assembly of the Front Face

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

- Torque screwdriver
- Decade resistance box (0 to 10 KOhms)
- Laminar flow hood Class 100 STD U209E (iso 5 STD 14644-3)

(2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)
- Red paint (Pa-pa-024)
- Gloves (Mi-mi-084)

(3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)), ([Refer to FIGURE-34-22-45-950-030-A01](#)), ([Refer to FIGURE-34-22-45-950-031-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

NOTE: It is mandatory to put on gloves and use a laminar air flow hood when you do the operations that follow.

SUBTASK 34-22-45-400-007-A01

(1) Preparation

- (a) Install the front face (3, 3A/60) on the BARO knob/pushbutton (3, 3A/130). Make sure that the wires go through the window.
- (b) Put a drop of loctite (Ad-cr-008) only on the two first fillets of the three screws (3, 3A/140).
- (c) Attach the BARO knob/pushbutton (3, 3A/130) on the front face (3, 3A/60) with the three screws (3, 3A/140).
- (d) Torque the screws (3, 3A/140) to 0.2 N.m.
- (e) Safety the heads of the screws (3, 3A/140) with red paint (Pa-pa-024).
- (f) Install the keys (3, 3A/80 to 3, 3A/120) in their housings on the front face (2/60).

NOTE: Make sure that the keys are in the correct positions ([Refer to Figure 13](#)).

- (g) Install the front face board A61 on the front face (3, 3A/60). Make sure that the wires of the BARO knob/pushbutton (3, 3A/130) go behind the front face board A61.
- (h) Attach the front face board A61 to the front face (3, 3A/60) with the seven screws (3, 3A/70).
- (i) Torque the screws (3, 3A/70) to 0.08 N.m.

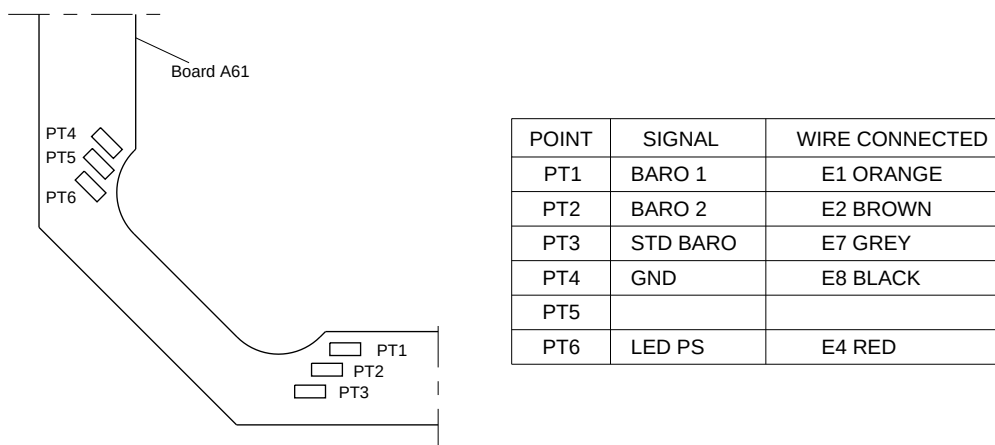
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- (j) Safety the heads of the screws (3, 3A/70) with red paint (Pa-pa-024).
- (k) Push the keys (3, 3A/80 to 3, 3A/120) and their switches to make sure that they operate correctly.
- (l) Cut the wires of the BARO knob/pushbutton (3, 3A/130) to a length of 2 cm (0.79 in).
- (m) Remove the insulation from the ends of the wires and apply tin to them.
- (n) Solder the five wires of the BARO knob/pushbutton (3, 3A/130) to the front face board A61
(Refer to Figure 7004).

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Figure 7004/Graphic 34-22-45-991-704-A01
Connection between BARO Knob/Pushbutton and Front Face Board A61
SUBTASK 34-22-45-400-008-A01

(2) Final Operations

- (a) Put the flex cable under the optical module A5.
- (b) Install the gasket (2 or 2A or 2B or 2C/160) on the board A61 of the front face A6.
- (c) Attach the front face A6 to the frame with the four screws (2 or 2A or 2B or 2C/150) safetied with bonding compound (Ad-cr-008).
- (d) Connect the flex cable of the front face to the connector of the motherboard TB1.
- (e) Torque the screws (2 or 2A or 2B or 2C/150) to 0.2 N.m.
- (f) Apply a bead of mastic (Ad-si-011) between the inertial measurement unit A9 and the front face A6 to prevent leakage of light.
- (g) Remove the unwanted mastic with a cloth (mi-mi-078) soaked with isopropyl alcohol (OI-cl-011) and let the mastic cure at ambient room temperature.

E. Close-up

- (1) Not applicable.

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TASK 34-22-45-400-813-A01

13. Installation of the Covers

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

- (1) Tools, Fixtures and Equipment

- Torque screwdriver
- Adjustment fixture (ST07)

- (2) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- Loctite (Ad-cr-008)
- Mastic (Ad-si-011)
- Adhesive foam (Mi-mi-062)
- Cloth (mi-mi-078)

- (3) Referenced Information

- (a) ([Refer to FIGURE-34-22-45-950-020-A01](#)), ([Refer to FIGURE-34-22-45-950-021-A01](#)), ([Refer to FIGURE-34-22-45-950-022-A01](#)), ([Refer to FIGURE-34-22-45-950-023-A01](#)) to identify the parts.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-400-018-A01

- (1) Installation of the Covers

CAUTION: USE THE SHORT SCREWS TO ATTACH THE BOTTOM COVER AT THE LEVEL OF THE FLEX CABLES ([Refer to Figure 7006](#)).

CAUTION: THE HOSE THAT DOES NOT HAVE TIES MUST GO ON THE INNER SIDE OF THE INSTRUMENT AND NOT ON THE PROGRAMMING MOTHERBOARD TB1.

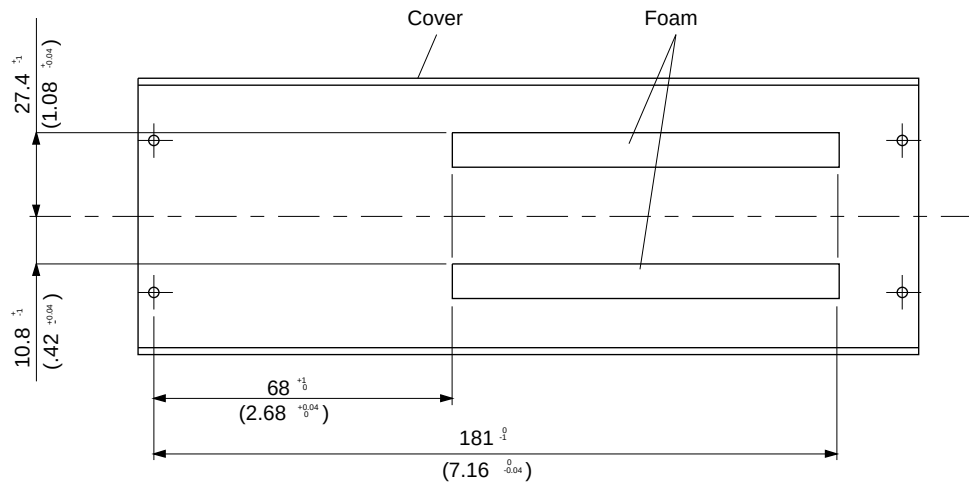
- (a) Remove the special hold tool (ST12).
- (b) Cut two adhesive foams (Mi-mi-062) to a length of 110 mm (4.33 in) and then remove the protection paper.
- (c) Bond the two foams to the upper cover ([Refer to Figure 7005](#)).

NOTE: The installation of the adhesive foams can be made easier through the use of the adjustment fixture (ST07).

- (d) Attach the covers (2 or 2A/110 or 2B or 2C/110) with the screws (2 or 2A or 2B or 2C/120) (in the correct positions) and safety them with bonding compound (Ad-cr-008).

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NOTE

ALL DIMENSIONS ARE GIVEN IN MILLIMETERS WITH
THEIR EQUIVALENTS IN INCHES IN BRACKETS

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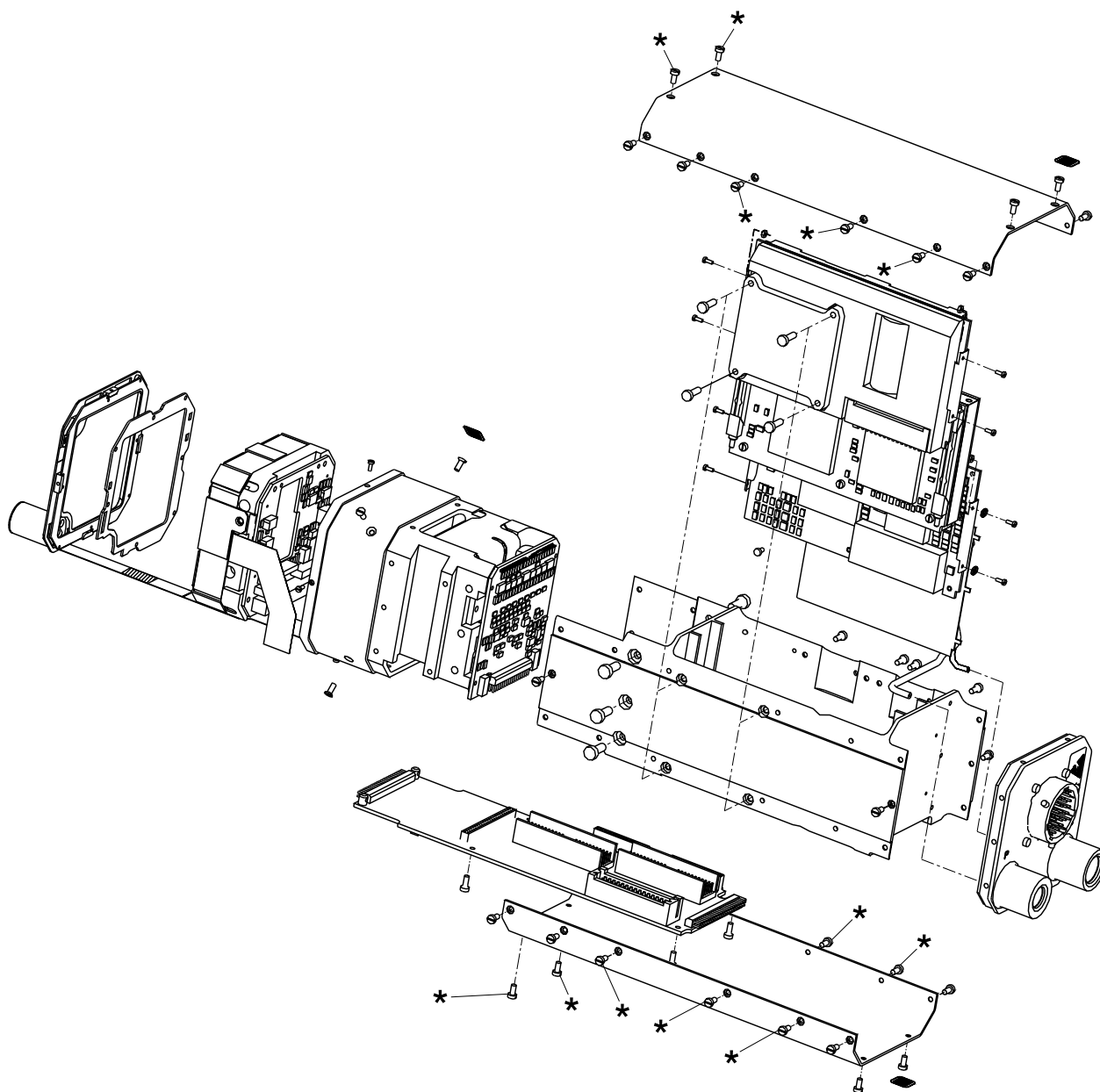
Figure 7005/Graphic 34-22-45-991-705-A01
Positions of Foams

- (e) Torque the screws (2 or 2A or 2B or 2C/130) to 0.4 N.m.
- (f) Torque the screws (2 or 2A or 2B or 2C/120) to 0.4 N.m
- (g) Do the downloading procedure ([Refer to PAGEBLOCK SPECIAL PROCEDURES](#)).

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Key : * Short screws

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Figure 7006/Graphic 34-22-45-991-706-A01
Positions of Short Screws

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- E. Close-up
 - (1) Not applicable.

FITS AND CLEARANCES

TASK 34-22-45-290-801-A01

1. Fits and Clearances
 - A. Introduction
 - (1) Not applicable

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TASK 34-22-45-998-001-A01

2. Torque Values

A. Torque Value Table

Table 8001 : Torque Value

IPL Figure-Item No.	Designation	Non-Standard Torque Value	
		N.m	lbf.in
2-410 2A-410 2B-430 2C-430	nuts	0.2	1.77
2-270 2A-270 2B-270 2C-270	screws	0.2	1.77
2-320 2A-320 2B-340 2C-340	screws	0.4	3.540
2-300 ; 2-307 2A-300 ; 2A-307 2B-300 ; 2B-307 2C-300 ; 2C-307	screws	0.4	3.540
5-90	screws	0.7	6.196
5-70	screws	0.4	3.540
2-340 2A-340 2B-360 2C-360	screws	0.4	3.540
2-430 2A-430 2B-450 2C-450	screws	0.4	3.540

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IPL Figure-Item No.	Designation	Non-Standard Torque Value	
		N.m	lbf.in
2-250 2A-250 2B-250 2C-250	screws	0.7	6.196
Control Processor Graphic Display Unit A2	fasteners	0.2	1.77
2-210 2A-210 2B-210 2C-210	screws	0.4	3.540
2-200 2A-200 2B-200 2C-200	screws	1.6	14.16
4-60	screws	0.4	3.540
4-170	screws	0.4	3.540
2-180 2A-180 2B-180 2C-180	screws	0.4	3.540
3-140 3A-140	screws	0.2	1.77
3-70 3A-70	screws	0.08	0.708
2-150 2A-150 2B-150 2C-150	screws	0.2	1.77
2-130 2A-130 2B-130 2C-130	screws	0.4	3.540

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IPL Figure-Item No.	Designation	Non-Standard Torque Value	
		N.m	lbf.in
2-120 2A-120 2B-120 2C-120	screws	0.4	3.540
2-350 2A-350 2B-370 2C-370	Quick unions	15 ± 1	132.75 ± 8.85
2-370 2A-370 2B-390 2C-390	Quick unions	15 ± 1	132.75 ± 8.85
10-20 10-40	screws	0.08 0.4	0.708 3.540

SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES

TASK 34-22-45-940-801-A01

1. General

A. Scope

- (1) This section gives the special tools, fixtures, equipment and consumables that are necessary for operations on the equipment (with their recommended suppliers).

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TASK 34-22-45-940-802-A01

2. Tools, Fixtures and Equipment.

A. Special Tools, Fixtures and Equipment

NOTE: It is possible to use equivalent alternatives for the items in the list.

Table 9001 : List of Special Tools

CMM CODE	PART NUMBER	DESIGNATION	USE							
			1	3	4	5	6	7	8	
ST02	F1276531	Shim						X		
ST04	F1276354	Extraction tool A5		X						
ST05	F1276198	Extraction tool A2		X			X			
ST06	F1273851	Extraction tool A3		X			X			
ST07	F1400352	Adjustment fixture						X		
ST08	F1275882	ZIF Package .Anvil .Female ZIF connector insertion tool .Male ZIF connector insertion tool .ZIF board support					X			
ST10	G3147AAA01	Software for Dataloading							X	
ST12	F1400351	Motherboard hold tool						X		
ST13	F1400055	End-fitting						X		
ST17	F1403026	Loading operational software media containing the loading operational software G3116AAG03, G3156AAB01, G5614AAK02 and G5615AAK02 for instrument C16221KA02							X	

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CMM CODE	PART NUMBER	DESIGNATION	USE							
			1	3	4	5	6	7	8	
ST18	F1403029	Loading operational software media containing the loading operational software G3116AAG03, G3156AAB01, G5614AAK02 and G5615AAK02 for instrument C16221LA03							X	
ST19	F1403675	Loading operational software media containing the loading operational software G3116AAG03, G3156AAB01, G5614AAK02 and G5615AAK02 for instrument C16221MA01							X	
ST20	F1404705	Loading operational software media containing the loading operational software G3116AAK01, G3156AAB01, G5614AAL02 and G5615AAL01 for instrument C16221QA01							X	
ST21	F1407891	Loading operational software media containing the loading operational software G3116AAG01, G3156AAB01, G5614AAK01 and G5615AAK01 for instrument C16221KA01							X	
ST22	F1407892	Loading operational software media containing the loading operational software G3116AAG01, G3156AAB01, G5614AAK01 and G5615AAK01 for instrument C16221LA01							X	

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CMM CODE	PART NUMBER	DESIGNATION	USE							
			1	3	4	5	6	7	8	
ST23	F1407893	Loading operational software media containing the loading operational software G3116AAG02 G3156AAB01, G5614AAK01 and G5615AAK01 for instrument C16221LA02							X	
ST24	F1408609	Loading software media containing the CPU software K1298AAA01, the GSM 2 software PNR K 1299AAA01 and the PC software PNR K1300AAA01	X							
ST25	E38100AA02	CPGDM tool containing the boot software G5616AAD01 for motherboard E16486JA01, the boot software G5616AAB01 for motherboards E16486BA01 and E16486BB01							X	
ST26	D16845AB01	Optical Assembly Test bench	X							
ST27	F1530837	Arinc link							X	
ST28	D45068AA	Hardware interface							X	
ST29	F1532013	Dataloading software medium F1532013 which contains the software K7902AAA03.							X	
ST30	F1279900	Stand							X	

NOTE: The meaning of the numbers in the USE column are as follows:
 1: TESTING 3: DISASSEMBLY 4: CLEANING 5: CHECK 6: REPAIR
 7: ASSEMBLY 8: SPECIAL PROCEDURES

B. Standard Tools, Fixtures and Equipment

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Table 9002 : Standard Tools

DESIGNATION	CHARACTERISTICS	USE						
		1	3	4	5	6	7	8
Standard electronics technician's toolkit			X				X	
Torque screwdriver	Torque: 0 to 1.6 N.m						X	
Torque wrench	Torque: 0 to 15 N.m						X	
Set of Torx screwdrivers	8x50mm		X					
Power supply	28VDC $\pm$ 0.5VDC, 5A minimum	X						X
Power supply	5V/400Hz,10mA AC, +/-0.25 Vrms	X						
Three ARINC 429 generators	One low speed	X						
	Two high speed	X						
Two ARINC 429 receivers	One high speed	X						
	One low speed	X						
One RS422 receiver	115,200 bauds	X						
One RS422 transmitter	115,200 bauds	X						
One Personal Computer	With ARINC and RS422 links	X						
Switches	Quantity 15, GND and OPEN positions	X						
DC voltmeter	50VDC scale, precision 1%, 2000. 000 points	X						
Brightness and Color meter	Minolta type CS-100 with No. 110 lens, or with filtered silicon photocell, acceptance angle of 1 degree, measurement range 0.001 to 14590 fL, precision $\pm$ 2% $\pm$ 1 on last digit	X						
Pulse generator	$\pm$ 240V $\pm$ 10V, 60 μ s $\pm$ 10 μ s	X						
Oscilloscope	100 MHz, precision 1%	X						
Discrete input	GND and OPEN positions	X						
Light source	Schott type KL1500, minimum lighting level 5000 Lux	X						
Cable	Compatible with unit under test	X						
Megohmmeter	Measurement range 1 to 2X10 (power 9) mohms, +/-2%	X						

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DESIGNATION	CHARACTERISTICS	USE							
		1	3	4	5	6	7	8	
Milliohmmeter	Measurement range 5 to 50 mohms, precision 0.05%, 10A	X							
Ammeter	Measurement range 5A, precision 1%	X							
Press	Capacity: 400 kgs					X			
Clinometer	Accuracy 3 minutes of arc	X							
Jeweller's magnifying glass	Commercially available				X				
One hyperterminal software (Emulation of VT type terminal)	Commercially available							X	
One personal computer (PC)	Microsoft Windows NT version 4.0, Service Pack 3 or later and one 115200 bauds RS232 port							X	
Laminar flow hood	Class 100 STD U209E (iso 5 STD 14644-3).						X		
Two-axis turntable	Remote-controlled, capable of 10-degree tilt in roll and pitch (position accuracy 0.1 degrees) with vertical support fixture (0-degree tilt, precision 0.1 degrees in roll and pitch).	X							
	<u>NOTE:</u> Or: – Single-axis turntable with minimum 90-degree rotation around the Z axis at a rate between 10 and 95 degrees/second (manual or automatic) – Support fixture with - 15 and -18 degree tilt angle in pitch, precision 0.1 degrees – Support fixtures with tilt angles of ±20 degrees in roll and pitch (precision 0.1 degrees).								
Time base	(can be included in the Personal Computer). Precision 0.2 seconds	X							

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DESIGNATION	CHARACTERISTICS	USE						
		1	3	4	5	6	7	8
Two calibrated pressure sources	Range 100 hPa to 1600 hPa, precision ± 0.2 hPa <u>NOTE:</u> Or: - Two pressure sources, range 100 hPa to 1600 hPa, precision ± 0.2 hPa - One pressure gauge, range 0 hPa to 1600 hPa, precision ± 0.2 hPa - One barometer, range 100 hPa to 1200 hPa, precision ± 0.2 hPa.	X						
Resistor	225 Ohms, 5%, 1W	X						
Resistor	220 KOhms, 5%, 1/4W	X						
Four resistors	10 KOhms, 5%, 1/4W	X						
Resistor	1 KOhm, 5%, 1/4W	X						
Decade resistance box	0 to 10 KOhms, 5%, 1/4W						X	
Digital multimeter	34410A USB , AGILENT	X						
Power supply	28VDC, 3A	X						
Photometer	CS200 MINOLTA	X						
One personal computer (PC)	Microsoft Windows XP with RS422 I/F and RS232 I/F	X						

NOTE: The meanings of the numbers in the USE column are as follows:
 1: TESTING 3: DISASSEMBLY 4: CLEANING 5: CHECK 6: REPAIR
 7: ASSEMBLY 8: SPECIAL PROCEDURES

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3. Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

Table 9003 : List of Consumables

DESIGNATION	TYPE	SUPPLIERS CODE OR NAME AND ADDRESS	USE							
			1	3	4	5	6	7	10	
Loctite 222	Ad-cr-008	HENKEL TECHNOLOGIES FRANCE DIVISION LOCTITE FSCM F7121						X		
Adhesive Rhodorsil CAF 3	Ad-si-003	RHODIA FSCM FAR98						X		
Adhesive Rhodorsil CAF 730S	Ad-si-006	BLUESTAR SILICONES FRANCE SAS FSCM FAK74						X		
Mastic Rhodorsil CAF 33 black	Ad-si-011							X		
Grease Rhodorsil Paste 7	Lu-gr-012	RHODIA FSCM FAR98						X		
Nylon thread white	Mi-mi-025	Commercialy available						X		
Adhesive foam 17401CLG057	Mi-mi-062	STAR-PACK SA FSCM F7235						X		
BOX (460x260x260) Ref: CAMS358									X	
Antistatic Foam 17401CLG086									X	
Stripper 1080A	Mi-mi-078	CHASE CORPORATION HUMISEAL DIV FSCM 0SR97						X		
Adhesive tape 5413-9-5mmx33m	Mi-ta-003	3M FRANCE FSCM F0347						X		
Silicone insulated Wire SIF 1*1mm black	Mi-wi-027	Commercialy available						X		
Isopropyl alcohol	OI-cl-011	Commercialy available			X		X	X		

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DESIGNATION	TYPE	SUPPLIERS CODE OR NAME AND ADDRESS	USE						
			1	3	4	5	6	7	10
Cleaner product ULS 400 D	OI-cl-016	ELECTROLUBE A DIVISION OF HK WENTWORTH LTD FSCM K2337						X	
Substrate cleaner 2	OI-cl-017	BALZERS MATERIAL FSCM S3187			X				
Acetone	OI-so-013	Commercially available		X					
Adhesive rubber 4058R	Mi-mi-064	NITTO FRANCE FSCM F9082							X
Red paint	Pa-pa-024	ELECTROLUBE A DIVISION OF HK WENTWORTH LTD FSCM K2337						X	
Precision wipe		Commercially available			X				
Silk paint brush		Commercially available			X				
Cotton gloves		Commercially available		X				X	
Tiflex XF160 black ink	Pa-in-004	TIFLEX FSCM F3190					X		
Solder Sn60Pb40 dia. 0.7mm	So-so-01	MULTICORE SOLDERS LTD FSCM K0729					X		
Solvent SND01L	OI-so-014	ABCHIMIE FSCM FASH4					X		
Araldite 2011	Ad-ep-001	HUNTSMAN ADVANCED MATERIALS EUROPE BVBA FSCM B4647					X		
Gloves	Mi-mi-084	MJB CO FSCM 87839						X	

NOTE:

The meaning of the numbers in the USE column are as follows:

1: TESTING 3: DISASSEMBLY 4: CLEANING 5: CHECK 6: REPAIR
7: ASSEMBLY 10: STORAGE

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SPECIAL PROCEDURES

TASK 34-22-45-650-801-A01

1. General

A. Reason for the job

- (1) The procedures in this pageblock give the instructions for the downloading of the operational software in the instrument, loading software programmed motherboard TB1, loading software. The downloading procedure is done through the rear connector of the instrument on the RS422 lines.

B. Job Setup Information

(1) Tools, Fixtures and Equipment

- One personal computer (PC) with Microsoft Windows NT version 4.0, Service Pack 3 or later and one 115200 bauds RS232 port with the conversion board RS232/RS422.
- Three 2-position switches.
- One 2-position double switch.
- One RS232 cable to connect the PC to the instrument.
- One power supply 28VDC ± 0.5 VDC, 5A minimum.
- One software for Dataloading G3147AAA01(ST10).
- One downloading operational software diskette for instrument C16221KA01 F1407891 (ST21) corresponding to softwares G3116AAG01, G3156AAB01, G5614AAK01 and G5615AAK01.
- One downloading operational software diskette for instrument C16221KA02 F1403026 (ST17) corresponding to softwares G3116AAG03, G3156AAB01, G5614AAK02 and G5615AAK02.
- One downloading operational software diskette for instrument C16221LA01 F1407892 (ST22) corresponding to softwares G3116AAG01, G3156AAB01, G5614AAK01 and G5615AAK01.
- One downloading operational software diskette for instrument C16221LA02 F1407893 (ST23) corresponding to softwares G3116AAG02, G3156AAB01, G5614AAK01 and G5615AAK01.
- One downloading operational software diskette for instrument C16221LA03 F1403029 (ST18) corresponding to softwares G3116AAG03, G3156AAB01, G5614AAK02 and G5615AAK02.
- One downloading operational software diskette for instrument C16221MA01 F1403675 (ST19) corresponding to softwares G3116AAG03, G3156AAB01, G5614AAK02 and G5615AAK02.
- One downloading operational software diskette for instrument C16221QA01 F1404705 (ST20) corresponding to softwares G3116AAK01, G3156AAB01, G5614AAL02 and G5615AAL01.
- One CPGDM tool E38100AA02 (ST25) containing the boot software G5616AAD01 for motherboard E16486JA01, the boot software G5616AAB01 for motherboards E16486BA01 and E16486BB01.
- One hyperterminal software (Emulation of VT type terminal).

(2) Consumables:

- None.

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(3) Referenced Information:

- ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions
- ([Refer to Figure 11001](#)), ([Refer to Figure 11002](#)) to identify the test setup and display.

C. Procedure

- (1) Not applicable

D. Close-up

- (1) Not applicable

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TASK 34-22-45-650-802-A01

2. Downloading of the Operating software

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

- One personal computer (PC) with Microsoft Windows NT version 4.0, Service Pack 3 or later and one 115200 bauds RS232 port.
- One RS232 cable to connect the PC to the connector J1of the equipment.
- One power supply 28VDC ± 0.5 VDC, 5A minimum.

(2) Consumables

- None

(3) Referenced Information

- (a) ([Refer to Figure 11001](#)), ([Refer to Figure 11002](#)) to identify the test setup and Shortcut Icon on Windows Desktop.

C. Job Set-up

- (1) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.

D. Procedure

SUBTASK 34-22-45-650-001-A01

(1) Hardware Setup

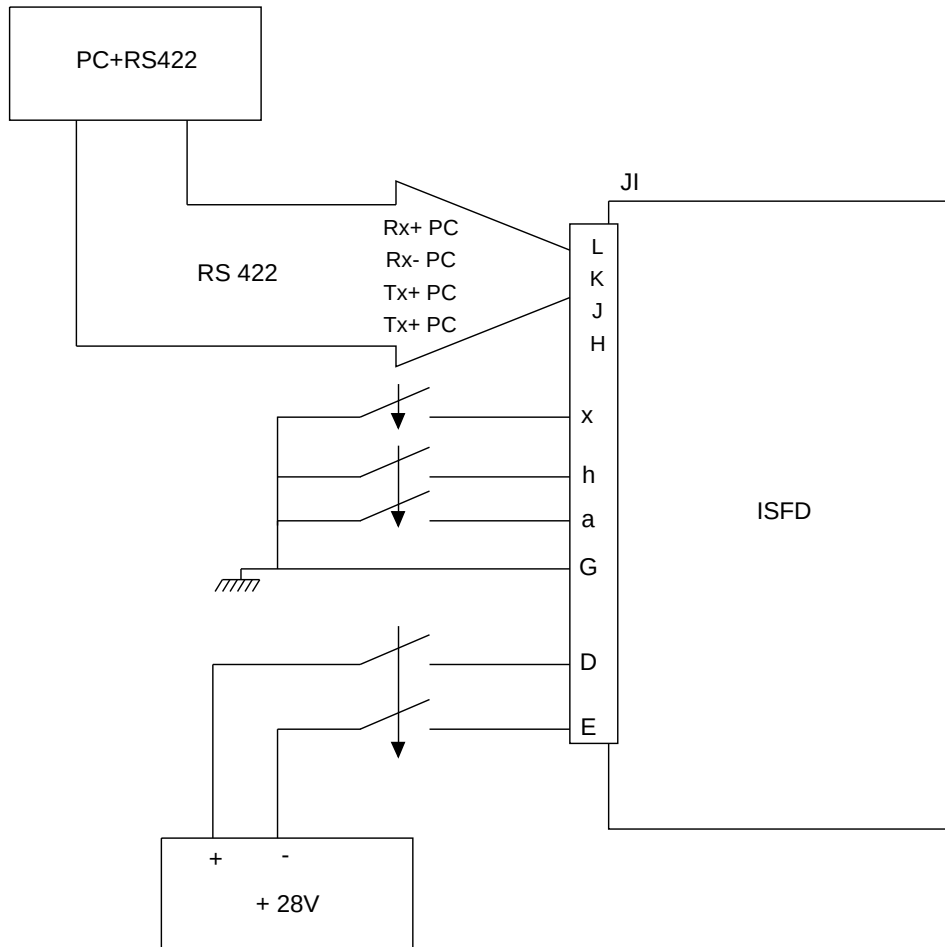
([Refer to Figure 11001](#))

- (a) Set the switches connected to inputs "X", "h" and "a" of the instument to the ground position.
- (b) Connect the RS422 line of the PC to the inputs "L", "K", "J" and "H" of the instument.
- (c) Connect the 28VDC power supply to the inputs "D"(+), "E"(-) of the instument with the double switch in the open position.
- (d) Close the double switch to energize the instrument.

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D_CM_TAV_342245_11_AAA0_01_00

Figure 11001/Graphic 34-22-45-991-851-A01
Hardware Setup

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SUBTASK 34-22-45-650-002-A01

(2) Hardware Setup

(a) Program Setup

NOTE: The download program is supplied on diskettes. It is necessary to have "administrator rights" to install the program on the PC.

- 1 Install the diskette 1 in the drive of the PC.
- 2 Select "run..." in the "Start" menu.
- 3 Enter "'a:/setup" in the dialog box and click on "OK" to start the setup procedure.
- 4 Obey the instructions shown on the screen.

NOTE: At the end of the downloading procedure, all the files of the program are copied on the hard disk and a shortcut icon is displayed on the Windows desktop ([Refer to Figure 11002](#)).

┌



D_CM_TAV_342245_11_AAB0_01_00

Figure 11002/Graphic 34-22-45-991-852-A01
Shortcut Icon on Windows Desktop

(b) Click on this icon to start the downloading procedure.

NOTE: The user can change the parameters that follow:

- The language
- The instrument version
- The serial communication port
- The operating mode.

SUBTASK 34-22-45-650-003-A01

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(3) Executable File and Possible Parameters

- (a) The name of the executable file for the downloading program is TELE_FPI_2ND_RS.EXE.
- (b) The command line used to run the downloading program is as follows, and has five variable parameters (%1, %2, %3, %4 and %5):
 - TELE_FPI_2ND_RS.exe %1 %2 %3 %4 %5
- (c) The variable parameters are as follows:
 - 1 %1: language of the user interface. This parameter is mandatory and the possible values are as follows:
 - FR for French
 - GB for English
 - 2 %2: name of the instrument. This parameter is mandatory and the possible values are as follows:
 - ICS for an integrated electronic standby instrument
 - EEI for an electronic engine indicator
 - EFI for an electronic flight indicator.
 - 3 %3: name of the serial communication port. This parameter is mandatory and the possible values are as follows:
 - COM1 for the serial port COM1
 - COMX for the serial port COMX (X is the number of a valid serial port on the PC)
 - *** for another serial port (***) is the name of the port).
 - 4 %4: operating mode. This parameter is optional and includes two modes and two submodes:
 - Free mode: the user can select each file to be downloaded into the instrument.
 - Locked mode: the files are selected by the program in relation to the part number of the instrument.
 - Locked maintenance submode: the user enters the part number in a dialog box.
 - Locked shop-floor submode: the part number is supplied in the fifth parameter of the command line.
 - The possible values for the parameter %4 are given in Table follows ([Refer to Table 11001](#)).

Table 11001 : Possible Values for 4%

Parameter	Mode
No Parameter	locked-maintenance
MODE_LIBRE	Free
C16XXXXX	locked shop-floor mode C16XXXXX is the instrument Part Number

- 5 %5: serial number of the instrument to be downloaded. This parameter is optional and has the form that follows:
 - XX: This is the serial number of the instrument and is only valid if the parameter %4 is C16XXXXX.

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(d) The procedure to change the program parameters is as follows:

- 1 Move the cursor to the shortcut icon.
- 2 Click on the left button of the mouse and select "Properties" in the scroll menu.
- 3 Select the "Shortcut" tabsheet.
- 4 Enter the new parameter in the "Target" box after the name of the executable file. For example: C:\Program Files\Telechargement\TELE_FPI-2ND-RS.exe GB_ICCS_COM1.
- 5 Click on "OK" to confirm the changes.

SUBTASK 34-22-45-650-004-A01

(4) Download Files

(a) For each new instrument part number (PN), it is necessary to copy the related files on the hard disk.

NOTE: The files are supplied by Sextant on validated diskettes.

(b) The procedure to copy the files is as follows:

- 1 Launch the program using the shortcut "TELE_FPI_2.....".
- 2 Click on the "New PN" button.
- 3 In response to the prompt shown on the screen, copy the diskette No. 1 that contains the file related to the new PN.
- 4 In response to the prompt shown on the screen, copy the diskette No. 2 that contains the file related to the new PN.
- 5 [\(Refer to Table 11002\)](#) for the correspondance between the part numbers of the disks No.1 and No.2 and instrument.

Table 11002 : Cross Reference Disk and Instrument Part Numbers

PART NUMBERS	C16221						
	KA01	KA02	LA01	LA02	LA03	MA01	QA01
Disk Part Numbers	F1407891	F1403026	F1407892	F1407893	F1403029	F1403675	F1404705

- 6 When the copy of all the files is completed, it is possible to download the new instrument.

NOTE: The program must be used in Locked Maintenance Mode.

- 7 Launch the program using the short cut "TELE_FPI_2.....", the "instrument selection page" opens up.
- 8 Select the new Part Number in the scrolling list.
- 9 Type in the instrument Serial Number.
- 10 Click on "OK" to validate, the main window " Downloading Interface for IESI " opens up.
- 11 Click on the "start" button to initiate the downloading process.
- 12 During the downloading do a check of the following message in the box "execution message":
 - DOWNLOADING THE EEPROM CONF MEMORY
 - DOWNLOADING THE EEPROM PNRSN MEMORY

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- DOWNLOADING THE CPU MEMORY
- DOWNLOADING THE GS MEMORY

13 After the downloading of all memory areas the message in the box “execution message “ must be: “ DOWNLOADING SUCCESSFULLY COMPLETED “.

14 Click on the tab “Checksum test “, click on “start “ and do a check of the CHECKSUM in the table “ Read checksum“. The correct CHECKSUM is:

Table 11003 : Cross Checksums and Instrument Part Numbers

PART NUMBERS	C16221						
	KA01	KA02	LA01	LA02	LA03	MA01	QA01
FEPROM CPU	2254 BC16	B867 33B7	2254 BC16	2254 BC16	B867 33B7	B867 33B7	70EE 3675
FEPROM GSM	0002 4865	0002 3270	0002 4865	0002 4865	0002 3270	0002 3270	0003 66EE
PARAMETERS EEPROM	AAA5 DBBD	1B23 A0EF	AAA5 DBBD	B003 E56C	1B23 A0EF	1B23 A0EF	56A9 7E86

E. Close-up

(1) Not applicable.

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TASK 34-22-45-650-803-A01

3. Downloading of the BOOT software

A. Reason for the job

- (1) Self explanatory

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

- One personal computer (PC) with Microsoft Windows NT version 4.0, Service Pack 3 or later and one 115200 bauds RS232 port.
- One RS232 cable to connect the PC to the connector J1of the equipment.
- One hyperterminal software (Emulation of VT type terminal).
- One CPGDM tool E38100AA02 (ST25).
- Three 2-position switches.
- One 2-position double switch.
- One power supply 28VDC ± 0.5 VDC, 5A minimum.

(2) Consumables

- None

(3) Referenced Information

- (a) ([Refer to Figure 11001](#)) to identify the test setup.

C. Job Set-up

SUBTASK 34-22-45-650-010-A01

(1) Job Set-up

- (a) ([Refer to TASK-34-22-45-400-801-A01](#)), Safety Precautions.
- (b) Remove the Control Processor Graphic Unit (A2) ([Refer to TASK-34-22-45-000-806-A01](#)).
- (c) Install the CPGDM tool E38100AA02 (ST25).

NOTE: The minimum configuration necessary for downloading is the structure (1-450), the power supply (1-240), the rear interconnection assy (1-330) and the Motherboard (1-420).

D. Procedure

SUBTASK 34-22-45-650-005-A01

(1) Hardware Setup

([Refer to Figure 11001](#))

- (a) Connect the CPGDM tool (ST25) to connector P2 of the motherboard TB1 ([Refer to Figure 2002](#)).
- (b) Connect the RS422 line of the PC to the inputs "L", "K", "J" and "H" of the instrument.
- (c) Connect the 28VDC power supply to the inputs "D"(+), "E"(-) of the instrument with the double switch in the open position.

SUBTASK 34-22-45-650-006-A01

(2) Software Running

- (a) Run the hyperterminal software.

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- (b) Enter a name and select a shortcut.
- (c) Click on OK to validate.
- (d) Select the serial communication port.
- (e) Click on OK to validate.
- (f) Enter the following data:

- Speed bits: 115200
- Data bits: 8
- Parity: none
- Stop bits: 1
- Flow control: hardware

- (g) Click on OK to validate.
- (h) Energize the instrument.

SUBTASK 34-22-45-650-007-A01

(3) Erasing memory

NOTE: Before the tests, it is necessary to do a full erase of the memory to always have the same initial conditions.

- (a) Enter after ">" the words "mob_erase_all".
- (b) Make sure that the indicated message is "erase EEPROM MOB".

SUBTASK 34-22-45-650-008-A01

(4) Memory test

- (a) Enter after ">" the word "eepst".
- (b) Make sure that the indicated message is "EEPROM OK".

SUBTASK 34-22-45-650-009-A01

(5) Software loading

- (a) Enter after ">" the words "mob_dwld XXX".

NOTE: The value XXX is related to the software part number for the motherboard
(Refer to Table 11004).

Table 11004 : Cross Reference XXX Value and Software Part Numbers for Motherboard

XXX VALUES	SOFTWARE PART NUMBERS FOR MOTHERBOARD
AAB	G5616AAB01
AAD	G5616AAD01

- (b) After the downloading of memory, the message must be "Checksum OK = checksum value".

NOTE: The value XXX is related to the software part number for the motherboard
(Refer to Table 11005).

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Table 11005 : Cross Reference Checksum Values and Software Part Numbers for Motherboard

CHECKSUM VALUES	SOFTWARE PART NUMBERS FOR MOTHERBOARD
0xCC9C1E9E	G5616AAB01
0xCC7CFBF0	G5616AAD01

(c) Make sure that the checksum values agree with the ones in the table.

E. Close-up

(1) Not applicable.

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TASK 34-22-45-650-804-A01

4. Three-hour test

A. Reason for the Job

(1) Three-Hour Test

- (a) This test is done to isolate the fault for the Inertial Measurement Unit and Pressure Sensors.
- (b) This test can be done if:
 - 1 The reason for removal is not clearly confirmed by the initial shop findings.
 - 2 The reason for removal is related to the attitude, altitude and/or airspeed functions.

B. Job Set-up Information

(1) Tools, Fixtures and Equipment

(a)

NOTE: It is possible to use equivalent alternatives for the items in the list.

Table 11006 : Tools, Fixtures and Equipment

PART NUMBER	QTY	NAME
Not specific	1	Personal Computer (PC) with OS: Microsoft Windows XP Service Pack 2 or later CPU: Pentium 2.8 GHz minimum RAM: 512 Mo Serial: one serial port (RS232, capable pf 115200 bauds) with a conversion board RS232/RS422
Not specific	1	Power supply 28VDC +/- 0.5VDC, 5A minimum
Ballard tech UA1431	1	One ARINC A429 USB card plugged into an USB port with drivers installed on computer.(Refer to card manufacturer manual)
ST27	1	Arinc link F1530837
ST28	1	Hardware interface D45068AA
ST29	1	Dataloading software medium F1532013 which contains the software K7902AAA03
ST30	1	Stand F1279900

(2) Referenced Information

- Not applicable

C. Job Set-Up

(1) Not applicable

D. Procedure

SUBTASK 34-22-45-650-011-A01

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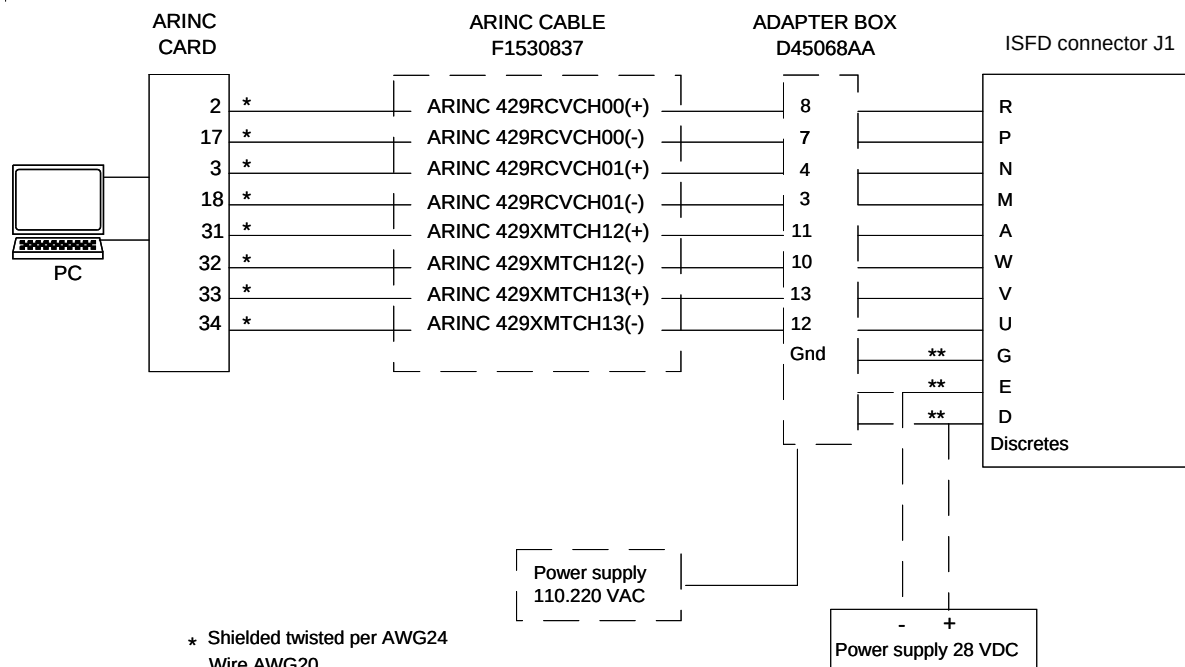
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(1) Hardware Setup

(Refer to Figure 11003)

- (a) Position of the instrument: horizontal on a stable surface at the work station.
- (b) Make sure that the last power-up of the equipment was more than 2 hours before the start of the Three-Hour test.
- (c) Set the instrument to 0 degrees tilt configuration on the stand (ST30).
- (d) Connect the 110 /220Vac power supply to the hardware interface (ST28).



D_CM_TAV_342245_11_AAC0_01_00

Figure 11003/Graphic 34-22-45-991-853-A01
Hardware Setup

SUBTASK 34-22-45-650-012-A01

(2) Software Setup

(a) Software prerequisites

- 1 Connect the USB-Arinc 429 card to a USB port.
- 2 Download the drivers on computer with the instructions given in the card manufacturer manual.

(b) Program set-up

- 1 Install the software medium (ST29) in the drive of the PC.
- 2 Extract the folder "K7902AAA03.zip" into the desired destination directory.
- 3 Go to this destination directory and click on the right button of the mouse on "IESI_diagnosis_Software.exe".
- 4 Select "send to".
- 5 Select "desktop".

NOTE: At the end of the downloading procedure, all the files of the program are

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copied on the hard disk and a shortcut icon is displayed on the Windows "desktop" .

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(c) Software/Hardware Configuration

- 1 Start the program with the shortcut "IESI_diagnosis_Software.exe".
- 2 Select "Tools" in the menu.
- 3 Select "Devices selector".
- 4 Select the Arinc card in the list by S/N.
- 5 Click on "Apply and Restart".

SUBTASK 34-22-45-650-013-A01

(3) Three-Hour Test

- (a) Start the program with the shortcut "IESI_diagnosis_Software.exe".
- (b) Select the name of the user.
- (c) Select the new Part Number in the scrolling list .
- (d) Type in the instrument Serial Number.
- (e) Type an amendment if necessary.
- (f) Click on the "Start" button.
- (g) Make sure that the instructions given in the message [\(Refer to Figure 11004\)](#) are obeyed.
- (h) Click on "OK" to confirm the message.
- (i) Click on "OK" to confirm the message [\(Refer to Figure 11005\)](#).
- (j) Connect the 28VDC power supply to the inputs "D" (+) and "E" (-) [\(Refer to Figure 11003\)](#).
- (k) Make sure that the display luminance is at maximum level.

SUBTASK 34-22-45-650-014-A01

(4) Result

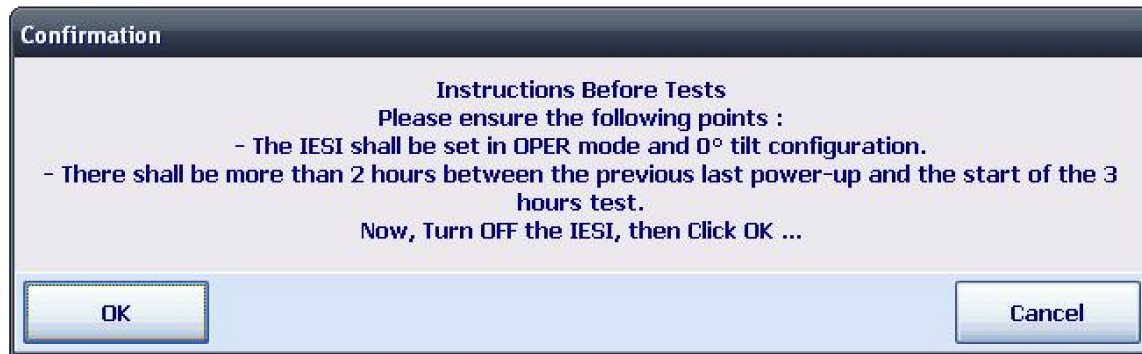
NOTE: The final test result is shown in the "verdict window" on the screen of the Personal Computer.

- (a) Click on the Report Editor to view the test results list.

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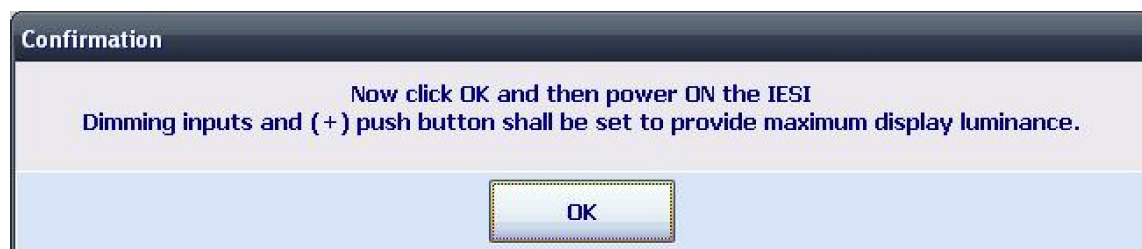
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D_CM_TAV_342245_11_AAD0_01_00j

Figure 11004/Graphic 34-22-45-991-854-A01
Message : Instructions Before Test



D_CM_TAV_342245_11_AAE0_01_00j

Figure 11005/Graphic 34-22-45-991-855-A01
Message : Instructions

E. Close-Up

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(1) Not applicable

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STORAGE (INCLUDING TRANSPORTATION)

TASK 34-22-45-500-801-A01

1. General

A. Introduction

- (1) The procedure in this pageblock contains instructions for storage and transportation of the equipment.
- (2) The special containers prevent damage to the equipment during transport and are used by Thales for the initial delivery of the equipment.

NOTE: To ensure safer transportation and storage of the ISFD, Thales has successfully tested a suitable packaging as per MIL-STD-810E (Environment Test Methods and Engineering Guidelines).

When returning the ISFD for repair or else to a Thales facility, or approved repair center, Thales recommends that you ship the item in a suitable ATA 300 re-usable cardboard box using a suitable inserted foam.

B. Reason for the Job

- (1) Self explanatory.

C. Job Set-up Information

- (1) Standard Tools, Fixtures and Equipment
 - (a) (Refer to [PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)).
 - LRU container.
- (2) Special Tools, Fixtures and Equipment
 - (a) STAR-PACK Box CAMS358
 - (b) STAR-PACK Antistatic foam 17401CLG086

- (3) Consumables

NOTE: It is possible to use equivalent alternatives for the items in the list.

- (a) Adhesive rubber (mi-mi-064).

- (4) Referenced Information

- (Refer to [PAGEBLOCK CLEANING](#)), Cleaning of the equipment.
- (Refer to [PAGEBLOCK CHECK](#)), Visual inspection.
- (Refer to [PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)), Special and standard tools.

D. Job Set-up

SUBTASK 34-22-45- 550-001- A01

- (a) Clean the equipment before the storage ([Refer to TASK-34-22-45-100-801-A01](#)).
- (b) Do a visual check of the equipment before the storage ([Refer to TASK-34-22-45-200-801-A01](#)).

E. Procedure

SUBTASK 34-22-45- 620-001- A01

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(1) Preservation

- (a) Put a protective cap (Refer to IPL Fig 99-10) on the electrical connector.
- (b) Put a protective cap on the pressure unions.

NOTE: For PNR C16221QA01 : Install the pressure caps (Refer to IPL Fig 2C-365 and 2C-385).

NOTE: For PNRs C16221KAxx/LAxx/MAxx : Install the caps (Refer to IPL Fig 99-20).

- (c) Put a adhesive rubber on the front face.

SUBTASK 34-22-45- 530-001- A01

(2) Packing

- (a) Put the equipment in a static shielding bag.
- (b) Fold the end of the bag on the equipment.
- (c) Put all items in the container.
- (d) Put the identification labels on the component and on the outside of the container. Make sure that:
 - The identification label has all the related data of the equipment.
 - You can read the labels easily.
- (e) Close the container.

SUBTASK 34-22-45- 550-002- A01

(3) Storage

CAUTION: MAKE SURE THAT THERE IS NOT TOO MUCH WEIGHT ON THE CONTAINER, IF YOU USE THE STACK STORAGE METHOD. TOO MUCH WEIGHT ON THE CONTAINER CAN CAUSE DAMAGE TO THE COMPONENT.

CAUTION: DO NOT KEEP THE CONTAINER NEAR FLUIDS THAT CAN CAUSE CORROSION OR DAMAGE, AND DONOT KEEP NEAR SOURCES THAT MAKE HEAT OR OZONE.

NOTE: During storage operations, make sure that the environmental conditions are as follows :

- Pressure : No specific environmental level other than the ambient pressure level is necessary.
- Relative humidity: Refer to SOPM 20-02-00, THAV-A-20-02-00-020F0-028A-C - Special instructions in relation to SSD - General procedure.

- (a) Keep the container in a clean, dry room with a good supply of air.
- (b) Keep the temperature of the room and relative humidity level and refer to Standard Overhaul Practices Manual reference SOPM 20-02-00, Storage (Including transportation), §2.A Storage and Transportation of assemblies.
- (c) Put the container where you can easily read the identification label.

SUBTASK 34-22-45- 500-001- A01

(4) Transportation

- (a) Due to cases of damage during transportation of instrument ,the manufacturer ,to ensure safer transportation and storage of the instrument between operators and ourselves, has successfully tested, a specific packaging as per MIL-STD-810E (Environmental Test Methods and Engineering Guidelines.
- (b) You wish to obtain full details of this STAR-PACK product as well as terms of sales, we kindly request that you contact directly the supplier ([Refer to PAGEBLOCK SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES](#)).

SUBTASK 34-22-45- 500-002- A01

(5) Handling

- (a) The instrument contains sensitive hardware such as the rate sensors, LCDs, accelerometers and PSM. It is necessary to prevent shocks to the instrument during handling.

F. Close-up

- (1) Not applicable.

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ILLUSTRATED PARTS LIST

TASK 34-22-45-990-811-A01

1. Introduction

A. General

(1) This ILLUSTRATED PARTS LIST (IPL) is prepared to ATA Specification 2200. It is divided as follows:

- INTRO: Introduction
- OPL: Optional Parts List
- VCL: Vendor Code List
- EDI: Equipment Designator Index
- NI: Numerical Index
- Detailed Parts List.

B. Function and Use

(1) The IPL contains a list of all the components used in the equipment. Its function is to help the procurement of parts and subassemblies. The parts given in the IPL must be procured by the Part Numbers and from the manufacturers given in the Parts List. If not, the warranty can be cancelled.

(2) It is possible to identify parts as follows:

(a) From the Manufacturer's Part Number

- 1 Find the related Part Number in the Numerical Index (alpha and num). The columns to the right give the figure and item number of the part in the Detailed Parts List.

(b) From the Equipment Designator (for electronic components)

- 1 Find the equipment designator of the component on the Electrical Diagram. Then refer to the EDI to find:
- The figure and item number of the component in the Detailed Parts List.
 - In some cases, there is an individual EDI for each PCB. In this case, the EDI has a GEO LOC column which shows the geographical location of the component on the figure (in relation to terminal 1 of the component).

(c) If the Part Number or Equipment Designator of the Part are not known

- 1 Look for the figure which shows the part and its item number. Then refer to the Detailed Parts List to find its Part Number.

C. How to use the Detailed Parts List

(1) The Detailed Parts List contains one or more figures which show the main assemblies of the equipment. Details of the parts are given on the opposite and subsequent pages.

(2) The list is divided into these columns:

- 1st column: Figure Item - Figure and Item number
- 2nd column: Part Number - Manufacturer's Part Number
- 3rd column: Airline Stock Number
- 4th column: Nomenclature
- 5th column: Effectivity Code

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- 6th column: Units per assy - Quantity for the next higher assembly.
- (3) Details of the columns:
- (a) Figure and Item Number
 - 1 The figure number is given on the first line at the top of each page. Each item in the parts list which has a part number also has an item number. If there is a dash in front of an item number, the item is not shown on the figure. A letter before the item number refers to the figure which shows a variant of the related part. A letter after the item number identifies a variant of the part.
 - (b) Part Number
 - 1 Each assembly, subassembly and detailed part has a manufacturer Part Number (even if it is not shown on the figures). Part Numbers beginning with "A" represent manufacturer Part Numbers that are not compliant to ATA requirements (over length). Original manufacturer Part Numbers are given in the NOMENCLATURE column, under "ORDER OVERLGTH MPN" indication, followed by the FSCM (Federal Supply Code for Manufacturer).
 - (c) Airline Stock Number
 - 1 This column is for airline use.
 - (d) Nomenclature
 - 1 The NOMENCLATURE column gives the names of the assemblies and parts. The indication "ATTACHING PARTS" shows the parts which attach an assembly or part. These attaching parts are given directly below the assembly or part they attach. The next line has three asterisks, the first of which is directly below the item which is attached.
 - 2 A Vendor Code is given for all the items which do not have a prime manufacturer's Part Number. This Vendor Code is given at the right of the NOMENCLATURE column. The addresses and codes of Vendors are given in the VENDOR CODE LIST. Optional Part Numbers and Vendors are given in the OPTIONAL PARTS LIST in the IPL. Please note that only the manufacturers who appear in the OPL have been accepted as a result of the equipment qualification test, and should therefore be used.
 - 3 If a part can come from a range of possible parts, the indication "SEL FROM" is given below the part. The indication "ESDS" is given in the NOMENCLATURE column for electronic components which are sensitive to electrostatic discharges.
 - 4 Part Numbers beginning with "A" are manufacturer assigned reference to represent manufacturer source-controlled and specification-controlled parts.
 - (e) Effectivity Code
 - 1 A single upper case (e.g. A,B,C,) legend/code is used in the "effectivity" column to show an operator's current effectivity between parts, and depicts only restricted interchangeability relationships. The term "operator's current effectivity" refers to the complete effectivity range for all parts, and includes the original item, any or all of its variants and/or their attaching parts. These parts, if affected, are be updated each time a new part (variant) is added, superseded, modified or deleted.
 - 2 Within a figure, a part number identified by an effectivity code (a single upper case alpha designation) denotes that the coded part must be used with other parts identified with the same alpha designation. Also, coded parts can be used with all other noncoded parts (no alpha designation).
 - 3 The effectivity code begins with A and continues with B, C, D, ... Z. If required, the succession follows BA, CA, DA ... ZA, CB, DB, EB ... DC, EC, FC, etc., always beginning with the next alpha set which succeeds the last set by one alpha.

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- 4 The effectivity code identifies affected parts within a single figure only. The same code can appear in other figures, but it is only used for those parts within the figure in which it is listed. Effectivity does not relate to more than one figure. Figure alphas are considered a separate figure.
 - 5 Item numbers with variants that have a common effectivity code or no effectivity code can be intermixed provided that it is not specifically prohibited by special notes in the nomenclature.
- (f) Units per Assy
- 1 The UNITS PER ASSY column shows the number of parts necessary for the next higher assembly. For some assemblies or parts, the number is replaced by the letters RF (for Reference) and AR (As Required).

D. Abbreviations used in the Detailed Parts List

(1) List of abbreviations

AR :	As Required
AMDMNT :	Amendment
DET :	Detail
EFF :	Effectivity
ESDS :	Electrostatic Discharge Sensitive
NHA :	Next Higher Assembly
NP :	Not Procurable
OPT :	Optional
ORDER OVERLGTH MPN :	Order Overlength Manufacturer's Part Number
POST SB :	Post Service Bulletin
PRE SB :	Pre Service Bulletin
R :	Revised
RF :	For Reference
SEL FROM :	Select From
SUPSD BY :	Superseded By
SUPSDS :	Supersedes

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OPTIONAL PART LIST

MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
0603Y016-0104KX	VU1579	0603YC104K4ZA	V04222
		C0603X104K4RAC	V31433
0603Y050-0102KX	VU1579	06035C102K4ZA	V04222
0603Y050-0103KX	VU1579	06035C103K4ZA	V04222
		C0603X103K5RAC	V31433
0603Y050-0153KX	VU1579	06035C153K4ZA	V04222
0603Y100-0222KX	VU1579	06031C222K4ZA	V04222
0805Y050-0104KX	VU1579	08055C104K4ZA	V04222
0805Y100-0103KX	VU1579	08051C103K4ZA	V04222
0P297GS	V24355	0P297GSZ	V24355
1-0102567-3	VF0286	6-0102567-3	VD8512
101Y02L	VF8378	201M02T03L	VKB441
		211M02T03L	VF0318
		211M02T04L	VF0318
102S02	VF8378	211F02C01U	VF0318
1206Y050-0224KX	VU1579	12065C224K4ZA	V04222
1206Y200-0223KX	VU1579	C1206X223K2RAC	V31433
211F02C03U	VF0318	201F02C03U	VKB441
		211F02C01U	VF0318
		211F02C04U	VF0318
22293HD016004PA	VF4796	22293HD016004PA	VF7411
		22293HD016004PA	VF9667
		22293HD016004PA	VFAT06
251806600Y3	V34899	BLM41PG750SN1	VS4431
2N7002	VK1196	2N7002	V17856
		2N7002-215	VH0H68
		2N7002E3	V17856
		2N7002L	V1MQ07

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
501-0TK50-1002F	VS7293	2N7002LG	V1MQ07
		MMA02040C1002F00	VD9539
		MS1010KFE	VD1018
		SMM02040C1002F	VD1018
		WRM0204-1002F	V57027
		ZC0204FE10K	VD9188
520251-70	VF0349	E27620-25UJT	VF4796
520301-24	VF0349	520301-31	VF0349
52207-2090	V27264	2-84953-0	V00779
		52207-2085	V27264
		F10-2902006	VK0536
		F10-2902046R	VK0536
58102F61-23	VF6131	58102F61-23LF	VF6131
		87049-4616	V27264
		87760-4616	V27264
74LVC1G14GW	VH0H68	SN74LVC1G14DCK	V01295
76308-112	VF6131	76308-112LF	VF6131
		SBQ12PS100TG30	V06776
86455-036	VF6131	63453-036	VF6131
		63453-036LF	VF6131
86455-046	VF6131	63453-046	VF6131
		63453-046LF	VF6131
		79107-0022	V27264
92086-122	VF6131	92086-122LF	VF6131
94277-044	VF6131	94277-044LF	VF6131
94277-046	VF6131	94277-046LF	VF6131
A1204438	VF9111	E25123B20Q3ZEAPA	VF0188
		E25123B20Q3ZEAPA	VF4796
		E25123B20Q3ZEAPA	VFAT06
A1229703	VF9111	E25111B30Q6ZHAPA	VF7411

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
A1232078	VF9111	E25111B30Q6ZHAPA	VF9667
		E25111B30Q6ZHAPA	VFAT06
		E25123B20Q5ZEAPA	VF0188
		E25123B20Q5ZEAPA	VF7411
		E25123B20Q5ZEAPA	VFAT06
A1235914	VF9111	E25111B25Q6ZHAPA	VF0188
		E25111B25Q6ZHAPA	VF7411
		E25111B25Q6ZHAPA	VF9667
		E25111B25Q6ZHAPA	VFAT06
A1235962	VF9111	E25127A25Q5ZHAPA	VF0188
		E25127A25Q5ZHAPA	VF7411
		E25127A25Q5ZHAPA	VFAT06
A1240230	VF9111	CTC4-475X9050D8	V2N936
		T491D475K050ATZQ01	V31433
		TAJD475K050FJ	V17554
A1242090	VF9111	CR0805FX6191	V32997
		CR0805FX6191LF	V32997
		CRCW08056K19FK	VD1018
		MCR10F6191	VS5518
		RC0805F6K19L	VS7293
A1242093	VF9111	CR0805FX1212	V32997
		CR0805FX1212LF	V32997
		CRCW080512K1FK	VD1018
		MCR10F121	VS5518
		RC0805F12K1L	VS7293
A1246365	VF9111	CR0805FX5621LF	V32997
		CRCW08055K62FK	VD1018
		MCR10F5621	VS5518
		RC0805F5K62L	VS7293
A1246497	VF9111	AE77H20-41PNM040E	VFAFF1

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A1247907	VF9111	FDBA50H20-41PNKA1264	VF1983
		223-983	VC0410
		3-5100141-0	VD8512
A1248932	VF9111	E25111B40Q10ZHAPA	VF0188
		E25111B40Q10ZHAPA	VF4796
		E25111B40Q10ZHAPA	VFAT06
AD654JRZ	V24355	AD654JR	V24355
AD822AR	V24355	AD822ARZ	V24355
		AD822BR	V24355
		AD822BRZ	V24355
AT28HC256-70SI	V1FN41	AT28HC256-70SU	V1FN41
AT28HC256F90SI	V1FN41	AT28HC256F90SU	V1FN41
BAS21	VF8859	BAS21	V07263
		BAS21	V31019
		BAS21	VC6489
		BAS21	VH0H68
		BAS21L	V1MQ07
		BAS21LG	V1MQ07
BAT54S	VF8859	BAT54S	V07263
		BAT54S	VH0H68
		BAT54SE3-8	VF5752
BAV70L	VFAHD5	BAV70	VC6489
		BAV70	VH0H68
		BAV70LG	V1MQ07
BAV99W	VC6489	BAV99W	VH0H68
CR0805-464RF	VF7178	CR0805-100-464UF	VD1018
		CR0805FX4640	V32997
		CR0805FX4640LF	V32997
		CRCW0805-464RF	V91637
		MCR10F4640	VS5518

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
CR0805-4K75F	VF2663	RC0805F464RL	VS7293
		WCR0805-464RF	VF7178
		CR0805FX4751	V32997
		CR0805FX4751LF	V32997
		D12-100-4K75F	VD1018
CR0805F100RF	VF7178	MCR10F4751	VS5518
		RC0805F4K75L	VS7293
		CR0805-100-100RF	VD1018
		CR0805FX1000	V32997
		CR0805FX1000LF	V32997
CR0805FX1002	V32997	CRCW0805100RFK	VD1018
		MCR10F1000	VS5518
		RC0805F100RL	VS7293
		WCR0805-100RF	VF7178
		CR0805-100-10KF	VD1018
CR0805FX1003LF	V32997	CR0805FX1002LF	V32997
		CRCW080510K0FK	VD1018
		MCR10F1002	VS5518
		RC0805F10KL	VS7293
		CRCW0805100KFK	VD1018
CR0805FX1471	V32997	MCR10F1003	VS5518
		RC0805F100KL	VS7293
		CR0805-100-1K47F	VD1018
		CR0805FX1471LF	V32997
		CRCW08051K47FK	VD1018
CR0805FX1782	V32997	MCR10F1471	VS5518
		CR0805-100-17K8F	VD1018
		CR0805FX1782LF	V32997
		CRCW080517K8FK	VD1018
		MCR10F1782	VS5518

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CR0805FX2610	V32997	RC0805FR17K8L	VS7293
		CR0805-100-261RF	VD1018
		CRCW0805261RFK	VD1018
		MCR10F2610	VS5518
CR0805FX2613	V32997	RC0805F261RL	VS7293
		CR0805-100-261KF	VD1018
		CR0805FX2613LF	V32997
		CRCW0805261KFK	VD1018
CR0805FX3160	V32997	MCR10F2613	VS5518
		RC0805FR261KL	VS7293
		CR0805-100-316RF	VD1018
		CRCW0805316RFK	VD1018
CR0805FX3832	V32997	MCR10F3160	VS5518
		RC0805F316RL	VS7293
		CR0805-100-38K3F	VD1018
		CRCW080538K3FK	VD1018
CR0805FX4642	V32997	MCR10F3832	VS5518
		RC0805F38K3L	VS7293
		CR0805-100-46K4F	VD1018
		CR0805FX4642LF	V32997
CR0805FX5623	V32997	CRCW080546K4FK	VD1018
		MCR10F4642	VS5518
		RC0805F46K4L	VS7293
		CR0805-100-562KF	VD1018
CR0805FX6811	V32997	CR0805FX5623LF	V32997
		MCR10F5623	VS5518
		RC0805F562KL	VS7293
		CR0805-100-6K81F	VD1018
		CR0805FX6811LF	V32997
		CRCW08056K81FK	VD1018

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
CR0805K33K2F	VF2663	MCR10F6811	VS5518
		RC0805F6K81L	VS7293
		CR0805FX3322	V32997
		CRCW080533K2FK	VD1018
		MCR10F3322	VS5518
CR0805K56K2F	VF2663	RC0805F33K2L	VS7293
		CR0805FX5621	V32997
		CR0805FX5622LF	V32997
		D12-100-56K2F	VD1018
		ERJ6ENF5622	VS5175
D12-100-1MF	VD1018	MCR10F5622	VS5518
		RC0805F56K2L	VS7293
		WCR0805-56K2F	VF7178
		CR0805-100-1MF	VD1018
		CR0805FX1004	V32997
E25401B16QZHAPA	VF4796	CR0805FX1004LF	V32997
		MRC10F1004	VS5518
		RC0805F1ML	VS7293
		E25401B16QZHAPA	VF0188
		E25401B16QZHAPA	VF7411
E25514B20ZTAPA	VF4796	E25401B16QZHAPA	VFAT06
		E25514B20ZTAPA	VF9667
		FCMS35-1001-200	VF1379
		IL205A	V07263
		IRFR320PBF	VF5752
KSR251J	VC0166	IRFR320BTM	V07263
		IRFR320TRPBF	VK4184
		SIHFR320GE3	VK4184
		KSR251GLFS	V09353
		KSR551GLFS	V09353

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
LM2904D	VF8859	KSR551GLFS	VF3132
		KSR551GSLFS	V09353
		KSR551JS	VC0166
		LM2904D	V01295
		LM2904D	V1MQ07
		LM2904D	VX0213
		LM2904DG	V1MQ07
MCR10F2612	VS5518	LM2904MNPB	V27014
		CR0805FX2612LF	V32997
		CRCW080526K1FK	VD1018
MCR10F3161	VS5518	RC0805F26K1L	VS7293
		CRCW08053K16FK	VD1018
		RC0805F3K16L	VS7293
MMSZ5234B	V1MQ07	MMSZ5234BG	V1MQ07
MMSZ5240BG	V1MQ07	MMSZ5240B	V1MQ07
MMSZ5243BG	V1MQ07	MMSZ5243B	V1MQ07
PD10808	V0U8A3	PD10808LF	V0U8A3
R459-002	V75915	0459-002	V75915
		3404-0012	VS3629
RC0805F46R4L	VS7293	CRCW080546R4FK	VD1018
		WR08X46R4FL	VSER54
RC1206J4R7L	VS7293	CRCW12064R70JN	VD1018
S02222A	VF8859	MMBT2222AL	V1MQ07
		MMBT2222ALG	V1MQ07
		PMBT2222A	VH0H68
		SMBT2222A	VC6489
SFH2400	VF0889	SFH2400Z	VF0889
SLW14S1C7	VF6131	52030-1410	V27264
		52030-1429	V27264
		SLW14S1C7LF	VF6131

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
SM14M12C	V0U8A3	SM14M12CLF	V0U8A3
		SMDA12C8	V14936
SMBJ15CA001	V75915	SMBJ15CA001TR	VX0218
SMBJ33CA	VK4184	SMBJ33CA001	V75915
		SMBJ33CAE3	VK4184
SMCJ15	VK4184	SMCJ15A	VK4184
		SMCJ15A001	V75915
		SMCJ15AE3	VK4184
SMCJ15A	VK4184	SMCJ15A001	V75915
		SMCJ15AE3	VK4184
SMCJ15CA001	V75915	SMCJ15CA001TR	VX0218
SMCJ43CA001	V75915	SMCJ43CA001TR	VX0218
SN54AHC1G32DCK	V01295	74AHC1G32GW	VH0H68
TAJD106K035FJ	V17554	CTC4-106X9035D2E3	V2N936
		CTC4-106X9035D8	V2N936
		T491D106K035ATZQ01	V31433
TSL0807-102KR23	VS5988	18402102K	VX0015
		18402102KLFR	VX0015
VJ0603A221JXBA	VF7843	06031A221JATA	VF8489
		0603J100-0221JC	VU1579
VJ0805A101JXBA	VF7843	0805J100-0101JC	VU1579
		0805Y100-0101JC	VU1579
		C0805C101J1GAC	V31433
VJ0805A330JXBA	VF7843	0805J100-0330JC	VU1579
		0805Y100-0330JC	VU1579
WCR0603-10KF	VF7178	CR0603FX1002LF	V32997
		CRCW060310K0F	VD1018
		CRCW060310KFK	VD1018
		MCR03FX1002	VS5518
		RC0603F10KL	VS7293

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
WCR0603-1KF	VF7178	CR0603FX1001LF	V32997
		CRCW06031K00FK	VD1018
		MCR03FX1001	VS5518
		RC0603F1KL	VS7293
WCR0603-26K1F	VF7178	BCR1-16-2612F	VD5682
		CR0603FX2612	V32997
		CR0603FX2612LF	V32997
		CRCW060326K1FK	VD1018
WCR0603R005J	VF7178	MCR03FX2612	VS5518
		RC0603F26K1L	VS7293
		CR0603J000LF	V32997
		CRCW06030000Z	VD1018
WCR0805-14K7F	VF7178	RC0603J0RL	VS7293
		CR0805-100-14K7F	VD1018
		CRCW080514K7FK	VD1018
		RC0805F14K7L	VS7293
WCR0805-1KF	VF7178	CR0805-100-1KF	VD1018
		CR0805FX1001	V32997
		CR0805FX1001LF	V32997
		CRCW08051K00FK	VD1018
WCR0805-21K5F	VF7178	MCR10F1001	VS5518
		CR0805FX2152LF	V32997
		MCR10F2152	VS5518
		RC0805F21K5L	VS7293
WCR0805-31R6F	VF7178	CR0805-100-31R6F	VD1018
		CR0805FX31R6	V32997
		CR0805FX31R6LF	V32997
		CRCW080531R6FK	VD1018
		MCR10F31R6	VS5518
		RC0805FR31R6L	VS7293

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MANUFACTURER'S PART NUMBER	MFR CODE	OPTIONAL MFR PART NUMBER	OPT MFR CODE
WCR0805-5K11F	VF7178	CR0805-100-5K11F	VD1018
		CR0805FX5111	V32997
		D12-100-5K11F	VD1018
		MCR10F5111	VS5518
		RC0805F5K11L	VS7293
WCR0805-681RF	VF7178	CR0805-100-681UF	VD1018
		CR0805FX6810	V32997
		CR0805FX6810LF	V32997
		CRCW0805681RFK	VD1018
		MCR10F6810	VS5518
WCR08050R00F	VF7178	RC0805F681RL	VS7293
		CR0805J000	V32997
		CR0805J000LF	V32997
		CRCW08050000Z0	VD1018
		CRCW08050000ZS	VD1018
		MCR10J000	VS5518
		RC0805J0RL	VS7293

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VENDOR CODE LIST

CODE	VENDOR'S ADDRESS
V00779	TE CONNECTIVITY CORPORATION 2800 FULLING MILL RD BLDG 38 MIDDLETOWN 17057-3142 USA
V01295	TEXAS INSTRUMENTS INCORPORATED DBA TEXAS INSTRUMENTS DIV SEMICONDUCTOR GROUP 12500 T I BLVD 75243-0592 DALLAS TX USA
V027H6	BERG ELECTRONICS INC superseded by V22526
V04222	AVX CORPORATION DBA OLEAN ADVANCED PRODUCTS 801 17TH AVE S MYRTLE BEACH SC 29577-4245 USA
V06776	ROBINSON NUGENT INC NEW ALBANY IN 47150-3264 USA
V07263	FAIRCHILD SEMICONDUCTOR CORP 333 WESTERN AVE SOUTH PORTLAND ME 04106 USA
V09353	C AND K COMPONENTS INC 57 STANLEY AVE WATERTOWN MA 02472-4802 USA
VOU8A3	PROTEK DEVICES LP 2929 S FAIR LN TEMPE AZ 85282-3153 USA
V14936	GENERAL SEMICONDUCTOR INC 10 MELVILLE PARK RD MELVILLE NY 11747 USA
V17554	AVX TANTALUM CORPORATION DBA A V X DIV AVX TANTALUM CORP. 401 HILL ST BIDDEFORD 04005-4335 USA
V17856	SILICONIX INCORPORATED DIV SILICONIX 2201 LAURELWOOD RD SANTA CLARA CA 95054-1516 USA
V18915	PENTAIR TECHNICAL PRODUCTS 7328 TRADE ST SAN DIEGO 92121-3410 USA
V1FN41	ATMEL CORPORATION 2325 ORCHARD PKWY SAN JOSE CA 95131-1034 USA

CODE	VENDOR'S ADDRESS
V1MQ07	SEMICONDUCTOR COMPONENTS INDUSTRIES LLC DBA ON SEMICONDUCTOR DIV CORPORATE HEADQUARTERS 5005 E. MCDOWELL ROAD 85008-4229 PHOENIX AZ USA
V22526	FCI USA INC 825 OLD TRAIL RD ETTERS PA 17319-9392 USA
V24355	ANALOG DEVICES, INC. DIV CORPORATE HEADQUARTERS 1 TECHNOLOGY WAY NORWOOD MA 02062-2634 USA
V24984	HYDRAFLOW 1881 W MALVERN CERRITOS CA 90703 FULLERTON 92833-2403 USA
V27014	NATIONAL SEMICONDUCTOR CORPORATION 2900 SEMICONDUCTOR DR 95051-0606 SANTA CLARA CA USA
V27264	MOLEX INCORPORATED DIV MOLEX CONNECTOR CORPORATION 60532-1682 LISLE IL USA
V2N936	VISHAY SPRAGUE SANFORD INC DIV SPRAGUE 1754 MAIN ST SANFORD 04073-3245 ME USA
V31019	ALLEGRO MICROSYSTEMS INC 115 NORTHEAST CUTOFF WORCESTER MA 01606 USA
V31433	KEMET ELECTRONICS CORPORATION DBA KEMET 2835 KEMET WAY SC 29681-6202 SIMPSONVILLE USA
V31441	AMERICAN SAFETY FLIGHT SYSTEMS INC GLENDALE CA 91201-2427 USA
V32997	BOURNS, INC. 1200 COLUMBIA AVE RIVERSIDE CA 92507-2129 USA
V34899	FAIR-RITE PRODUCTS CORP. 1 COMMERCIAL ROW WALLKILL NY 12589-4438 USA

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CODE	VENDOR'S ADDRESS
V55322	SAMTEC INC./DBA SAMTEC 520 PARK EAST BLVD 47150-7251 NEW ALBANY IN USA
V57027	INTERNATIONAL RESISTIVE COMPANY OF TEXAS, LP DBA IRC DIV ADVANCED FILM DIVISION 4222 S STAPLES ST 78411-2702 CORPUS CHRISTI TX USA
V75915	LITTELFUSE, INC. 8755 W HIGGINS RD STE 500 60631-2701 CHICAGO IL USA
V81541	AIRPAX CORPORATION LLC DBA AIRPAX CORPORATION DIV POWER PROTECTION PRODUCTS 807 WOODS ROAD CAMBRIDGE MD 21613-0520 USA
V91637	VISHAY DALE ELECTRONICS, LLC DBA VISHAY INTERTECHNOLOGY DIV VISHAY AMERICAS 1122 23RD ST COLUMBUS NE 68601-3647 USA
V99017	CAPPLUGS LLC DBA MOKON DIV CAPPLUGS 2150 ELMWOOD AVE 14207-7198 BUFFALO NY USA
VC0166	C&K COMPONENTS HOLZHAUSER STRASSE 26-32 13509 BERLIN GERMANY
VC0410	ERNI ELEKTRONICS GMBH SEESTRASSE 9 73099 ADELBERG GERMANY
VC6489	INFINEON TECHNOLOGIES AG AM CAMPEON 1-12 80 09 49 85579 NEUBIBERG GERMANY
VD1018	VISHAY ELECTRONIC GMBH DR.-FELIX-ZANDMAN-PLATZ 95100 SELB 1180 GERMANY
VD5682	EMERSON PROCESS MANAGEMENT GMBH & CO.OHG ARGELSRIEDER FELD 7 82234 WESSLING GERMANY

CODE	VENDOR'S ADDRESS
VD8512	TYCO ELECTRONICS AMP GMBH AMPERESTRASSE 7-11 12 40 63225 LANGEN GERMANY
VD9188	YAGEO EUROPE GMBH DR. BERND SCHROEDER RAMSKAMP 70 25337 ELSHORN GERMANY
VD9539	VISHAY BC COMPONENTS BEYSCHLAG GMBH RUNGHOLTSTRASSE 8-10 25732 HEIDE 1220 GERMANY
VF0188	BLANC AERO INDUSTRIES - GROUPE LISI AEROSPACE 42 QUAI DE LA RAPEE 75012 PARIS 12 FRANCE
VF0225	SOURIAU 9 RUE PORTE DE BUC 78000 VERSAILLES FRANCE
VF0286	TYCO ELECTRONICS FRANCE SAS 29 CHS JULES CESAR BP 39 95301 CERGY PONTOISE CEDEX FRANCE
VF0318	ATI ELECTRONIQUE 6 RUE JEAN MERMOZ 91080 COURCOURONNES FRANCE
VF0349	LISI AUTOMOTIVE NOMEL LD FORET DU CHATEAU 61550 LA FERTE FRENEL FRANCE
VF0862	PHILIPS FRANCE 33 RUE DE VERDUN 92150 SURESNES FRANCE
VF0889	OSRAM SA 5 RUE D ALTORF BP 109 67120 MOLSHEIM FRANCE
VF1379	EXXELIA TECHNOLOGIES 93 RUE OBERKAMPF 75011 PARIS 11 FRANCE
VF1621	VISHAY SA 199 BD DE LA MADELEINE BP 1159 06000 NICE FRANCE

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CODE	VENDOR'S ADDRESS
VF1983	CONNECTEURS ELECTRIQUES DEUTSCH DIV CONNECTEURS AEROMILITAIRES 17 RUE LAVOISIER BP 135 27091 EVREUX CEDEX 9 FRANCE
VF2663	SFERNICE DIVISION SOVCOR superseded by VF1621
VF3132	C AND K COMPONENTS SAS 2 RUE BERTHOLLET 39100 DOLE FRANCE
VF4796	ETABLISSEMENTS BD 31 RUE DES CAYENNES 78700 CONFLANS SAINTE HONORINE FRANCE
VF5752	VISHAY SA DIVISION VISHAY COMPOSANTS superseded by VF1621
VF6131	FRAMATOME CONNECTORS INTERNATIONAL JUPITER SA superseded by VF0225
VF7178	TT ELECTRONICS 4 PL LOUIS ARMAND 75012 PARIS FRANCE
VF7411	MECAERO ARCONIC FASTENING SYSTEMS AND RINGS TOULOUSE BD DU GRAND CASTAING BP 86 31120 ROQUES FRANCE
VF7843	VISHAY SA DIVISION VITRAMON superseded by VF1621
VF8378	NICOMATIC ZONE INDUSTRIELLE 74890 BONS-EN-CHABLAIS FRANCE
VF8489	AVX SAS 2 RUE DU CHEMIN DES FEMMES 91300 MASSY FRANCE
VF8841	SJM EUROSTAT 45 ROUTE D'ORGELET 39130 PONT DE POITTE FRANCE
VF8859	STMICROELECTRONICS SA 29 BD ROMAIN ROLLAND 92120 MONTRouGE FRANCE
VF9111	THALES AVS FRANCE SAS 75-77 AV MARCEL DASSAULT CAMPUS THALES BORDEAUX 33700 MERIGNAC FRANCE

CODE	VENDOR'S ADDRESS
VF9667	SAS SATINOX ASSEMBLAGE 2 RUE DES COTES D'ORVAL BP 49 78550 HOUDAN FRANCE
VFAFF1	CONESYS EUROPE 24 AV JEAN MONNET 31770 COLOMIERS FRANCE
VFAHD5	ON SEMICONDUCTOR FRANCE SAS 132 CHE DE BASSO CAMBO BP 53512 31035 TOULOUSE CEDEX 1 FRANCE
VFAT06	DECOLLETAGE MALOUVIS 479 RUE GEN JACQUES DE MONFORT 74700 SALLANCHES FRANCE
VH0H68	NXP SEMICONDUCTORS NETHERLANDS B.V. GERSTWEG 2 PO BOX 30008 6534 AE NIJMEGEN NEDERLAND
VK0536	HARWIN PLC FITZHERBERT RD PO6 1RT FARLINGTON PORTSMOUTH HAMPSHIRE UK
VK1196	DIODES ZETEX SEMICONDUCTORS LIMITED ZETEX TECHNOLOGY PARK OL9 9LL CHADDERTON, OLDHAM LANCS UK
VK4184	VISHAY LTD PALLION INDUSTRIAL ESTATE SUNDERLAND TYNE AND WEAR SR4 6SU UK
VKB441	ITW MCMURDO CONNECTORS superseded by VKD399
VKD399	DANIAMANT LIMITED UNIT 3, THE ADMIRAL PARK, AIRPORT SERVICE ROAD PO3 5RQ PORTSMOUTH HAMPSHIRE UK
VS3629	SCHURTER AG WERKHOFSTRASSE 8 CH-6002 LUZERN SWITZERLAND
VS4431	MURATA MFG CO. LTD T 617 NAGAKAYO-SHI 617-8555 KYOTO JAPAN
VS4719	HAMAMATSU PHOTONICS K.K. 1126 1 ICHINO CHOME HAMAMATSU 430-8652 JAPAN

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CODE	VENDOR'S ADDRESS
VS5175	PANASONICS 3-5-1 ARIAKE, KOTO-KU 99999 TOKYO JAPAN
VS5518	ROHM LTD 21 SAIIN MIZOSAKI CHO UKYO KU KYOTO 615 JAPAN
VS5988	TDK-LAMBDA CORPORATION 1-13-1 NIHONBASHI CHUO-KU TOKYO 103 JAPAN
VS7293	YAGEO CORPORATION 3F, NO. 233-1PAO CHIAO ROAD, HSIN TIEN 99999 SINDIAN CITY TAIWAN
VSER54	WALSIN TECHNOLOGY CORPORATION 566-1, KAO-SHI ROAD, YANG- MEI, 99999 TAO YUAN TAIWAN
VSS993	NICHIA CORP. 491-100, OKA, KAMINAKACHO ANAN 774-0044 JAPAN
VU1579	KNOWLES UK LIMITED HETHEL ENGINEERING CENTRE CHAPMAN WAY NR14 8FB HETHEL, NORWICH NORFOLK UK
VX0015	FRONTIER ELECTRONICS CO. LTD 231 TAIPEI HSIEN TAIWAN
VX0213	PHILIPS SEMICONDUCTEURS superseded by VF0862
VX0218	LUCKY FORESTS CORPORATION LTD NO-2 LANE-549 YANGPING RD YANGMEI CITY TAO-YUAN TAIWAN

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EQUIPMENT DESIGNATOR INDEX

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	EQUIPMT DESIG	GEO LOC	FIGURE ITEM	EQUIPMT DESIG	GEO LOC	FIGURE ITEM
A1		2 330	TB1		2 420A	PS1		2A 240A
A1		2 330A	TB1		2 420B	PS1		2A 240B
A2		2 230	TB1		2 420C	TB1		2A 420
A2		2 230A	A1		2A 330	TB1		2A 420A
A2		2 230B	A1		2A 330A	TB1		2A 420B
A3		2 220	A2		2A 230	TB1		2A 420C
A3		2 220A	A2		2A 230A	A1		2B 350
A3		2 220B	A2		2A 230B	A2		2B 230
A3		2 220C	A3		2A 220	A2		2B 230A
A3		2 220D	A3		2A 220A	A3		2B 220
A5		2 170	A3		2A 220B	A3		2B 220A
A5		2 170A	A3		2A 220C	A5		2B 170
A6		2 140	A3		2A 220D	A5		2B 170A
A6		2 140A	A5		2A 170	A6		2B 140
A6		2 140B	A5		2A 170A	A6		2B 140A
A7		2 290	A6		2A 140	A7		2B 290
A7		2 290A	A6		2A 140A	A7		2B 290A
A7		2 290B	A6		2A 140B	A8		2B 310
A8		2 305	A7		2A 290	A8		2B 310A
A8		2 305A	A7		2A 290A	A8		2B 310B
A8		2 305B	A7		2A 290B	A8		2B 310C
A8		2 305C	A8		2A 305	A9		2B 190
A9		2 190	A8		2A 305A	PS1		2B 240
A9		2 190A	A8		2A 305B	PS1		2B 240A
PS1		2 240	A8		2A 305C	TB1		2B 440
PS1		2 240A	A9		2A 190	TB1		2B 440A
PS1		2 240B	A9		2A 190A	A1		2C 350
TB1		2 420	PS1		2A 240	A1		2C 350A

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EQUIPMT DESIG	GEO LOC	FIGURE ITEM
A2		2C 230
A2		2C 230A
A3		2C 220
A3		2C 220A
A3		2C 220B
A5		2C 170
A5		2C 170A
A5		2C 190A
A6		2C 140
A6		2C 140A
A7		2C 290
A7		2C 290A
A7		2C 290B
A8		2C 310
A8		2C 310A
A8		2C 310B
A8		2C 310C
A8		2C 310D
A9		2C 190
PS1		2C 240
PS1		2C 240A
TB1		2C 440
TB1		2C 440A
TB1		2C 440B
A61		3 150
A61		3 150A
A61		3A 150
A61		3A 150A
A51		4 160
A51		4 160A

EQUIPMT DESIG	GEO LOC	FIGURE ITEM
DS1		4 50
DS1		4 50A
DS2		4 140
DS2		4 140A
DS3		4 140
DS3		4 140A
DS4		4 140
DS4		4 140A
J4		4 110
J4		4 110A
A11		5 130
J1		5 110
R1		5 50
R2		5 50
C1	4B	6 40
C2	4B	6 40
CR1	1B	6 45
CR2	1B	6 45
CR3	1B	6 45
CR4	1A	6 45
CR5	1A	6 45
CR6	6C	6 50
CR7	6B	6 50
CR8	6B	6 50
CR9	6B	6 50
P1	3CC	6 55
P2	5AA	6 60
P3	1AA	6 60
P4	1CC	6 65
P5	8AA	6 70

EQUIPMT DESIG	GEO LOC	FIGURE ITEM
P6	7C	6 75
P7	1AA	6 80
P8	3AA	6 80
P9	6BB	6 85
R1	4C	6 90
R2	4B	6 90
R3	4B	6 90
R4	1B	6 90
R5	1B	6 95
R6	1B	6 90
R7	1B	6 95
R8	1B	6 90
R9	1B	6 95
U1	3B	6 100
U2	4B	6 100
C1	4B	6A 45
C2	4B	6A 45
C3	6B	6A 50
C4	6B	6A 50
C5	6B	6A 50
C6	6B	6A 50
C7	8A	6A 55
C8	8A	6A 55
C9	7A	6A 55
CR1	1B	6A 60
CR2	1B	6A 60
CR3	1B	6A 60
CR4	1A	6A 60
CR5	1A	6A 60
CR6	6C	6A 65

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EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
CR7	6B	6A	65
CR8	6B	6A	65
CR10	6B	6A	70
P1	3CC	6A	75
P2	5AA	6A	80
P3	1AA	6A	80
P4	1CC	6A	85
P4	1CC	6A	85A
P5	8AA	6A	90
P6	7C	6A	95
P7	1AA	6A	100
P8	3AA	6A	100
P9	6BB	6A	105
R1	4C	6A	110
R2	4B	6A	110
R3	4B	6A	110
R4	1B	6A	110
R5	1B	6A	115
R6	1B	6A	110
R7	1B	6A	115
R8	1B	6A	110
R9	1B	6A	115
R10	8A	6A	120
R11	8A	6A	120
R12	7A	6A	120
U1	4B	6A	125
U2	3B	6A	125
C1	1C	6B	45
C2	1C	6B	45
C3	8B	6B	45

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
C4	7B	6B	45
C5	6C	6B	45
C6	6C	6B	45
C7	6B	6B	50
C8	3B	6B	50
C9	6B	6B	50
C10	6C	6B	45
C11	2A	6B	45
C12	2A	6B	45
C13	8B	6B	45
C14	3A	6B	45
C15	3C	6B	45
C16	3C	6B	45
C17	4A	6B	45
C18	6B	6B	45
C19	6B	6B	45
C20	8B	6B	45
C21	8C	6B	45
C22	6B	6B	45
C23	7B	6B	45
C24	6B	6B	45
C27	6B	6B	50
C28	6C	6B	50
C29	5C	6B	50
C31	7B	6B	45
C32	6B	6B	45
C34	7B	6B	45
C35	7A	6B	45
C36	6B	6B	45
C38	8B	6B	45

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
C39	8A	6B	45
C40	6C	6B	45
C41	7B	6B	45
C43	6C	6B	45
C44	8B	6B	45
C46	6C	6B	45
C48	8B	6B	45
C49	6B	6B	45
C50	8B	6B	45
C51	6B	6B	45
C52	7B	6B	45
C53	6B	6B	45
C54	8A	6B	45
C55	6B	6B	45
C57	6B	6B	45
C59	7A	6B	45
C61	8A	6B	45
C1	2C	7	10
D3	1B	7	15
D4	1A	7	15
D5	1A	7	15
D6	2A	7	15
D7	2A	7	15
D8	3C	7	15
D9	3A	7	15
D10	3B	7	15
D11	3A	7	15
D12	3B	7	15
Q1	3C	7	20
R2	1A	7	25

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EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
R3	1A	7	25
R4	3A	7	25
R5	3A	7	25
R6	3A	7	25
R7	2C	7	30
R8	2C	7	35
R9	2C	7	30
R10	2C	7	40
S1	1A	7	45
S1	1A	7	45A
S2	2A	7	45
S2	2A	7	45A
S3	3A	7	45
S3	3A	7	45A
S4	3A	7	45
S4	3A	7	45A
S5	3C	7	45
S5	3C	7	45A
U1	3D	7	50
C1	2AA	8	35
C3	1A	8	40
C4	2B	8	45
C5	2BB	8	35
C6	2BB	8	35
C7	2B	8	50
C8	3B	8	55
C8	3B	8	55A
C10	2C	8	60
C11	2C	8	60
C12	2C	8	60

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
C13	1BB	8	35
C14	1AA	8	35
C15	1BB	8	35
C16	1AA	8	35
C17	1BB	8	35
C18	1BB	8	35
C19	1BB	8	35
C20	1BB	8	65
C21	1AA	8	35
C22	1B	8	35
C23	1B	8	70
C24	3AA	8	55
C24	3AA	8	55A
C25	3B	8	73
C26	3AA	8	73
CR1	1AA	8	75
CR2	1BB	8	80
CR3	2A	8	85
CR4	2B	8	90
CR5	2AA	8	95
CR6	2BB	8	80
CR7	3BB	8	100
CR8	2BB	8	80
L1	2B	8	105
L2	2BB	8	110
L3	2B	8	105
P2	1A	8	115
P3	1B	8	115
P4	2A	8	120
P5	1C	8	126

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
P5	1C	8	126A
P6	1A	8	125
Q1	1B	8	130
Q2	1B	8	135
Q3	1A	8	130
Q4	2AA	8	140
Q5	2A	8	135
Q6	2BB	8	130
Q7	2BB	8	130
Q8	2A	8	145
Q9	3BB	8	145
Q10	3BB	8	145
Q11	3BB	8	145
Q12	3BB	8	145
Q13	2AA	8	130
R1	2A	8	150
R2	2AA	8	155
R3	2A	8	160
R4	2A	8	165
R5	2A	8	170
R6	1BB	8	175
R7	1A	8	180
R8	1BB	8	185
R9	1A	8	180
R10	2AA	8	190
R11	1BB	8	155
R12	1A	8	195
R13	1A	8	200
R14	2AA	8	205
R15	2BB	8	210

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EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
R16	2B	8	215
R17	2B	8	220
R18	1AA	8	225
R19	2AA	8	230
R20	2BB	8	180
R21	2BB	8	235
R22	2AA	8	240
R23	2BB	8	245
R25	2BB	8	250
R26	2BB	8	253
R26	2BB	8	253A
R27	2BB	8	200
R28	3BB	8	255
R29	3BB	8	255
R30	2BB	8	260
R31	3BB	8	260
R32	2A	8	225
R33	3C	8	260
R34	1BB	8	265
R35	1BB	8	255
R36	3A	8	270
R37	1B	8	150
R38	1B	8	275
R39	2BB	8	175
R40	3A	8	270
R41	2AA	8	280
R42	2AA	8	285
R43	2AA	8	235
R44	2AA	8	180
R45	2A	8	225

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
R46	1BB	8	290
S1	3A	8	295
T1	3B	8	300
U1	2B	8	305
U2	1B	8	310
U3	2A	8	315
C1	3CC	9	15
C2	1A	9	20
C3	1A	9	20
C9	2C	9	30
CR1	2BB	9	35
CR2	2BB	9	35
CR3	3BB	9	40
CR4	2AA	9	35
CR5	2AA	9	35
CR6	3AA	9	45
CR7	3BB	9	45
CR8	2CC	9	50
CR9	2CC	9	50
CR10	2CC	9	50
CR12	3BB	9	50
CR13	3CC	9	50
CR14	3CC	9	50
CR18	2A	9	40
CR19	1C	9	55
CR21	3BB	9	40
CR22	3A	9	40
CR24	3A	9	40
FL2	1B	9	60
FL3	2C	9	60

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
FL4	3AA	9	65
FL6	2A	9	65
J2	1AA	9	70
R1	3BB	9	75
R2	3AA	9	75
R3	3B	9	80
R4	3A	9	80
R5	3BB	9	75
R6	3BB	9	75
R7	3BB	9	80
R8	3BB	9	80
R9	3A	9	80
R10	3A	9	80
R11	3BB	9	80
R12	3BB	9	75
R13	3BB	9	75
R14	3BB	9	75
R15	3BB	9	75
R16	2C	9	75
R17	2B	9	75
R18	2C	9	75
R19	3A	9	75
R20	2A	9	75
R21	2CC	9	75
R22	2B	9	75
R23	2C	9	75
R24	2C	9	75
R25	3B	9	75
R26	3B	9	75
R27	3BB	9	80

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EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
R28	2A	9	75
R29	3C	9	75
R30	3B	9	75
R31	2A	9	75
R32	2C	9	75
R33	2A	9	75
R34	2CC	9	75
R35	3BB	9	85
U1	1BB	9	90
U2	2CC	9	90
U4	2BB	9	95
U5	2BB	9	95
U6	2BB	9	95
U7	2AA	9	95
U9	3C	9	100
U10	2C	9	100
U16	2BB	9	95
U17	2BB	9	95
U18	2B	9	95
U19	2B	9	95
C1	1AA	9A	35
C2	1AA	9A	35
C3	1BB	9A	30
C4	1BB	9A	30
C5	1BB	9A	30
C6	1BB	9A	30
C7	1AA	9A	35
C8	1AA	9A	35
C9	1A	9A	35
C10	1A	9A	35

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
C11	1B	9A	30
C12	1B	9A	30
C13	1BB	9A	30
C14	1BB	9A	30
C16	3C	9A	25
C17	2C	9A	25
C18	1A	9A	35
C19	1A	9A	35
C20	2C	9A	50
C21	2C	9A	50
C22	1C	9A	40
C23	1CC	9A	40
C24	1C	9A	40
C25	1C	9A	40
C26	1C	9A	40
C27	1CC	9A	40
C28	1CC	9A	40
C29	1CC	9A	40
C30	1CC	9A	40
C31	1CC	9A	40
C32	1CC	9A	40
C33	1C	9A	40
C34	1C	9A	40
C35	1B	9A	45
C36	1BB	9A	45
C37	1BB	9A	45
C38	1B	9A	40
C39	1B	9A	40
C40	1B	9A	40
CR1	3AA	9A	55

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
CR2	2AA	9A	55
CR3	3AA	9A	65
CR4	3AA	9A	65
CR5	2AA	9A	55
CR6	2AA	9A	55
CR7	3BB	9A	65
CR8	3BB	9A	65
CR9	3A	9A	65
CR10	3A	9A	65
CR11	2AA	9A	55
CR12	2AA	9A	55
CR13	3C	9A	75
CR14	3C	9A	75
CR15	3C	9A	75
CR16	3BB	9A	65
CR17	3BB	9A	65
CR18	2C	9A	70
CR19	2CC	9A	70
CR20	2BB	9A	70
CR21	2BB	9A	70
CR22	2A	9A	70
CR23	2A	9A	70
CR24	2BB	9A	70
CR25	2BB	9A	70
CR26	3CC	9A	70
CR27	3CC	9A	70
CR28	3CC	9A	70
CR29	2BB	9A	70
CR30	3CC	9A	70
CR31	2A	9A	65

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EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
CR32	2CC	9A	65
CR33	2CC	9A	65
CR34	3CC	9A	70
CR35	3A	9A	70
CR36	2CC	9A	70
CR41	1B	9A	60
CR42	1B	9A	60
CR43	1B	9A	60
F1	2AA	9A	80
F2	2AA	9A	80
F3	2AA	9A	80
F4	2AA	9A	80
J2	1BB	9A	85
L1	1AA	9A	95
L2	1AA	9A	95
L3	1AA	9A	95
L4	1AA	9A	95
L5	1BB	9A	95
L6	1BB	9A	95
L7	1AA	9A	95
L8	1AA	9A	95
L9	1A	9A	95
L10	1B	9A	95
L11	1B	9A	95
L12	1B	9A	95
L13	2C	9A	90
L14	2C	9A	90
L16	2C	9A	100
L19	1A	9A	95
L21	1A	9A	95

EQUIPMT DESIG	GEO LOC	FIGURE ITEM	
L22	1C	9A	95
L23	1CC	9A	95
L24	1C	9A	95
L25	1C	9A	95
L26	1C	9A	95
L27	1CC	9A	95
L28	1CC	9A	95
L29	1CC	9A	95
L30	1CC	9A	95
L31	1CC	9A	95
L32	1CC	9A	95
L33	1C	9A	95
L34	1C	9A	95
L35	1C	9A	95
L36	1BB	9A	95
L37	1BB	9A	95
L38	1B	9A	95
L39	1B	9A	95
L40	1B	9A	95
R1	3AA	9A	105
R2	2AA	9A	105
R3	3AA	9A	105
R4	2AA	9A	105
R11	1BB	9A	110
A7		10	50
A8		10	50A

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
0603Y016-0104KX		6B	55 2
		7	10 1
0603Y050-0102KX		9A	30 8
0603Y050-0103KX		6B	50 6
0603Y050-0153KX		9A	40 16
0603Y100-0120JC		6B	60 11
0603Y100-0222KX		9A	45 3
0603Y100-0470JC		6B	45 44
0805Y050-0104KX		6	40 2
		6A	45 2
		8	35 12
0805Y100-0103KX		9	15 1
0805Y100-0331JC		8	50 1
0P297GS		8	310 1
1-0102567-3		6	55 1
		6A	75 1
		6B	75 1
101Y02L		8	120 1
102S02		4 - 110A	1
1206Y050-0102JX		6A	50 4
1206Y050-0224KX		9A	50 2
1206Y200-0223KX		9A	25 2
1210Y025-0105KX		8	65 1
1812Y100-0472JC		9	30 1
1812Y200-0472JC		8	55 2
		8	73 2
		8	60 3
1812Y3K0-0150JC		8	60 3
1QF3-2		2C	365 1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
1QF3-3		2C 385	1
20-873-0025		10 - 60	
		10 - 80	
		99 - 30	12
211F02C03U		4 110	1
2220Y100-0474KX		9 20	2
22293HD016004PA		2 400	8
		2A 400	8
		2B 420	8
		2C 420	8
251806600Y3		6B 70	4
		9A 90	2
2N7002		8 130	5
3224J1-103		8 270	2
35-7B2-3-3		2 390	4
		2A 390	4
		2B 410	4
		2C 410	4
39986632		9 25	1
40006-1A44		2 350	1
		2A 350	1
		2B 370	1
		2C 370	1
40006-1B46		2 370	1
		2A 370	1
		2B 390	1
		2C 390	1
45740023		2 180	4
		2A 180	4
		2B 180	4

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
501-0TK50-1002F		2C 180	4
		9 80	8
520251-00		5 100	7
520251-70		2 440	4
		2A 440	4
		4 180	4
520301-24		5 80	4
52207-2090		6A 95	1
		6B 95	1
57000-491-1		3 - 70A	7
		3A - 70A	7
57000-520-1		3 70	7
		3A 70	7
57000-551-1		10 20	4
57000-972-1		2 270	2
		2A 270	2
		2B 270	2
		2C 270	2
57001-711-1		2 250	4
		2A 250	4
		2B 250	4
		2C 250	4
57008-370-1		2 130	16
		2 210	2
		2 340	2
		2 430	4
		2A 130	16
		2A 210	2
		2A 340	2
		2A 430	4

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
57040-430-1		2B 130	16
		2B 210	2
		2B 360	2
		2C 130	16
		2C 210	2
		2C 360	2
		4 60	3
		2 360	1
		2A 360	1
		2B 380	1
57230-948-1		2C 380	1
		2 380	1
		2A 380	1
		2B 400	1
		2C 400	1
57231-240-1		99 20	2
		6A - 85A	1
		6B 85	1
		8 295	1
57990-127-1		6B 135	1
58102F61-23		6 80	2
66L040-0247		6A 100	2
		6B 100	2
		6 85	1
		6A 105	1
74LVC1G14GW		6B 105	1
		9 70	1
		9A 85	1
76308-112		6 75	1
		8 125	1
86455-036			
86455-046			
88384-130			
92086-122			

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
94277-044		8 126	1
		6 70	1
94277-046		6 65	1
		6A 85	1
A1204438		3 140	3
		3A 140	3
A1229703		5 70	4
A1232078		3 - 140A	3
		3A - 140A	3
A1235914		4 170	4
		5 90	7
A1235962		2 300	4
		2 307	4
		2 320	8
		2A 300	4
		2A 307	4
		2A 320	8
		2B 300	4
		2B 320	4
		2B 340	8
		2C 300	4
		2C 320	4
		2C 340	8
		8 70	1
		8 190	1
		8 265	1
A1240230		8 290	1
A1242090		5 110	1
A1242093		6 60	2
A1246365		6A 80	2
A1246497			
A1247907			

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
A1248932		6B 80	2
		2 200	6
		2A 200	6
		2B 200	6
		2C 200	6
A1249088		9 85	1
A1249089		9A 105	4
A1250582		9 75	26
AD654JRZ		8 305	1
AD822AR		7 50	1
AT28HC256 - 70SI		6 100	2
		6A - 126A	2
		6B 130	2
AT28HC256F90SI		6A - 126	2
BAS21		9A 60	3
BAT54S		6 45	5
		6A 60	5
		9 55	1
BAV70L		8 100	1
BAV99W		6B 65	5
BLM21AJ601SN1		8 110	1
BLM21BD601SN1		9A 95	33
BZD27C91		8 90	1
C16221KA01		1 1	RF
		2 - 1	RF
C16221KA01CND1		2 - 700	
		99 - 1	
C16221KA02		1 - 1A	RF
		2 - 1A	RF
C16221KA02CND1		2 - 700A	

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
C16221LA01		99 - 1A	
		1 - 1B	RF
		2A - 1	RF
C16221LA01CND1		2A - 700	
		99 - 1B	
C16221LA02		1 - 1C	RF
		2A - 1A	RF
C16221LA02CND1		2A - 700A	
		99 - 1C	
C16221LA03		1 - 1D	RF
		2A - 1B	RF
C16221LA03CND1		2A - 700B	
		99 - 1D	
C16221MA01		1 - 1E	RF
		2B - 1	RF
C16221MA01CND1		2B - 700	
		99 - 1E	
C16221QA01		1 - 1F	RF
		2C - 1	RF
CEC20		99 10	1
CEC6C200V10KK		8 - 55A	2
CR0805-464RF		8 250	1
		8 253	1
CR0805-4K75F		8 165	1
CR0805F100RF		8 260	3
CR0805FX1002		8 150	2
CR0805FX1003LF		7 35	1
		8 180	4
		8 - 253A	1
CR0805FX1471		8 205	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
CR0805FX1782		8 160	1
CR0805FX2610		8 185	1
CR0805FX2613		8 235	2
CR0805FX3160		7 25	5
CR0805FX3832		8 230	1
CR0805FX4642		8 195	1
CR0805FX5623		7 30	2
		8 285	1
CR0805FX6811		8 255	3
CR0805K33K2F		8 170	1
CR0805K56K2F		8 210	1
D12-100-1MF		6A 120	3
		8 155	2
DLW5BSN191SQ2		9A 100	1
E16364JA		2 140	1
		3 - 1	RF
E16364JB		2 - 140A	1
		2B 140	1
		3 - 1A	RF
E16364JC		2 - 140B	1
		2B - 140A	1
		3 - 1B	RF
E16364KA		2A 140	1
		3A - 1	RF
E16364KB		2A - 140A	1
		2C 140	1
		3A - 1A	RF
E16364KC		2A - 140B	1
		2C - 140A	1
		3A - 1C	RF

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
E16368CA		5 130	1
		9 - 1	RF
E16368FA		5 - 130A	1
		9A - 1	RF
E16371DA		2A 190	1
		2C 190	1
E16371EA		2 190	1
E16371GA		2B 190	1
E16387EA		2 330	1
		2A 330	1
		2B 350	1
		2C 350	1
E16387JA		5 - 1	RF
		2 - 330A	1
		2A - 330A	1
		2C - 350A	1
E16389JA		5 - 1A	RF
		3 150	1
		3A 150	1
E16389JB		7 - 1	RF
		3 - 150A	1
		3A - 150A	1
E16441BA		7 - 1A	RF
		3 130	1
		3A 130	1
E16441DA		3 - 130A	1
E16441GA		3A - 130A	1
E16441HA		3 - 130B	1
E16441JA		3 - 130C	1
E16441JB		3A - 130B	1
E16441MA			

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
E16441MB		3A - 130C	1
E16453HA01		2 230	1
		2A 230	1
E16453HA02		2 - 230A	1
		2A - 230A	1
		2B 230	1
E16453HB03		2 - 230B	1
		2A - 230B	1
		2B - 230A	1
E16453MA01		2C 230	1
E16453MA02		2C - 230A	1
E16485BB01		2 220	1
		2A 220	1
E16485EA01		2 - 220A	1
		2A - 220A	1
		2B 220	1
E16485GA01		2C 220	1
E16485GA02		2C - 220A	1
E16485HA01		2C - 220B	1
E16485KA01		2 - 220B	1
		2A - 220B	1
		2B - 220A	1
E16485RA01		2 - 220C	1
		2A - 220C	1
E16485UA01		2 - 220D	1
		2A - 220D	1
E16486BA01		2 420	1
		2A 420	1
		6 - 1	RF
E16486BB01		2 - 420A	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
E16486JA01		2A - 420A	1
		2B 440	1
		2C 440	1
		6A - 1	RF
		2 - 420B	1
		2A - 420B	1
		2B - 440A	1
		2C - 440A	1
		6A - 1A	RF
		2 - 420C	1
E16981HA01		2A - 420C	1
		2C - 440B	1
		6B - 1	RF
		2 240	1
E17395AA		2A 240	1
		2 - 240A	1
E17395DA		2A - 240A	1
		2B 240	1
		2C 240	1
		2 - 240B	1
		2A - 240B	1
		2B - 240A	1
E17395EA		2C - 240A	1
		2 290	1
		2 305	1
		2A 290	1
		2A 305	1
		2B 290	1
E17397AA		2B 310	1
		2C 290	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
E17397BA		2C 310	1
		2B - 310A	1
E17397CA		2C - 310A	1
		10 50	1
		10 - 70	
		2 - 290A	1
		2 - 305A	1
		2A - 290A	1
		2A - 305A	1
		2B - 290A	1
		2B - 310B	1
		2C - 290A	1
E17397DA		2C - 310B	1
		10 - 50B	1
		10 - 70A	
		2 - 305B	1
		2A - 305B	1
		2B - 310C	1
		2C - 310C	1
		2 - 190A	1
		2A - 190A	1
		2C - 190A	1
E18520EA		4 50	1
E18520FA		4 - 50A	1
		4 160	1
E19947BA		8 - 1	RF
E19947GA		4 - 160A	1
E24100BB		8 - 1A	RF
E24100EA		2 170	1
E24119CC		2A 170	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
E24119FA		2B 170	1
		2C 170	1
		4 - 1	RF
		2 - 170A	1
		2A - 170A	1
		2B - 170A	1
		2C - 170A	1
		4 - 1A	RF
E25401B16QZHAPA		2 410	8
		2A 410	8
		2B 430	8
		2C 430	8
E25514B20ZTAPA		2 280	2
		2A 280	2
		2B 280	2
		2C 280	2
E38062AA		2 - 140C	1
		3 - 1C	RF
F1252154		4 140	3
F1252667		4 150	4
F1252893		4 130	6
F1252952		8 300	1
F1273136		5 120	1
F1275310		2 310	8
		2A 310	8
		2B 330	8
		2C 330	8
F1276069		3 100	1
		3A 100	1
F1276070		3 110	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
F1276474		3A 110	1
		2 - 450A	1
		2A - 450A	1
		2B - 460A	1
F1276479		2C - 460A	1
		2 - 110A	2
		2A - 110A	2
		2C - 110A	2
F1276484		2 120	16
		2A 120	16
		2B 120	16
		2C 120	16
F1276493		2 450	1
		2A 450	1
		2B 460	1
		2C 460	1
F1276554		2 100	1
		2A 100	1
		2B 100	1
		2C 100	1
F1276705		2 110	2
		2A 110	2
		2B 110	2
		2C 110	2
F1276812		2 160	1
		2A 160	1
		2B 160	1
		2C 160	1
F1276834		3 120	1
		3A 120	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
F1278453		3	90 1
		3A	90 1
F1279045		2	260 1
		2A	260 1
		2B	260 1
		2C	260 1
F1279326		3	60 1
F1279528		3	80 1
		3A	80 1
F1279712		3A	60 1
F1380513		4	80 1
F1380515		4	70 1
F1390783		4	120 1
F1391805		4	90 1
F1391806		4	100 1
F1400012		2	150 4
		2B	150 4
F1400013		2A	150 4
		2C	150 4
F1400249		2	60 1
		2A	60 1
F1401164		6A	40 1
F1401165		6A	130 1
F1401166		2 - 430A	4
		2A - 430A	4
		2B	450 4
		2C	450 4
F1401882		2A - 60A	1
F1402858		2 - 60A	1
		2A - 60B	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
		2B 60	1
		2C 60	1
F1403026		2 - 8A	1
F1403029		2A - 8B	1
F1403314		6A 125	2
F1403611		5 60	1
F1403666		6A - 40A	1
F1403667		6A - 130A	1
F1403675		2B - 8	1
F1404705		2C - 8	1
F1404945		10 - 50A	
F1405513		4 - 140A	3
F1406777		3 - 60A	1
F1406778		3A - 60A	1
F1406953		10 30	4
F1406974		10 40	16
F1407405		6B 40	1
F1407829		10 10	1
F1407830		10 - 1	RF
		2 - 290B	1
		2A - 290B	1
		2C - 290B	1
F1407831		10 - 1A	RF
		2 - 305C	1
		2A - 305C	1
		2C - 310D	1
F1407891		2 - 8	1
F1407892		2A - 8	1
F1407893		2A - 8A	1
F1408017		2 - 120A	16

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
F2000008		2A - 120A	16
		2B - 120A	16
		2C - 120A	16
		6B - 15	1
		3 - 10	1
		3A - 10	1
		4 - 10	1
		5 - 10	1
		6 - 15	1
		6A - 15	1
F2000068		8 - 10	1
		9A - 10	1
		3 - 20	1
		3A - 20	1
		4 - 20	1
		5 - 20A	1
		5 - 40	
		6 - 20	1
		6A - 20	1
		6B - 20	1
F2000071		8 - 15	1
		9A - 15	1
		3 - 30	1
		3A - 30	1
		4 - 30	1
		5 - 30	1
		6 - 25	1
		6A - 25	1
		6B - 25	1
		8 - 19	1
F2000100			

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
F2000103		3 - 40A	1
		3A - 40A	1
		4 - 40A	1
		5 - 40B	1
		6A 30	1
		6B 30	1
		8 20	1
F2000116		2 - 70	1
		2A - 70	1
		2B - 70	1
		2C - 70	1
F2000119		2 80	1
		2A 80	1
		2B 80	1
		2C 80	1
F2000175		2 90	1
		2A 90	1
		2B 90	1
		2C 90	1
F2000190		3 50	1
		3A 50	1
		6 35	1
		6A 35	1
		6B 35	1
		8 30	1
		9 10	1
		9A 20	1
F2000247		3 40	1
		3A 40	1
		4 40	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
F2000383		5 - 20	
		5 40A	1
		6 30	1
		2 50	1
		2A 50	1
		2B 50	1
FCMS35-1001-200		2C 50	1
		9 100	2
G3116AAG01		2 - 10	1
		2A - 10	1
G3116AAG02		2A - 10A	1
G3116AAG03		2 - 10A	1
		2A - 10B	1
		2B - 10	1
G3116AAK01		2C - 10	1
G3156AAB01		2 - 40	1
		2A - 40	1
		2B - 40	1
		2C - 40	1
G5614AAK01		2 - 20	1
		2A - 20	1
G5614AAK02		2 - 20A	1
		2A - 20A	1
		2B - 20	1
G5614AAL02		2C - 20	1
G5615AAK01		2 - 30	1
		2A - 30	1
G5615AAK02		2 - 30A	1
		2A - 30A	1
		2B - 30	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
G5615AAL01		2C - 30	1
G5616AAB01		6 - 10	1
		6A - 10	1
G5616AAD01		6A - 10A	1
		6B - 10	1
IL205A		8 135	2
IRFR320PBF		8 145	5
KSR251J		7 45	5
KSR551GSLFS		7 - 45A	5
LM2904D		8 315	1
MCR10F2612		6 90	6
		6A 110	6
MCR10F3161		8 280	1
MMSZ5234B		8 80	3
MMSZ5240BG		8 85	1
MMSZ5243BG		8 95	1
MMT122-02SDHAK		6A 90	1
		6B 90	1
NFAC1CC221S1H8		9 65	2
NFAC1CC223R1H8		9 60	2
NFE61HT332Z2A9		9 95	8
NSCW100		7 15	10
PD10796LF		6A 70	1
PD10808		6 50	4
		6A 65	3
		9 45	2
R459-002		9A 80	4
RC0805F46R4L		7 40	1
RC1206J4R7L		8 220	1
S02222A		8 140	1

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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
S5493-01		8 75	1
SFH2400		7 20	1
SLW14S1C7		8 115	2
SM14M12C		9 90	2
SMBJ15CA001		9A 55	6
SMBJ33CA		9 40	5
		9A 70	16
SMCJ15		9 50	6
SMCJ15A		9A 75	3
SMCJ15CA001		9 35	4
SMCJ43CA001		9A 65	11
SN54AHC1G32DCK		6B 140	1
STORAGEC16221K		2 - 999	1
		2A - 999	1
		2B - 999	1
		99 - 1F	RF
STORAGEC16221Q		2C - 999	1
		99 - 1G	RF
TAJD106K035FJ		8 45	1
TLE122-01GDV		8 - 126A	1
TSL0807-102KR23		8 105	2
VJ0603A221JXBA		9A 35	8
VJ0805A101JXBA		8 40	1
VJ0805A330JXBA		6A 55	3
WCR0603-10KF		6B 110	8
WCR0603-1KF		6B 115	3
WCR0603-26K1F		6B 125	7
WCR0603R005J		6B 120	10
		9A 110	1
WCR0805-14K7F		8 245	1

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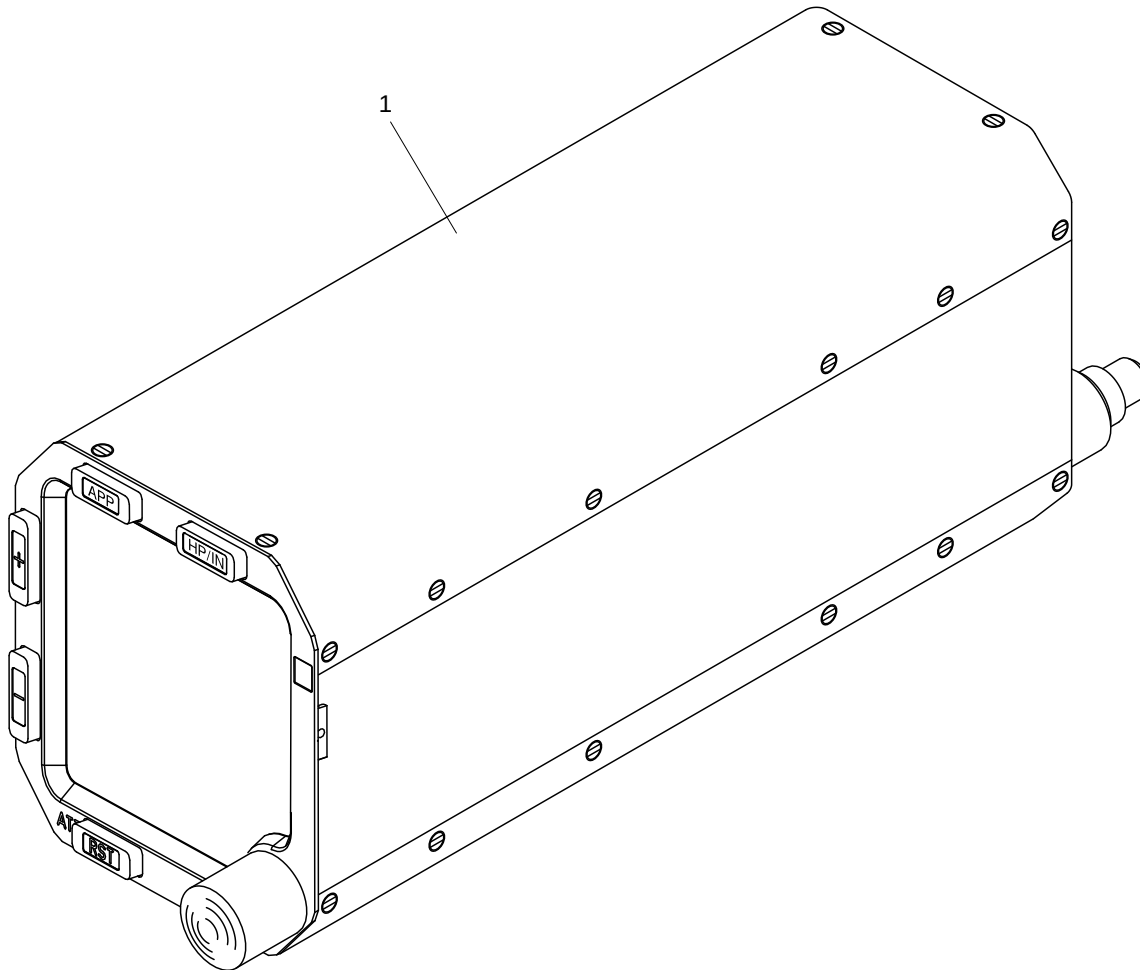
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PART NUMBER	AIRLINE STOCK NO.	FIGURE ITEM	TLA REQ
WCR0805-1KF		6	95
		6A	115
		8	200
WCR0805-21K5F		8	240
WCR0805-31R6F		8	275
WCR0805-5K11F		8	175
WCR0805-681RF		8	215
WCR08050R00F		5	50
		8	225

ISFD (INTEGTD STBY FLGHT DISPLAY)

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D_CM_TAV_342245_10_AAA0_01_00_

Figure 10001/Graphic 34-22-45-991-901-A01
(Figure 1) ISFD

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
1	1	C16221KA01	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) PRE SB C16221K-34-001 SEE 342245-2-1 FOR DET		RF
-	1A	C16221KA02	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO H,J TO N POST SB C16221K-34-001 SEE 342245-2-1A FOR DET		RF
-	1B	C16221LA01	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) PRE SB C16221L-34-001 SEE 342245-2A-1 FOR DET		RF
-	1C	C16221LA02	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) POST SB C16221L-34-001 PRE SB C16221L-34-002 SEE 342245-2A-1A FOR DET		RF
-	1D	C16221LA03	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO H,J TO N POST SB C16221L-34-002 SEE 342245-2A-1B FOR DET		RF
-	1E	C16221MA01	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO G SEE 342245-2B-1 FOR DET		RF
-	1F	C16221QA01	ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO H,J,K SEE 342245-2C-1 FOR DET		RF

- ITEM NOT ILLUSTRATED

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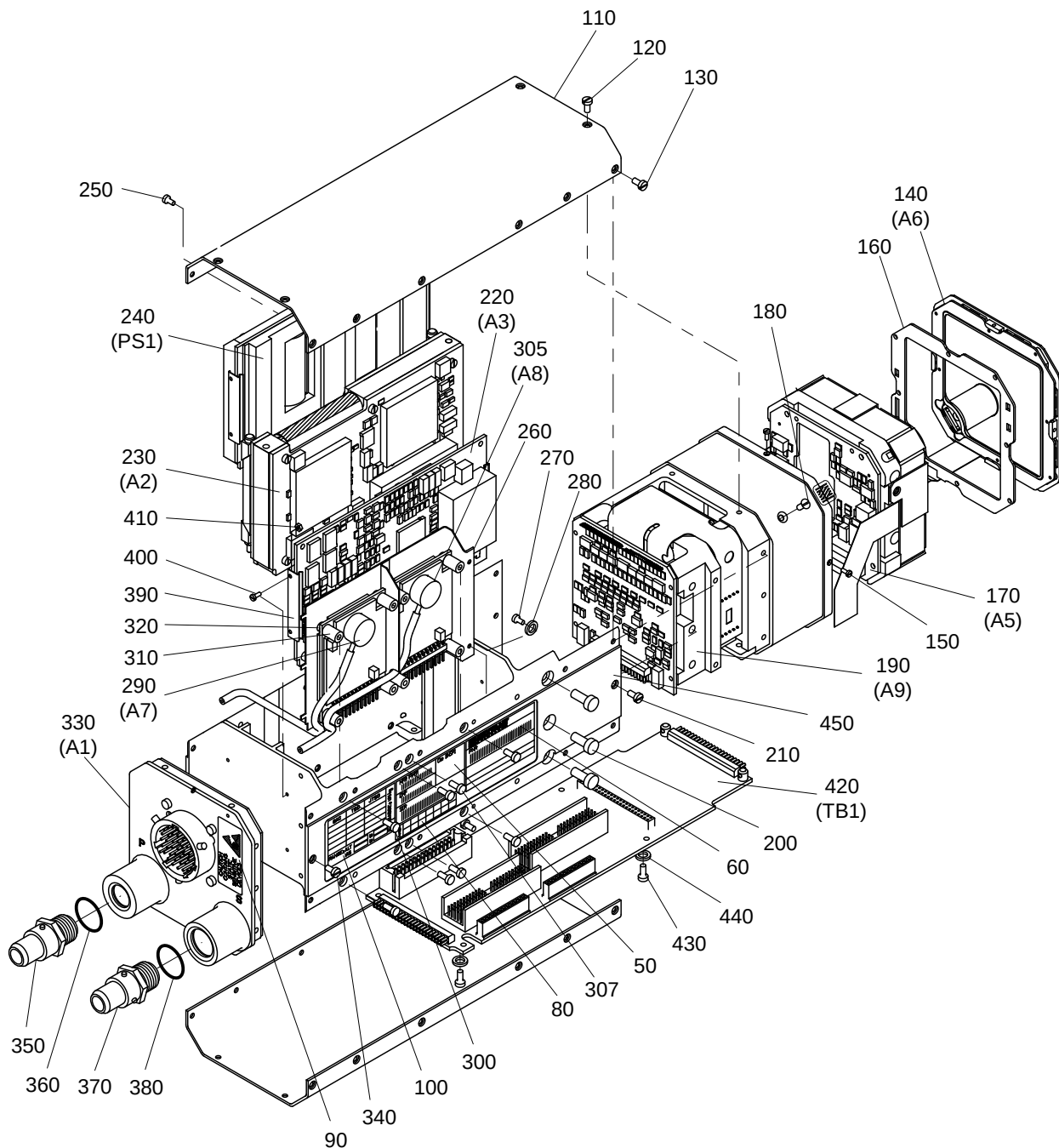
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ISFD (INTEGTD STBY FLGHT DISPLAY)

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Figure 10002/Graphic 34-22-45-991-902-A01
(Figure 2) ISFD

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2 - 1	C16221KA01		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) PRE SB C16221K-34-001 SEE 342245-1-1 FOR NHA	A	RF
- 1A	C16221KA02		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO H,J TO N POST SB C16221K-34-001 SEE 342245-1-1A FOR NHA	B	RF
- 8	F1407891		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221K-34-001	A	1
- 8A	F1403026		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221K-34-001	B	1
- 10	G3116AAG01		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221K-34-001	A	1
- 10A	G3116AAG03		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221K-34-001	B	1
- 20	G5614AAK01		..SOFTWARE CPU OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221K-34-001	A	1
- 20A	G5614AAK02		..SOFTWARE CPU OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221K-34-001	B	1
- 30	G5615AAK01		..SOFTWARE GSM OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221K-34-001	A	1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2 - 30A	G5615AAK02		..SOFTWARE GSM OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221K-34-001	B	1
- 40	G3156AAB01		..SOFTWARE IDENT REFER TO SPECIAL TOOLS PAGEBLOCK	NP	1
50	F2000383		.LABEL IDENTIFICATION PERMANENT BARCODE		1
60	F1400249		.LABEL PNR BARCODE PRE SB C16221K-34-001	A	1
- 60A	F1402858		.LABEL PNR BARCODE POST SB C16221K-34-001	B	1
- 70	F2000116		.PROTECTOR LABEL		1
80	F2000119		.LABEL MODIFICATION SUFFIX		1
90	F2000175		.LABEL STATIC SENSITIVE DEVICES		1
100	F1276554		.LABEL TSO/QAC		1
110	F1276705		.COVER PRE SB C16221K-34-015		2
-110A	F1276479		.COVER POST SB C16221K-34-015 ATTACHING PARTS	B	2
120	F1276484		.SCREW		8
-120A	F1408017		.SCREW COOKED	B	8
130	57008-370-1		.SCREW ***		8
140	E16364JA		.FACE FRONT ASSY (ESDS) PRE SB C16221K-34-005 PRE SB C16221K-34-010 SEE 342245-3-1 FOR DET		1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2 -140A	E16364JB		.FACE FRONT ASSY (ESDS) POST SB C16221K-34-005 PRE SB C16221K-34-010 SEE 342245-3-1A FOR DET	B	1
-140B	E16364JC		.FACE FRONT ASSY (ESDS) POST SB C16221K-34-010 SEE 342245-3-1B FOR DET	B	1
-140C	E38062AA		.FACE FRONT ASSY (ESDS) POST SB C16221K-34-010 SEE 342245-3-1C FOR DET	B	1
150	F1400012		ATTACHING PARTS .SCREW ***		4
160	F1276812		.GASKET		1
170	E24119CC		.OPTICAL ASSY (ESDS) PRE SB C16221K-34-007 SEE 342245-4-1 FOR DET		1
-170A	E24119FA		.OPTICAL ASSY (ESDS) POST SB C16221K-34-007 SEE 342245-4-1A FOR DET	B	1
180	45740023		ATTACHING PARTS .SCREW ***		4
190	E16371EA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS) PRE SB C16221K-34-015		1
-190A	E18520EA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS) POST SB C16221K-34-015	B	1

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2 200	A1248932		ATTACHING PARTS .SCREW MACH SST FIL H SOCK HEX-LOBE M4X10 ORDER OVERLGTH MPN E25111B40Q10ZHAPA MADE BY VF7411 OPT SEE TABLE		6
210	57008-370-1		.SCREW ***		2
220	E16485BB01		.PCB INTERFACE PROG (ESDS) PRE SB C16221K-34-002 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1
- 220A	E16485EA01		.PCB INTERFACE PROG (ESDS) POST SB C16221K-34-002 PRE SB C16221K-34-008 PRE SB C16221K-34-012 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	B	1
- 220B	E16485KA01		.PCB INTERFACE PROG (ESDS) POST SB C16221K-34-008 PRE SB C16221K-34-012 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	B	1
- 220C	E16485RA01		.PCB INTERFACE PROG (ESDS) PRE AMDMNT L PRE SB C16221K-34-014 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	B	1
- 220D	E16485UA01		.PCB INTERFACE PROG (ESDS) POST AMDMNT L POST SB C16221K-34-012 POST SB C16221K-34-014 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	B	1

- ITEM NOT ILLUSTRATED

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2 230	E16453HA01		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) PRE SB C16221K-34-002 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)		1
- 230A	E16453HA02		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) POST SB C16221K-34-002 PRE SB C16221K-34-009 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)	B	1
- 230B	E16453HB03		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) POST SB C16221K-34-009 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)	B	1
240	E17395AA		.PCB POWER SUPPLY (ESDS) PRE SB C16221K-34-004 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)		1
- 240A	E17395DA		.PCB POWER SUPPLY (ESDS) POST SB C16221K-34-004 PRE SB C16221K-34-011 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)	B	1
- 240B	E17395EA		.PCB POWER SUPPLY (ESDS) POST SB C16221K-34-011 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)	B	1
250	57001-711-1		ATTACHING PARTS .SCREW ***		4
260	F1279045		.SHIELDING ATTACHING PARTS		1
270	57000-972-1		.SCREW		2

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2 280	E25514B20ZTAPA		.WASHER FLAT SST DIA 2 OPT SEE TABLE (A1236272) ***	VF4796	2
290	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221K-34-006		1
- 290A	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221K-34-006 PRE SB C16221K-34-015	B	1
- 290B	F1407830		.SENSOR PRESSURE EQUIPPED (ESDS) POST SB C16221K-34-015 SEE 342245-10-1 FOR DET	B	1
300	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
305	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221K-34-006		1
- 305A	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221K-34-006 PRE SB C16221K-34-015	B	1
- 305B	E17397DA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221K-34-006 PRE SB C16221K-34-015	B	1
- 305C	F1407831		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221K-34-015 SEE 342245-10-1A FOR DET	B	1

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2 307	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
310	F1275310		.SPACER		8
320	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		1
330	E16387EA		.INTERCONNECTION REAR ASSY (ESDS) PRE SB C16221K-34-012 SEE 342245-5-1 FOR DET		1
- 330A	E16387JA		.INTERCONNECTION REAR ASSY (ESDS) POST SB C16221K-34-012 SEE 342245-5-1A FOR DET	B	1
340	57008-370-1		ATTACHING PARTS .SCREW ***		2
350	40006-1A44		.UNION QUICK DISCONNECT (A1204291)	V31441	1
360	57230-948-1		.RING SEALING		1
370	40006-1B46		.UNION QUICK DISCONNECT (A1179250)	V31441	1
380	57231-240-1		.RING SEALING		1
390	35-7B2-3-3		.GUIDE (A1250210)	V18915	4

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2 400	22293HD016004PA		ATTACHING PARTS .SCREW VF4796 MACH SST FIL H FLAT M1.6X4 OPT SEE TABLE (A1235516)		2
410	E25401B16QZHAPA		.NUT VF4796 HEX BRSS M1.6 OPT SEE TABLE (A1236144)		2
420	E16486BA01		*** .PCB MOTHER PROG (ESDS) PRE SB C16221K-34-005 SEE 342245-6-1 FOR DET		1
- 420A	E16486BB01		.PCB MOTHER PROG (ESDS) PRE AMDMNT G POST SB C16221K-34-005 SEE 342245-6A-1 FOR DET	B	1
- 420B	E16486JA01		.PCB MOTHER PROG (ESDS) POST AMDMNT G PRE SB C16221K-34-015 SEE 342245-6A-1A FOR DET	B	1
- 420C	E16981HA01		.PCB MOTHER PROG (ESDS) POST SB C16221K-34-015 SEE 342245-6B-1 FOR DET	B	1
430	57008-370-1		ATTACHING PARTS .SCREW PRE SB C16221K-34-005		4
- 430A	F1401166		.SCREW POST SB C16221K-34-005	B	4
440	520251-70		.WASHER VF0349 SPRING TENSION BRSS DIA 2.5 OPT SEE TABLE (A1236581)	A	4
450	F1276493		*** .STRUCTURE		1
- 450A	F1276474		.STRUCTURE	B	1
- 700	C16221KA01CND1		.DELETED (INCORRECT PART NUMBER)		

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2 - 700A	C16221KA02CND1		.DELETED (INCORRECT PART NUMBER)		
- 999	STORAGEEC16221K		.STORAGE ISFD SEE 342245-99-1F FOR DET	NP	1

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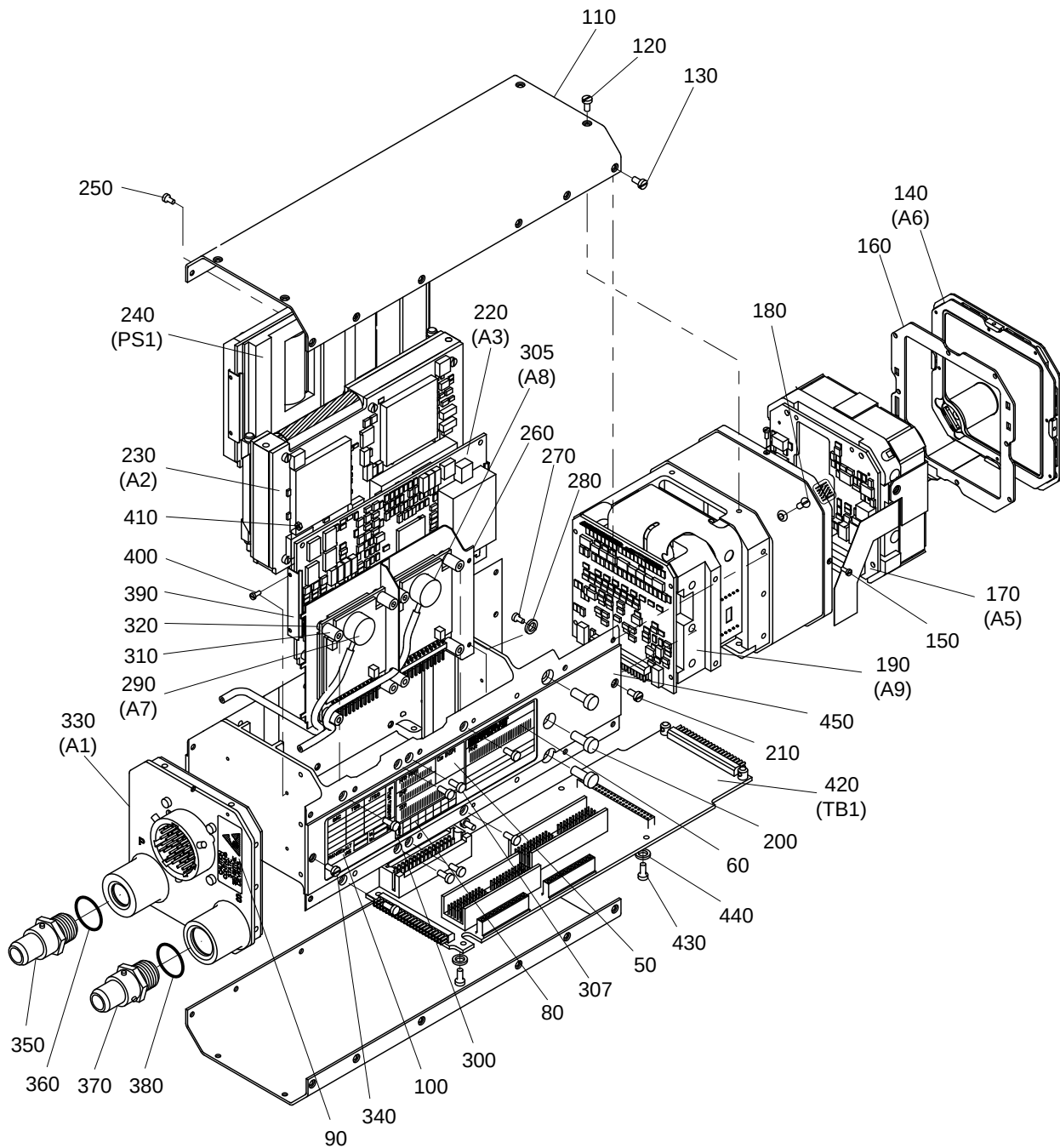
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ISFD (INTEGTD STBY FLGHT DISPLAY)

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Figure 10003/Graphic 34-22-45-991-903-A01
(Figure 2A) ISFD

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2A - 1	C16221LA01		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) PRE SB C16221L-34-001 SEE 342245-1-1B FOR NHA	A	RF
- 1A	C16221LA02		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) POST SB C16221L-34-001 PRE SB C16221L-34-002 SEE 342245-1-1C FOR NHA	B	RF
- 1B	C16221LA03		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO H,J TO N POST SB C16221L-34-002 SEE 342245-1-1D FOR NHA	C	RF
- 8	F1407892		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221L-34-001	A	1
- 8A	F1407893		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221L-34-001 PRE SB C16221L-34-002	B	1
- 8B	F1403029		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221L-34-002	C	1
- 10	G3116AAG01		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221L-34-001	A	1
- 10A	G3116AAG02		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221L-34-001 PRE SB C16221L-34-002	B	1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2A - 10B	G3116AAG03		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221L-34-002	C	1
- 20	G5614AAK01		..SOFTWARE CPU OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221L-34-002	AB	1
- 20A	G5614AAK02		..SOFTWARE CPU OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221L-34-002	C	1
- 30	G5615AAK01		..SOFTWARE GSM OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK PRE SB C16221L-34-002	AB	1
- 30A	G5615AAK02		..SOFTWARE GSM OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK POST SB C16221L-34-002	C	1
- 40	G3156AAB01		..SOFTWARE IDENT REFER TO SPECIAL TOOLS PAGEBLOCK	NP	1
50	F2000383		.LABEL IDENTIFICATION PERMANENT BARCODE		1
60	F1400249		.LABEL PNR BARCODE PRE SB C16221L-34-001	A	1
- 60A	F1401882		.LABEL PNR BARCODE POST SB C16221L-34-001 PRE SB C16221L-34-002	B	1
- 60B	F1402858		.LABEL PNR BARCODE POST SB C16221L-34-002	C	1
- 70	F2000116		.PROTECTOR LABEL		1
80	F2000119		.LABEL MODIFICATION SUFFIX		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2A 90	F2000175		.LABEL STATIC SENSITIVE DEVICES		1
100	F1276554		.LABEL TSO/QAC		1
110	F1276705		.COVER PRE SB C16221L-34-015		2
-110A	F1276479		.COVER POST SB C16221L-34-015	C	2
120	F1276484		ATTACHING PARTS		
-120A	F1408017		.SCREW COOKED	C	8
130	57008-370-1		.SCREW ***		8
140	E16364KA		.FACE FRONT ASSY (ESDS) PRE SB C16221L-34-005 SEE 342245-3A-1 FOR DET		1
-140A	E16364KB		.FACE FRONT ASSY (ESDS) POST SB C16221L-34-005 PRE SB C16221L-34-010 SEE 342245-3A-1A FOR DET	C	1
-140B	E16364KC		.FACE FRONT ASSY (ESDS) POST SB C16221L-34-010 SEE 342245-3A-1C FOR DET	C	1
150	F1400013		ATTACHING PARTS .SCREW ***		4
160	F1276812		.GASKET		1
170	E24119CC		.OPTICAL ASSY (ESDS) PRE SB C16221L-34-007 SEE 342245-4-1 FOR DET		1
-170A	E24119FA		.OPTICAL ASSY (ESDS) POST SB C16221L-34-007 SEE 342245-4-1A FOR DET	C	1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2A 180	45740023		ATTACHING PARTS .SCREW ***		4
190	E16371DA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS) PRE SB C16221L-34-015		1
-190A	E18520FA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS) POST SB C16221L-34-015	C	1
200	A1248932		ATTACHING PARTS .SCREW MACH SST FIL H SOCK HEX-LOBE M4X10 ORDER OVERLGTH MPN E25111B40Q10ZHAPA MADE BY VF7411 OPT SEE TABLE		6
210	57008-370-1		.SCREW ***		2
220	E16485BB01		.PCB INTERFACE PROG (ESDS) PRE SB C16221L-34-003 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1
-220A	E16485EA01		.PCB INTERFACE PROG (ESDS) POST SB C16221L-34-003 PRE SB C16221L-34-008 PRE SB C16221L-34-012 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	C	1
-220B	E16485KA01		.PCB INTERFACE PROG (ESDS) POST SB C16221L-34-008 PRE SB C16221L-34-012 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	C	1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2A - 220C	E16485RA01		.PCB INTERFACE PROG (ESDS) PRE AMDMNT L PRE SB C16221L-34-014 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	C	1
- 220D	E16485UA01		.PCB INTERFACE PROG (ESDS) POST AMDMNT L POST SB C16221L-34-012 POST SB C16221L-34-014 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)	C	1
230	E16453HA01		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) PRE SB C16221L-34-003 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)		1
- 230A	E16453HA02		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) POST SB C16221L-34-003 PRE SB C16221L-34-009 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)	C	1
- 230B	E16453HB03		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) POST SB C16221L-34-009 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)	C	1
240	E17395AA		.PCB POWER SUPPLY (ESDS) PRE SB C16221L-34-004 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)		1
- 240A	E17395DA		.PCB POWER SUPPLY (ESDS) POST SB C16221L-34-004 PRE SB C16221L-34-011 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)	C	1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2A - 240B	E17395EA		.PCB POWER SUPPLY (ESDS) POST SB C16221L-34-011 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)	C	1
250	57001-711-1		ATTACHING PARTS .SCREW ***		4
260	F1279045		.SHIELDING ATTACHING PARTS		1
270	57000-972-1		.SCREW		2
280	E25514B20ZTAPA		.WASHER FLAT SST DIA 2 OPT SEE TABLE (A1236272) *** VF4796		2
290	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221L-34-006		1
- 290A	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221L-34-006 PRE SB C16221L-34-015	C	1
- 290B	F1407830		.SENSOR PRESSURE EQUIPPED (ESDS) POST SB C16221L-34-015 SEE FIG 10 FOR DET	C	1
300	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
305	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221L-34-006		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2A - 305A	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221L-34-006 PRE SB C16221L-34-015	C	1
- 305B	E17397DA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221L-34-006 PRE SB C16221L-34-015	C	1
- 305C	F1407831		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221L-34-015 SEE FIG 10 FOR DET	C	1
307	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
310	F1275310		.SPACER ATTACHING PARTS		8
320	A1235962		.SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		1
330	E16387EA		.INTERCONNECTION REAR ASSY (ESDS) PRE SB C16221L-34-012 SEE 342245-5-1 FOR DET		1
- 330A	E16387JA		.INTERCONNECTION REAR ASSY (ESDS) POST SB C16221L-34-012 SEE 342245-5-1A FOR DET	C	1
340	57008-370-1		ATTACHING PARTS .SCREW ***		2

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2A 350	40006-1A44		.UNION QUICK DISCONNECT (A1204291)	V31441	1
360	57230-948-1		.RING SEALING		1
370	40006-1B46		.UNION QUICK DISCONNECT (A1179250)	V31441	1
380	57231-240-1		.RING SEALING		1
390	35-7B2-3-3		.GUIDE (A1250210)	V18915	4
400	22293HD016004PA		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M1.6X4 OPT SEE TABLE (A1235516)	VF4796	2
410	E25401B16QZHAPA		.NUT HEX BRSS M1.6 OPT SEE TABLE (A1236144) ***	VF4796	2
420	E16486BA01		.PCB MOTHER PROG (ESDS) PRE SB C16221L-34-005 SEE 342245-6-1 FOR DET		1
- 420A	E16486BB01		.PCB MOTHER PROG (ESDS) PRE AMDMNT G POST SB C16221L-34-005 SEE 342245-6A-1 FOR DET	C	1
- 420B	E16486JA01		.PCB MOTHER PROG (ESDS) POST AMDMNT G PRE SB C16221L-34-015 SEE 342245-6A-1A FOR DET	C	1
- 420C	E16981HA01		.PCB MOTHER PROG (ESDS) POST SB C16221L-34-015 SEE FIG 6B FOR DET	C	1

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2A 430	57008-370-1		ATTACHING PARTS		4
- 430A	F1401166		.SCREW PRE SB C16221L-34-005	C	4
440	520251-70		.SCREW POST SB C16221L-34-005	AB	4
			.WASHER VF0349 SPRING TENSION BRSS DIA 2.5 OPT SEE TABLE (A1236581)		

450	F1276493		.STRUCTURE		1
- 450A	F1276474		.STRUCTURE	C	1
- 700	C16221LA01CND1		.DELETED (INCORRECT PART NUMBER)		
- 700A	C16221LA02CND1		.DELETED (INCORRECT PART NUMBER)		
- 700B	C16221LA03CND1		.DELETED (INCORRECT PART NUMBER)		
- 999	STORAGEC16221K		.STORAGE NP ISFD SEE 342245-99-1F FOR DET		1

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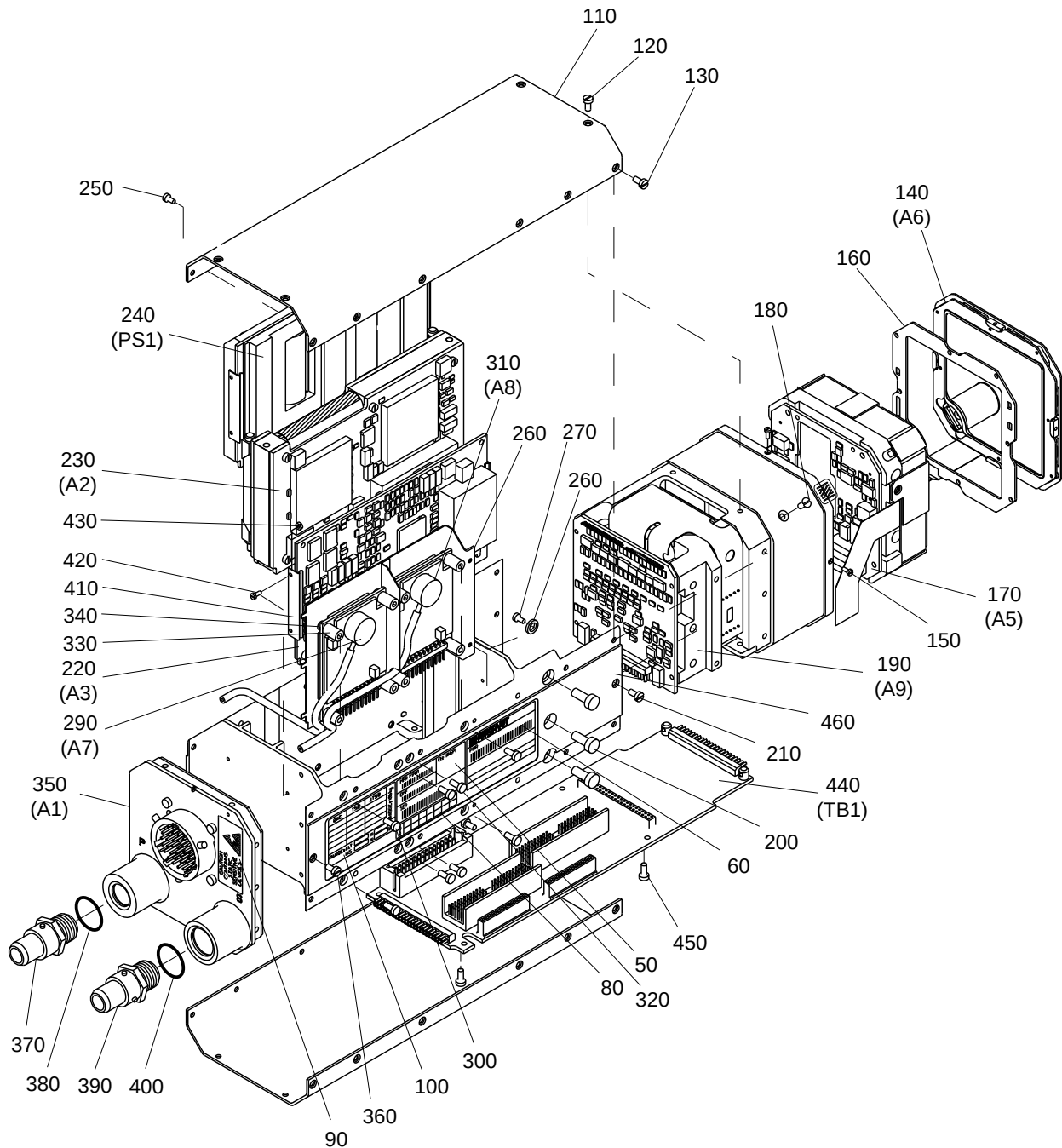
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ISFD (INTEGTD STBY FLGHT DISPLAY)

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Figure 10004/Graphic 34-22-45-991-904-A01
(Figure 2B) ISFD

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2B- 1	C16221MA01		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO G SEE 342245-1-1E FOR NHA		RF
- 8	F1403675		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 10	G3116AAG03		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 20	G5614AAK02		..SOFTWARE CPU OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 30	G5615AAK02		..SOFTWARE GSM OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 40	G3156AAB01		..SOFTWARE IDENT REFER TO SPECIAL TOOLS PAGEBLOCK	NP	1
50	F2000383		.LABEL IDENTIFICATION PERMANENT BARCODE		1
60	F1402858		.LABEL PNR BARCODE		1
- 70	F2000116		.PROTECTOR LABEL		1
80	F2000119		.LABEL MODIFICATION SUFFIX		1
90	F2000175		.LABEL STATIC SENSITIVE DEVICES		1
100	F1276554		.LABEL TSO/QAC		1
110	F1276705		.COVER ATTACHING PARTS		2
120	F1276484		.SCREW		8
- 120A	F1408017		.SCREW COOKED		8
130	57008-370-1		.SCREW ***		8

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2B 140	E16364JB		.FACE FRONT ASSY (ESDS) PRE SB C16221M-34-005 SEE 342245-3-1A FOR DET		1
- 140A	E16364JC		.FACE FRONT ASSY (ESDS) POST SB C16221M-34-005 SEE 342245-3-1B FOR DET		1
150	F1400012		ATTACHING PARTS .SCREW ***		4
160	F1276812		.GASKET		1
170	E24119CC		.OPTICAL ASSY (ESDS) PRE SB C16221M-34-002 SEE 342245-4-1 FOR DET		1
- 170A	E24119FA		.OPTICAL ASSY (ESDS) POST SB C16221M-34-002 SEE 342245-4-1A FOR DET		1
180	45740023		ATTACHING PARTS .SCREW ***		4
190	E16371GA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS)		1
200	A1248932		ATTACHING PARTS .SCREW MACH SST FIL H SOCK HEX-LOBE M4X10 ORDER OVERLGTH MPN E25111B40Q10ZHAPA MADE BY VF7411 OPT SEE TABLE		6
210	57008-370-1		.SCREW ***		2
220	E16485EA01		.PCB INTERFACE PROG (ESDS) PRE SB C16221M-34-003 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2B - 220A	E16485KA01		.PCB INTERFACE PROG (ESDS) POST SB C16221M-34-003 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1
230	E16453HA02		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) PRE SB C16221M-34-004 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)		1
- 230A	E16453HB03		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) POST SB C16221M-34-004 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)		1
240	E17395DA		.PCB POWER SUPPLY (ESDS) PRE SB C16221M-34-006 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)		1
- 240A	E17395EA		.PCB POWER SUPPLY (ESDS) POST SB C16221M-34-006 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)		1
250	57001-711-1		ATTACHING PARTS .SCREW ***		4
260	F1279045		.SHIELDING ATTACHING PARTS		1
270	57000-972-1		.SCREW		2
280	E25514B20ZTAPA		.WASHER FLAT SST DIA 2 OPT SEE TABLE (A1236272) ***	VF4796	2
290	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221M-34-001		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2B - 290A	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221M-34-001 ATTACHING PARTS		1
300	A1235962		.SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
310	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221M-34-001		1
- 310A	E17397BA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221M-34-001		1
- 310B	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221M-34-001		1
- 310C	E17397DA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221M-34-001 ATTACHING PARTS		1
320	A1235962		.SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
330	F1275310		.SPACER ATTACHING PARTS		8
340	A1235962		.SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2B 350	E16387EA		.INTERCONNECTION REAR ASSY (ESDS) SEE 342245-5-1 FOR DET		1
360	57008-370-1		ATTACHING PARTS .SCREW ***		2
370	40006-1A44		.UNION QUICK DISCONNECT (A1204291)	V31441	1
380	57230-948-1		.RING SEALING		1
390	40006-1B46		.UNION QUICK DISCONNECT (A1179250)	V31441	1
400	57231-240-1		.RING SEALING		1
410	35-7B2-3-3		.GUIDE (A1250210)	V18915	4
420	22293HD016004PA		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M1.6X4 OPT SEE TABLE (A1235516)	VF4796	2
430	E25401B16QZHAPA		.NUT HEX BRSS M1.6 OPT SEE TABLE (A1236144) ***	VF4796	2
440	E16486BB01		.PCB MOTHER PROG (ESDS) PRE AMDMNT D SEE 342245-6A-1 FOR DET		1
- 440A	E16486JA01		.PCB MOTHER PROG (ESDS) POST AMDMNT D SEE 342245-6A-1A FOR DET		1
450	F1401166		ATTACHING PARTS .SCREW ***		4
460	F1276493		.STRUCTURE		1
- 460A	F1276474		.STRUCTURE		1
- 700	C16221MA01CND1		.DELETED (INCORRECT PART NUMBER)		

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2B- 999	STORAGEEC16221K		.STORAGE ISFD SEE 342245-99-1F FOR DET	NP	1

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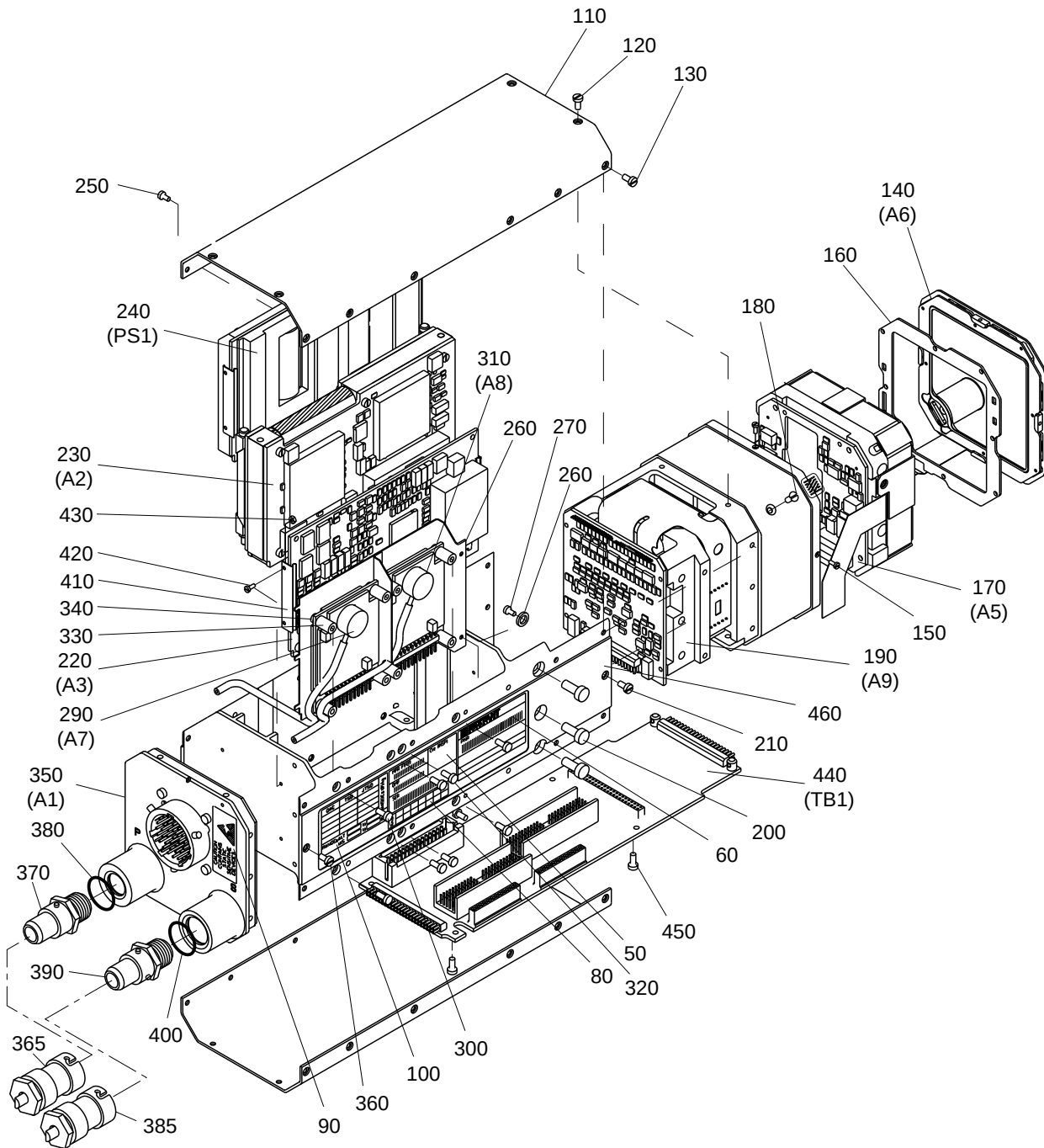
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ISFD (INTEGTD STBY FLGHT DISPLAY)

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Figure 10005/Graphic 34-22-45-991-905-A01
(Figure 2C) ISFD

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2C - 1	C16221QA01		ISFD (INTEGTD STBY FLGHT DISPLAY) (ESDS) AMDMNT A TO H,J,K SEE 342245-1-1F FOR NHA		RF
- 8	F1404705		.DISKETTE SOFTWARE ISFD REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 10	G3116AAK01		..SOFTWARE CONFIG REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 20	G5614AAL02		..SOFTWARE CPU OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 30	G5615AAL01		..SOFTWARE GSM OPERATL REFER TO SPECIAL TOOLS PAGEBLOCK		1
- 40	G3156AAB01		..SOFTWARE IDENT REFER TO SPECIAL TOOLS PAGEBLOCK	NP	1
50	F2000383		.LABEL IDENTIFICATION PERMANENT BARCODE		1
60	F1402858		.LABEL PNR BARCODE		1
- 70	F2000116		.PROTECTOR LABEL		1
80	F2000119		.LABEL MODIFICATION SUFFIX		1
90	F2000175		.LABEL STATIC SENSITIVE DEVICES		1
100	F1276554		.LABEL TSO/QAC		1
110	F1276705		.COVER PRE SB C16221Q-34-009		2
- 110A	F1276479		.COVER POST SB C16221Q-34-009 ATTACHING PARTS		2
120	F1276484		.SCREW		8
- 120A	F1408017		.SCREW COOKED		8

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2C 130	57008-370-1		.SCREW ***		8
140	E16364KB		.FACE FRONT ASSY (ESDS) PRE SB C16221Q-34-005 SEE 342245-3A-1A FOR DET		1
-140A	E16364KC		.FACE FRONT ASSY (ESDS) POST SB C16221Q-34-005 SEE 342245-3A-1B FOR DET		1
150	F1400013		ATTACHING PARTS .SCREW ***		4
160	F1276812		.GASKET		1
170	E24119CC		.OPTICAL ASSY (ESDS) PRE SB C16221Q-34-002 SEE 342245-4-1 FOR DET		1
-170A	E24119FA		.OPTICAL ASSY (ESDS) POST SB C16221Q-34-002 SEE 342245-4-1A FOR DET		1
180	45740023		ATTACHING PARTS .SCREW ***		4
190	E16371DA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS) PRE SB C16221Q-34-009		1
-190A	E18520FA		.INERTIAL MEASUREMENT UNIT (IMU) (ESDS) POST SB C16221Q-34-009		1
200	A1248932		ATTACHING PARTS .SCREW MACH SST FIL H SOCK HEX-LOBE M4X10 ORDER OVERLGTH MPN E25111B40Q10ZHAPA MADE BY VF7411 OPT SEE TABLE		6

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2C 210	57008-370-1		.SCREW ***		2
220	E16485GA01		.PCB INTERFACE PROG (ESDS) PRE SB C16221Q-34-003 PRE SB C16221Q-34-007 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1
- 220A	E16485GA02		.PCB INTERFACE PROG (ESDS) POST SB C16221Q-34-003 PRE SB C16221Q-34-007 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1
- 220B	E16485HA01		.PCB INTERFACE PROG (ESDS) POST SB C16221Q-34-007 REFER TO CMM 34-22-26 (VENDOR CODE VF9111)		1
230	E16453MA01		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT (ESDS) PRE SB C16221Q-34-004 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)		1
- 230A	E16453MA02		.CONTROL PROCESSOR GRAPHIC DISPLAY UNIT POST SB C16221Q-34-004 REFER TO CMM 34-22-27 (VENDOR CODE VF9111)		1
240	E17395DA		.PCB POWER SUPPLY (ESDS) PRE SB C16221Q-34-006 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)		1
- 240A	E17395EA		.PCB POWER SUPPLY (ESDS) POST SB C16221Q-34-006 REFER TO CMM 34-22-28 (VENDOR CODE VF9111)		1

- ITEM NOT ILLUSTRATED

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2C 250	57001-711-1		ATTACHING PARTS .SCREW ***		4
260	F1279045		.SHIELDING		1
270	57000-972-1		ATTACHING PARTS .SCREW		2
280	E25514B20ZTAPA		.WASHER FLAT SST DIA 2 OPT SEE TABLE (A1236272) ***	VF4796	2
290	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221Q-34-001		1
-290A	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221Q-34-001 PRE SB C16221Q-34-009		1
-290B	F1407830		.SENSOR PRESSURE EQUIPPED (ESDS) POST SB C16221Q-34-009 SEE FIG 10 FOR DET		1
300	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
310	E17397AA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221Q-34-001		1
-310A	E17397BA		.SENSOR PRESSURE ASSY (ESDS) PRE SB C16221Q-34-001		1
-310B	E17397CA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221Q-34-001 PRE SB C16221Q-34-009		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
2C-310C	E17397DA		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221Q-34-001 PRE SB C16221Q-34-009		1
-310D	F1407831		.SENSOR PRESSURE ASSY (ESDS) POST SB C16221Q-34-009 SEE FIG 10 FOR DET		1
320	A1235962		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		4
330	F1275310		.SPACER ATTACHING PARTS		8
340	A1235962		.SCREW MACH SST FIL H FLAT M2.5X5 ORDER OVERLGTH MPN E25127A25Q5ZHAPA MADE BY VF4796 OPT SEE TABLE ***		1
350	E16387EA		.INTERCONNECTION REAR ASSY (ESDS) PRE SB C16221Q-34-007 SEE 342245-5-1 FOR DET		1
-350A	E16387JA		.INTERCONNECTION REAR ASSY (ESDS) POST SB C16221Q-34-007 SEE 342245-5-1A FOR DET		1
360	57008-370-1		ATTACHING PARTS .SCREW ***		2
365	1QF3-2		.CAP PRESSURE (A2002934)	V24984	1
370	40006-1A44		.UNION QUICK DISCONNECT (A1204291)	V31441	1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
2C 380	57230-948-1		.RING SEALING		1
385	1QF3-3		.CAP PRESSURE (A2002933)	V24984	1
390	40006-1B46		.UNION QUICK DISCONNECT (A1179250)	V31441	1
400	57231-240-1		.RING SEALING		1
410	35-7B2-3-3		.GUIDE (A1250210)	V18915	4
420	22293HD016004PA		ATTACHING PARTS .SCREW MACH SST FIL H FLAT M1.6X4 OPT SEE TABLE (A1235516)	VF4796	2
430	E25401B16QZHAPA		.NUT HEX BRSS M1.6 OPT SEE TABLE (A1236144) ***	VF4796	2
440	E16486BB01		.PCB MOTHER PROG (ESDS) PRE AMDMNT E SEE 342245-6A-1 FOR DET		1
- 440A	E16486JA01		.PCB MOTHER PROG (ESDS) POST AMDMNT E PRE SB C16221Q-34-009 SEE 342245-6A-1A FOR DET		1
- 440B	E16981HA01		.PCB MOTHER PROG (ESDS) POST SB C16221Q-34-009 SEE FIG 6B FOR DET		1
450	F1401166		ATTACHING PARTS .SCREW ***		4
460	F1276493		.STRUCTURE		1
- 460A	F1276474		.STRUCTURE		1
- 999	STORAGEC16221Q		.STORAGE ISFD SEE 342245-99-1G FOR DET	NP	1

- ITEM NOT ILLUSTRATED

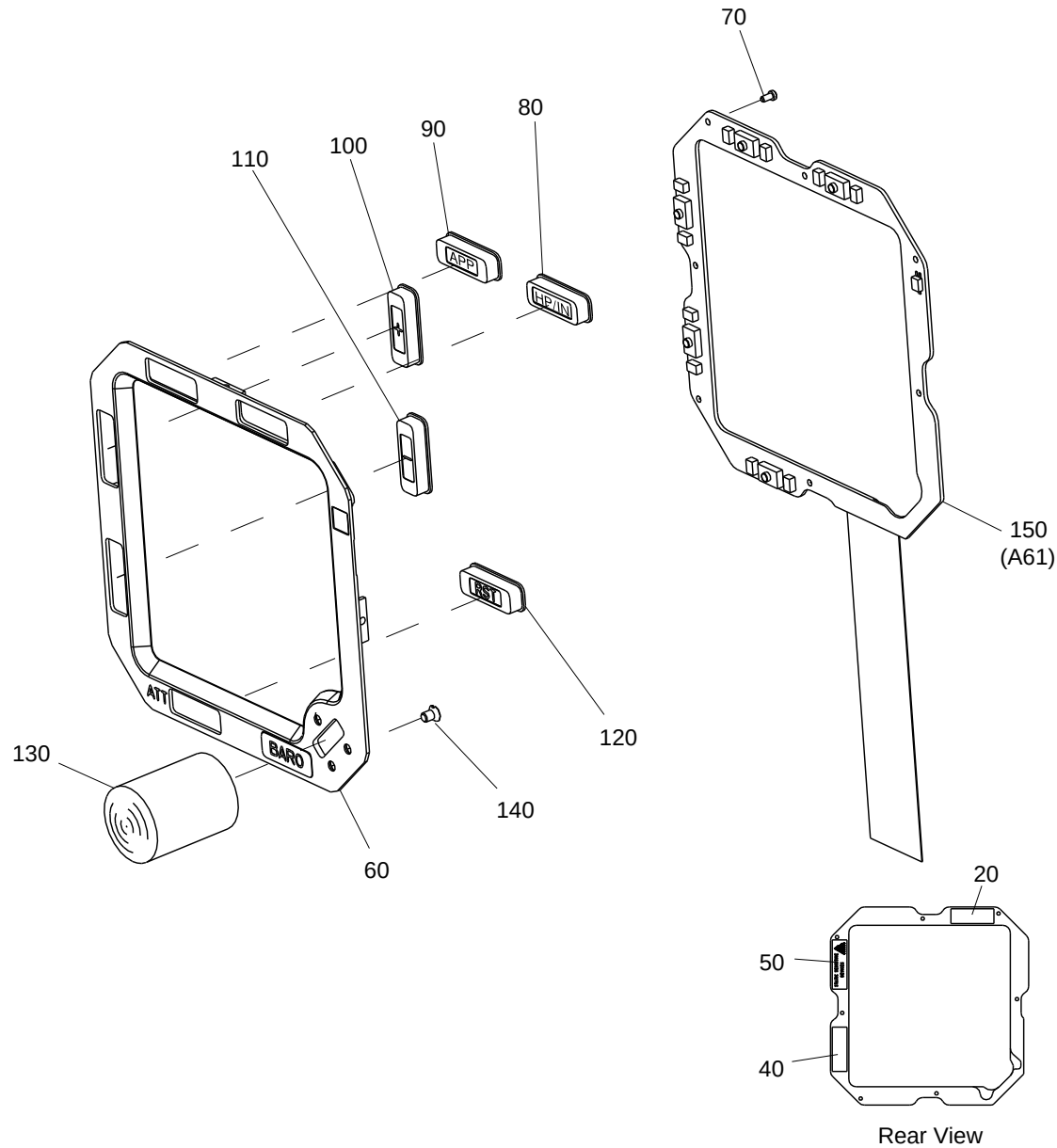
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FACE FRONT ASSY

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D_CM_TAV_342245_10_AAF0_01_00

Figure 10006/Graphic 34-22-45-991-906-A01
(Figure 3) FACE, FRONT FACE (E16364J.)

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
3 - 1	E16364JA		FACE FRONT ASSY (ESDS) AMDMNT A PRE SB C16221K-34-010 PRE SB C16221K-34-005 SEE 342245-2-140 FOR NHA	A	RF
- 1A	E16364JB		FACE FRONT ASSY (ESDS) AMDMNT A POST SB C16221K-34-005 PRE SB C16221K-34-010 PRE SB C16221M-34-005 SEE 342245-2-140A FOR NHA SEE 342245-2B-140 FOR NHA	B	RF
- 1B	E16364JC		FACE FRONT ASSY (ESDS) AMDMNT A,B POST SB C16221K-34-010 POST SB C16221M-34-005 SEE 342245-2-140B FOR NHA SEE 342245-2B-140A FOR NHA	C	RF
- 1C	E38062AA		FACE FRONT ASSY (ESDS) AMDMNT A POST SB C16221K-34-010 SEE 342245-2-140C FOR NHA	D	RF
- 10	F2000068		.PROTECTOR LABEL		1
20	F2000071		.LABEL IDENTIFICATION		1
- 30	F2000100		.PROTECTOR LABEL		1
40	F2000247		.LABEL MODIFICATION SUFFIX SUPSD BY ITEM 40A	AB	1
- 40A	F2000103		.LABEL MODIFICATION SUFFIX SUPSDS ITEM 40		1
50	F2000190		.LABEL STATIC SENSITIVE DEVICES		1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
3	60	F1279326	.FACE FRONT SUPSD BY ITEM 60A	ABD	1
-	60A	F1406777	.FACE FRONT SUPSDS ITEM 60	BC	1
	70	57000-520-1	ATTACHING PARTS .SCREW SUPSD BY ITEM 70A	ABD	7
-	70A	57000-491-1	.SCREW SUPSDS ITEM 70 ***	BC	7
	80	F1279528	.KEY HP/IN		1
	90	F1278453	.KEY APP		1
	100	F1276069	.KEY PLUS		1
	110	F1276070	.KEY MINUS		1
	120	F1276834	.KEY RST		1
	130	E16441BA	.ENCODER PRE AMDMNT A PRE SB C16221K-34-010 PRE SB C16221M-34-005	AB	1
-	130A	E16441GA	.ENCODER POST AMDMNT A PRE SB C16221K-34-010 PRE SB C16221M-34-005	AB	1
-	130B	E16441JA	.ENCODER PRE SB C16221K-34-010 PRE SB C16221M-34-005	C	1
-	130C	E16441JB	.ENCODER POST SB C16221K-34-010 POST SB C16221M-34-005	CD	1
	140	A1204438	ATTACHING PARTS .SCREW MACH SST FLAT H 90CSK M2X3 ORDER OVERLGTH MPN E25123B20Q3ZEAPA MADE BY VF7411 OPT SEE TABLE SUPSD BY ITEM 140A		3

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
3 -140A	A1232078		.SCREW MACH SST FLAT H 90CSK M2X5 ORDER OVERLGTH MPN E25123B20Q5ZEAPA MADE BY VF4796 OPT SEE TABLE SUPSDS ITEM 140 ***		3
150	E16389JA		.PCB FRONT FACE PRE SB C16221K-34-005 SEE 342245-7-1 FOR DET	AD	1
-150A	E16389JB		.PCB FRONT FACE POST SB C16221K-34-005 SEE 342245-7-1A FOR DET	BC	1

- ITEM NOT ILLUSTRATED

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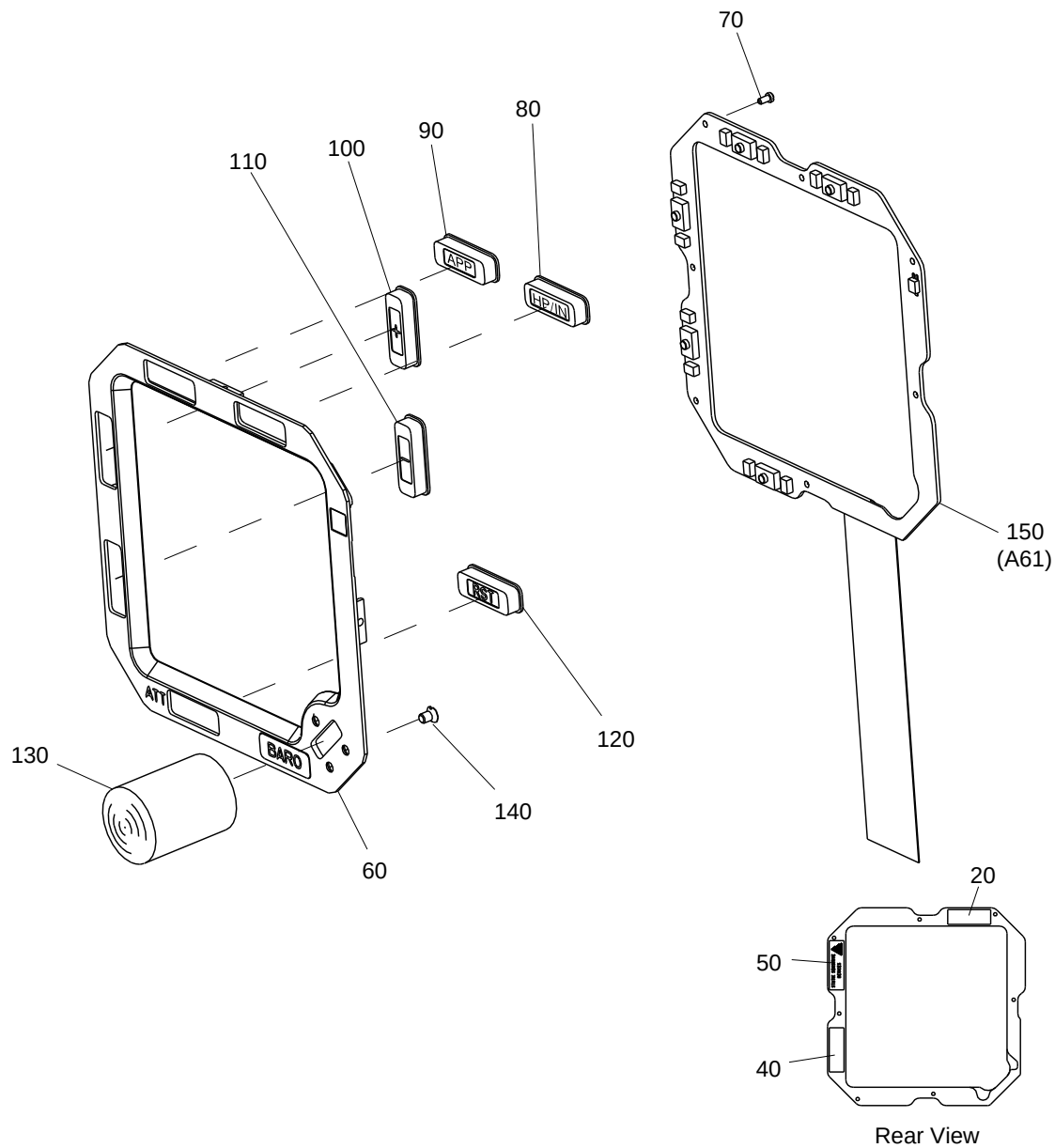
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FACE FRONT ASSY

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C16221K



D_CM_TAV_342245_10_AAG0_01_00

Figure 10007/Graphic 34-22-45-991-907-A01
(Figure 3A) FACE, FRONT FACE (E16364K.)

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
3A - 1	E16364KA		FACE FRONT ASSY (ESDS) AMDMNT A PRE SB C16221L-34-005 SEE 342245-2A-140 FOR NHA	A	RF
- 1A	E16364KB		FACE FRONT ASSY (ESDS) AMDMNT A POST SB C16221L-34-005 PRE SB C16221L-34-010 PRE SB C16221Q-34-005 SEE 342245-2A-140A FOR NHA SEE 342245-2C-140 FOR NHA	B	RF
- 1C	E16364KC		FACE FRONT ASSY (ESDS) AMDMNT A,B POST SB C16221L-34-010 POST SB C16221Q-34-005 SEE 342245-2A-140B FOR NHA SEE 342245-2C-140A FOR NHA	C	RF
- 10	F2000068		.PROTECTOR LABEL		1
20	F2000071		.LABEL IDENTIFICATION		1
- 30	F2000100		.PROTECTOR LABEL		1
40	F2000247		.LABEL MODIFICATION SUFFIX SUPSD BY ITEM 40A	AB	1
- 40A	F2000103		.LABEL MODIFICATION SUFFIX SUPSDS ITEM 40		1
50	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
60	F1279712		.FACE FRONT SUPSD BY ITEM 60A		1
- 60A	F1406778		.FACE FRONT SUPSDS ITEM 60 ATTACHING PARTS	BC	1
70	57000-520-1		.SCREW SUPSD BY ITEM 70A		7

- ITEM NOT ILLUSTRATED

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
3A - 70A	57000-491-1		.SCREW SUPSDS ITEM 70 ***	BC	7
80	F1279528		.KEY HP/IN		1
90	F1278453		.KEY APP		1
100	F1276069		.KEY PLUS		1
110	F1276070		.KEY MINUS		1
120	F1276834		.KEY RST		1
130	E16441DA		.ENCODER PRE AMDMNT A PRE SB C16221L-34-010	AB	1
- 130A	E16441HA		.ENCODER POST AMDMNT A PRE SB C16221L-34-010 PRE SB C16221Q-34-005	AB	1
- 130B	E16441MA		.ENCODER PRE SB C16221L-34-010 PRE SB C16221Q-34-005	C	1
- 130C	E16441MB		.ENCODER POST SB C16221L-34-010 POST SB C16221Q-34-005	C	1
140	A1204438		ATTACHING PARTS .SCREW MACH SST FLAT H 90CSK M2X3 ORDER OVERLGTH MPN E25123B20Q3ZEAPA MADE BY VF7411 OPT SEE TABLE SUPSD BY ITEM 140A		3
- 140A	A1232078		.SCREW MACH SST FLAT H 90CSK M2X5 ORDER OVERLGTH MPN E25123B20Q5ZEAPA MADE BY VF4796 OPT SEE TABLE SUPSDS ITEM 140 ***		3
150	E16389JA		.PCB FRONT FACE PRE SB C16221L-34-005 SEE 342245-7-1 FOR DET	A	1

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
3A - 150A	E16389JB		.PCB FRONT FACE POST SB C16221L-34-005 SEE 342245-7-1A FOR DET	BC	1

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OPTICAL ASSY

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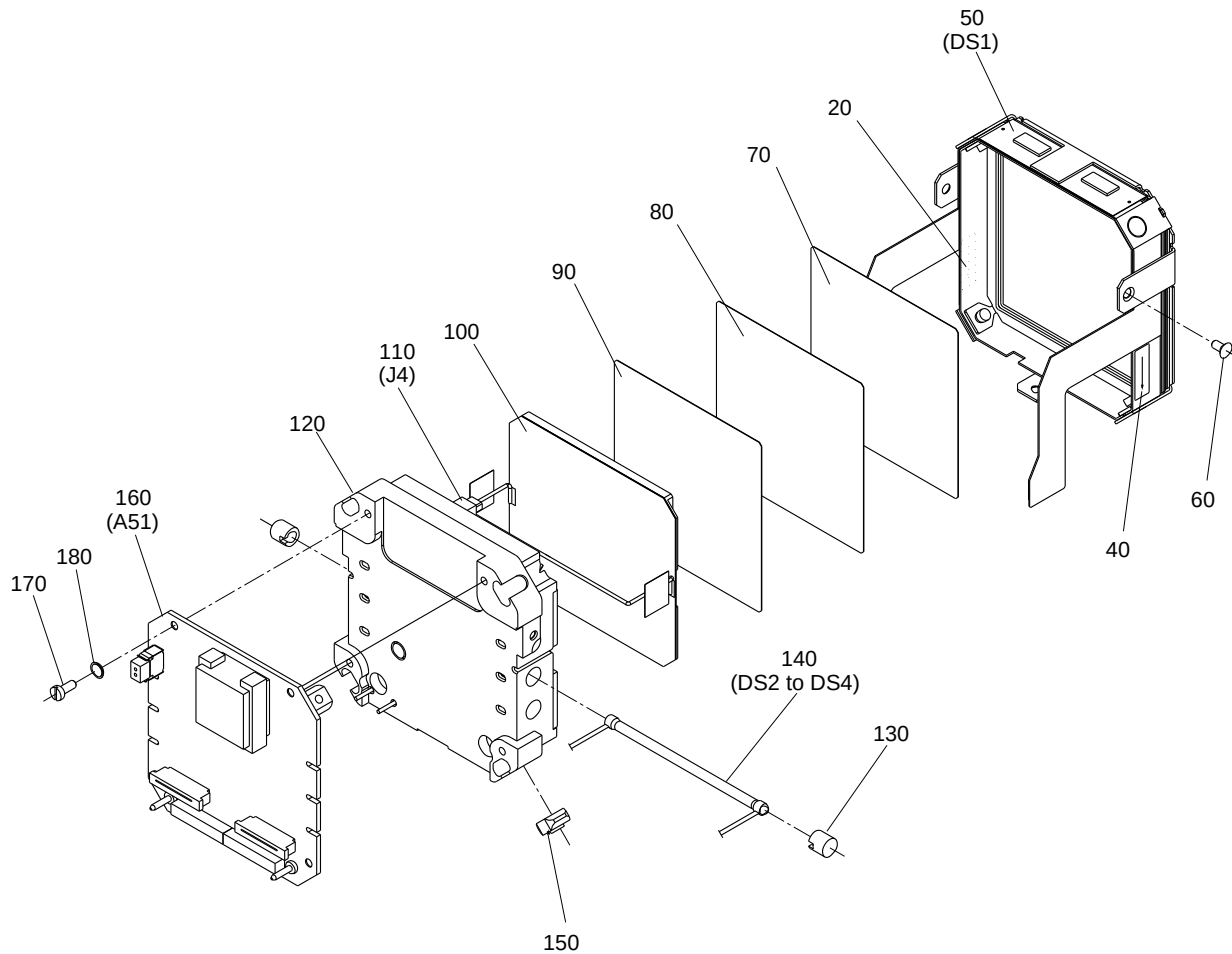
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Figure 10008/Graphic 34-22-45-991-908-A01
(Figure 4) OPTICAL, ASSY

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
4 - 1	E24119CC		OPTICAL ASSY (ESDS) PRE SB C16221K-34-007 PRE SB C16221L-34-007 PRE SB C16221M-34-002 PRE SB C16221Q-34-002 SEE 342245-2-170 FOR NHA SEE 342245-2A-170 FOR NHA SEE 342245-2B-170 FOR NHA SEE 342245-2C-170 FOR NHA	A	RF
- 1A	E24119FA		OPTICAL ASSY (ESDS) AMDMNT A POST SB C16221K-34-007 POST SB C16221L-34-007 POST SB C16221M-34-002 POST SB C16221Q-34-002 SEE 342245-2-170A FOR NHA SEE 342245-2A-170A FOR NHA SEE 342245-2B-170A FOR NHA SEE 342245-2C-170A FOR NHA	B	RF
- 10	F2000068		. PROTECTOR LABEL		1
20	F2000071		. LABEL IDENTIFICATION		1
- 30	F2000100		. PROTECTOR LABEL		1
40	F2000247		. LABEL MODIFICATION SUFFIX PRE SB C16221K-34-007 PRE SB C16221L-34-007 PRE SB C16221M-34-002 PRE SB C16221Q-34-002	A	1
- 40A	F2000103		. LABEL MODIFICATION SUFFIX POST SB C16221K-34-007 POST SB C16221L-34-007 POST SB C16221M-34-002 POST SB C16221Q-34-002	B	1
50	E19947BA		. DISPLAY PRE SB C16221K-34-013 PRE SB C16221L-34-013 PRE SB C16221M-34-007 PRE SB C16221Q-34-008		1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
4 - 50A	E19947GA		.DISPLAY POST SB C16221K-34-013 POST SB C16221L-34-013 POST SB C16221M-34-007 POST SB C16221Q-34-008 ATTACHING PARTS	B	1
60	57040-430-1		.SCREW ***		3
70	F1380515		.POLARIZER DELETED BY SB C16221K-34-013 DELETED BY SB C16221L-34-013 DELETED BY SB C16221M-34-007 DELETED BY SB C16221Q-34-008		1
80	F1380513		.DIFFUSER DELETED BY SB C16221K-34-013 DELETED BY SB C16221L-34-013 DELETED BY SB C16221M-34-007 DELETED BY SB C16221Q-34-008		1
90	F1391805		.DIFFUSER		1
100	F1391806		.HEATER		1
110	211F02C03U		.CONNECTOR VF0318 OPT SEE TABLE (A1171874)	A	1
- 110A	102S02		.CONNECTOR VF8378 OPT SEE TABLE OLD PNR 201F02C01U MADE BY VF0318 (A1242587)	B	1
120	F1390783		.UNIT LIGHTING		1
130	F1252893		.CAP		6
140	F1252154		.TUBE FLUORESCENT PRE SB C16221K-34-007 PRE SB C16221L-34-007 PRE SB C16221M-34-002 PRE SB C16221Q-34-002	A	3
- 140A	F1405513		.TUBE FLUORESCENT POST SB C16221K-34-007 POST SB C16221L-34-007 POST SB C16221M-34-002 POST SB C16221Q-34-002	B	3
150	F1252667		.NUT		4

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
4 160	E24100BB		.PCB CBL (ESDS) PRE SB C16221K-34-007 PRE SB C16221L-34-007 PRE SB C16221Q-34-002 PRE SB C16221M-34-002 SEE 342245-8-1 FOR DET	A	1
- 160A	E24100EA		.PCB CBL (ESDS) POST SB C16221K-34-007 POST SB C16221L-34-007 POST SB C16221M-34-002 POST SB C16221Q-34-002 SEE 342245-8-1A FOR DET	B	1
170	A1235914		ATTACHING PARTS .SCREW MACH SST FIL H SOCK HEX-LOBE M2.5X6 ORDER OVERLGTH MPN E25111B25Q6ZHAPA MADE BY VF4796 OPT SEE TABLE		4
180	520251-70		.WASHER VF0349 SPRING TENSION BRSS DIA 2.5 OPT SEE TABLE (A1236581) ***		4

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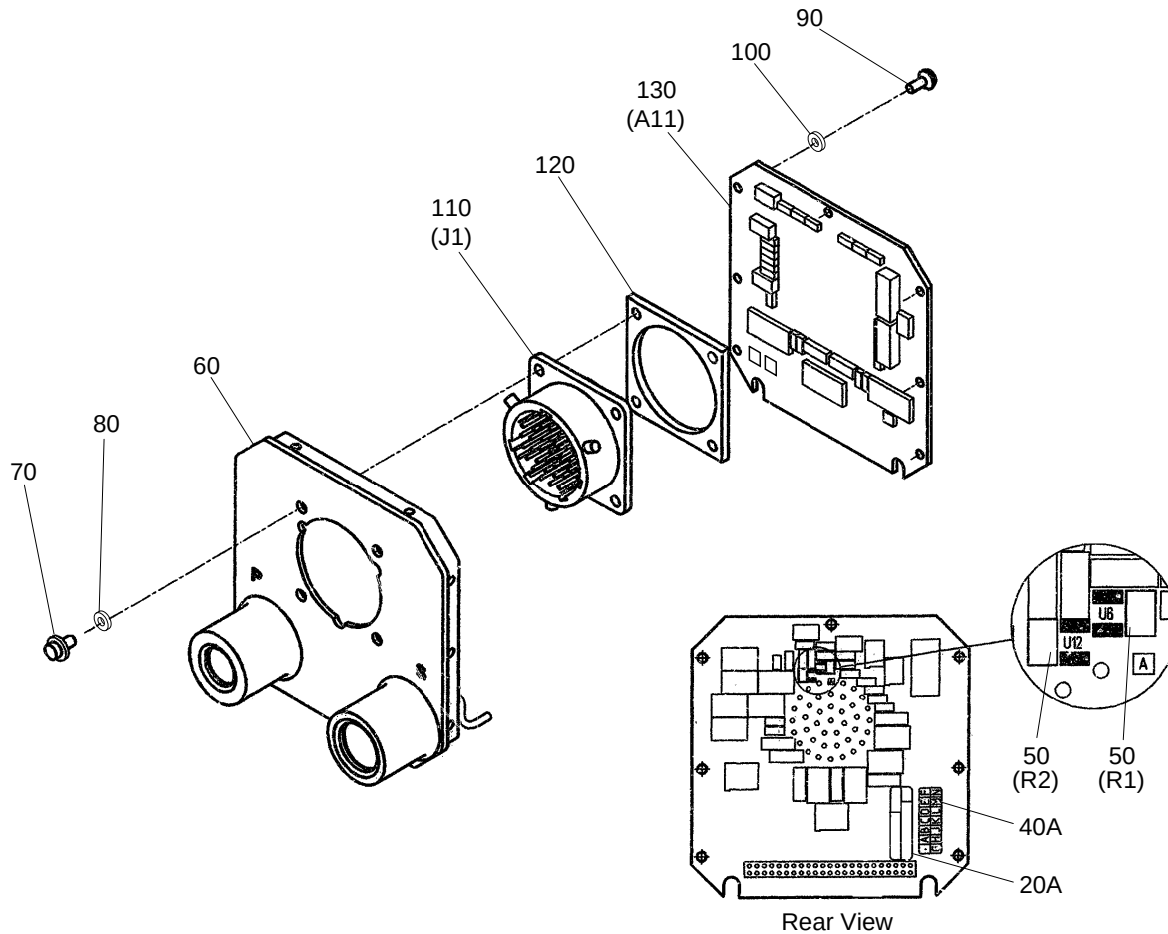
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INTERCONNECTION REAR ASSY

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Figure 10009/Graphic - Sheet 01 / 34-22-45-991-909-A01
(Figure 5) INTERCONNECTION, REAR ASSY (sheet 1/2) (E16387E.)

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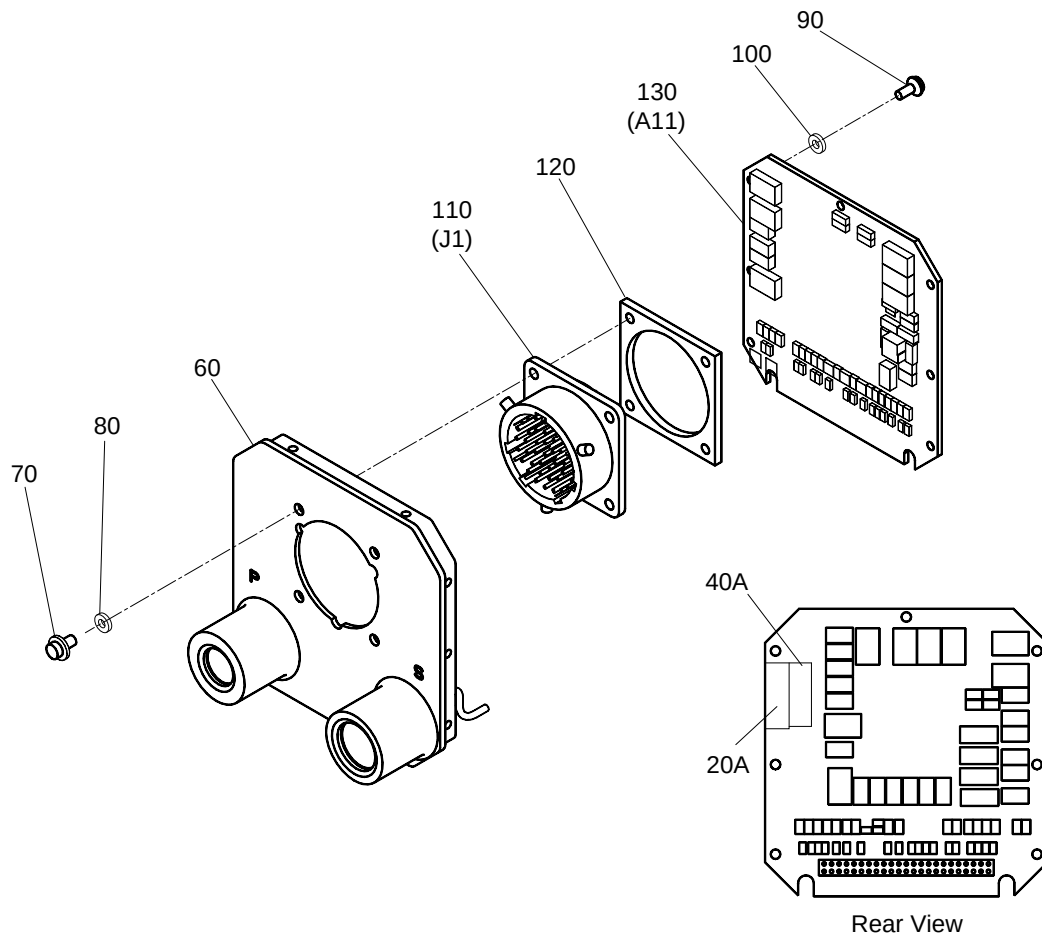
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Figure 10009/Graphic - Sheet 02 / 34-22-45-991-909-A01
(Figure 5) INTERCONNECTION, REAR ASSY (sheet 2/2) (E16387J.)

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
5 - 1	E16387EA		INTERCONNECTION REAR ASSY (ESDS) PRE SB C16221K-34-012 PRE SB C16221L-34-012 PRE SB C16221Q-34-007 SEE 342245-2-330 FOR NHA SEE 342245-2A-330 FOR NHA SEE 342245-2B-350 FOR NHA SEE 342245-2C-350 FOR NHA	A	RF
- 1A	E16387JA		INTERCONNECTION REAR ASSY (ESDS) POST SB C16221K-34-012 POST SB C16221L-34-012 POST SB C16221Q-34-007 SEE 342245-2-330A FOR NHA SEE 342245-2A-330A FOR NHA SEE 342245-2C-350A FOR NHA	B	RF
- 10	F2000068		.PROTECTOR LABEL		1
- 20	F2000247		.DELETED (INCORRECT PART NUMBER)		
20A	F2000071		.LABEL IDENTIFICATION		1
- 30	F2000100		.PROTECTOR LABEL		1
- 40	F2000071		.DELETED (INCORRECT PART NUMBER)		
40A	F2000247		.LABEL MODIFICATION SUFFIX	A	1
- 40B	F2000103		.LABEL MODIFICATION SUFFIX	B	1
50	WCR08050R00F		.RESISTOR VF7178 FIXED 45.3 KOHM PM0.1PCT 1/30W OPT SEE TABLE OLD PNR OCR0805-ORJ MADE BY VD1018 (A1240078)	A	2
60	F1403611		.BACKPLATE EQUIPPED		1

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
5	70	A1229703	ATTACHING PARTS .SCREW MACH SST FIL H SOCK HEX-LOBE M3X6 ORDER OVERLGTH MPN E25111B30Q6ZHAPA MADE BY VF4796 OPT SEE TABLE		4
	80	520301-24	.WASHER SPRING TENSION SST DIA 3 OPT SEE TABLE (A1236538)	VF0349	4
	90	A1235914	.SCREW MACH SST FIL H SOCK HEX-LOBE M2.5X6 ORDER OVERLGTH MPN E25111B25Q6ZHAPA MADE BY VF4796 OPT SEE TABLE		7
	100	520251-00	.WASHER SPRING TENSION SST DIA 2.5 (A1236539) ***	VF4796	7
	110	A1246497	.CONNECTOR ORDER OVERLGTH MPN 8525-02H20B41PNH045 MADE BY VF0225 OPT SEE TABLE		1
	120	F1273136	.FLANGE		1
	130	E16368CA	.PCB FIB (ESDS) SEE 342245-9-1 FOR DET	A	1
	-130A	E16368FA	.PCB FIB (ESDS) SEE 342245-9A-1 FOR DET	B	1

- ITEM NOT ILLUSTRATED

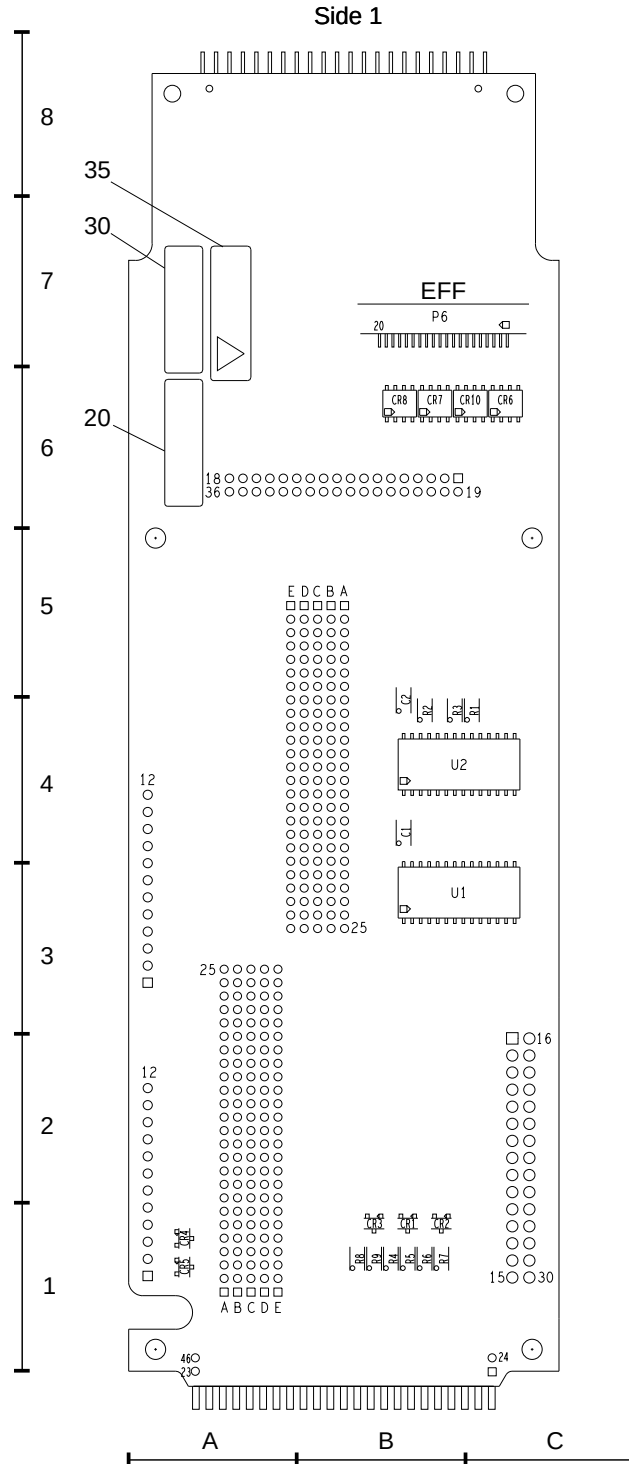
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PCB MOTHER PROG

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D_CM_TAV_342245_10_AAJO_01_00

Figure 10010/Graphic - Sheet 01 / 34-22-45-991-910-A01
(Figure 6) PCB, MOTHER PROG (sheet 1/2) (E16486BA..)

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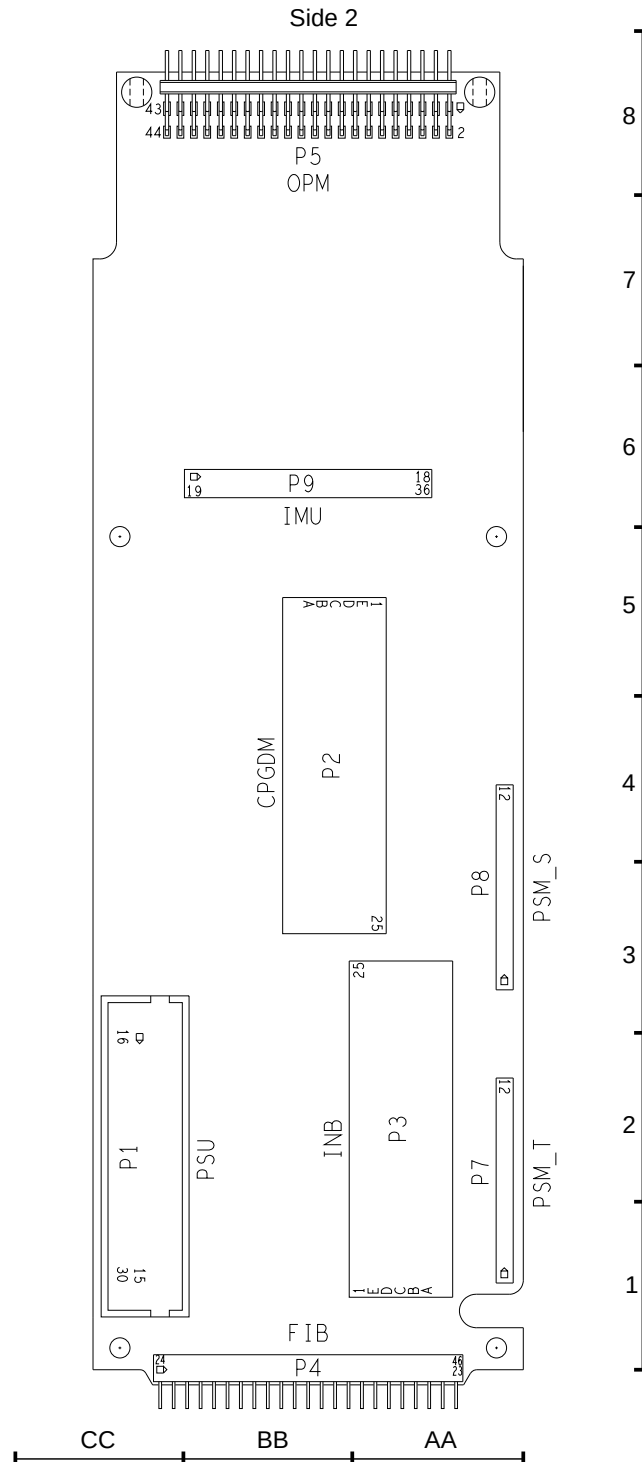
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Figure 10010/Graphic - Sheet 02 / 34-22-45-991-910-A01
(Figure 6) PCB, MOTHER PROG (sheet 2/2) (E16486BA..)

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
6 - 1	E16486BA01		PCB MOTHER PROG (ESDS) PRE SB C16221K-34-005 PRE SB C16221L-34-005 SEE 342245-2-420 FOR NHA SEE 342245-2A-420 FOR NHA		RF
- 10	G5616AAB01		.SOFTWARE BOOT REFER TO SPECIAL TOOLS PAGEBLOCK	NP	1
- 15	F2000068		.PROTECTOR LABEL		1
20	F2000071		.LABEL IDENTIFICATION		1
- 25	F2000100		.PROTECTOR LABEL		1
30	F2000247		.LABEL MODIFICATION SUFFIX		1
35	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
40	0805Y050-0104KX		.CAPACITOR FIXED 100NF PM10PCT 50V OPT SEE TABLE OLD PNR VJ0805Y104KXA MADE BY VF7843 (A1253142)	VU1579	2
45	BAT54S		.DIODE OPT SEE TABLE (A1238186)	VF8859	5
50	PD10808		.DIODE OPT SEE TABLE OLD PNR SMDA12CTH MADE BY V13409 (A1252420)	V0U8A3	4
55	1-0102567-3		.CONNECTOR OPT SEE TABLE (A1253529)	VF0286	1
60	A1247907		.CONNECTOR ORDER OVERLGTH MPN HM2P08PA5110AALF MADE BY VF6131 OPT SEE TABLE OLD PNR 0-0100141-1 MADE BY VF0286		2

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
6 65	94277-046		.STRIP TERMINAL OPT SEE TABLE (A1253577)	VF6131	1
70	94277-044		.STRIP TERMINAL OPT SEE TABLE (A1253649)	VF6131	1
75	88384-130		.CONNECTOR (A1252177)	V027H6	1
80	76308-112		.STRIP TERMINAL OPT SEE TABLE (A1253855)	VF6131	2
85	86455-036		.STRIP TERMINAL OPT SEE TABLE (A1246210)	VF6131	1
90	MCR10F2612		.RESISTOR FIXED 26.1KOHM PM1PCT 1/10W OPT SEE TABLE (A1245938)	VS5518	6
95	WCR0805-1KF		.RESISTOR FIXED 1KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805K1KF MADE BY VF2663 (A1242085)	VF7178	3
100	AT28HC256-70SI		.INTEGRATED CIRCUIT MEMORY EEPROM 32KX8 70NS H-CMOS (ESDS) OPT SEE TABLE (A1253576)	V1FN41	2

- ITEM NOT ILLUSTRATED

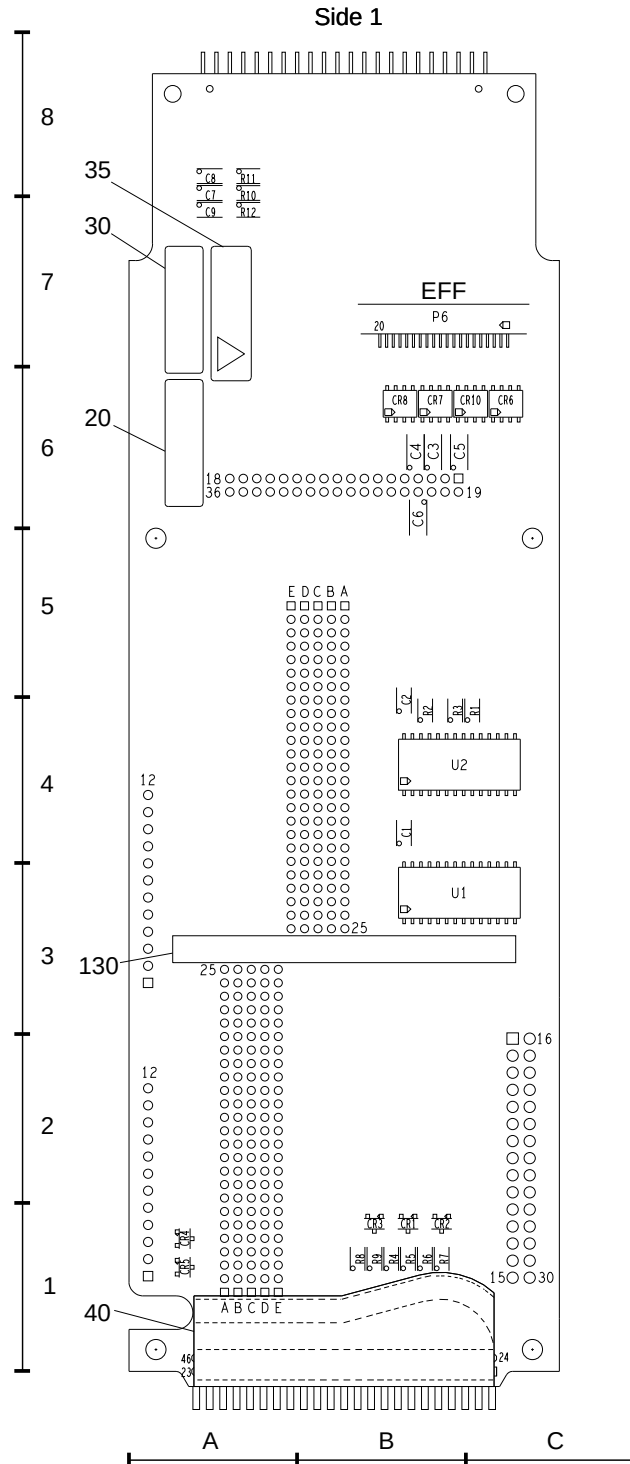
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PCB MOTHER PROG

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D_CM_TAV_342245_10_AAK0_01_00

Figure 10011/Graphic - Sheet 01 / 34-22-45-991-911-A01
(Figure 6A) PCB, MOTHER PROG (sheet 1/2) (E16486BB../JA..)

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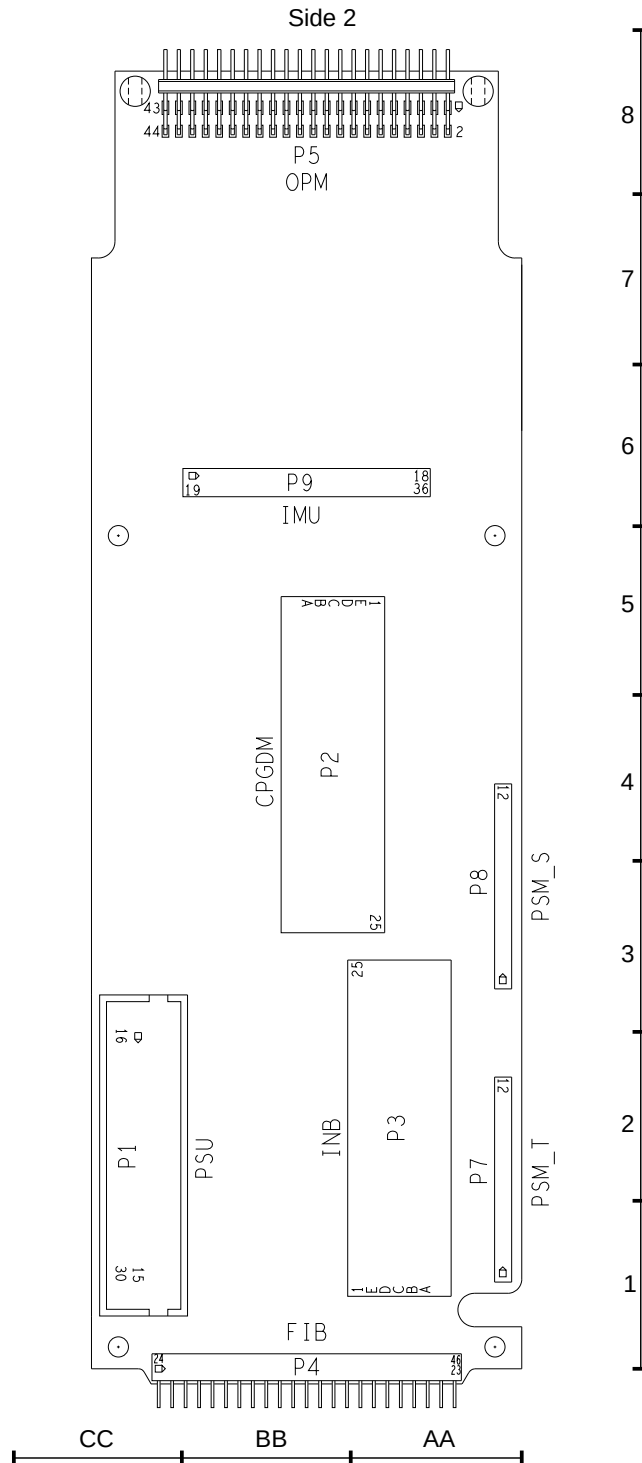
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D_CM_TAV_342245_10_AAK0_02_00

Figure 10011/Graphic - Sheet 02 / 34-22-45-991-911-A01
(Figure 6A) PCB, MOTHER PROG (sheet 2/2) (E16486BB../JA..)

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
6A - 1	E16486BB01		PCB MOTHER PROG (ESDS) POST SB C16221K-34-005 POST SB C16221L-34-005 SEE 342245-2-420A FOR NHA SEE 342245-2A-420A FOR NHA SEE 342245-2B-440 FOR NHA SEE 342245-2C-440 FOR NHA	A	RF
- 1A	E16486JA01		PCB MOTHER PROG (ESDS) PRE SB C16221K-34-015 PRE SB C16221L-34-015 PRE SB C16221Q-34-009 SEE 342245-2-420B FOR NHA SEE 342245-2A-420B FOR NHA SEE 342245-2B-440A FOR NHA SEE 342245-2C-440A FOR NHA	B	RF
- 10	G5616AAB01		.SOFTWARE BOOT REFER TO SPECIAL TOOLS PAGEBLOCK	NP A	1
- 10A	G5616AAD01		.SOFTWARE BOOT REFER TO SPECIAL TOOLS PAGEBLOCK	NP B	1
- 15	F2000068		.PROTECTOR LABEL		1
20	F2000071		.LABEL IDENTIFICATION		1
- 25	F2000100		.PROTECTOR LABEL		1
30	F2000103		.LABEL MODIFICATION SUFFIX		1
35	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
40	F1401164		.GUIDE SUPSD BY ITEM 40A		1
- 40A	F1403666		.GUIDE SUPSDS ITEM 40		1
45	0805Y050-0104KX		.CAPACITOR FIXED 100NF PM10PCT 50V OPT SEE TABLE OLD PNR VJ0805Y104KXA MADE BY VF7843 (A1253142)	VU1579	2

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
6A 50	1206Y050-0102JX		.CAPACITOR VU1579 FIXED 1NF PM5PCT 50V OLD PNR VJ1206A102JXA MADE BY VF7843 (A1235243)		4
55	VJ0805A330JXBA		.CAPACITOR VF7843 FIXED 33PF PM5PCT 100V OPT SEE TABLE OLD PNR 0805Y100-0330JC MADE BY VU1579 (A1235252)		3
60	BAT54S		.DIODE VF8859 OPT SEE TABLE (A1238186)		5
65	PD10808		.DIODE VOU8A3 OPT SEE TABLE OLD PNR SMDA12CTH MADE BY V13409 (A1252420)		3
70	PD10796LF		.DIODE VOU8A3 OLD PNR SMDA24CTH MADE BY V13409 (A1251433)		1
75	1-0102567-3		.CONNECTOR VF0286 OPT SEE TABLE (A1253529)		1
80	A1247907		.CONNECTOR ORDER OVERLGTH MPN HM2P08PA5110AALF MADE BY VF6131 OPT SEE TABLE OLD PNR 0-0100141-1 MADE BY VF0286		2
85	94277-046		.STRIP VF6131 TERMINAL OPT SEE TABLE (A1253577) SUPSD BY ITEM 85A		1
- 85A	58102F61-23		.STRIP VF6131 TERMINAL OPT SEE TABLE (A2002990) SUPSDS ITEM 85		1
90	MMT122-02SDHAK		.STRIP V55322 TERMINAL (A2000559)		1

- ITEM NOT ILLUSTRATED

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
6A 95	52207-2090		.CONNECTOR V27264 OPT SEE TABLE (A2000013)		1
100	76308-112		.STRIP VF6131 TERMINAL OPT SEE TABLE (A1253855)		2
105	86455-036		.STRIP VF6131 TERMINAL OPT SEE TABLE (A1246210)		1
110	MCR10F2612		.RESISTOR VS5518 FIXED 26.1KOHM PM1PCT 1/10W OPT SEE TABLE (A1245938)		6
115	WCR0805-1KF		.RESISTOR VF7178 FIXED 1KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805K1KF MADE BY VF2663 (A1242085)		3
120	D12-100-1MF		.RESISTOR VD1018 FIXED 1MOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR MCR10F1004 MADE BY VS5118 (A1242101)		3
125	F1403314		.SELECTION INTEGRATED CIRCUIT		2
- 126	AT28HC256F90SI		..INTEGRATED V1FN41 CIRCUIT MEMORY EEPROM 32KX8 90NS H-CMOS (ESDS) OPT SEE TABLE (A1254323)		1
- 126A	AT28HC256-70SI		..INTEGRATED V1FN41 CIRCUIT MEMORY EEPROM 32KX8 70NS H-CMOS (ESDS) OPT SEE TABLE (A1253576)		1
130	F1401165		.SHIM SUPSD BY ITEM 130A		1
- 130A	F1403667		.SHIM SUPSDS ITEM 130		1

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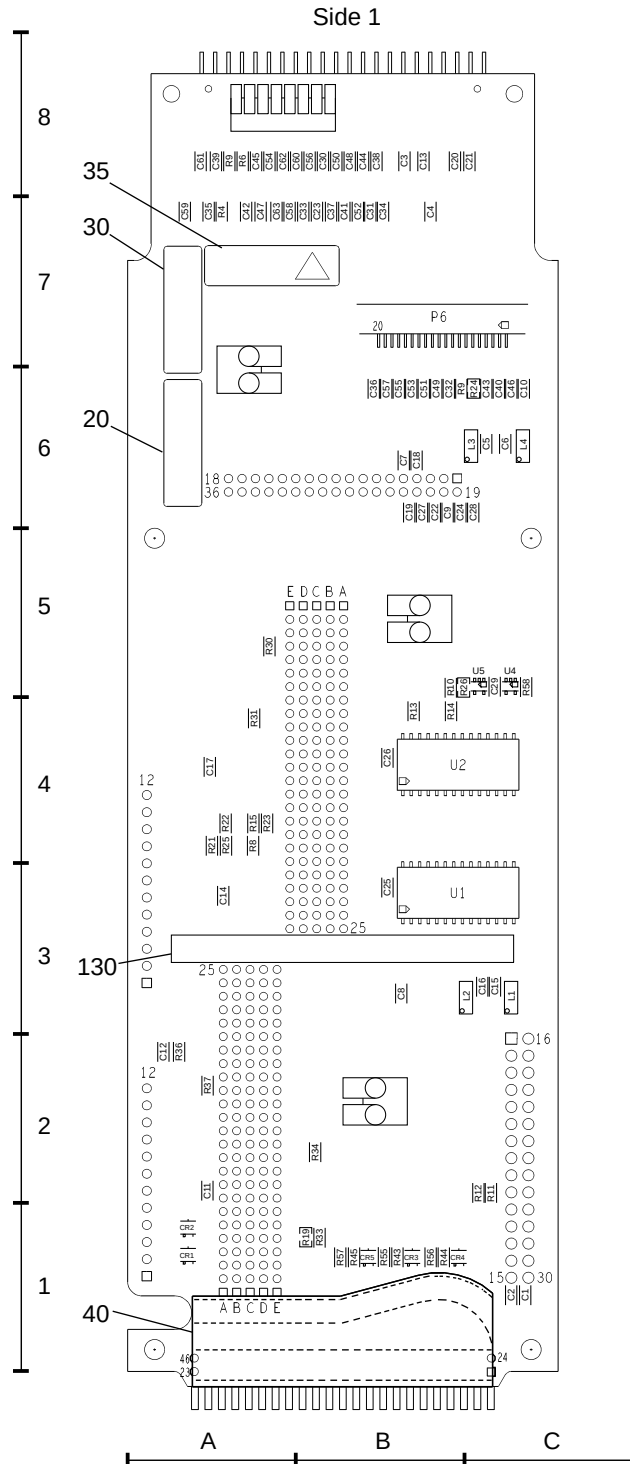
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PCB MOTHER PROG

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Figure 10012/Graphic - Sheet 01 / 34-22-45-991-912-A01
(Figure 6B) PCB, MOTHER PROG (sheet 1/2)

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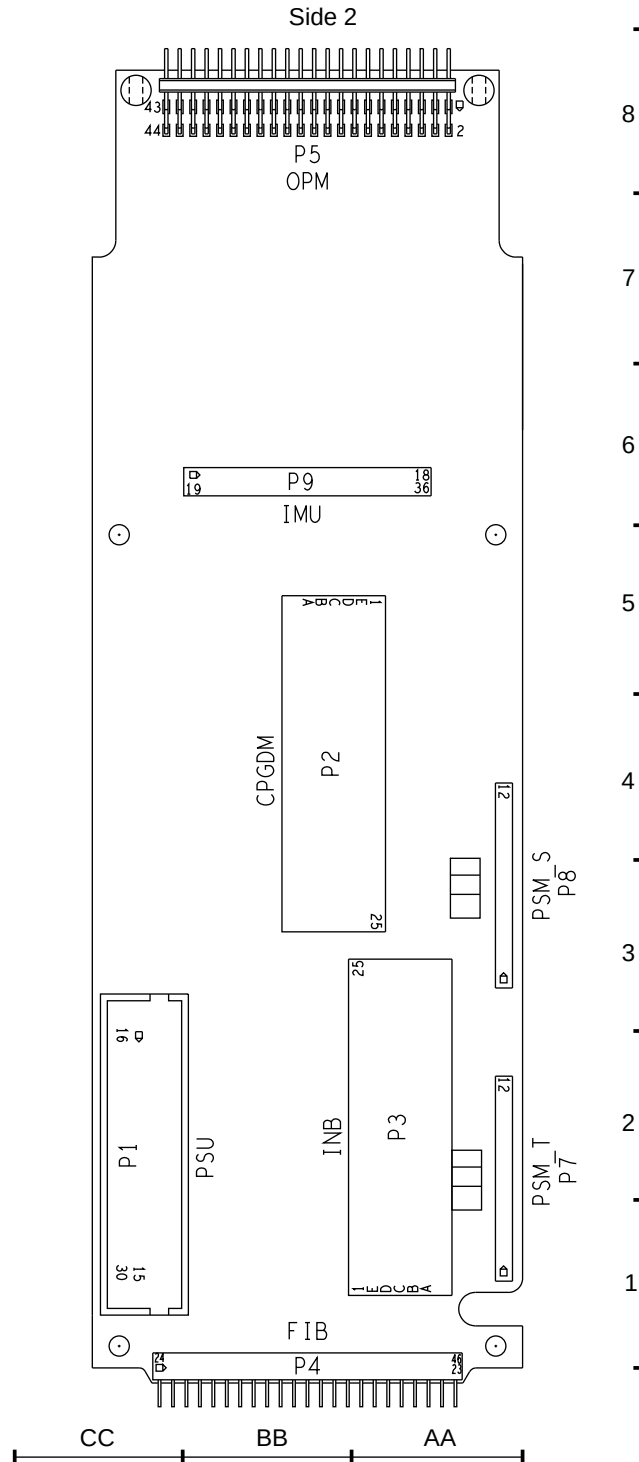
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Figure 10012/Graphic - Sheet 02 / 34-22-45-991-912-A01
(Figure 6B) PCB, MOTHER PROG (sheet 2/2)

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
6B - 1	E16981HA01		PCB MOTHER PROG (ESDS) POST SB C16221K-34-015 POST SB C16221L-34-015 POST SB C16221Q-34-009 SEE 342245-2-420C FOR NHA SEE FIG 2A FOR NHA SEE FIG 2C FOR NHA		RF
- 10	G5616AAD01		.SOFTWARE BOOT	NP	1
- 15	F2000008		.LABEL IDENTIFICATION		1
20	F2000071		.LABEL IDENTIFICATION		1
- 25	F2000100		.PROTECTOR LABEL		1
30	F2000103		.LABEL MODIFICATION SUFFIX		1
35	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
40	F1407405		.GUIDE		1
45	0603Y100-0470JC		.CAPACITOR FIXED 47PF PM5PCT 100V (A1256796)	VU1579	44
50	0603Y050-0103KX		.CAPACITOR FIXED 10NF PM10PCT 50V OPT SEE TABLE (A1252503)	VU1579	6
55	0603Y016-0104KX		.CAPACITOR FIXED 100NF PM10PCT 16V OPT SEE TABLE (A1252504)	VU1579	2
60	0603Y100-0120JC		.CAPACITOR FIXED 12PF PM5PCT 100V (A1257311)	VU1579	11
65	BAV99W		.DIODE OPT SEE TABLE (A1253697)	VC6489	5
70	251806600Y3		.COIL 0 MICROHENRY OPT SEE TABLE (A1252594)	V34899	4
75	1-0102567-3		.CONNECTOR OPT SEE TABLE (A1253529)	VF0286	1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
6B 80	A1247907		.CONNECTOR ORDER OVERLGTH MPN HM2P08PA5110AALF MADE BY VF6131 OPT SEE TABLE OLD PNR 0-0100141-1 MADE BY VF0286		2
85	58102F61-23		.STRIP TERMINAL OPT SEE TABLE (A2002990)	VF6131	1
90	MMT122-02SDHAK		.STRIP TERMINAL (A2000559)	V55322	1
95	52207-2090		.CONNECTOR OPT SEE TABLE (A2000013)	V27264	1
100	76308-112		.STRIP TERMINAL OPT SEE TABLE (A1253855)	VF6131	2
105	86455-036		.STRIP TERMINAL OPT SEE TABLE (A1246210)	VF6131	1
110	WCR0603-10KF		.RESISTOR FIXED 10KOHM PM1PCT 1/16W OPT SEE TABLE OLD PNR WCR0603-1002F MADE BY VF7178 (A1251833)	VF7178	8
115	WCR0603-1KF		.RESISTOR FIXED 1KOHM PM1PCT 1/16W OPT SEE TABLE OLD PNR WCR0603-1001F MADE BY VF7178 (A1251826)	VF7178	3
120	WCR0603R005J		.RESISTOR FIXED 0 OHM PM5PCT 1/16W OPT SEE TABLE OLD PNR WCR0603R005JI MADE BY VF7178 (A1251847)	VF7178	10

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
6B 125	WCR0603-26K1F		.RESISTOR VF7178 FIXED 26.1KOHM PM1PCT 1/16W OPT SEE TABLE OLD PNR WCR0603-2612F MADE BY VF7178 (A1253319)		7
130	AT28HC256-70SI		.INTEGRATED V1FN41 CIRCUIT MEMORY EEPROM 32KX8 70NS H-CMOS (ESDS) OPT SEE TABLE (A1253576)		2
135	74LVC1G14GW		.INTEGRATED VH0H68 CIRCUIT INVERTER CMOS (ESDS) OPT SEE TABLE OLD PNR SN74LVC1G14DCK MADE BY V01295 (A2001607)		1
140	SN54AHC1G32DCK		.INTEGRATED V01295 CIRCUIT 1 GATE OR 2 INPUT CMOS (ESDS) OPT SEE TABLE (A1256119)		1

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PCB FRONT FACE

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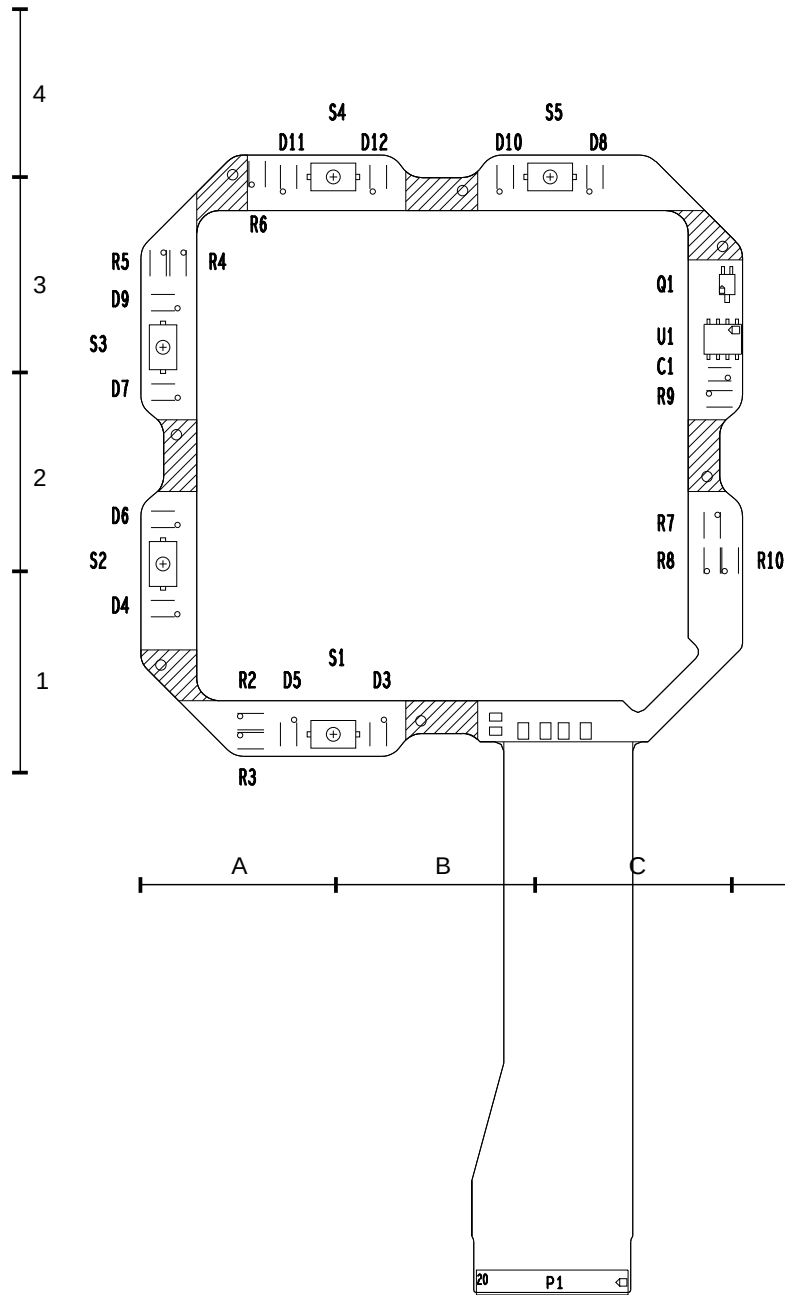
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Figure 10013/Graphic 34-22-45-991-913-A01
(Figure 7) PCB, FRONT FACE

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
7 - 1	E16389JA		PCB FRONT FACE PRE SB C16221K-34-005 PRE SB C16221L-34-005 SEE 342245-3-150 FOR NHA SEE 342245-3A-150 FOR NHA	A	RF
- 1A	E16389JB		PCB FRONT FACE POST SB C16221K-34-005 POST SB C16221L-34-005 SEE 342245-3-150A FOR NHA SEE 342245-3A-150A FOR NHA	B	RF
10	0603Y016-0104KX		.CAPACITOR VU1579 FIXED 100NF PM10PCT 16V OPT SEE TABLE OLD PNR VJ0603Y104KXJ MADE BY VF7843 (A1252504)		1
15	NSCW100		.LED VSS993 (A1253274)		10
20	SFH2400		.PHOTODIODE VF0889 OPT SEE TABLE (A1252671)		1
25	CR0805FX3160		.RESISTOR V32997 FIXED 316 OHM PM1PCT 1/10W OPT SEE TABLE (A1247403)		5
30	CR0805FX5623		.RESISTOR V32997 FIXED 562KOHM PM1PCT 1/10W OPT SEE TABLE (A1248947)		2
35	CR0805FX1003LF		.RESISTOR V32997 FIXED 100KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR D12-100-100KF MADE BY VD1018 (A1240119)		1
40	RC0805F46R4L		.RESISTOR VS7293 FIXED 46.4 OHM PM1PCT 1/10W OPT SEE TABLE (A1246354)		1
45	KSR251J		.SWITCH VC0166 OPT SEE TABLE (A1252521)	A	5
- 45A	KSR551GSLFS		.SWITCH VF3132 OPT SEE TABLE (A2000442)	B	5

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
7 50	AD822AR		1234567 .INTEGRATED CIRCUIT OPERATL AMPLIFIER OPT SEE TABLE (A1250476)	V24355	1

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PCB CBL

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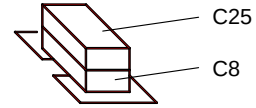
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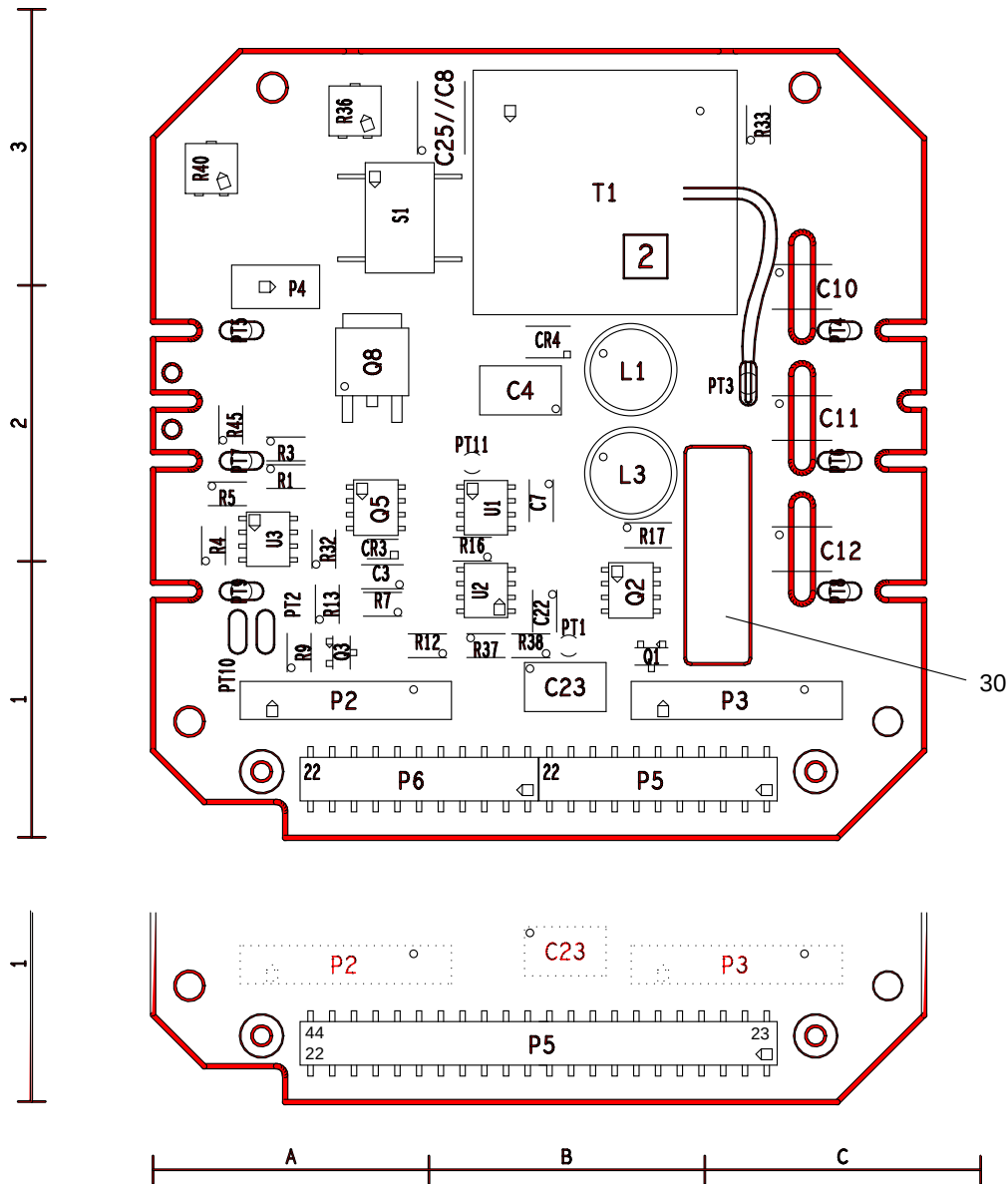
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Mounting Detail
C25 - C8



Side 1



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Figure 10014/Graphic - Sheet 01 / 34-22-45-991-914-A01
(Figure 8) PCB, CBL (sheet 1/2)

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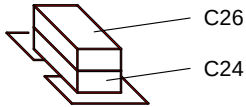
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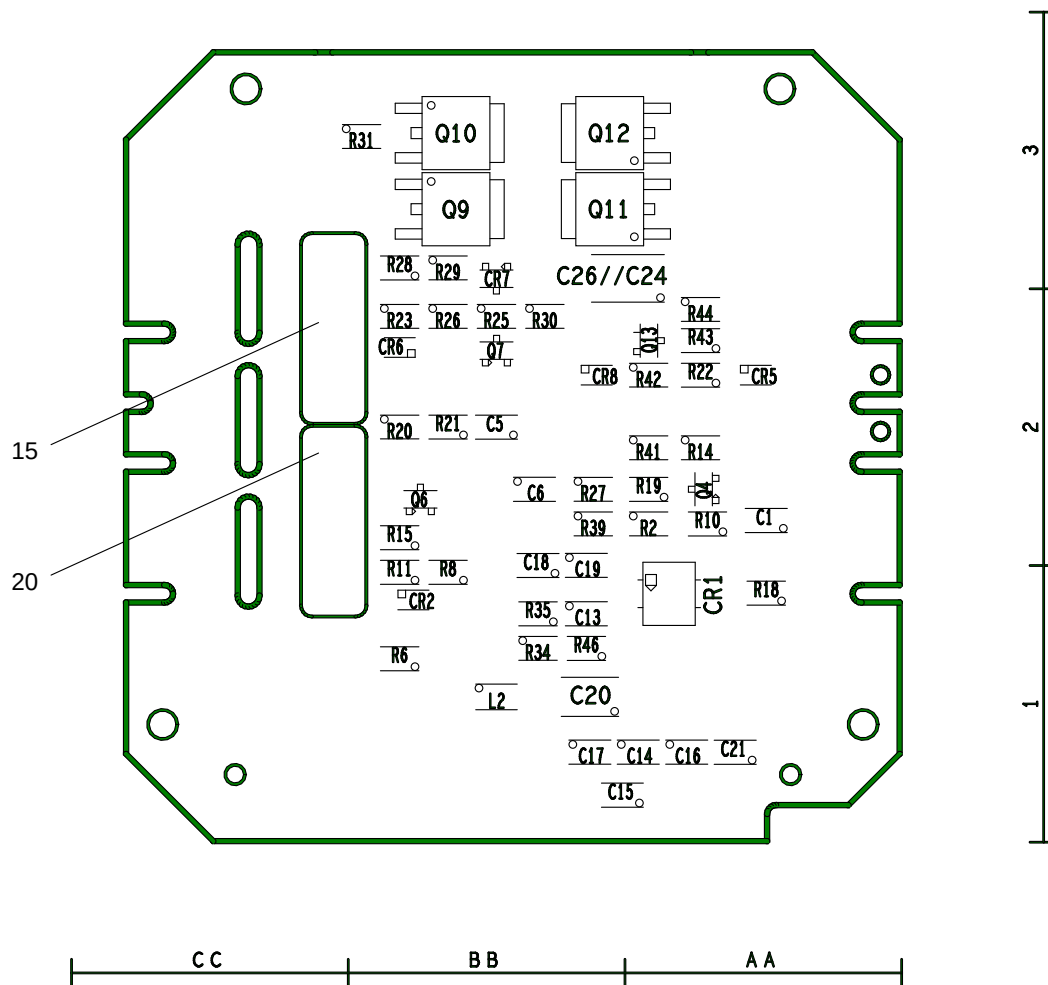
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Mounting Detail
C26 - C24



Side 2



D_CM_TAV_342245_10_AAN0_02_00

Figure 10014/Graphic - Sheet 02 / 34-22-45-991-914-A01
(Figure 8) PCB, CBL (sheet 2/2)

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PARTS LIST

FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
8 - 1	E24100BB		PCB CBL (ESDS) PRE SB C16221K-34-007 PRE SB C16221L-34-007 PRE SB C16221M-34-002 PRE SB C16221Q-34-002 SEE 342245-4-160 FOR NHA	A	RF
- 1A	E24100EA		PCB CBL (ESDS) AMDMNT A POST SB C16221K-34-007 POST SB C16221L-34-007 POST SB C16221M-34-002 POST SB C16221Q-34-002 SEE 342245-4-160A FOR NHA	B	RF
- 10	F2000068		.PROTECTOR LABEL		1
15	F2000071		.LABEL IDENTIFICATION		1
- 19	F2000100		.PROTECTOR LABEL	B	1
20	F2000103		.LABEL MODIFICATION SUFFIX ADDED BY SB C16221L-34-007 ADDED BY SB C16221L-34-013 ADDED BY SB C16221M-34-002 ADDED BY SB C16221M-34-007 ADDED BY SB C16221Q-34-002 ADDED BY SB C16221Q-34-008 ADDED BY SB C16221K-34-007 ADDED BY SB C16221K-34-013	B	1
30	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
35	0805Y050-0104KX		.CAPACITOR FIXED 100NF PM10PCT 50V OPT SEE TABLE OLD PNR VJ0805Y104KXA MADE BY VF7843 (A1253142)	VU1579	12
40	VJ0805A101JXBA		.CAPACITOR FIXED 100PF PM5PCT 100V OPT SEE TABLE OLD PNR VJ0805A101JXB MADE BY VF7843 (A1235254)	VF7843	1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
8	45	TAJD106K035FJ	.CAPACITOR V17554 FIXED 10MUF PM10PCT 35V OPT SEE TABLE (A1246869)		1
	50	0805Y100-0331JC	.CAPACITOR VU1579 FIXED 330PF PM5PCT 100V OLD PNR VJ0805A331JXB MADE BY VF7843 (A1235259)		1
	55	1812Y200-0472JC	.CAPACITOR VU1579 FIXED 4.7NF PM5PCT 200V OLD PNR VJ1812A472JXC MADE BY VF7843 (A1252595) PRE SB C16221K-34-007 PRE SB C16221K-34-013 PRE SB C16221L-34-007 PRE SB C16221L-34-013 PRE SB C16221M-34-002 PRE SB C16221Q-34-002 PRE SB C16221Q-34-008 PRE SB C16221M-34-007		2
- 55A		CEC6C200V10KK	.CAPACITOR VF1379 FIXED 10NF PM10PCT 200V (A2007748) POST SB C16221L-34-007 POST SB C16221L-34-013 POST SB C16221M-34-002 POST SB C16221M-34-007 POST SB C16221Q-34-002 POST SB C16221Q-34-008 POST SB C16221K-34-007 POST SB C16221K-34-013	B	2
	60	1812Y3K0-0150JC	.CAPACITOR VU1579 FIXED 15PF PM5PCT 3000V OLD PNR SPT928-400V15UJ MADE BY VF1379 (A1249578)		3
	65	1210Y025-0105KX	.CAPACITOR VU1579 FIXED 1MUF PM10PCT 25V OLD PNR VJ1210Y105KXXT MADE BY VF7841 (A1248383)		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
8	70	A1240230	.CAPACITOR FIXED 4.7MUF PM10PCT 50V ORDER OVERLGTH MPN CTC4-475X9050D2E3 MADE BY V2N936 OPT SEE TABLE		1
	73	1812Y200-0472JC	.CAPACITOR FIXED 4.7NF PM5PCT 200V DELETED BY SB C16221K-34-013 DELETED BY SB C16221L-34-013 DELETED BY SB C16221M-34-007 DELETED BY SB C16221Q-34-008 (A1252595) ADDED BY SB C16221K-34-007 ADDED BY SB C16221L-34-007 ADDED BY SB C16221M-34-002 ADDED BY SB C16221Q-34-002	VU1579 B	2
	75	S5493-01	.PHOTODIODE (A1249232)	VS4719	1
	80	MMSZ5234B	.DIODE OPT SEE TABLE (A1251219)	V1MQ07	3
	85	MMSZ5240BG	.DIODE OPT SEE TABLE OLD PNR MMSZ5240B MADE BY V1MQ07 (A1250942)	V1MQ07	1
	90	BZD27C91	.DIODE (A1206629)	VX0213	1
	95	MMSZ5243BG	.DIODE OPT SEE TABLE (A1250943)	V1MQ07	1
	100	BAV70L	.DIODE OPT SEE TABLE (A1237914)	VFAHD5	1
	105	TSL0807-102KR23	.COIL OPT SEE TABLE (A1206445)	VS5988	2
	110	BLM21AJ601SN1	.COIL 0 MICROHENRY (A1249840)	VS4431	1
	115	SLW14S1C7	.CONNECTOR OPT SEE TABLE (A1242163)	VF6131	2

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
8 120	101Y02L		.CONNECTOR OPT SEE TABLE OLD PNR 201M02T03L MADE BY VF0318 (A1242588)	VF8378	1
125	92086-122		.STRIP TERMINAL OPT SEE TABLE (A1254579)	VF6131	1
126	92086-122		.STRIP TERMINAL OPT SEE TABLE (A1254579)	VF6131	1
-126A	TLE122-01GDV		.STRIP TERMINAL (A2005369)	V55322	B 1
130	2N7002		.TRANSISTOR MOSFET N (ESDS) OPT SEE TABLE (A1238017)	VK1196	5
135	IL205A		.PHOTOCOUPLER OPT SEE TABLE OLD PNR MOC205 MADE BY V07263 (A1211540)	VK4184	2
140	S02222A		.TRANSISTOR OPT SEE TABLE (A1239417)	VF8859	1
145	IRFR320PBF		.TRANSISTOR MOSFET N (ESDS) OPT SEE TABLE (A1242157)	VF5752	5
150	CR0805FX1002		.RESISTOR FIXED 10KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805K10KF MADE BY VF2663 (A1242092)	V32997	2
155	D12-100-1MF		.RESISTOR FIXED 1MOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR MCR10F1004 MADE BY VS5518 (A1242101)	VD1018	2

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
8 160	CR0805FX1782		.RESISTOR V32997 FIXED 17.8KOHM PM1PCT 1/10W OPT SEE TABLE (A1245936)		1
165	CR0805-4K75F		.RESISTOR VF2663 FIXED 4.75KOHM PM1PCT 1/10W OPT SEE TABLE (A1242458)		1
170	CR0805K33K2F		.RESISTOR VF2663 FIXED 33.2KOHM PM1PCT 1/10W OPT SEE TABLE (A1247067)		1
175	WCR0805-5K11F		.RESISTOR VF7178 FIXED 5.11KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR A1242089 (A1242089)		2
180	CR0805FX1003LF		.RESISTOR V32997 FIXED 100KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR D12-100-100KF MADE BY VD1018 (A1240119)		4
185	CR0805FX2610		.RESISTOR V32997 FIXED 261 OHM PM1PCT 1/10W OPT SEE TABLE (A1245931)		1
190	A1242090		.RESISTOR FIXED 6.19KOHM PM1PCT 1/10W ORDER OVERLGTH MPN CR0805-100-6K19F MADE BY VD1018 OPT SEE TABLE		1
195	CR0805FX4642		.RESISTOR V32997 FIXED 46.4KOHM PM1PCT 1/10W OPT SEE TABLE (A1246373)		1
200	WCR0805-1KF		.RESISTOR VF7178 FIXED 1KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805K1KF MADE BY VF2663 (A1242085)		2
205	CR0805FX1471		.RESISTOR V32997 FIXED 1.47KOHM PM1PCT 1/10W OPT SEE TABLE (A1245933)		1

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
8	210	CR0805K56K2F	.RESISTOR VF2663 FIXED 56.2KOHM PM1PCT 1/10W OPT SEE TABLE (A1246374)		1
	215	WCR0805-681RF	.RESISTOR VF7178 FIXED 681 OHM PM1PCT 1/10W OPT SEE TABLE (A1245932)		1
	220	RC1206J4R7L	.RESISTOR VS7293 FIXED 4.7 OHM PM5PCT 1/4W OPT SEE TABLE OLD PNR D25-200-4R7J MADE BY VD1018 (A1234992)		1
	225	WCR08050R00F	.RESISTOR VF7178 FIXED 45.3 KOHM PM0.1PCT 1/30W OPT SEE TABLE OLD PNR OCR0805-0RJ MADE BY VD1018 (A1240078)		3
	230	CR0805FX3832	.RESISTOR V32997 FIXED 38.3KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR MCR10F3832 MADE BY VS5518 (A1242095)		1
	235	CR0805FX2613	.RESISTOR V32997 FIXED 261KOHM PM1PCT 1/10W OPT SEE TABLE (A1250700)		2
	240	WCR0805-21K5F	.RESISTOR VF7178 FIXED 21.5KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805FX2152 MADE BY V32997 (A1246370)		1
	245	WCR0805-14K7F	.RESISTOR VF7178 FIXED 14.7KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805FX1472 MADE BY V32997 (A1246368)		1
	250	CR0805-464RF	.RESISTOR VF7178 FIXED 464 OHM PM1PCT 1/10W OPT SEE TABLE (A1242463)		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
8 253	CR0805-464RF		.RESISTOR VF7178 FIXED 464 OHM PM1PCT 1/10W OPT SEE TABLE OLD PNR WCR0805-464RF MADE BY VF7178 (A1242463) PRE SB C16221K-34-007 PRE SB C16221L-34-007 PRE SB C16221M-34-002 PRE SB C16221Q-34-002	A	1
- 253A	CR0805FX1003LF		.RESISTOR V32997 FIXED 100KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR D12-100-100KF MADE BY VD1018 (A1240119) POST SB C16221K-34-007 POST SB C16221L-34-007 POST SB C16221M-34-002 POST SB C16221Q-34-002	B	1
255	CR0805FX6811		.RESISTOR V32997 FIXED 6.81KOHM PM1PCT 1/10W OPT SEE TABLE (A1246008)		3
260	CR0805F100RF		.RESISTOR VF7178 FIXED 100 OHM PM1PCT 1/10W OPT SEE TABLE OLD PNR CR0805K100UF MADE BY VF2663 (A1245930)		3
265	A1242093		.RESISTOR FIXED 12.1KOHM PM1PCT 1/10W ORDER OVERLGTH MPN CR0805-100-12K1F MADE BY VD1018 OPT SEE TABLE		1
270	3224J1-103		.RESISTOR V32997 VARIABLE 10KOHM PM10PCT 1/4W OLD PNR 3224J1-103E MADE BY VS3767 (A1254791)		2
275	WCR0805-31R6F		.RESISTOR VF7178 FIXED 31.6 OHM PM1PCT 1/10W OPT SEE TABLE (A1246566)		1

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
8	280	MCR10F3161	.RESISTOR VS5518 FIXED 3.16KOHM PM1PCT 1/10W OPT SEE TABLE OLD PNR D12-100-3K16F MADE BY VD1018 (A1246363)		1
	285	CR0805FX5623	.RESISTOR V32997 FIXED 562KOHM PM1PCT 1/10W OPT SEE TABLE (A1248947)		1
	290	A1246365	.RESISTOR FIXED 5.62KOHM PM1PCT 1/10W ORDER OVERLGTH MPN CR0805-100-5K62F MADE BY VD1018 OPT SEE TABLE OLD PNR CR0805FX5621 MADE BY V32997		1
	295	66L040-0247	.SWITCH V81541 THERMOSTATIC (A1254580)		1
	300	F1252952	.TRANSFORMER		1
	305	AD654JRZ	.INTEGRATED V24355 CIRCUIT CONVERTER A/N VOLT/FREQ OPT SEE TABLE (A1248256)		1
	310	OP297GS	.INTEGRATED V24355 CIRCUIT OPERATL AMPLIFIER OPT SEE TABLE (A1208335)		1
	315	LM2904D	.INTEGRATED VF8859 CIRCUIT OPERATL AMPLIFIER OPT SEE TABLE (A1246229)		1

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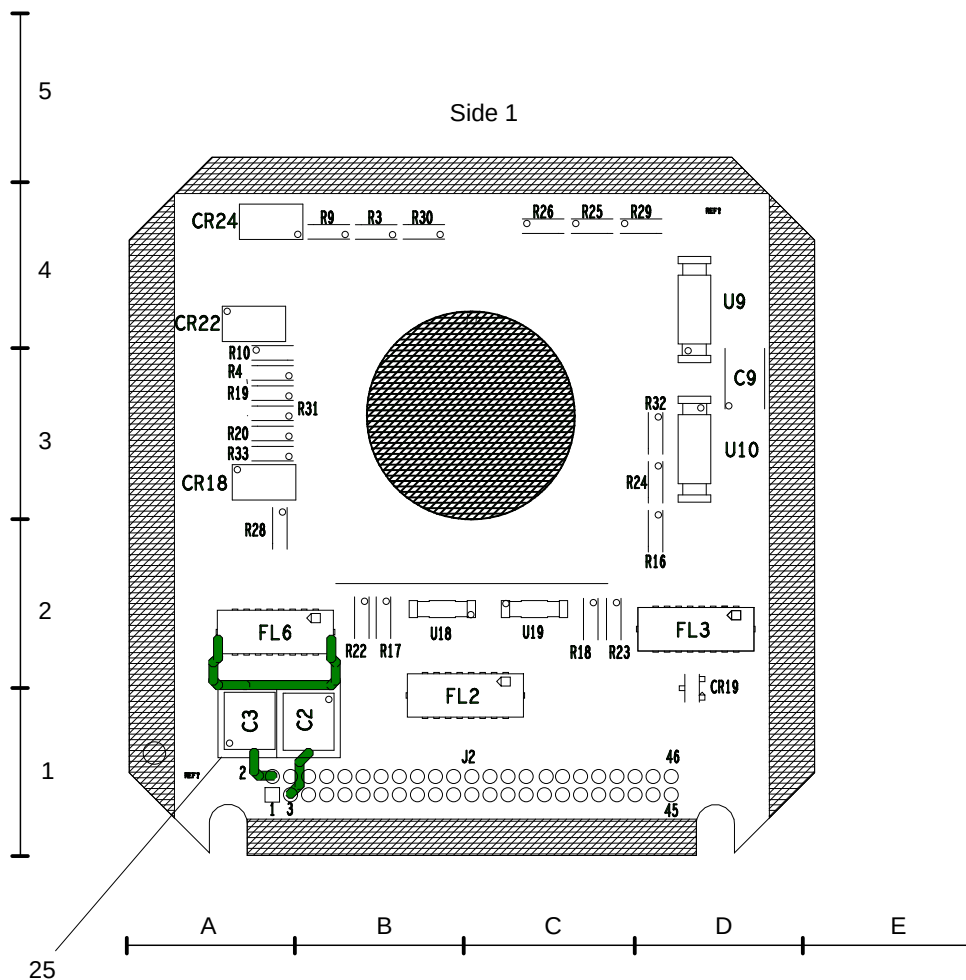
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Figure 10015/Graphic - Sheet 01 / 34-22-45-991-915-A01
(Figure 9) PCB, FIB (sheet 1/2) (E16368C.)

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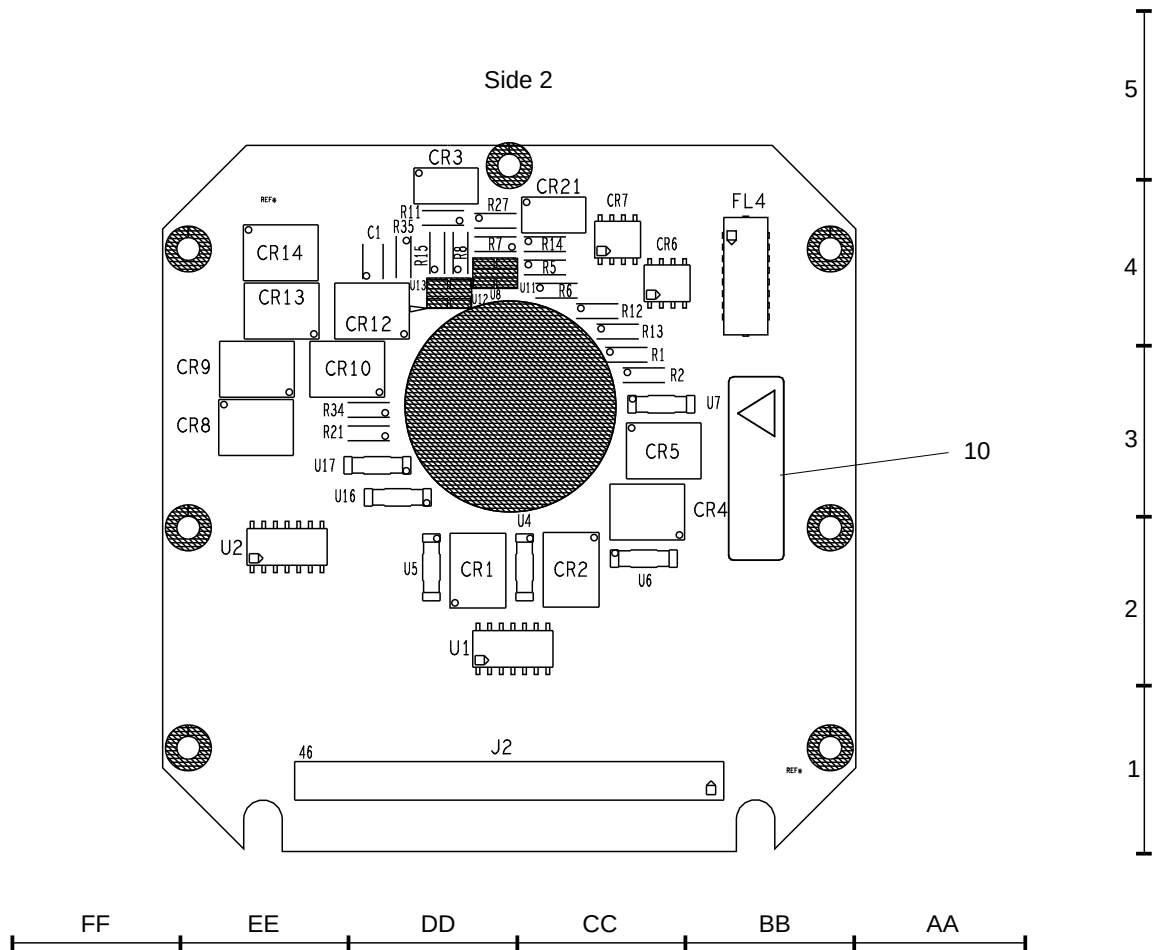
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Figure 10015/Graphic - Sheet 02 / 34-22-45-991-915-A01
(Figure 9) PCB, FIB (sheet 2/2) (E16368C.)

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
9 - 1	E16368CA		PCB FIB (ESDS) SEE 342245-5-130 FOR NHA		RF
10	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
15	0805Y100-0103KX		.CAPACITOR FIXED 10NF PM10PCT 100V OPT SEE TABLE (A1235408)	VU1579	1
20	2220Y100-0474KX		.CAPACITOR FIXED 470NF PM10PCT 100V (A1235429)	VU1579	2
25	39986632		.SUPPORT		1
30	1812Y100-0472JC		.CAPACITOR FIXED 4.7NF PM5PCT 100V OLD PNR VJ1812A472JXB MADE BY VF7843 (A1235265)	VU1579	1
35	SMCJ15CA001		.DIODE OPT SEE TABLE (A1245866)	V75915	4
40	SMBJ33CA		.DIODE OPT SEE TABLE (A1251606)	VK4184	5
45	PD10808		.DIODE OPT SEE TABLE OLD PNR SMDA12CTH MADE BY V13409 (A1252420)	VOU8A3	2
50	SMCJ15		.DIODE OPT SEE TABLE (A1237276)	VK4184	6
55	BAT54S		.DIODE OPT SEE TABLE (A1238186)	VF8859	1
60	NFAC1CC223R1H8		.FILTER 50V 300MA (A1190103)	VS4431	2
65	NFAC1CC221S1H8		.FILTER 50V 300MA OLD PNR NFA81R00C221 MADE BY VS4431 (A1251435)	VS4431	2

- ITEM NOT ILLUSTRATED

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
9	70	86455-046	.STRIP TERMINAL OPT SEE TABLE (A1253517)	VF6131	1
	75	A1250582	.RESISTOR FIXED 1KOHM PM2PCT 1/4W ORDER OVERLGTH MPN CMA02040X1001G00 MADE BY VD9539 OLD PNR CMA0204-1001G MADE BY VK0004		26
	80	501-0TK50-1002F	.RESISTOR FIXED 10KOHM PM1PCT 1/4W OPT SEE TABLE OLD PNR A1234873 (A1234873)	VS7293	8
	85	A1249088	.RESISTOR FIXED 39 OHM PM2PCT 1/4W ORDER OVERLGTH MPN CMA02040X3909G00 MADE BY VD9539 OLD PNR CMA0204-39RG MADE BY VK0004		1
	90	SM14M12C	.DIODE OPT SEE TABLE (A1250693)	V0U8A3	2
	95	NFE61HT332Z2A9	.FILTER 100V 2A (A1207593)	VS4431	8
	100	FCMS35-1001-200	.FILTER 200V 10A OPT SEE TABLE (A1249841)	VF1379	2

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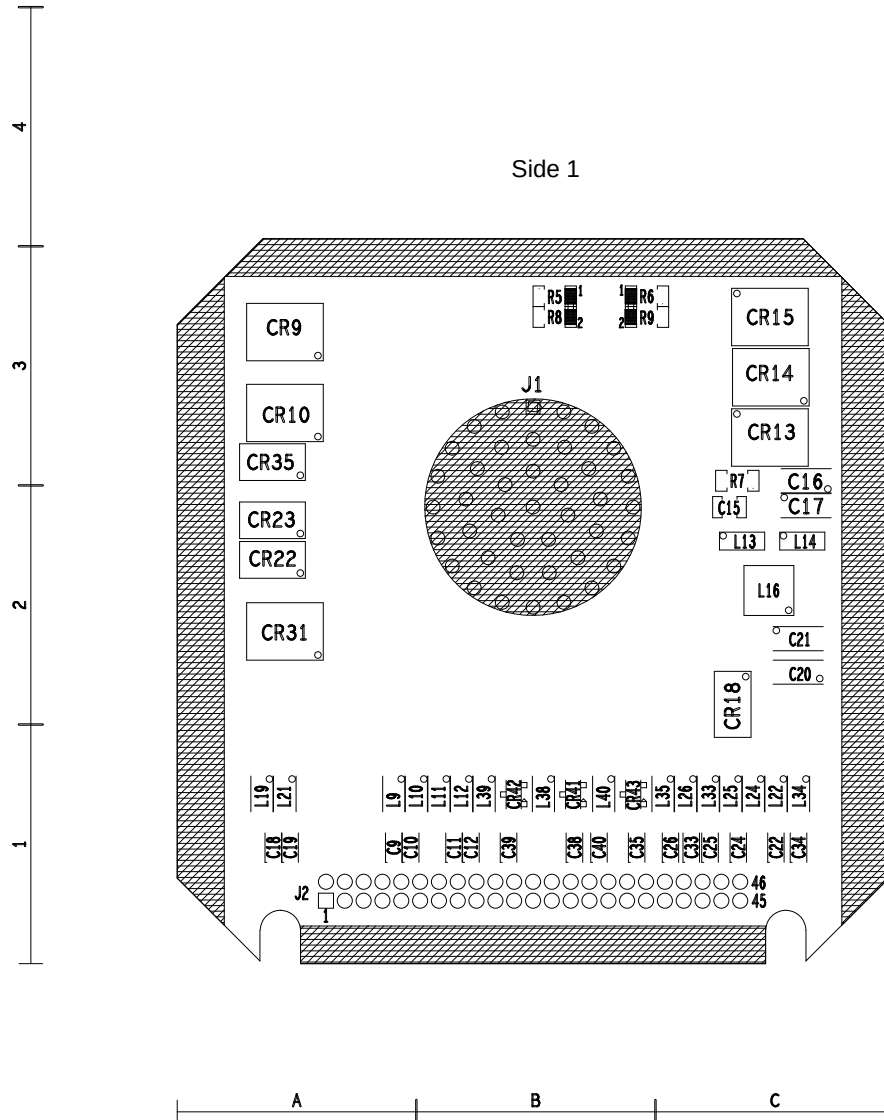
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Figure 10016/Graphic - Sheet 01 / 34-22-45-991-916-A01
(Figure 9A) PCB, FIB (sheet 1/2) (E16368F.)

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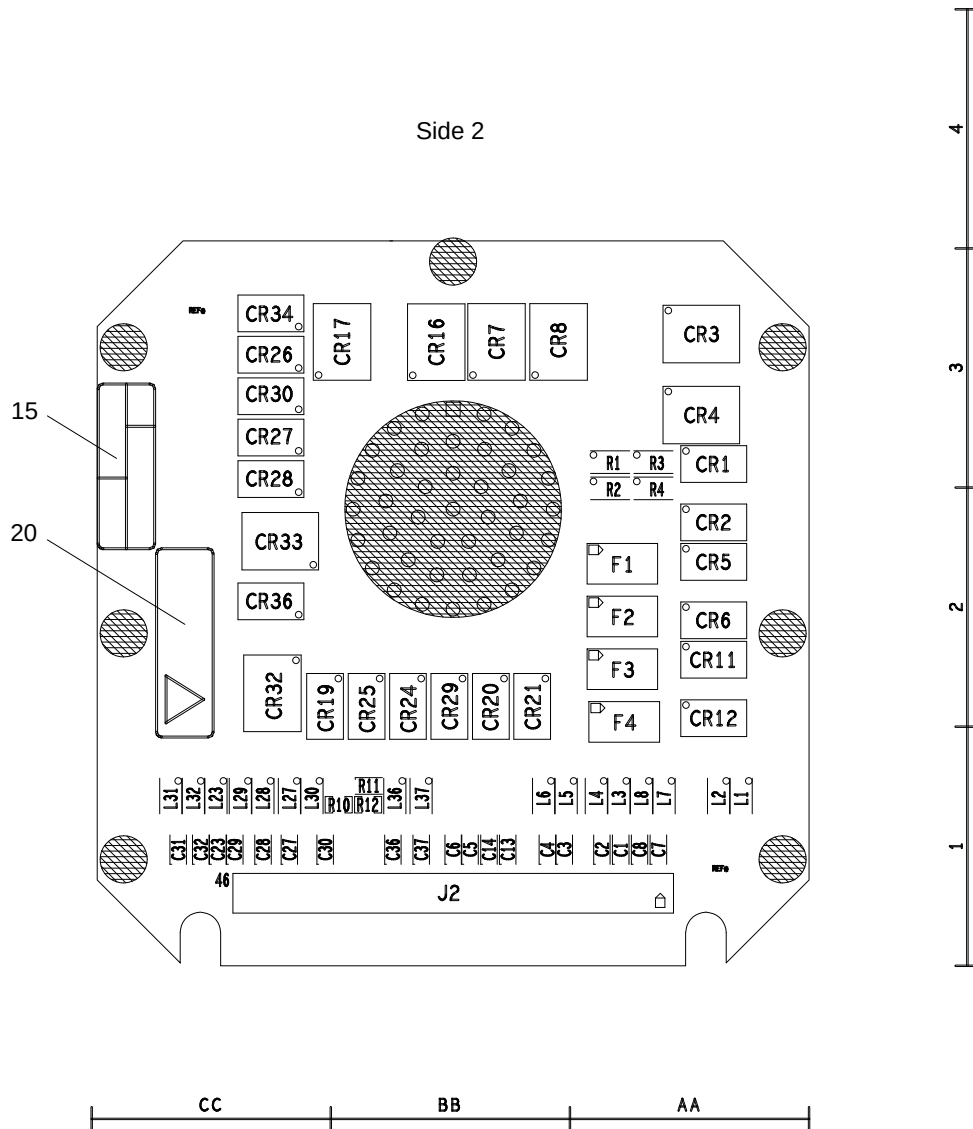
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Figure 10016/Graphic - Sheet 02 / 34-22-45-991-916-A01
(Figure 9A) PCB, FIB (sheet 2/2) (E16368F.)

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
9A - 1	E16368FA		PCB FIB (ESDS) SEE 342245-5-130A FOR NHA		RF
- 10	F2000068		.PROTECTOR LABEL		1
15	F2000071		.LABEL IDENTIFICATION		1
20	F2000190		.LABEL STATIC SENSITIVE DEVICES		1
25	1206Y200-0223KX		.CAPACITOR FIXED 22NF PM10PCT 200V OPT SEE TABLE (A1248323)	VU1579	2
30	0603Y050-0102KX		.CAPACITOR FIXED 1NF PM10PCT 50V OPT SEE TABLE (A1252502)	VU1579	8
35	VJ0603A221JXBA		.CAPACITOR FIXED 220PF PM5PCT 100V OPT SEE TABLE (A1252519)	VF7843	8
40	0603Y050-0153KX		.CAPACITOR FIXED 15NF PM10PCT 50V OPT SEE TABLE (A1253551)	VU1579	16
45	0603Y100-0222KX		.CAPACITOR FIXED 2.2NF PM10PCT 100V OPT SEE TABLE (A1257133)	VU1579	3
50	1206Y050-0224KX		.CAPACITOR FIXED 220NF PM10PCT 50V OPT SEE TABLE (A2004126)	VU1579	2
55	SMBJ15CA001		.DIODE OPT SEE TABLE (A1237367)	V75915	6
60	BAS21		.DIODE OPT SEE TABLE (A1237913)	VF8859	3
65	SMCJ43CA001		.DIODE OPT SEE TABLE (A1251599)	V75915	11
70	SMBJ33CA		.DIODE OPT SEE TABLE (A1257081)	VK4184	16

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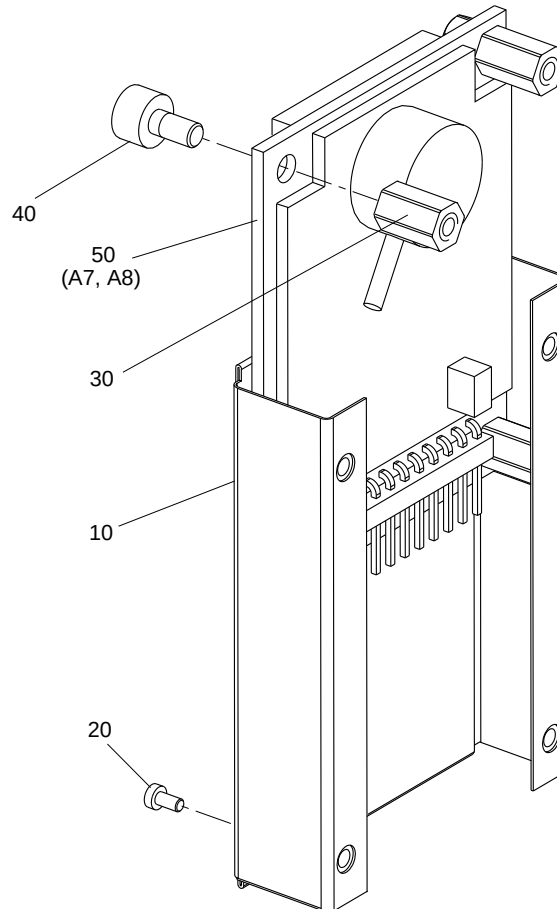
FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
9A 75	SMCJ15A		.DIODE VK4184 OPT SEE TABLE OLD PNR SMCJ15A001 MADE BY VX0102 (A2002766)		3
80	R459-002		.FUSE V75915 2A OPT SEE TABLE (A1257214)		4
85	86455-046		.STRIP VF6131 TERMINAL OPT SEE TABLE (A1253517)		1
90	251806600Y3		.COIL V34899 0 MICROHENRY OPT SEE TABLE OLD PNR BLM41PG750SN1 MADE BY VS4431 (A1252594)		2
95	BLM21BD601SN1		.COIL VS4431 0 MICROHENRY (A1254200)		33
100	DLW5BSN191SQ2		.COIL VS4431 (A2003043)		1
105	A1249089		.RESISTOR FIXED 10KOHM PM2PCT 1/4W ORDER OVERLGTH MPN CMA02040X1002G00 MADE BY VD9539 OLD PNR CMA020410KG MADE BY VK0004		4
110	WCR0603R005J		.RESISTOR VF7178 FIXED 0 OHM PM5PCT 1/16W OPT SEE TABLE OLD PNR WCR0603R005JI MADE BY VF7178 (A1251847)		1

- ITEM NOT ILLUSTRATED

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SENSOR PRESSURE EQUIPPED

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Figure 10017/Graphic 34-22-45-991-917-A01
(Figure 10) SENSOR

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
10 - 1	F1407830		SENSOR PRESSURE EQUIPPED (ESDS) POST SB C16221K-34-015 POST SB C16221L-34-015 POST SB C16221Q-34-009 SEE 342245-2-290B FOR NHA SEE FIG 2A FOR NHA SEE FIG 2C FOR NHA	A	RF
- 1A	F1407831		SENSOR PRESSURE ASSY (ESDS) POST SB C16221K-34-015 POST SB C16221L-34-015 POST SB C16221Q-34-009 SEE 342245-2-305C FOR NHA SEE FIG 2A FOR NHA SEE FIG 2C FOR NHA	B	RF
10	F1407829		.SHIELDING		1
20	57000-551-1		ATTACHING PARTS .SCREW ***		4
30	F1406953		.PILLAR		4
40	F1406974		ATTACHING PARTS .SCREW ***		4
50	E17397CA		.SENSOR PRESSURE ASSY (ESDS)		1
- 50A	F1404945		.DELETED		
- 50B	E17397DA		.SENSOR PRESSURE ASSY (ESDS)	B	1
- 60	20-873-0025		STORAGE PARTS .DELETED (A1169633) ***	VF8841	
- 70	E17397CA		..DELETED		
- 70A	E17397DA		..DELETED		
- 80	20-873-0025		STORAGE PARTS ..DELETED (A1169633) ***	VF8841	

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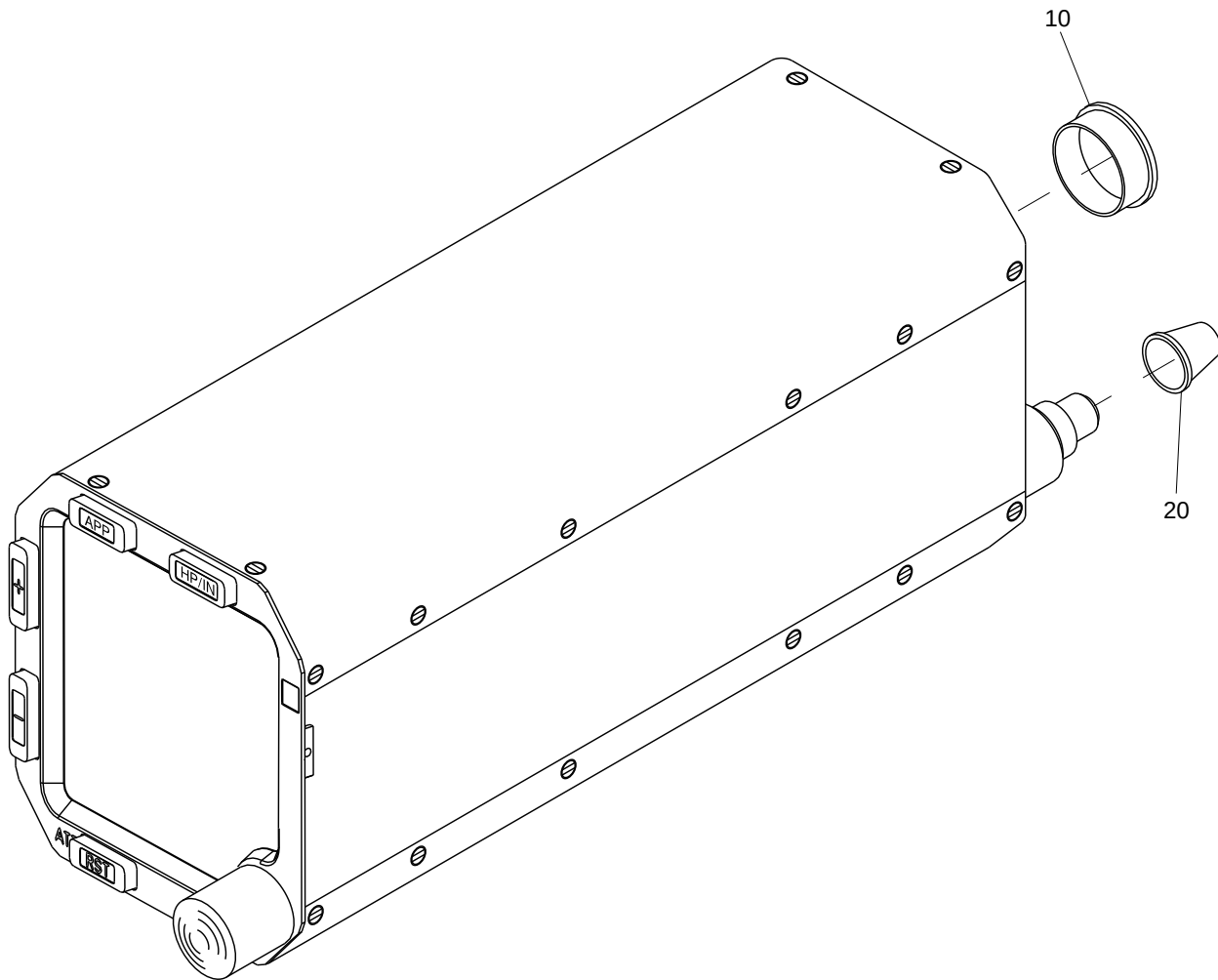
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Figure 10018/Graphic 34-22-45-991-918-A01
(Figure 99) ISFD

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FIGURE ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE	EFF. CODE	UN. PER ASS
			1234567		
99 - 1	C16221KA01CND1		DELETED (INCORRECT PART NUMBER)		
- 1A	C16221KA02CND1		DELETED (INCORRECT PART NUMBER)		
- 1B	C16221LA01CND1		DELETED (INCORRECT PART NUMBER)		
- 1C	C16221LA02CND1		DELETED (INCORRECT PART NUMBER)		
- 1D	C16221LA03CND1		DELETED (INCORRECT PART NUMBER)		
- 1E	C16221MA01CND1		DELETED (INCORRECT PART NUMBER)		
- 1F	STORAGEEC16221K		STORAGE ISFD SEE 342245-2-999 FOR NHA SEE 342245-2A-999 FOR NHA SEE 342245-2B-999 FOR NHA	NP A	RF
- 1G	STORAGEEC16221Q		STORAGE ISFD SEE 342245-2C-999 FOR NHA	NP B	RF
10	CEC20		STORAGE PARTS .CAP V99017 PROTECTIVE OLD PNR CEC-20 MADE BY V99017 (A1248865)		1
20	57990-127-1		.CAP PROTECTIVE	A	2
- 30	20-873-0025		.BAG VF8841 ANTISTATIC FOR PCB A1,A2,A3,A5,A6,A7, A8,A9,A11,A51,PS1,TB1 (A1169633) ***		12

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